

Moisture-induced stresses in cross-laminated wood panels

Doctoral Thesis

Author(s):

Gereke, Thomas

Publication date:

2009

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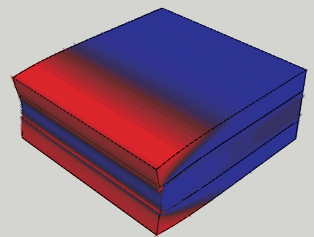
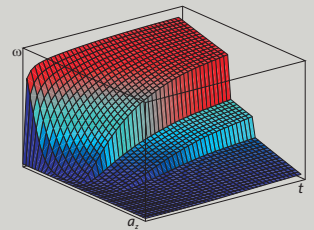
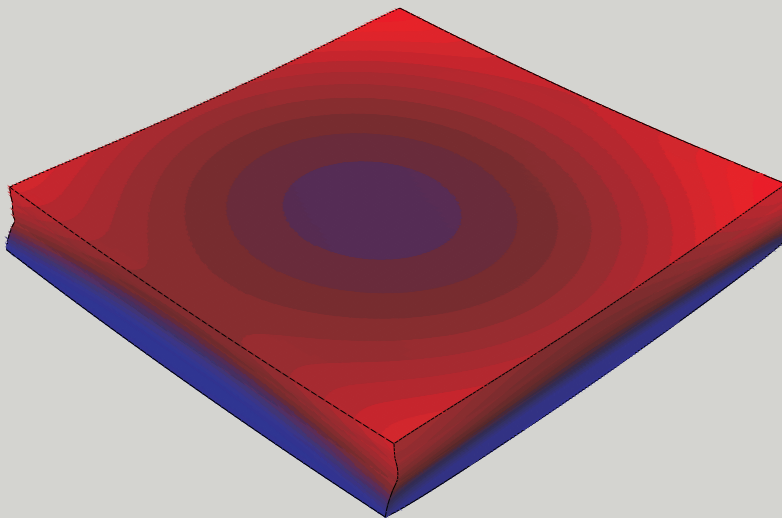
<https://doi.org/https://doi.org/10.3929/ethz-a-005898668>

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Thomas Gereke

Moisture-induced stresses in cross-laminated wood panels



DISS. ETH No. 18427

MOISTURE-INDUCED STRESSES IN CROSS-LAMINATED WOOD PANELS

A dissertation submitted to
ETH ZURICH

for the degree of
Doctor of Sciences

presented by

THOMAS GEREKE

Dipl.-Ing., University of Leipzig
born October 8, 1980 in Magdeburg (Germany)
citizen of Germany

accepted on the recommendation of

Prof. Dr.-Ing. habil. P. Niemz, examiner

Prof. Dr. J.G.M. van Mier, co-examiner

Prof. Dr. P.J. Gustafsson, co-examiner

Prof. Dr.-Ing. habil. M. Kaliske, co-examiner

2009

*"The grand aim of all science is to cover the greatest
number of empirical facts by logical deduction from
the smallest number of hypotheses or axioms." ¹*

Albert Einstein (1879 - 1955)

¹Quoted in: Barnett, L.: The meaning of Einstein's new theory. *Life*, p. 22, January 9, 1950.

Acknowledgements

The present PhD thesis "Moisture-induced stresses in cross-laminated wood panels" has been carried out from January 2006 to June 2009 at the Institute for Building Materials (Wood Physics group), ETH Zurich. The work has been financed in parts by the Swiss Forest and Wood Research Fund (Schweizerischer Fonds zur Förderung der Wald- und Holzforschung) and the European Cooperation in Science and Technology (COST, Action E49 "Processes and Performances of Wood-Based Panels"). Two Short Term Scientific Missions (STSM) at the Lund University in Sweden during 2007 and 2008 were also financed by COST. Financial support is gratefully acknowledged.

First of all I wish to thank my supervisor, Prof. Dr.-Ing. habil. Peter Niemz, for sharing his broad overview of research on wood and for his excellent guidance and support throughout this study.

Likewise I would like to thank Prof. Dr. Per Johan Gustafsson and Dr. Kent Persson for their inspiring cooperation during my two stays at the Division of Structural Mechanics, Lund University. Special thanks to Prof. Gustafsson for examining the thesis.

I am also grateful to Prof. Dr.-Ing. habil. Michael Kaliske from the Institute for Structural Analysis, Dresden University of Technology for his suggestions in the final phase of my study and also for examining the thesis.

I would like to express special thanks to Prof. Dr. Jan G. M. van Mier for being part of the examination committee.

In addition, I want to thank my colleagues at the Wood Physics Group for supporting me during my studies and for giving me the social distraction from my work. For carrying out parts of the experimental tests during his diploma thesis, for cutting a lot of samples and for proof-reading the manuscript I want to thank Philipp Hass. Thanks also to Matus

Joscak for helping with parts of the experiments. I wish to express my gratefulness to Thomas Schnider and Edi Risi for cutting a lot of wood which principally made the experimental tests possible. For his help with the manuscript and the fruitful discussions I also thank Stefan Hering.

Furthermore, I want to thank my family and friends for their support. Finally, special thanks go to Antje for her encouragements, great patience and for being the wonderful and lovely person she is.

Thomas Gereke

June 2009

Abstract

When cross-laminated wood panels are exposed to moisture variations, the crosswise bonding of the layers may cause problems. Shape distortions may reduce the serviceability. Drying stresses cause cracks on the surface, which act as a potential target for water accumulation and disrupt the natural wood surface. Moisture-induced deformations may lead to problems in areas where two panels are connected, such as gap opening or failure due to compression. It was the objective of the present study to determine and to describe moisture-induced stresses and deformations and to find possibilities to reduce them.

The moisture field throughout laminated wood panels, which is substantially characterised by the glue line diffusivity, is an essential basis for the investigation of panel response on moisture impact. Thus, the moisture behaviour of laminated spruce panels was investigated by means of sorption and cup measurements and finite element simulations of the moisture transport. The diffusion coefficient of the glue line was evaluated from the measurements. A high resistance to moisture diffusion of the glue lines was detected. From the results it was concluded that the governing process of moisture transport through wood adhesive joints is water-vapour diffusion.

In laboratory tests, the shape stability of cross-laminated spruce panels was investigated by applying a moisture difference on both large panel faces. The internal stress state was studied in experiments where panels were exposed to a single moistening step. The stresses were determined by release of strain and dynamic determination of the modulus of elasticity. Numerical simulations completed these tests. A three-dimensional material model considering elastic deformation, moisture-induced swelling and mechano-sorptive deformation was applied. The results show significant influences of annual ring angle, layer ratio, middle layer material and pre-stresses on both hygroscopic warping and internal stresses.

In parameter studies, the influences of input parameters on the results of numerical sim-

ulations were studied. The mechano-sorptive effect was found to have a strong influence on the results. Furthermore, stiffness and coefficients of hygroexpansion of the wood and the diffusivity of the glue lines affect the results of the simulations.

From the findings of this thesis it was concluded to apply vertical annual rings and to avoid horizontal annual rings in the preparation of lamellas. Pre-stresses in the outer layers may be a valid method to reduce the consequences of moisture variations but production costs and relaxation of the pre-stresses are critical factors. When applying wood composites in the middle layer, then oriented strand board is most preferable. A large moisture diffusivity of the connection system is capable to reduce warping.

Zusammenfassung

Werden Brettsper Holzplatten Feuchteveränderungen ausgesetzt, können sich Probleme aus der kreuzweisen Verklebung der Lagen ergeben. Verformungen und Plattenverzug können die Gebrauchstauglichkeit reduzieren. Trocknungsspannungen verursachen Risse an der Oberfläche, die dann als möglicher Angriffspunkt für Wasseransammlungen agieren und die natürliche Holzoberfläche zerstören. Feuchteinduzierte Verformungen können in Verbindungsstößen zweier Platten zu Problemen führen, d.h. Fugenöffnung oder Druckversagen. Das Ziel der vorliegenden Arbeit war es Eigenspannungen zu bestimmen und Möglichkeiten zu finden diese zu reduzieren.

Eine wichtige Grundlage für die Untersuchungen ist dabei eine genaue Kenntnis des Feuchtefeldes innerhalb der Platten, welches wesentlich durch die Klebfugendiffusivität bestimmt wird. Daher wurde das Feuchteverhalten in Sorptions- und Diffusionsmessungen bestimmt und der Feuchtetransport mittels Finite Element Simulationen betrachtet. Der Diffusionskoeffizient der Klebfugen wurde aus den Messungen abgeschätzt. Es konnte dabei ein grosser Diffusionswiderstand der Klebfugen festgestellt werden. Aus den Ergebnissen konnte geschlossen werden, dass der Feuchtetransport durch Klebfugen durch Wasserdampfdiffusion erfolgt.

In Laboruntersuchungen wurde die Formstabilität von Brettsper Holzplatten aus Fichtenholz im Differenzklima getestet. Durch eine Steigerung der relativen Luftfeuchte der Umgebungsluft wurden Eigenspannungen hervorgerufen. Diese Spannungen konnten dann durch Freischneiden der Dehnungen und dynamische Bestimmung des Elastizitätsmoduls bestimmt werden. Diese Messungen wurden durch numerische Simulationen komplettiert. Dazu wurde ein dreidimensionales Materialmodell angewandt, das elastische Deformationen, feuchteinduziertes Quellen und mechano-sorptive Verformungen berücksichtigt. Die Ergebnisse zeigen einen signifikanten Einfluss des Jahrringwinkels, des Lagenverhältnisses, des Mittellagenmaterials und von Vorspannungen.

In Parameterstudien wurde der Einfluss der Eingangsgrößen auf die Ergebnisse der numerischen Simulationen untersucht. Es ergab sich ein starker Einfluss des mechano-sorptiven Effekts sowohl bei der Wahl der Materialparameter als auch des Berechnungsmodells. Weiterhin haben die Steifigkeit und die Quellung von Holz und der Diffusionswiderstand der Klebfugen einen grossen Einfluss.

Aus den Erkenntnissen der Studie wurde geschlussfolgert, dass stehende Jahrringe bei der Lamellenherstellung zu bevorzugen sind. Liegende Jahrringe wirkten sich dagegen negativ auf Spannungen und Verformungen aus und sollten daher vermieden werden. Vorgespannte Decklagen haben einen positiven Effekt auf die Folgen von Feuchteänderungen, allerdings sind höhere Produktionskosten und die Relaxation der Vorspannungen kritisch zu betrachtende Faktoren. Bei der Substitution der Holzmittellagen durch Holzwerkstoffe erwies sich Oriented Strand Board (OSB) als die beste Variante. Eine hohe Feuchtedurchlässigkeit des Verbindungssystems ist eine weitere geeignete Methode um Plattenverzug zu verringern.

Notations

Roman upper case letters

A_1, A_2, A_3	shape factors of the $D_{adh} - \omega$ relationship
$\mathbf{C}$	stiffness matrix
C	moduli (generic term for moduli of elasticity and shear moduli)
C_0, C_m	shape factors of the relation between E_1, E_3, G_{13} and ω of MDF
$\mathbf{D}$	matrix of diffusion coefficients
D_a	coefficient of interdiffusion of water vapour in bulk air
D_{adh}	diffusion coefficient of adhesive
D_T	diffusion coefficient of wood above t^* in the transverse direction
D_T^*	diffusion coefficient of wood below t^* in the transverse direction
E_1, E_2, E_3	moduli of elasticity of wood-based panels in the directions 1, 2 and 3
E_L, E_R, E_T	moduli of elasticity of wood in the orthotropic directions
E_{dyn}	modulus of elasticity measured by means of dynamic methods
F	fractional mass change
G_{12}, G_{13}, G_{23}	shear moduli of wood based panels
G_{LR}, G_{LT}, G_{RT}	shear moduli in the orthotropic planes
G_ω^w	specific gravity of the cell wall at moisture content ω
$\mathbf{J}$	moisture diffusion flux vector
J	moisture flux
J_h	heat flux
$\mathbf{J}_m$	moisture flux vector at the surface
K	thermal conductivity
LR	layer ratio
NOE	total number of elements
N_x, N_y, N_z	number of elements in the appropriate panel axis
R	specific gas constant
$\mathbf{S}$	compliance matrix
$\dot{\mathbf{S}}$	compliance rate matrix
T	temperature

V_0	ovendry volume of wood
$\overline{X}$	mean value of appropriate physical value X

Roman lower case letters

a	dimension in thickness direction
a_0, a_1, a_2, a_3	shape factors of the $s_{ij} - \omega$ relationship
a_{ML}	middle layer thickness
a_{OL}	outer layer thickness
a_{tot}	total panel thickness
$a_{k,dry}$	panel dimension in the initial (dry) climate, $k = x, y, z$
$a_{k,wet}$	panel dimension in the test (wet) climate, $k = x, y, z$
b	width
c	concentration of water based on the ovendry volume of wood
c	sound velocity
c	specific heat
c_a	concentration of moisture in wood in equilibrium with ambient air
c_s	concentration of moisture in wood at the wood surface
cup_{xz}, cup_{yz}	cup deformations in the xz- and the yz-plane
e_h, e_v	distances characterising lamella position in timber log
$f_1 \dots f_{11}$	shape factors of the $c - \varphi - \rho_0$ relationship
h_c	surface-emission coefficient based on the concentration of moisture in wood as the driving potential
h_m	surface-emission coefficient based on the moisture content as the driving potential
l	length
l_0	initial length
l_1	final length
$\mathbf{m}$	mechano-sorption property matrix
m_L, m_R, m_T	mechano-sorption coefficients for normal stresses in the orthotropic directions
m_{LR}, m_{LT}, m_{RT}	mechano-sorption coefficients for shear stresses in the orthotropic planes
m_0	weight of the ovendry sample
m_∞	final weight of a sample
$m(t)$	weight at time t
n	number of specimens tested
$\mathbf{n}$	normal vector (perpendicular to surface)
p	pressure
q_k	linear swelling of the panel in k -direction, $k = x, y, z$

r_1, r_2	parameters of the relation between moduli and oven-dry density
s_{ij}	elastic coefficients
t	time
t^*	reduction time of D_T^*
$twist$	twist deformation
u_k	displacement in k -direction, $k = x, y, z$

Greek letters

α	vector of swelling coefficients
α_k	swelling coefficient of panels in k -direction, $k = x, y, z$
$\alpha_L, \alpha_R, \alpha_T$	swelling coefficients in the orthotropic directions of wood
$\alpha_1, \alpha_2, \alpha_3$	swelling coefficients of wood-based panels in the directions 1, 2, 3
$\beta_1, \beta_2, \beta_3$	shrinkage coefficients of wood-based panels in the directions 1, 2, 3
$\delta_1, \delta_2, \delta_3$	parameters of the regression functions of cup_{xz} , cup_{yz} and $twist$
ε	vector of total strains
$\dot{\varepsilon}$	vector of total strain rates
$\dot{\varepsilon}_e$	vector of elastic strain rates
$\dot{\varepsilon}_\omega$	vector of moisture strain rates
$\dot{\varepsilon}_{\omega\sigma}$	vector of mechano-sorptive strain rates
$\varepsilon_L, \varepsilon_R, \varepsilon_T$	total normal strains in the orthotropic directions of wood
$\gamma_{LR}, \gamma_{LT}, \gamma_{RT}$	shear strains
φ	relative humidity
κ	reduction factor of the diffusion coefficient of wood
$\mu_{LR}, \mu_{RL}, \mu_{LT},$ $\mu_{TL}, \mu_{RT}, \mu_{TR}$	coupling terms between stresses and mechano-sorptive strains
$\nu_{12}, \nu_{21}, \nu_{13},$ $\nu_{31}, \nu_{23}, \nu_{32}$	Poisson's ratios of wood-based panels
$\nu_{LR}, \nu_{RL}, \nu_{LT},$ $\nu_{TL}, \nu_{RT}, \nu_{TR}$	Poisson's ratios of wood
θ	annual ring angle
ρ_0	oven-dry density
σ	vector of stresses
$\sigma_L, \sigma_R, \sigma_T$	normal stresses in the orthotropic directions
$\tau_{LR}, \tau_{LT}, \tau_{RT}$	shear stresses in the orthotropic planes
ω	moisture content
ω_0	initial moisture content
ω_∞	equilibrium moisture content
ξ	aspect ratio of dynamic and static modulus
ψ_1, ψ_2, ψ_3	angles between geometric and orthotropic coordinate system

Coordinate systems

1, 2, 3	geometric coordinate sytem of wood-based panels synonyms: 1 - production direction 2 - perpendicular to production direction 3 - thickness direction
x, y, z	geometric panel coordinate system synonyms: x - major axis, fibre direction of the outer layers y - minor axis, perpendicular to the fibre direction of the outer layers z - thickness direction
x_1, x_2, x_3	geometric coordinate system of a wooden block or lamella
L, R, T	orthotropic coordinate system of wood, $L=1$, $R=2$, $T=3$

Abbreviations

1K PUR	one-component polyurethane
BL	bottom layer
CWP	cross-laminated wood panel
DIC	digital image correlation
EXT	external loaded specimens
FSP	fibre saturation point
L	longitudinal direction
MC	moisture content
MDF	medium density fibre board
MFA	micro fibril angle
ML	middle layer
MUF	melamine-urea-formealdehyde resin
OL	outer/surface layers
OSB	oriented strand board
PB	particleboard
R	radial direction
RH	relative humidity
STD	standard deviation
T	tangential direction
TL	top layer

UF urea-formealdehyde resin

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Chapter 1

Introduction

1.1 Background

Ecology of constructions is more important today than it has been in previous years. Increasing energy costs has forced building owners to increase the energy efficiency of their buildings. Wood material plays an important role in this process. Due to its hygroscopicity, wood influences the room climate in a positive manner [108]. Furthermore, it is a natural and renewable resource. Wood, as a building material, also plays an important role in regard to the climate as it stores carbon dioxide.

Cross-laminated solid wood panels have been applied in the building sector, initially as formliners for concrete constructions but today load-bearing constructions are built from large scale panels. Fig. 1.1 shows an example, where an existing building was extended by a wooden construction using cross-laminated solid wood panels. Cross-laminated panels are also applied in the furniture industry.

The panels are built up symmetrically, and consist of an uneven number of layers. The longitudinal direction of wood (grain direction) is rotated 90° from one layer to the adjacent layer (Fig. 1.2). The advantages are evident:

- improvement of the dimensional stability due to the orientation of fibres in both in-plane directions (x- and y-direction, Fig. 1.2),
- an increase in the plane isotropy and, in particular, adjustment of the low stiffness in the direction perpendicular to the grain of the raw material.



Figure 1.1 Application of cross-laminated solid wood panels, project "Supertanker" in Zurich, Switzerland, lefthand side: exterior view (www.supertanker.ch), middle and right-hand side: interior views, panels of Pius Schuler AG (Switzerland)

However, the crosswise structure yields problems when the relative humidity of the ambient air changes. The free swelling and shrinkage of adjacent layers differs by a factor of 10 (radial/longitudinal) to 20 (tangential/longitudinal). As a consequence of this difference, stresses and even cracks may develop. Cracks act as a potential target for water accumulation, which may partly destroy the wood by fungi decay. Moreover, the natural wood surface, which is regarded as a positive factor of this kind of panel, is disrupted. In full scale buildings, panel warping has been observed as a troublesome result due to different climates at the two panel faces. This negatively affects the usability.

Fig. 1.2 introduces the panel coordinate system. The x-axis, which is also termed the major direction, denotes the direction of the fibres of the outer layers. The y-axis denotes the fibre direction of the middle layer and is, thus, termed the minor direction. The z-axis points to the thickness direction perpendicular to the panel plane.

Generally, panels are industrially produced from lamellas. After drying, these lamellas

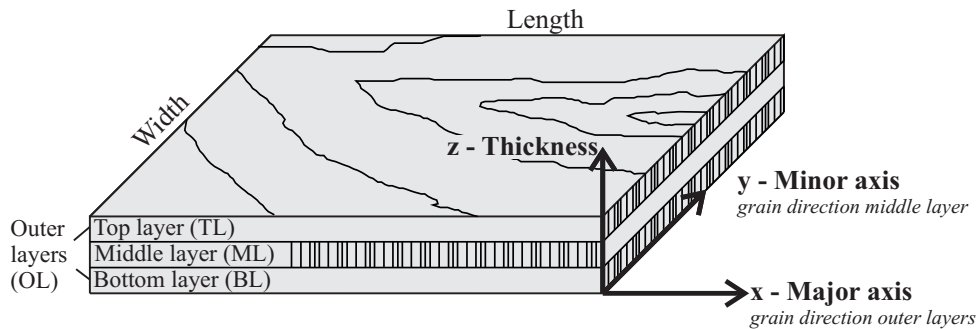


Figure 1.2 Notations applied in this thesis

are planed to the final dimension. Structural errors, such as knots and cracks, may now be detected at the surface and removed from the raw material depending on the quality requirements. The lamellas are then glued either by finger joints or butt joints. Afterwards, the lamellas are assembled to endless wood carpets by gluing at the small edges. These carpets are sawn to the final dimension and glued together to panels. Hydraulic or vacuum presses are used for this process. The middle layer may have a special characteristic. The lamellas of the middle layer may not be glued at the edge-to-edge surfaces or may be sawn from wooden blocks that were initially glued together from lamellas. The applied adhesive may be one-component polyurethane (1K PUR), urea formaldehyde resin (UF) or melamine-urea-formaldehyde resin (MUF).

1.2 Objective

The purpose of this study was to determine and to describe moisture-induced stresses and deformations in cross-laminated wood panels, with the final aim to reduce appearing stresses. The basic goal was to find methods that are suitable to measure moisture-induced stresses and the hygroscopic warping of cross-laminated panels. Furthermore, the current method used to measure swelling properties in the three panel axes should be improved, since the swelling differs in different layers. The simulation model for the calculation of stresses and deformations in wood, which was tested earlier by other researchers, should be validated with cross-laminated wood panels. Both experimental tests and finite element calculations are the basics for the description of moisture-induced stresses in wooden panels.

1.3 Outline of the thesis

To clarify how wooden panels react to various environmental humidity conditions, knowledge about the moisture distribution within the panel is required. For this purpose, investigations on the moisture profile were performed. The complex behaviour of moisture diffusion in glued wooden samples made computer simulation necessary. Finite element simulations were performed to investigate the moisture profile, which affects the shape stability of wooden panels. The time variation of the moisture distribution is important and has been investigated as well.

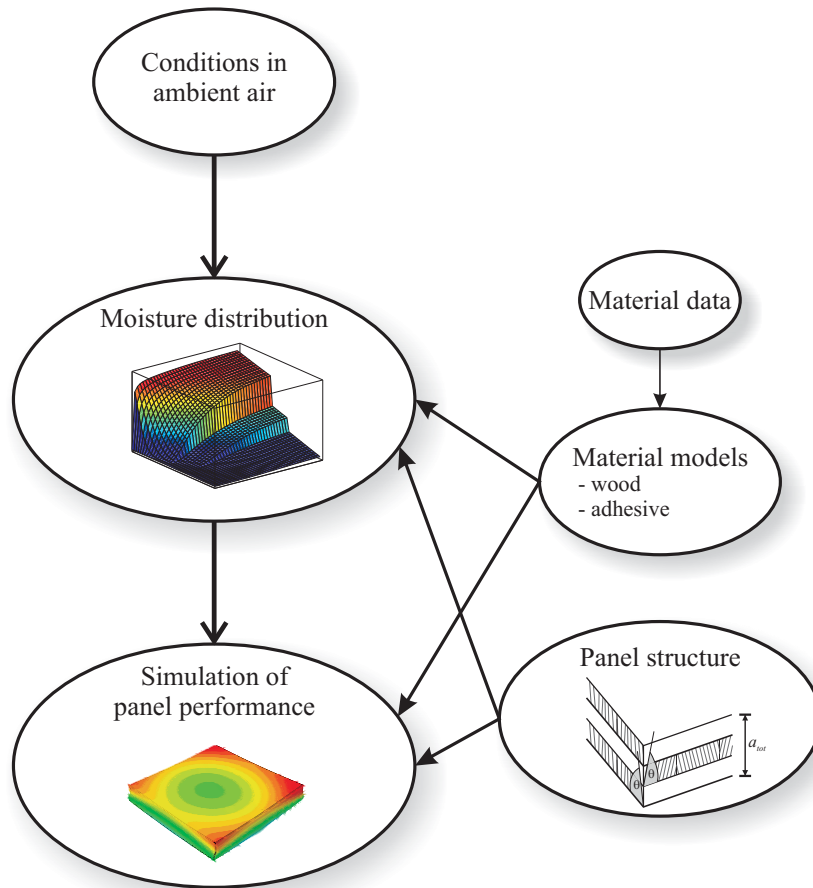


Figure 1.3 The principal of the simulation of the mechanical response of wood panels

An essential quality requirement of cross-laminated wood panels is the shape stability under changing environmental conditions, which was investigated in laboratory tests. Again, finite element simulations were necessary. The simulation of the mechanical response on moisture variations needs to be based on three input fields:

- the moisture distribution in the panel based on the time and the conditions in ambient air,
- the material data of the applied materials (wood, adhesive, wood-based materials) and
- the panel specifications.

Fig. 1.3 illustrates the principal of the simulation of moisture-induced stresses and deformations in wood panels. The mechanical modelling is based on the elastic, moisture-induced and mechano-sorptive components of the strain tensor. Phenomena such as creep or plastification are not regarded.

To make a statement on moisture-induced stresses, the measurement of these stresses and simulations are necessary. The internal stress state was determined by release of strain and dynamic determination of the Young's modulus.

The outline in detail is as follows:

In Chapter 2 the state of research is provided. Firstly, the relationship between the structure of wood and its properties is drawn. The hygroscopicity and orthotropy of wood are derived from the anatomical structure. The rheological behaviour is indicated. Then, a literature overview over measuring moisture-induced stresses in wood is given and investigations on cross-laminated solid wood panels from other researchers are provided. The material modelling is introduced and, finally, properties of wood-based composites are presented.

In Chapter 3, results from measurements on sorption and moisture profiles are shown. The material model for diffusion processes in glued wood samples is presented. Additionally, results of modelling diffusion in spruce-MDF laminates are given.

Chapter 4 deals with the hygroscopic warping of cross-laminated wood panels. Experimental tests are completed by finite element simulations.

Chapter 5 provides investigations on the internal stress state of cross-laminated wood panels.

In Chapter 6, various numerical studies of the panel performance are presented.

Chapter 7 summarises the thesis and provides ideas on future research.

1.4 Restrictions of the thesis

- Since a 1% change in the moisture content has a ten times greater effect on wood dimensions than a 1K change in the temperature [75], the investigations were confined to the influence of moisture on stresses and deformations. The temperature was kept constant at 20°C during most experiments (and is stated if otherwise). For this reason, the influence of the temperature on the moisture content and, consequentially, the influence of temperature on the material parameters of wood is neglected in this thesis.

- Unless otherwise stated, one-component polyurethane (Purbond[®] HB110) was used for assembling panels and other specimens. The pressing set-up was as follows:
 - Adhesive application: 200 g m^{-2} , one-sided,
 - Forming pressure: $p = 0.8 \text{ N mm}^{-2}$,
 - Pressing time: $t = 3\text{h}$.
- Only moisture content below the fibre saturation point (no free water) is regarded.
- All specimens were made of Norway spruce (*Picea abies* [L.] Karst.).
- The terms "modulus of elasticity" and "Young's modulus" describe the same material parameter. The use of this termination is not accurate if the direction differs from one of the main axes L, R and T, but is nevertheless applied.
- The terms "moisture-induced stresses" and "eigenstresses" cover the same topic, namely stresses induced by variations in moisture. In the present thesis the term "internal stresses" is used equally to describe these stresses. However, stresses may also be induced by external loading.

Chapter 2

State of research

2.1 The wood structure

A tree cross section consists of different parts. The stem carries food to the living parts of the tree (the inner living part). It is covered with a layer of bark (the outer dead part) that protects the stem from outer influences. In the inner part of the bark, the cambium produces wood cells in its inner part and bark cells in its outer part. It can be seen only under the microscope. Depending on the species the wood may be differentiated into sapwood, heartwood and pith (from the outside to the centre of the stem). Heartwood is often darker in colour and consists of dead cells that are inactive.

In temperate climates, the wood that is produced early in the season differs from that produced late in the season. Earlywood fibres being responsible for transport and latewood fibres mechanically support the stem. Due to thicker cell walls and smaller lumens, latewood cells have a higher density which leads to a darker colour and makes the annual (growth) ring visible to the naked eye (Fig. 2.1). Differences in the radial width of growth rings are due to environmental conditions during the growth period, such as soil and climate.

The material in the stem consists of wood cells. In softwoods, such as spruce or pine, 90-95% of the volume is composed of wood cells termed tracheids, fibres or grain. The main functions of the tracheids are mechanical support and fluid transport. They are longitudinal aligned in the direction of the stem. Another type of cells, termed parenchyma, is aligned in the radial direction, which is the direction from the pith, at the centre of the



Figure 2.1 Lefthand side: *Pinus radiata*, Walker [121, chapter 1], righthand side: Norway spruce (*Picea abies*) clearly showing the thicker-walled latewood tracheids and the thinner-walled earlywood tracheids

stem, to the bark. They are the basic elements of the wood rays. Their function in the stem is that of nutrient storage. It is assumed that rays reinforce the stem in the radial direction, which has been shown for hardwoods by Burgert and Eckstein [11].

A tracheid consists of a cell wall enclosing a lumen. The cell wall is the structural part of the cell, the cell lumen being the cavity of the cell where fluids are transported. The cell wall is composed of the primary wall (P) and the compound secondary wall (S). The secondary wall consists of three layers: the outer layer S_1 , the middle layer S_2 and the inner layer S_3 (Fig. 2.2). These layers and the primary wall differ in the orientation of their microfibrils, which are threads-like units of cellulose. In the thin P-wall ($0.1 \mu\text{m}$) the microfibrils are randomly oriented. In the S-wall the microfibrils run round the cell in a helical pattern at a certain angle to the longitudinal cell axis, the microfibril angle (MFA). The outermost S-layer, S_1 , is 0.1 to $0.3 \mu\text{m}$ thick and its MFA ranges from 50° to 70° depending on the wood species [24, 59]. The thickness of the dominant S_2 -layer differs between earlywood (1 - $4 \mu\text{m}$) and latewood (3 - $8 \mu\text{m}$). The S_2 making up about 70 - 80% of the thickness of the cell wall. The MFA can vary from 10° to 30° depending on the species, the age and the source of the cell, earlywood or latewood [24, 59]. The structure of the S_2 has a strong influence on such cell properties as stiffness and shrinkage [81] and even on the mechanical properties of wood [e.g., 12, 14, 53, 67, 86]. The $0.1 \mu\text{m}$ thick S_3 as the inner part of the S-wall has a MFA of 60° to 90° [122]. The cells are interconnected by the middle lamella which is free of cellulose fibrils. Wood cells are further discussed in

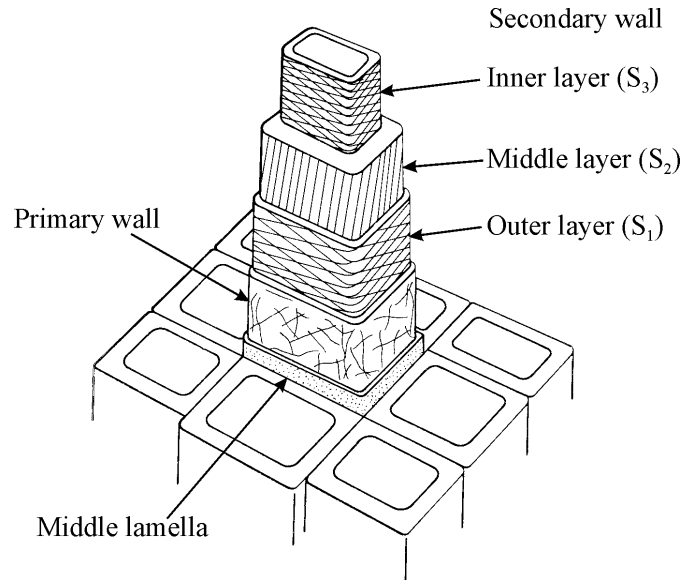


Figure 2.2 The cell wall structure, Dinwoodie [24]

Bodig and Jayne [8], Dinwoodie [25] and Kollmann and Côté [59].

The growth rings create a cylindrical structure. This causes a strong direction dependency of the material parameters such as stiffness, strength and shrinkage. The longitudinal to the stem axis arrangement of the fibres leads to, for example a large stiffness in the fibre direction and a considerably low stiffness perpendicular to it. Properties even vary significantly between the transverse directions. Typically, in the radial direction (from pith to bark) the stiffness is twice that of the tangential to the growth rings direction. Furthermore, the moisture-induced swelling and shrinkage is considerably smaller in the longitudinal direction than in the directions radial and tangential to the growth rings. Typical softwoods swell and shrink twice as much tangentially than radially. This plays a significant role, e.g., in the drying of sawn timber. Due to the cylindrical structure and the radial/tangential differences, the deformation of boards after drying depends on the position from where in the stem it was sawn [79, Fig. 2.3].

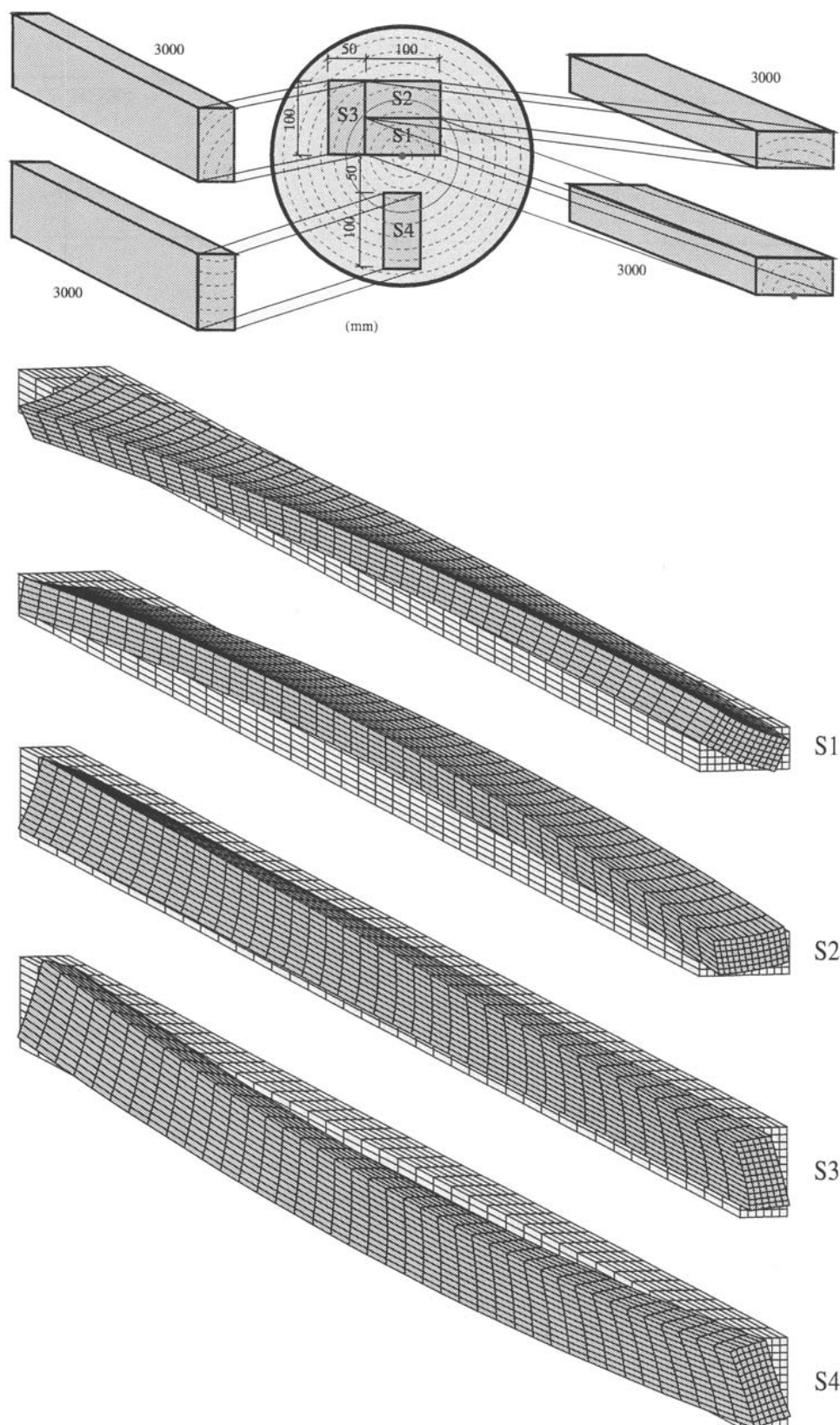


Figure 2.3 Simulated drying deformation of boards positioned in different locations in the stem, Ormarsson [79]

2.2 Wood as a hygroscopic material

2.2.1 General introduction

Wood is a hygroscopic material. It takes up water either as water vapour by sorption from the surrounding air or by capillary forces from liquid water. A wide range of investigations has been performed in the field of moisture transport. General works dealing with transport phenomena in wood are authored by Kollmann [58], Siau [93; 94], Skaar [99] and Stamm [103; 104].

Moisture in wood exists as bound water in the hygroscopic cell walls or as free water (liquid or water vapour) in the cell lumen. The wood moisture content ω [%] is defined as the mass of moisture in wood expressed as a fraction of the oven-dry mass m_0 [kg]:

$$\omega(t) = \frac{m(t) - m_0}{m_0} \times 100 \quad (2.1)$$

The mass of moisture is the mass of the wet sample $m(t)$ [kg] at time t [s] minus the oven-dry mass of the sample. The relation between concentration c [kg m⁻³] and moisture content is

$$c(t) = \frac{m(t) - m_0}{V_0} = \omega(t) \rho_0 . \quad (2.2)$$

The concentration of water is stated with the dry volume of wood V_0 [m³] as the reference. Thus, effects of volume changes due to swelling are omitted from the mass balance. Moisture content and concentration are in equilibrium with the relative humidity φ [%] of the ambient air. This equilibrium moisture content increases with increasing relative humidity (RH) from basically 0% in dry air to the fibre saturation point of 28-32% at about 98% RH (saturated cell walls) and theoretically full saturation of the lumen at 100% RH. Sorption isotherms displaying the $\omega - \varphi$ relation for constant temperatures. The effect of the equilibrium moisture content of wood being higher in desorption than in absorption is termed sorption hysteresis. Thus, the moisture content at a given relative humidity depends on the moisture content history.

2.2.2 Fick's laws of diffusion

Moisture transport in wood below the fibre saturation point is a process governed by diffusion: bound-water diffusion in cell walls and water-vapour diffusion in the lumen [57, 94].

According to Fick's first law of diffusion, the flux vector $\mathbf{J}$ [kg m⁻² s⁻¹] is proportional to the gradient in concentration towards the direction of diffusion $\partial c / \partial x$ [kg m⁻⁴]. The constant of proportionality is the diffusion coefficient D [m² s⁻¹]. Thus, in a three dimensional description of diffusion under steady-state conditions (i.e. constant concentration, $\partial c / \partial t = 0$) the flux is

$$\mathbf{J} = -\mathbf{D} \nabla c . \quad (2.3)$$

The matrix $\mathbf{D}$ contains the temperature- and moisture-dependent diffusion coefficients for the different material directions in the diagonal. ∇ is the spatial gradient operator. In a one-dimensional case with x as the diffusion direction, Eq. 2.3 simplifies to

$$J = -D \frac{\partial c}{\partial x} . \quad (2.4)$$

There are several alternative potentials apart from the concentration, e.g., vapour pressure p , chemical or water potential Ψ [94]. From the continuity equation (mass conservation)

$$\nabla \cdot (\mathbf{J}) = -\frac{\partial c}{\partial t} \quad (2.5)$$

and Eq. 2.3 the diffusion for unsteady-state conditions

$$\frac{\partial c}{\partial t} = \nabla \cdot (\mathbf{D} \nabla c) \quad (2.6)$$

can be found. In a one dimensional case it again simplifies to

$$\frac{\partial c}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial c}{\partial x} \right) \quad (2.7)$$

and for a constant diffusion coefficient to

$$\frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial x^2} . \quad (2.8)$$

2.2.3 Analogy between moisture and heat diffusion

The phenomena of moisture diffusion and heat flow are commonly regarded as the same [16]. Corresponding to Eqs. 2.4 and 2.8, the equations of heat flow are

$$J_h = -K \frac{\partial T}{\partial x} , \quad (2.9)$$

$$\frac{\partial T}{\partial t} = \left(\frac{K}{c\rho} \right) \frac{\partial^2 T}{\partial x^2} , \quad (2.10)$$

where T [K] is the temperature, K [$\text{J m}^{-1} \text{K}^{-1} \text{s}^{-1}$] is the thermal conductivity, c [$\text{J kg}^{-1} \text{K}^{-1}$] is the specific heat and ρ [kg m^{-3}] is the density. In Eq. 2.9, J_h [$\text{J m}^{-2} \text{s}^{-1}$] is the amount of heat flowing in the direction x . Identifying concentration c with temperature T and taking $D = K$ in Eqs. 2.4 and 2.9 and $D = K/(c\rho)$ in Eqs. 2.8 and 2.10 yield

$$c\rho = 1 . \quad (2.11)$$

This analogy is needed in order to simulate moisture flux in common finite element software, since only heat flow is included in the material modelling of such programs.

2.2.4 Measurements of diffusion coefficients

Basically, there are two methods for measuring diffusion coefficients in wood: the steady-state cup method and the transient (unsteady-state) sorption method.

In steady-state measurements a sample of wood is fastened on a cup and a moisture gradient is induced with silica gel, saturated salt solutions or water in the cup [94]. If the relative humidity is higher inside the cup than outside, then moisture will flow through the sample and out of the cup and vice versa. The diffusion coefficient may then be calculated from the function of weight change of the cup versus time.

In transient measurements the sample is initially conditioned in a relative humidity φ_1 until equilibrium and then the relative humidity is rapidly changed to a value φ_2 . The wood sample will absorb or desorb water depending on whether φ_2 is higher or lower than φ_1 . The fractional mass change F [-] of the sample is defined as

$$F(t) = \frac{m(t) - m_0}{m_\infty - m_0} , \quad (2.12)$$

whereas m_∞ [kg] is the final weight of the sample. F may then be plotted as a function of the square root of time, $t^{1/2}$. From the initial slope of such a sorption curve it is possible to calculate the diffusion coefficient [118].

2.2.5 Non-Fickian behaviour

When Eqs. 2.3 - 2.8 hold true for a transport process, it is called "Fickian". When surface resistance is neglected, sorption measurements have the following characteristics [118]:

- the initial slope of the $F - t^{1/2}$ - curve is linear,
- curves for samples of different thicknesses are superimposed when plotted as a function of time divided by the square root of the thickness of the sample,
- diffusivity calculated from cup and sorption measurements should be the same.

Otherwise it is referred to as "non-Fickian". This anomalous behaviour has been reported by Crank and Park [17] for polymers and by Wadsö [118] for wood.

Fig. 2.4 shows the differences between the two types of diffusion for the sorption in wood samples. It has been found by Wadsö [118; 117; 119; 120] that diffusion in particular at high relative humidity ranges is significantly non-Fickian. It is thought that cup measurements are not greatly influenced by non-Fickian behaviour, since it is an unsteady-state phenomenon related to sorption.

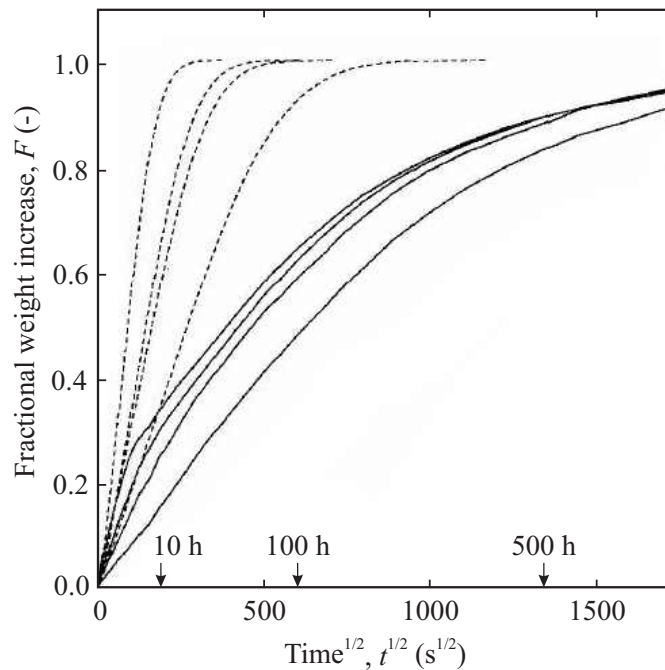


Figure 2.4 Sorption measurements from $\varphi_1 = 75\%$ to $\varphi_2 = 84\%$ on pine sapwood specimens (*Pinus sylvestris*) compared with simulations by means of Fick's law (dashed lines), Wadsö [119]

2.2.6 Boundary conditions

If airflow at the wood surface is slow, then one has to consider the surface resistance to diffusion. The following boundary condition can be stated:

$$\mathbf{n}\mathbf{J}_m = h_c (c_a - c_s) . \quad (2.13)$$

The surface-emission coefficient h_c [$\text{m}_w \text{s}^{-1}$] is expressed on the basis of concentration of moisture in wood, c . c_a and c_s are the concentrations in equilibrium with ambient air and at the wood surface, respectively. The mass flow $\mathbf{J}_m$ [$\text{kg m}^{-2} \text{s}^{-1}$] with the normal vector $\mathbf{n}$ is driven by the difference in both concentrations. The boundary is schematically drawn in Fig. 2.5.

At higher air velocities the resistance is negligible [87] and the concentration of moisture in wood at the surface can be assumed as identical to the concentration corresponding to the ambient air:

$$c_a = c_s . \quad (2.14)$$

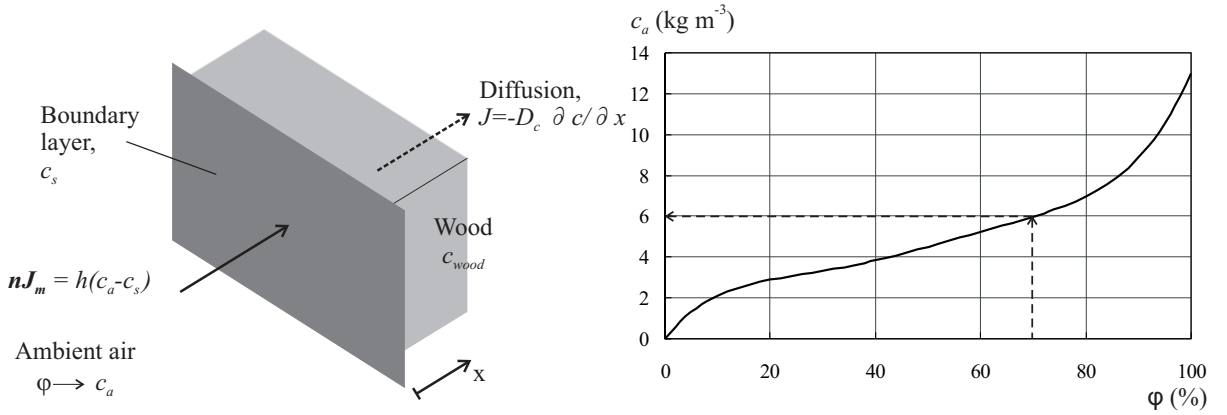


Figure 2.5 Principal of the boundary layer

2.3 Orthotropic elasticity

As described in Section 2.1, the micro-structure and the cylindrical shape of wood imply anisotropic mechanical properties. These, referred to the axes of symmetry and described with cylindrical coordinates (Fig. 2.6a), are commonly simplified as orthotropic. In Fig. 2.6b a wooden block is illustrated, where the geometric and the orthotropic axes are

misaligned. Two angles are required to relate the two coordinate systems. They are commonly known as grain angle ψ_1 and ring angle ψ_3 [8]². The grain angle is measured between the x_1 - and the L-axis. The ring angle is included between the tangential axis and the x_3 -axis of the block.

The interrelationship between load and deformation is described by Hooke's law of elasticity. The strain tensor ε_{ij} is linked to the stress tensor σ_{kl} by introducing a tensor of proportionality.

$$\varepsilon_{ij} = S_{ijkl} \sigma_{kl} \quad i, j, k, l = 1, 2, 3 \quad (2.15)$$

With the relation $\underline{\underline{C}} = \underline{\underline{S}}^{-1}$ between stiffness and compliance, Eq. 2.15 merges into

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl} \quad i, j, k, l = 1, 2, 3. \quad (2.16)$$

The fourth order compliance tensor $\underline{\underline{S}}$ contains 81 elastic coefficients. Due to the symmetry of the stress and the strain tensor only 36 elastic coefficients are independent. Thus, Eq. 2.15 may be written in Voigt notation as:

$$\begin{pmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \gamma_{12} \\ \gamma_{13} \\ \gamma_{23} \end{pmatrix} = \begin{bmatrix} s_{11} & s_{12} & s_{13} & s_{14} & s_{15} & s_{16} \\ s_{21} & s_{22} & s_{23} & s_{24} & s_{25} & s_{26} \\ s_{31} & s_{32} & s_{33} & s_{34} & s_{35} & s_{36} \\ s_{41} & s_{42} & s_{43} & s_{44} & s_{45} & s_{46} \\ s_{51} & s_{52} & s_{53} & s_{54} & s_{55} & s_{56} \\ s_{61} & s_{62} & s_{63} & s_{64} & s_{65} & s_{66} \end{bmatrix} \begin{pmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \tau_{12} \\ \tau_{13} \\ \tau_{23} \end{pmatrix}. \quad (2.17)$$

The compliance matrix is symmetric, i.e. $s_{ij} = s_{ji}$. Only 21 independent elastic coefficients remain available. In an orthotropic material, such as wood, normal stresses in the direction of the main axes do not evoke shear strains, while shear stresses do not evoke normal strains. It is common practice to refer the three main axes L , R and T (Fig. 2.6) to the indices 1, 2 and 3 [8]. Eq. 2.17 simplifies to

$$\begin{pmatrix} \varepsilon_L \\ \varepsilon_R \\ \varepsilon_T \\ \gamma_{LR} \\ \gamma_{LT} \\ \gamma_{RT} \end{pmatrix} = \begin{bmatrix} s_{11} & s_{12} & s_{13} & 0 & 0 & 0 \\ s_{21} & s_{22} & s_{23} & 0 & 0 & 0 \\ s_{31} & s_{32} & s_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & s_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & s_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & s_{66} \end{bmatrix} \begin{pmatrix} \sigma_L \\ \sigma_R \\ \sigma_T \\ \tau_{LR} \\ \tau_{LT} \\ \tau_{RT} \end{pmatrix}. \quad (2.18)$$

The material behaviour may then be described by 12 (9) independent parameters. The diagonal coefficients s_{ij} ($i = j = 1 \dots 3$) relate the elastic strains to the stresses and are

²Bodig and Jayne [8] define the three angles as θ (grain angle), ϕ (ring angle) and ψ .

reciprocals of the moduli of elasticity or Young's moduli E :

$$s_{11} = E_L^{-1} \quad s_{22} = E_R^{-1} \quad s_{33} = E_T^{-1} . \quad (2.19)$$

The elastic coefficients s_{ij} ($i = j = 4 \dots 6$) relate the shear strains γ to the shear stresses τ by the shear moduli G as

$$s_{44} = G_{LR}^{-1} \quad s_{55} = G_{LT}^{-1} \quad s_{66} = G_{RT}^{-1} . \quad (2.20)$$

The Poisson's ratio ν_{ij} defines the ratio of the transverse strain ε_i (normal to the applied load) to the axial strain ε_j (in the direction of the applied load) in uniaxial tension or compression:

$$\nu_{ij} = \frac{\varepsilon_i}{\varepsilon_j} . \quad (2.21)$$

They define the non-diagonal coefficients s_{ij} ($i \neq j$) as follows:

$$\begin{aligned} s_{12} &= -\nu_{LR} \cdot s_{22} & s_{13} &= -\nu_{LT} \cdot s_{33} & s_{23} &= -\nu_{RT} \cdot s_{33} \\ s_{21} &= -\nu_{RL} \cdot s_{11} & s_{31} &= -\nu_{TL} \cdot s_{11} & s_{32} &= -\nu_{TR} \cdot s_{22} \end{aligned} \quad (2.22)$$

Unlike the elastic coefficients s_{ij} the indices of the Poisson's ratios ν_{ij} are not interconvertible ($s_{ij} = s_{ji}$ but $\nu_{ij} \neq \nu_{ji}$). Consequently, the compliance matrix is defined as

$$\mathbf{S} = \begin{bmatrix} E_L^{-1} & -\nu_{LR}E_R^{-1} & -\nu_{LT}E_T^{-1} & 0 & 0 & 0 \\ -\nu_{RL}E_L^{-1} & E_R^{-1} & -\nu_{RT}E_T^{-1} & 0 & 0 & 0 \\ -\nu_{TL}E_L^{-1} & -\nu_{TR}E_R^{-1} & E_T^{-1} & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{LR}^{-1} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{LT}^{-1} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{RT}^{-1} \end{bmatrix} . \quad (2.23)$$

The compliance matrix (Eq. 2.23) is only valid as long as stresses and strains are related to the main axes L , R and T . If stresses and strains deviate from these axes, they have to be transformed. A detailed description of the transformation procedure can be found in Grimsel [37]. Spatial illustrations, termed "deformation bodies", are shown for different wood species in Grimsel [37] and Keunecke et al. [54]. Polar diagrams of the LR, LT and RT planes, i.e. 2-dimensional sections through these bodies, have already been published in earlier works [44, 56]. Applying the elastic coefficients s_{ij} measured by Neuhaus [74] on Norway spruce (see Appendix) to Grimsel's transformation rules for tension gives the

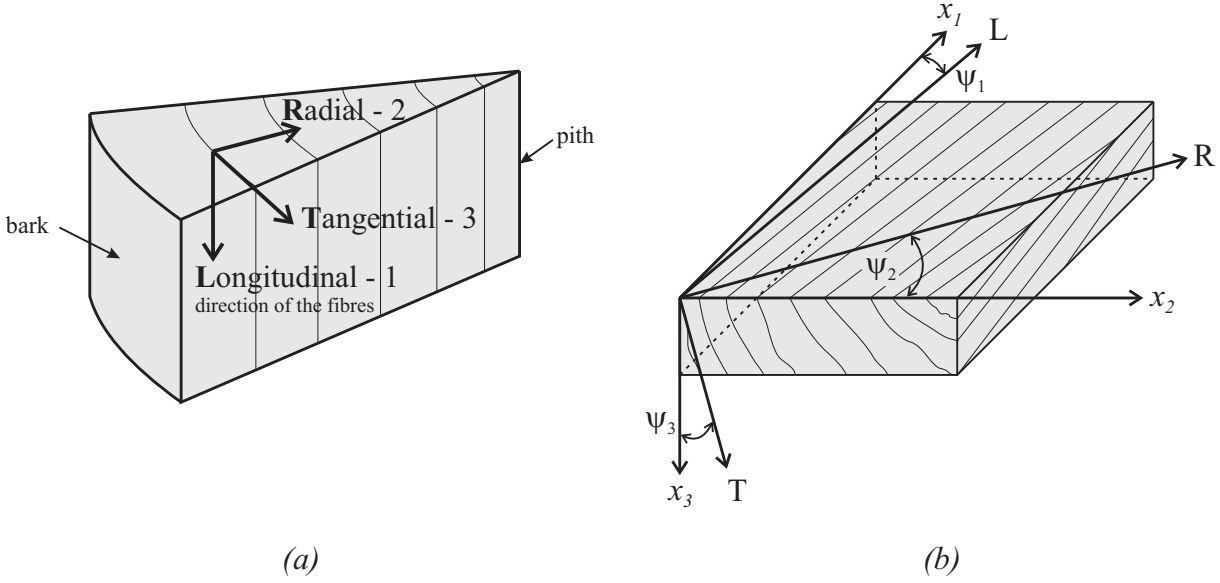


Figure 2.6 (a) The orthotropic wood directions, (b) Geometric (x_1, x_2, x_3) and orthotropic (L, R, T) axes of a wooden block (according to Bodig and Jayne [8])

load-directional dependence as illustrated in Fig. 2.7. The anisotropy is clearly seen. The compliance is twice as large in the tangential direction than in the radial direction. The largest compliance, i.e. the lowest stiffness, in tension under an angle of 45° in the RT plane is a result of the considerably small shear modulus of Norway spruce (and softwoods in general), G_{RT} , which is due to the large variations in density between early- and latewood. The compliance and, thus, the stiffness clearly depend on the moisture content in the wood. With increasing moisture content the stiffness decreases.

2.4 The rheological behaviour of wood

Creep is the time-dependent strain that develops when wood for a long or short period of time is exposed to constant stress under constant environmental conditions. It has been studied by many researchers, see e.g., Hanhijärvi [42], Hartnack [43], Morlier [73] and Toratti [113]. The loading with a constant stress at time t_0 causes an instantaneous elastic deformation (Fig. 2.8a). A creep deformation develops when the constant stress is retained. If the load is removed at time t_1 , the elastic part of the deformation will disappear. A certain amount of the creep strain will recover. However, this recovery strain is only a part of the total creep strain since total recovery of the creep strain doesn't occur. The two creep components are often called elastic and plastic creep.

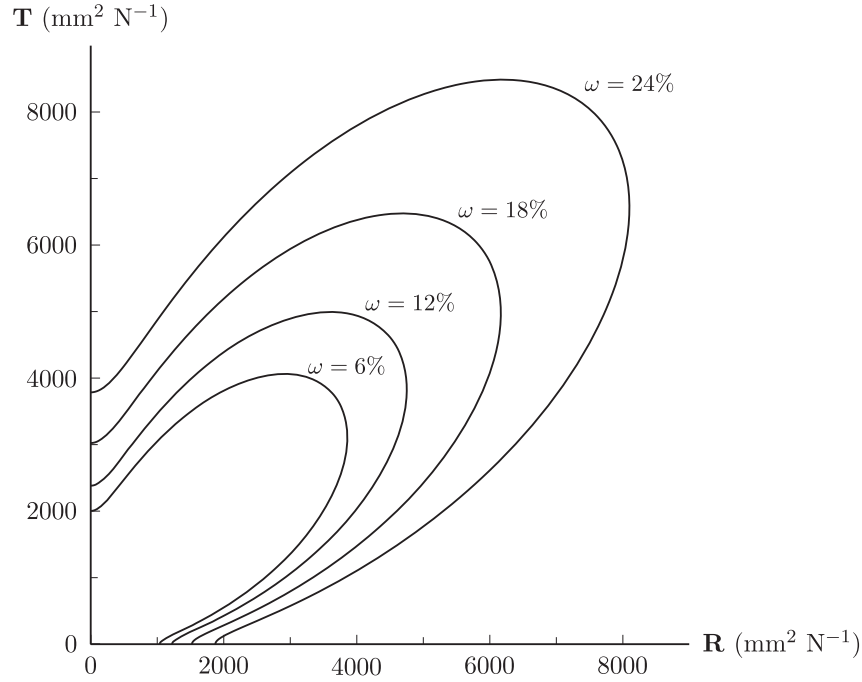


Figure 2.7 Load-directional dependence of spruce compliance in the RT plane at different moisture contents, elastic coefficients s_{ij} taken from Neuhaus [74]

If a constant stress is applied for a long time, the creep deformation may be divided into three parts (Fig. 2.8b). Firstly, creep increases appreciably during the first hours after loading. This is referred to as primary creep. Thereafter, creep deformation is characterised by a decreasing deformation velocity. A long period of slowly increasing strain follows (secondary creep). Just prior to failure, the deformation increases rapidly (tertiary creep). A loaded sample may fail due to tertiary creep, depending on the amount of the applied load.

The effect that the deformation of a loaded drying specimen is significantly greater than the sum of the deformation of a loaded specimen under constant environmental conditions and the deformation of a non-loaded drying specimen is called mechano-sorptive effect (Fig. 2.9). It has been observed experimentally, e.g., by Mårtensson [71], Ranta-Maunus [85] and Svensson [105]. Mechano-sorption has been observed for all types of loading (tension, compression, bending, shear, torsion) and loading direction (parallel and perpendicular to the grain). However, the amount of mechano-sorptive strain differs. Ranta-Maunus [84] found no mechano-sorption below 8-10% wood moisture content. This is in agreement to observations of Liu [68] and Toratti [113], who detected a stronger mechano-sorptive effect at higher moisture contents. If mechano-sorption depends on the velocity of the moisture content change is vague. Hanhijärvi [42] and Toratti [113]

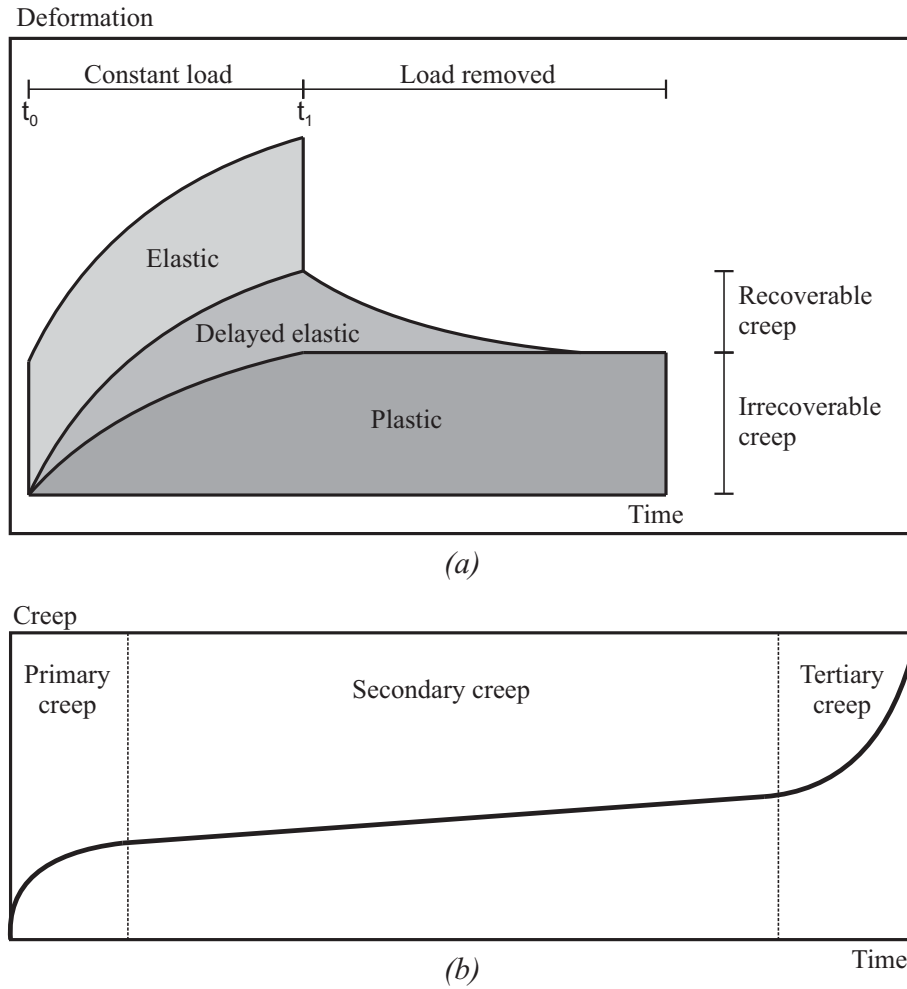


Figure 2.8 Illustration of creep

(a) Deformations due to constant load and after unloading, Dinwoodie [25]

(b) Creep due to constant uniaxial stress and the phases of creep, Niemz [75]

regarded the mechano-sorptive strain as independent to the rate of change in moisture content $\dot{\omega}$. However, later Toratti [114] admitted that mechano-sorption depends on the velocity of wood moisture change. This is in agreement with conclusions drawn by Hunt [46]. It can be summarised that time is negligible compared to $\dot{\omega}$.

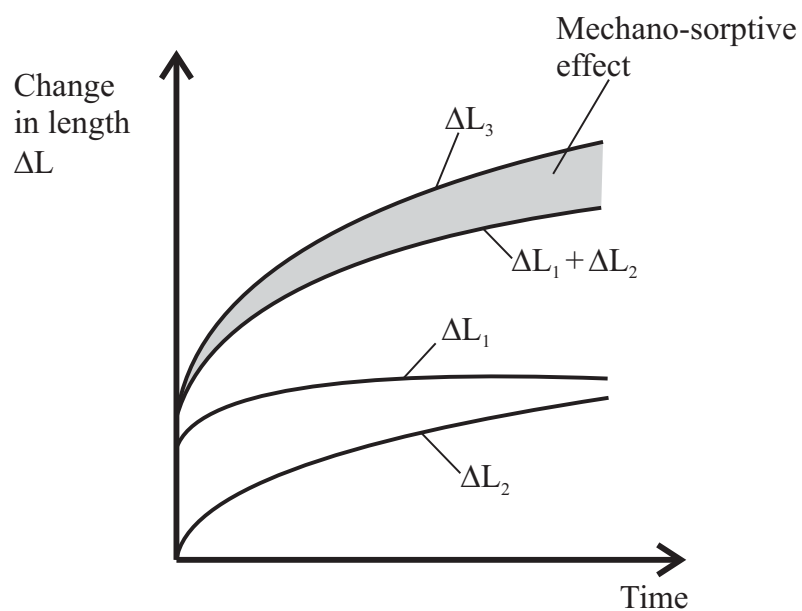
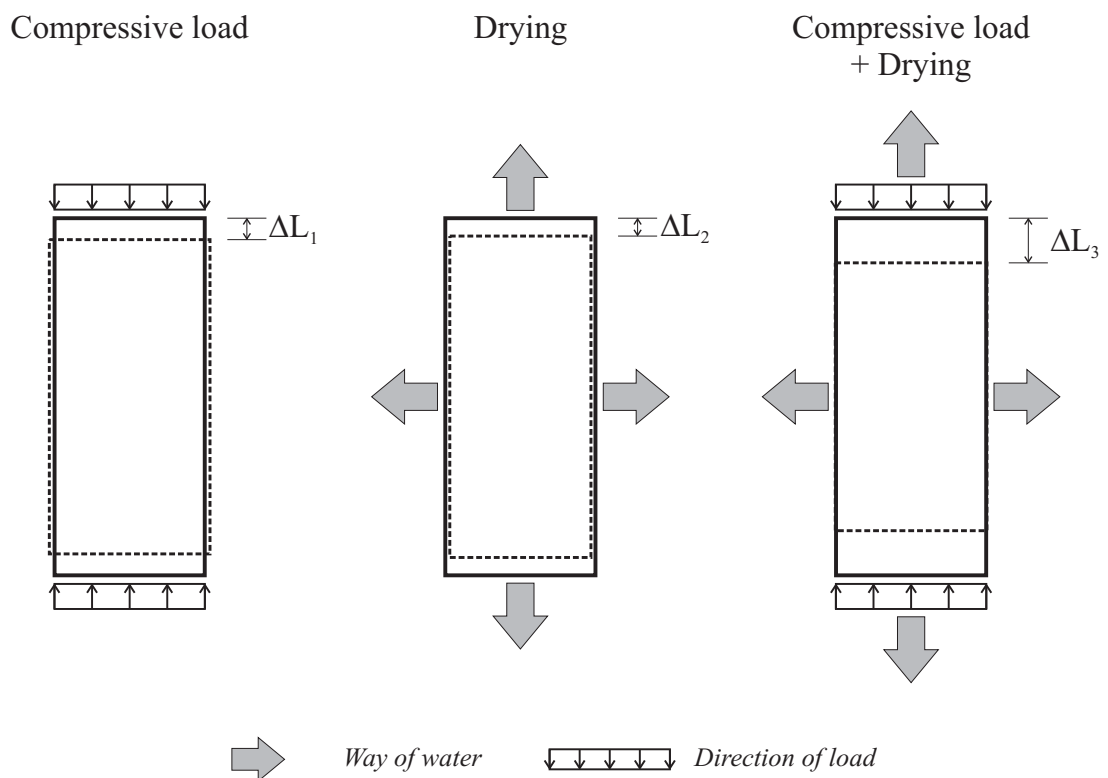


Figure 2.9 Illustration of the mechano-sorptive effect

2.5 Moisture-induced stresses in wood

Change in ambient relative humidity leads to a moisture change in the cross section. A non-uniform moisture distribution leads to different swelling/shrinkage. The fact that areas are restrained by adjacent areas leads to internal stresses in wood. Since the mechanical strength is much lower perpendicular to the grain than in the fibre direction due to the wood structure (see Section 2.1), stresses in the transverse direction play an important role in the application of wood (risk of cracks). In recent years, many works on the mechanical response of wood on humidity changes have been carried out [e.g., 20, 42, 71]. In these works the stress response to a specific climate history has been predicted but not confirmed by measurements.

A so-called slicing technique was employed by Behnke et al. [6], Jönsson [51], McMillen [72], Niemz and Wang [78] and Svensson and Toratti [106]. It was originally developed by Peck [80]. Stresses are determined by cutting the specimens into slices and measuring the change in length. Material in tension contracts on slicing and material in compression expands on slicing.

Svensson and Toratti [106] determined stresses in clear wood samples of *Picea abies*. The equipment is shown in Fig. 2.10. Released deformations were measured by five transducers positioned against the end of the specimen. Jönsson [51] [see also 49, 50, 52] applied a contact-free technique. The released deformations of glulam samples were measured by a digital camera. He compared the position of markers that were photographed before and just after sawing into slices.

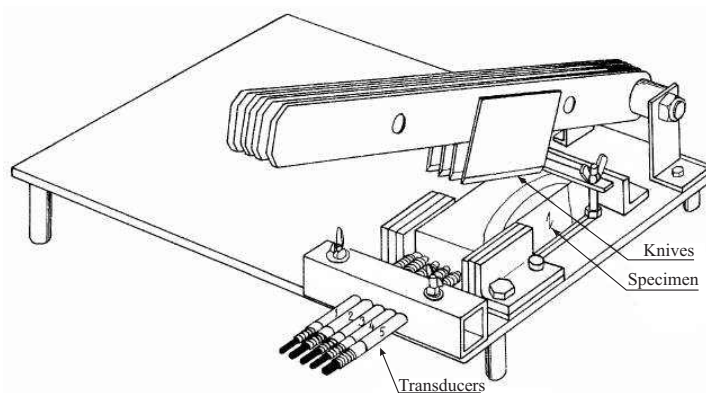


Figure 2.10 Equipment for slicing and measuring released deformations in wood, Svensson and Toratti [106]

McMillen [72] tested stresses in red oak (*Quercus rubra*) during air drying. The results, plotted in Fig. 2.11, show that, firstly, tensile stresses develop in the outer parts and compression stresses in the inner part. The shrinkage of the outer parts is hindered by the inner part, which is not yet affected by drying. Later, stresses decrease due to a more balanced moisture distribution. Further into the drying process, the drying inner part is restrained by the overstretched outer parts [124]. Stress distribution changes sign. This phenomenon is known as stress reversal or casehardening.

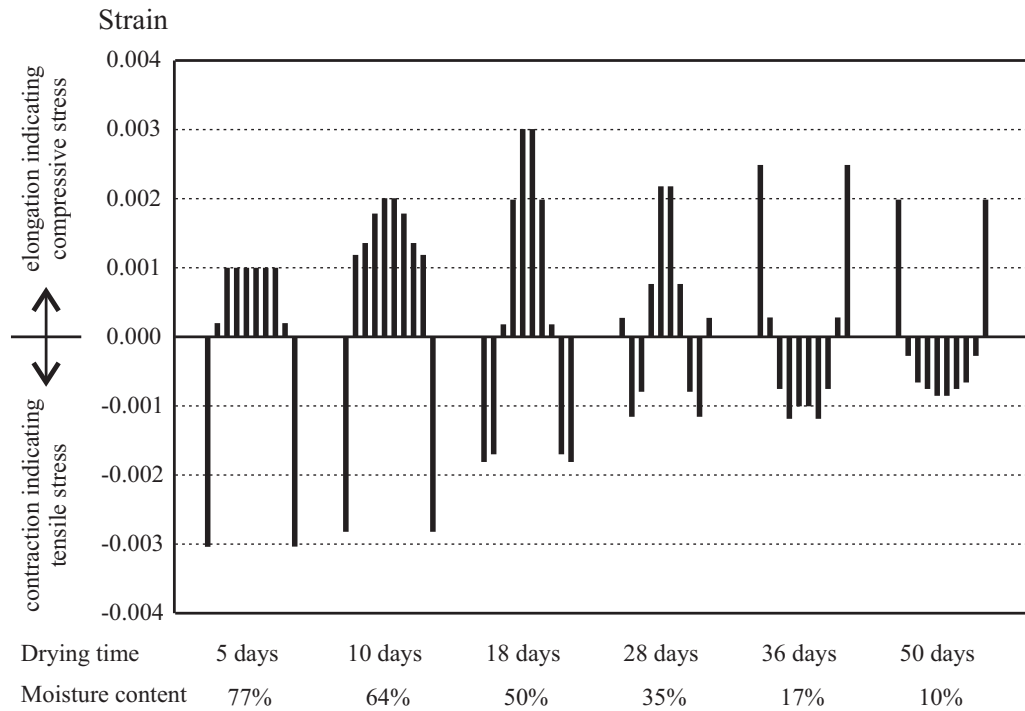


Figure 2.11 Strain distribution in 50mm red oak (*Quercus rubra*) during air-drying, graphic according to McMillen [72]

2.6 Mechanical material modelling

"In the last four decades, the finite element method, FEM, has become the prevalent technique used for analyzing physical phenomena in the field of structural, solid and fluid mechanics as well as for the solution of field problems." [69]

Mackerle [69] gives a general overview about finite element analysis in wood research. Literature dealing with the finite element method are, e.g., Bathe [3] and Zienkiewicz and Taylor [127], where the finite element formulations are given. In the present study, the finite element code ABAQUS was applied. Two material subroutines were used to analyse

wood during relative humidity changes, namely UMATHT and UMAT. The subroutines were used to calculate the diffusion process (UMATHT) and the mechanical constitutive behaviour of wood (UMAT). The basic equations for modelling diffusion in wood were presented in Section 2.2. The mechanical material modelling, which is stated below, is based on a strain rate formulation. Although plastic deformations may occur during moistening (compression) and cracks may develop during drying of wood (tension), these will not be considered in the present study. Since the short term behaviour of cross-laminates is investigated, creep strain is not included into the model.

The constitutive model assumes the total strain rate $\dot{\boldsymbol{\epsilon}}$ to be the sum of the elastic strain rate $\dot{\boldsymbol{\epsilon}}_{el}$, the moisture-induced strain rate $\dot{\boldsymbol{\epsilon}}_{\omega}$, the mechano-sorptive strain rate $\dot{\boldsymbol{\epsilon}}_{\omega\sigma}$ and the creep strain rate $\dot{\boldsymbol{\epsilon}}_c$:

$$\dot{\boldsymbol{\epsilon}} = \dot{\boldsymbol{\epsilon}}_{el} + \dot{\boldsymbol{\epsilon}}_{\omega} + \dot{\boldsymbol{\epsilon}}_{\omega\sigma} + \dot{\boldsymbol{\epsilon}}_c . \quad (2.24)$$

The dot denotes the derivative with respect to time. Ormarsson [79] applied that model to simulate deformations and stresses in wood during drying. Ormarsson used moisture and temperature dependent material properties that deviate in the orthotropic directions and took into account the influence of growth rings, the spiral grain and the conical shape of the log as well as variations in the wood properties with distance from pith to bark. Santaoja et al. [88] included mechano-sorptive creep in the three dimensional model.

The elastic strain rate is obtained by differentiation of Eq. 2.15, i.e.

$$\dot{\boldsymbol{\epsilon}}_{el} = \mathbf{S}\dot{\boldsymbol{\sigma}} + \dot{\mathbf{S}}\boldsymbol{\sigma} . \quad (2.25)$$

The compliance matrix $\mathbf{S}$ is given in Eq. 2.23. The rate of change of compliance $\dot{\mathbf{S}}$ is given in Eq. 4.8. The moisture-induced strain rate is assumed to depend on the rate of change in moisture content as

$$\dot{\boldsymbol{\epsilon}}_{\omega} = \boldsymbol{\alpha}\dot{\omega} , \quad (2.26)$$

where $\dot{\omega}$ is the rate of change of moisture content. The matrix $\boldsymbol{\alpha}$ contains the swelling or shrinking material parameters in the orthotropic directions α_T , α_L and α_R and is defined as

$$\boldsymbol{\alpha}^T = \begin{pmatrix} \alpha_L & \alpha_R & \alpha_T & 0 & 0 & 0 \end{pmatrix} . \quad (2.27)$$

The mechano-sorptive strain rate was proposed by Takemura [107] and Leicester [66] (see also Ranta-Maunus [85] and Santaoja et al. [88]) as

$$\dot{\boldsymbol{\epsilon}}_{\omega\sigma} = \mathbf{m}\boldsymbol{\sigma} |\dot{\omega}| . \quad (2.28)$$

The matrix $\mathbf{m}$ is a mechano-sorption property matrix given by

$$\mathbf{m} = \begin{bmatrix} m_L & -\mu_{RL}m_R & -\mu_{TL}m_T & 0 & 0 & 0 \\ -\mu_{LR}m_L & m_R & -\mu_{TR}m_T & 0 & 0 & 0 \\ -\mu_{LT}m_L & -\mu_{RT}m_R & m_T & 0 & 0 & 0 \\ 0 & 0 & 0 & m_{LR} & 0 & 0 \\ 0 & 0 & 0 & 0 & m_{LT} & 0 \\ 0 & 0 & 0 & 0 & 0 & m_{RT} \end{bmatrix}. \quad (2.29)$$

m_L , m_R and m_T are mechano-sorption coefficients for the orthotropic directions and m_{LR} , m_{LT} and m_{RT} are mechano-sorption coefficients for the orthotropic planes. The mechano-sorptive strain is coupled between the different directions by μ_{LR} , μ_{RL} , μ_{LT} , μ_{TL} , μ_{RT} and μ_{TR} . The matrix $\mathbf{m}$ is assumed to be symmetric, which results in

$$\mu_{RL} = \frac{m_L}{m_R}\mu_{LR}, \quad \mu_{TL} = \frac{m_L}{m_T}\mu_{LT}, \quad \mu_{TR} = \frac{m_R}{m_T}\mu_{RT}.$$

2.7 Previous research studies on cross-laminated wood panels

2.7.1 General literature

Various research studies concerning cross-laminated solid wood panels have been carried out in recent years in the German-speaking part of Central Europe. A general overview is given in Dunky and Niemz [27]. Design recommendations have been developed by Blass and Görlacher [7] and Jöbstl and Schickhofer [48]. Various studies concerned the determination of mechanical properties, mainly in bending tests of beams: Donzé et al. [26], Krug et al. [63; 64], Niemz and Bencat [76], Tobisch and Krug [111] and Tobisch and Plattes [112]. Large-sized samples were tested: Czaderski et al. [18] performed panel bending tests, Gsell et al. [38], Gülzow [39] and Gülzow et al. [40] evaluated elastic parameters non-destructively by means of eigenfrequencies. Other studies concerned the testing of joints and fastening devices [95, 115] and of fire performance [30].

2.7.2 Moisture

Diffusion processes, thermal conductivity and the equilibrium moisture content of cross-laminated wood panels were tested by Bader et al. [2] and Popper et al. [83; 82]. Bader et al. [2] detected an influence of hollow spaces on the thermal conductivity and water vapour resistance. With an increased amount of cavities, the water vapour resistance in cup tests was decreased. Popper et al. [82] also performed diffusion tests by means of the cup method in climate differences of 65/100% RH as well as 65/0% RH. The authors investigated the water vapour resistance of industrially produced three- and five-layered cross-laminated solid wood panels. The results indicate that the water vapour resistance is significantly a function of glue lines per panel thickness. With increased ratio of glue lines per panel thickness, the water vapour resistance is increased. Water vapour resistance was not found to depend on the type of adhesive. Furthermore, the authors concluded a linear distribution of moisture within the panels from the mean moisture content of the panels of about 20% and the theoretical moisture contents of 12% ($\varphi = 65\%$) and 28% ($\varphi = 100\%$) at the top and bottom face, respectively.

The moisture content in 60mm thick panels exposed to outdoor weathering under a roof and in a climate of 2°C/90% RH was measured as a function of time by Niemz et al. [77]. While stored in an outdoor climate, the moisture changes took place mainly in the outer zones. Simulations showed good agreement with the measurements.

2.7.3 Moisture-related distortions

Schwab et al. [91] found swelling coefficients α (percentaged change in length per 1% change in moisture content) of $\alpha_x = 0.012 \dots 0.020 \text{ \% \%}^{-1}$ and $\alpha_y = 0.034 \dots 0.042 \text{ \% \%}^{-1}$ in three-layered cross-laminated panels from Norway spruce with a thickness of 20mm. This is higher compared to spruce wood in the longitudinal direction (Dahlblom et al. [19]: $\alpha_L = 0.003 \dots 0.007 \text{ \% \%}^{-1}$) but negligible compared to spruce wood perpendicular to the grain (Sell [92]: $\alpha_R = 0.15 \dots 0.19 \text{ \% \%}^{-1}$, $\alpha_T = 0.27 \dots 0.36 \text{ \% \%}^{-1}$). Similar results were determined by Bader et al. [2]. Tobisch [110] investigated the dependency of the swelling and shrinkage on the layer ratio, which is the thickness of both outer layers a_{OL} referred to the total thickness a_{tot} of the panel:

$$LR = \frac{2a_{OL}}{a_{tot}} . \quad (2.30)$$

Tobisch [110] could not detect any significant influence of the layer ratio but swelling in the minor axis was larger than in the major axis (Fig. 2.12).

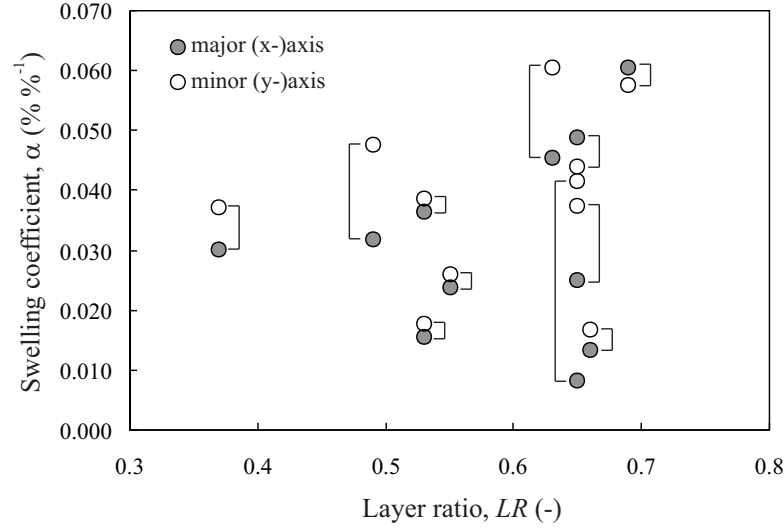


Figure 2.12 Swelling coefficient vs. layer ratio in the major and minor axes, values of different axes of one panel type are marked, according to Tobisch [110]

Tobisch [110] tested the dimensional stability of 500mm × 500mm large panels exposed to a moisture gradient in a double climate chamber (Fig. 2.13). The author calculated the warp deformation in the diagonal, and in the major and minor axes in mm m⁻¹ as

$$x_{1000} = r - \cos\left(\frac{180b}{2\pi r}\right) r \quad (2.31)$$

with the radius r of the circle segment (see Fig. 2.13c) as

$$r = \frac{x_{400}}{2} + \frac{(400/2)^2}{2x_{400}}. \quad (2.32)$$

x_{400} is the measured deformation perpendicular to the plane in mm at a distance of 400mm and x_{1000} is the calculated deformation perpendicular to the plane in mm at a length of the circle segment of $b = 1000$ mm. The results showed an increase in warp deformation in the major axis and in the diagonal, and a decrease in warp deformation in the minor axis with an increase in the relative thickness of the outer layers, LR . The warping of panels assembled by means of polyvinylacetate (PVAc) was approximately 20% smaller than the warping of panels assembled by means of MUF. This was explained by the higher elasticity of the PVAc glue lines which resulted in reduced transmission of shear stresses and, thus, a smaller warping. The tests of different middle layer materials showed that spruce panels were more stable to warp than panels where the middle layer was substituted

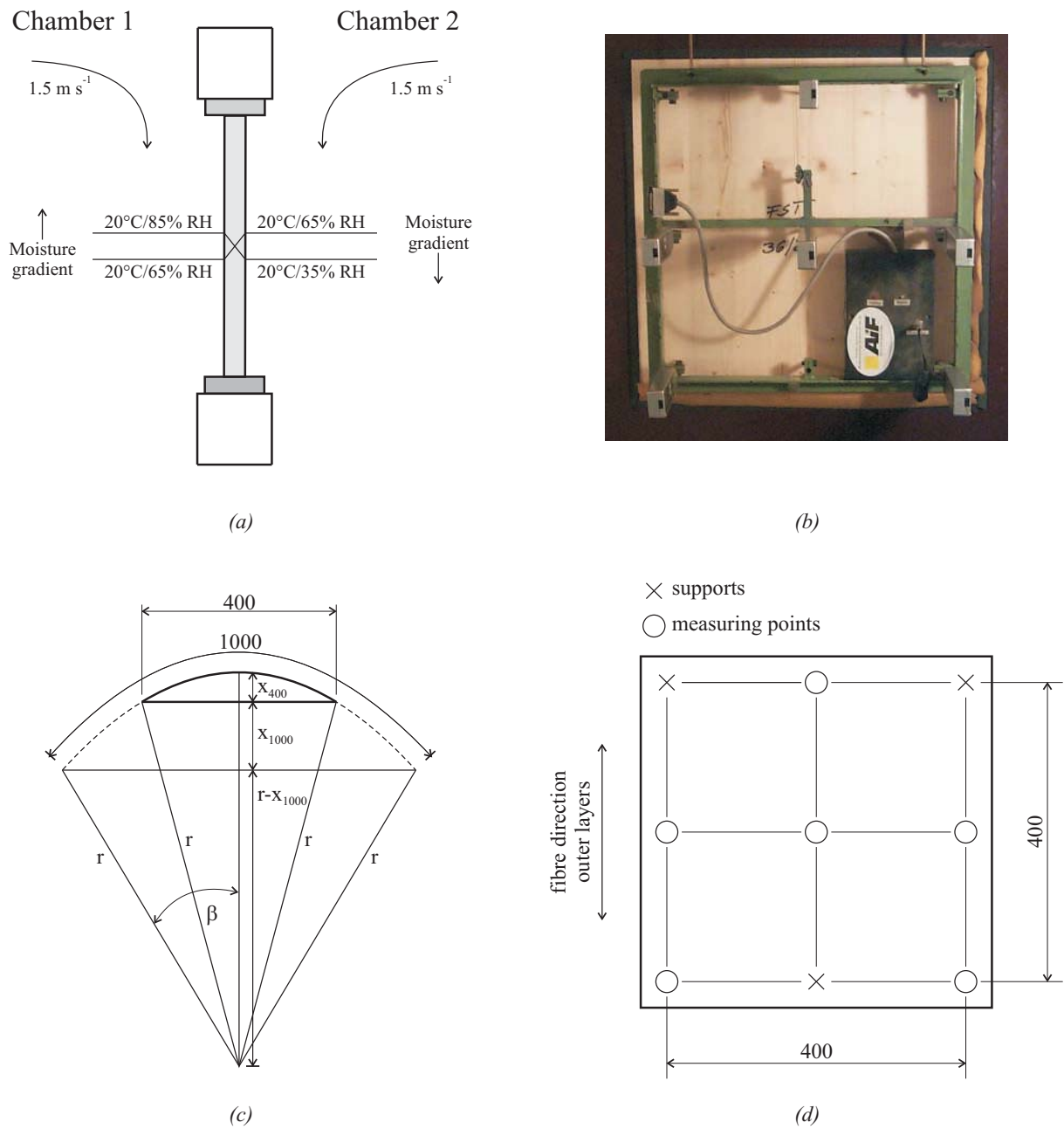


Figure 2.13 Illustration of the double climate experiment according to Tobisch [110]

- (a) Illustration of the double climate chamber and acting state variables
- (b) Measurement of deformations on the panels
- (c) Principal of the calculation of warp
- (d) Points of measurement and supports

(all dimensions in mm)

by particleboard or OSB (Fig. 2.14). A substitution with plywood showed better results but is too expensive for practical applications. Even here, the warp deformation in the minor axis decreased with increased layer ratio (Fig. 2.14) while it slightly increased in the major axis. Systematic cross-laminated layers were more stable to warp than panels without cross lamination. Cross lamination describes the fact that the 1-axis of the wood composite middle layer (Fig. 2.17) is aligned parallel to the y-axis of the panel coordinate system (Fig. 1.2).

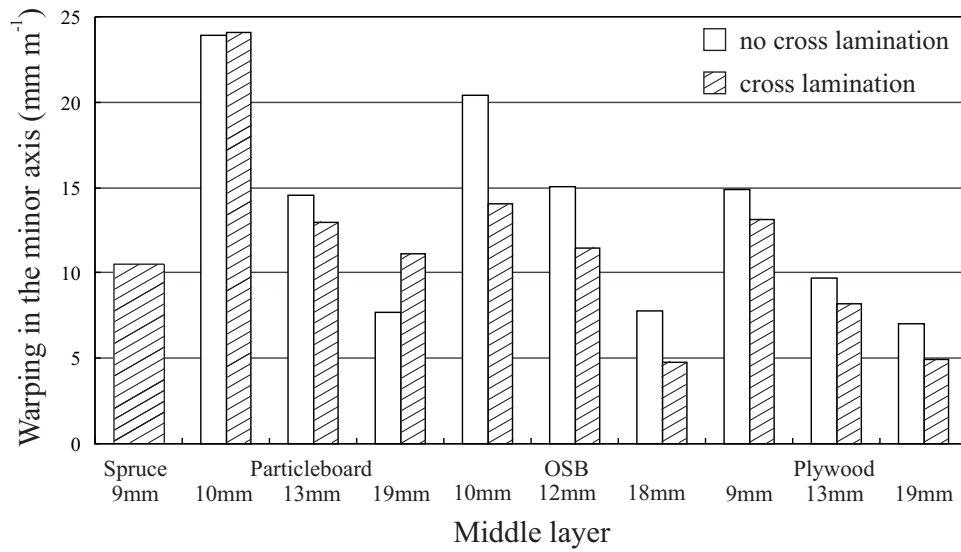


Figure 2.14 Warping in the minor axis of three-layered panels with substituted middle layers of different middle layer thicknesses compared to cross-laminated solid wood panel (CWP), test and analysis according to Fig. 2.13, $a_{tot} = 27\text{mm}$, $t = 48\text{h}$, results determined by Jensen and Krug [47] (CWP) and Tobisch [110]

2.7.4 Internal stress state

Weber [123] performed investigations on the internal stress state of three-layered cross-laminated solid wood panels from Norway spruce. He applied a method where the moisture-induced strains were released by sawing the panels, firstly, into stripes and, finally, into slices of the individual layers. The method is illustrated more closely in Section 5.2.2. Weber [123] studied panels with different outer layer thicknesses ($a_{OL,1} = 7.5\text{mm}$, $a_{OL,2} = 10\text{mm}$ and $a_{OL,3} = 15\text{mm}$) with the middle layer thickness kept constant at 10mm. This gives layer ratios of $LR_1 = 0.60$, $LR_2 = 0.67$ and $LR_3 = 0.75$. From $480\text{mm} \times 480\text{mm}$ panels, eight samples measuring $120\text{mm} \times 240\text{mm}$ were produced. A specific growth ring orientation was not considered. The samples were initially conditioned in $20^\circ\text{C}/35\%$ RH and were then transferred to varying climates: $\varphi_1 = 65\%$, $\varphi_2 = 82\%$ and $\varphi_3 = 86\%$, each with constant conditions of $T = 20^\circ\text{C}$. The tests were carried out for 14 days.

The results showed a high variation in stresses parallel to the grain due to the very small measurable strain. The measured stresses perpendicular to the grain, however, showed acceptable results (Fig. 2.15). The figure indicates the fact that moistening leads to compressive stresses perpendicular to the fibres in both, the outer layers and the middle layer. The minimum stress, i.e. the maximum compressive stress, was reached after a short period of three days in the outer layers. The stresses in the middle layer reached their maximum value after seven days. The decrease in the moisture difference between the outer layers and the middle layer led to a decrease in the compressive stresses after the maximum. The maximum compressive stress in the middle layers was higher than the maximum compressive stress in the outer layers. This is due to the fact that the middle layer was hindered from both sides, while the outer layers were able to swell unhindered at the surface. The comparison between different climates showed an increase of the compressive stresses with increasing relative humidity. The study of different outer layer thicknesses showed increased maximum compressive stresses with a decrease in the outer layer thickness.

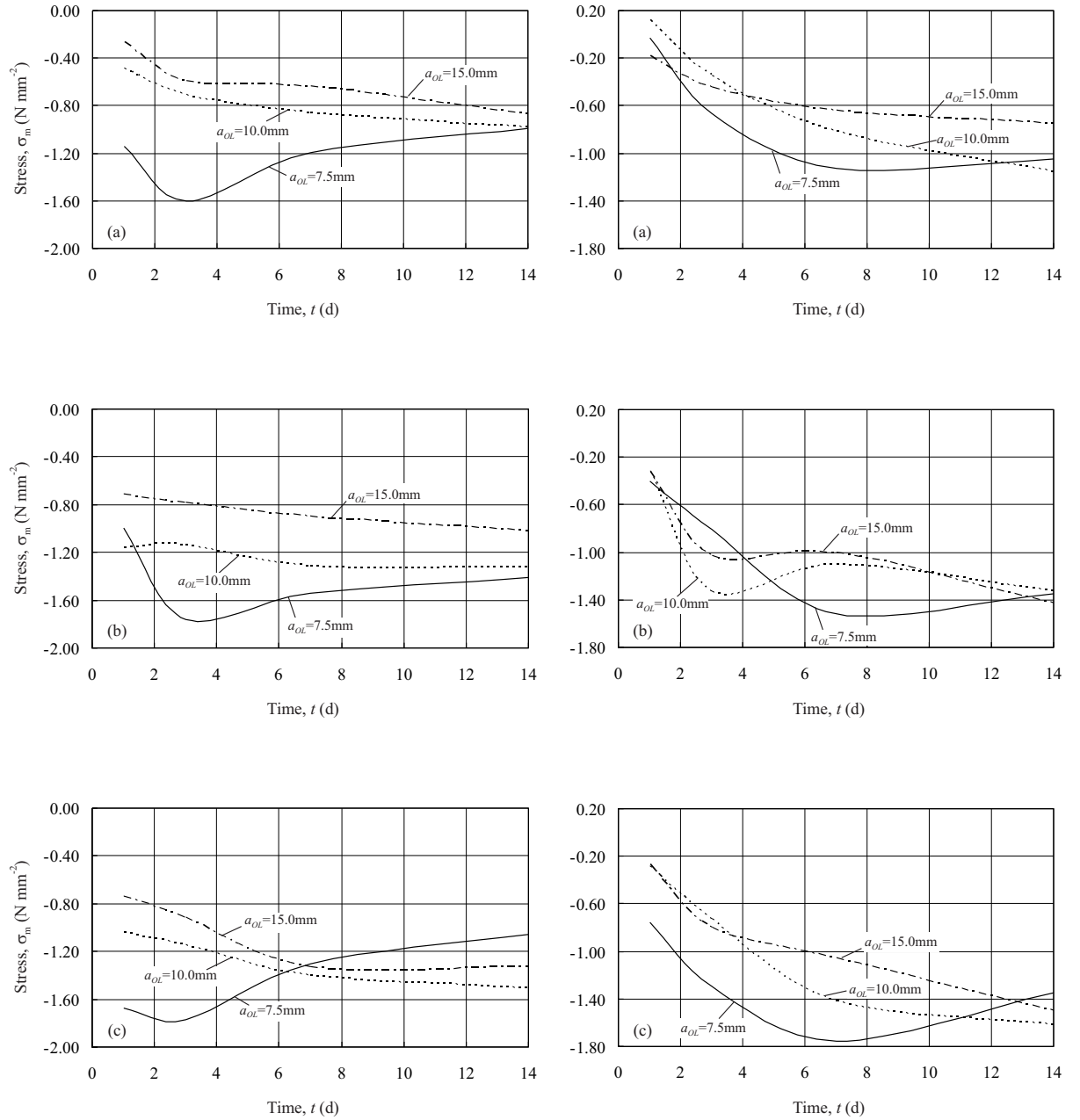


Figure 2.15 History plots of mean stresses in the outer layers (lefthand side) and the middle layers (righthand side) perpendicular to the grain exposed to different climates: (a) 65% RH, (b) 82% RH, (c) 86% RH and with different outer layer thicknesses a_{OL} , middle layer thickness $a_{ML} = 10$ mm, initial climate $\varphi = 35\%$, according to Weber [123]

2.8 Wood-based panels

Within the framework of this thesis only a short introduction into wood-based panels is given. A more detailed overview is drawn, e.g., in Chapman [15], Dunky and Niemz [27], Maloney [70] and Youngquist [126].

Wood-based panels such as oriented strand board (OSB), particleboard (PB) and medium density fibreboard (MDF) consist of elements of widely varying shapes and sizes bonded together with an adhesive system (Fig. 2.16). With the size of the structural elements, i.e. strands, particles or fibres, the properties of the produced panels significantly vary [27]. On the one hand, the strength decreases with increasing disintegration of the raw material, but on the other hand, the homogeneity, the isotropy, the thermal insulation and the surface quality increase.



Figure 2.16 Types of wood-based panels, from left to right: oriented strand board (OSB), particleboard (PB) and medium density fibreboard (MDF)

Depending on the manufacturing, the properties of fibreboard and particleboard slightly differ between the direction of the conveyor-belt and perpendicular to it (compare Tab. 4.5). Larger differences between both of the in-plane directions exist in OSB. Due to the three-layered structure and the alignment of the surface layer strands in the production direction and, thus, the arrangement of the longitudinal wood axis in that direction, the modulus of elasticity is considerably larger in the production direction than perpendicular to it. However, medium density fibreboard can be assumed as plane isotropic [8] with identical properties in the 1- and 2-direction (Fig. 2.17): $E_1 = E_2$, $G_{23} = G_{13}$, $\nu_{12} = \nu_{21}$, $\nu_{23} = \nu_{32} = \nu_{13} = \nu_{31}$ and $\alpha_1 = \alpha_2$. The shear modulus G_{12} can be obtained from E_1 and ν_{12} :

$$G_{12} = \frac{E_1}{2(1 + \nu_{12})} \quad (2.33)$$

Consequently, seven independent parameters need to be determined to reproduce the

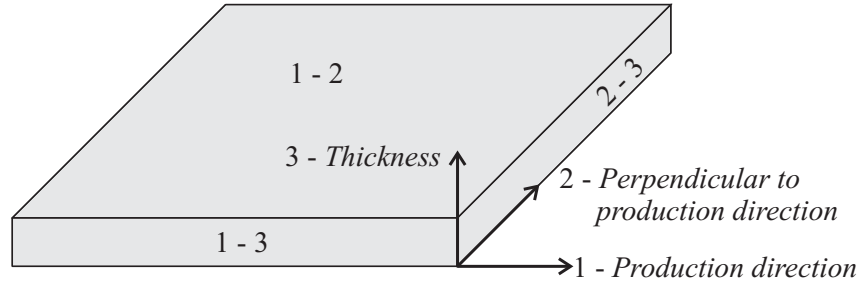


Figure 2.17 Schematic drawing of a MDF specimen with definition of the coordinate system

mechanical response of MDF to a change in moisture: E_1 , E_3 , ν_{12} , ν_{13} , G_{13} , α_1 and α_3 (swelling coefficients) or β_1 and β_3 (shrinkage coefficients).

A broad investigation on the effect of density and moisture content in MDF was performed by Ganey et al. [31; 32; 33; 34]. Therefore, panels were produced from Green black spruce (*Picea mariana*). A uniform vertical density distribution was achieved by removing the denser surface layers in a planer and subsequent sanding of the remaining core layer to a thickness of 6mm. Panels with different nominal densities were produced: 540 kg m⁻³, 650 kg m⁻³ and 800 kg m⁻³. From this material, sorption isotherms, diffusion coefficients in adsorption and desorption [31], mechanical properties (moduli of elasticity, shear moduli, Poisson's ratios) [34] and swelling and shrinkage coefficients [32] were determined.

Table 2.1 Mechanical properties of MDF, moduli: $C_{ij} = C_0 - C_m\omega$, $\omega = 0.069 \dots 0.135$, Ganey et al. [33; 34]

ρ_0 (kg m ⁻³)		E_1 (N mm ⁻²)	E_3 (N mm ⁻²)	G_{13} (N mm ⁻²)	ν_{12} (-)	ν_{13} (-)
540	C_0	1017.0	31.47	60.5	0.3	0.2
	C_m	4250.2	220.1	113.4		
650	C_0	1590.9	59.8	188.5	0.3	0.2
	C_m	4720.1	381.3	1080.9		
800	C_0	3081.9	119.4	237.4	0.3	0.2
	C_m	12563.2	842.5	1497.8		

Table 2.2 In-plane (1) and thickness (3) swelling (α) and shrinkage (β) coefficients of MDF, Ganev et al. [32]

ρ_0 (kg m ⁻³)	α_1	α_3 (% % ⁻¹)	β_1	β_3
540	0.017	0.74	0.034	0.54
650	0.023	0.77	0.048	0.62
800	0.036	0.84	0.077	0.95

The results are summarised in Tabs. 2.1 and 2.2. The mechanical properties and the expansion/contraction significantly differ between the in-plane directions and the thickness direction:

$$E_1 \gg E_3; \alpha_1 \ll \alpha_3; \beta_1 \ll \beta_3$$

Such as in solid wood, Young's and shear moduli decrease with increasing moisture content and increase with increasing density. The swelling and shrinkage coefficients also increase when the density is increased.

The coefficient of diffusion of MDF was found to depend on the moisture content [31]. In desorption D [m² s⁻¹] increases with ω [-]:

$$\begin{aligned} D_{540}(\omega) &= 2 \cdot 10^{-16} e^{1.5705\omega} \\ D_{650}(\omega) &= 2 \cdot 10^{-13} e^{0.7468\omega} \\ D_{800}(\omega) &= 3 \cdot 10^{-14} e^{0.8990\omega} \end{aligned} \quad (2.34)$$

while in adsorption D decreases with ω [31]:

$$\begin{aligned} D_{540}(\omega) &= 2 \cdot 10^{-6} e^{-1.1466\omega} \\ D_{650}(\omega) &= 2 \cdot 10^{-8} e^{-0.7894\omega} \\ D_{800}(\omega) &= 2 \cdot 10^{-8} e^{-0.7483\omega} \end{aligned} \quad (2.35)$$

This decrease is in agreement with results obtained by Wu and Suchsland [125] for particleboard. Ganev et al. [31] explained this phenomenon with the fact that water-vapour diffusion through air-filled pore spaces is the dominant moisture transfer mechanism in fibre- and particleboard. Siau [94] also presented decreasing diffusion coefficients with an increase in moisture content for water-vapour transport in wood. The diffusion coefficient was found to be larger when the density was reduced. The unexpected fact that D_{800} is

larger than D_{650} in adsorption is not discussed by the authors. Sonderegger and Niemz [100] also reported increasing diffusion coefficients with decreased density for fibre- and particleboards, which were studied in cup measurements. With the size of the structural elements the diffusion coefficient declines.

Chapter 3

Moisture transport

3.1 Introduction

To model stresses and deformations, e.g., by means of a finite element simulation, the moisture distribution in the panel is needed. The diffusion coefficients of Norway spruce are well documented in the literature: Hukka [45], Siau [94], Toratti [113] and Vanek and Teischinger [116]. The information in literature about the moisture diffusion coefficient of adhesives is, on the other hand, very limited. In this study, the diffusion coefficient of the adhesive joints was determined by a combination of experimental measurements and numerical simulations of the moisture distribution in the panels.

Adhesive joints are applied in a plenty of modern solid wood based materials. Besides cross-laminated solid wood panels, glulam (glued laminated timber) and glued cross beams should be mentioned. Nowadays, the diffusion coefficient of adhesives is widely unknown. The newer literature only concerns with extreme values for the adhesive [102], i.e. $D_{adh} = D_{wood}$ and $D_{adh} = 0$ (Fig. 3.1). The results show that both assumptions do not fulfil the experimental results. However, correct knowledge about the moisture behaviour of adhesive joints is fundamental in the simulation of moisture flux through wooden structural elements and, consequently, moisture-induced stresses and deformations.

Firstly, sorption measurements were performed at 20°C and relative humidities between 25% and 93%. The results, on the one hand, provide input data for modelling diffusion and, on the other hand, the test clarified whether moisture sorption depends on the

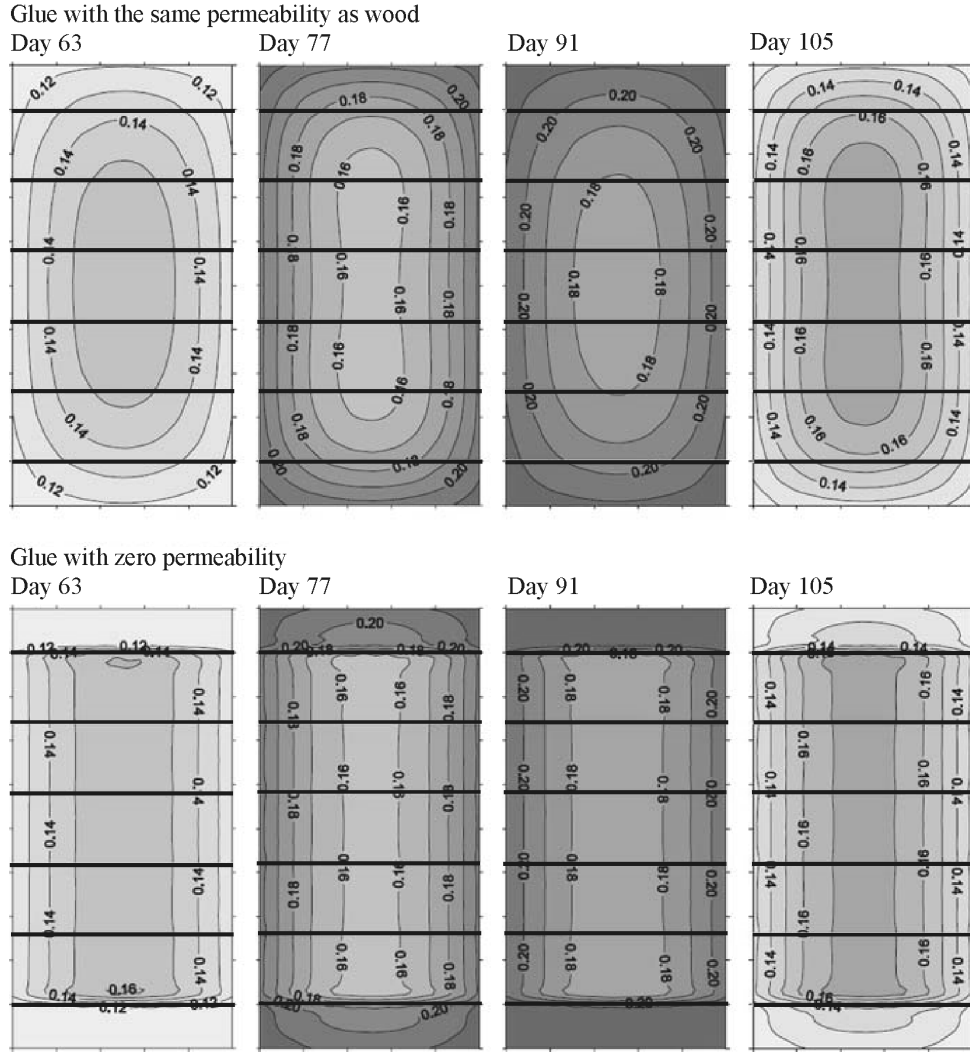


Figure 3.1 Simulated moisture distributions in a glulam sample exposed to climate variations between $\varphi_1 = 65\%$ and $\varphi_2 = 95\%$ at $T = 21^\circ\text{C}$, Srpčič et al. [102]

stress state or not. Furthermore, moisture profiles of three-layered solid wood panels were measured and simulated. The simulation was adapted to the moisture contents that were obtained in the measurements. This resulted in recent knowledge about the diffusion resistance of the glue lines. The moisture model evaluated in this chapter provides the possibility to calculate the moisture distribution within a specific panel and to calculate the history moisture development.

3.2 Sorption

3.2.1 Material and method

Former investigations of the ω - φ - relation at constant temperatures T , termed sorption isotherm, showed specific characteristics of cross-laminated solid wood panels [2, 83]:

- higher equilibrium moisture contents than spruce at low relative humidities ($\varphi < 65\%$)
- lower equilibrium moisture contents than spruce at high relative humidities.

To confirm these results and to detect reasons for these phenomena, sorption tests on three different types of specimens were conducted (Fig. 3.2):

- type A - clear samples of Norway spruce, $350 \leq \rho_0 \leq 550$ [kg m⁻³],
- type B - three-layered samples with parallel oriented layers,
- type C - three-layered samples with crosswise oriented layers.

Specimen types B and C were assembled by means of 1K PUR. Samples B and C compared to A should clarify if the assembling process (i.e. applying a pressure) or the adhesive affect the equilibrium moisture content. The pressing process may densify the wood, which may reduce the accessibility to sorption sites. Furthermore, it is possible that the penetration of the adhesive into the wood structure blocks sorption sites in the wood. The comparison between B and C should clarify if stresses due to the crosswise bonding in type C influence the equilibrium moisture content. Due to tensile stresses the area for sorption of water in the wood structure may increase and, thus, the equilibrium moisture content is higher compared to normal wood.

Additionally, specimens of type B and C were loaded by iron blocks with $m = 5\text{kg}$ and dimensions of $300\text{mm} \times 70\text{mm} \times 30\text{mm}$. These specimens are denoted EXT. In each case this external load hindered the samples in swelling in one direction (x, y or z). Two samples were loaded at a time, i.e. samples were loaded with 16.7 N mm^{-2} in x- and y-direction and with 9.8 N mm^{-2} in z-direction, respectively. This fixing may also affect the sorption curve.

Sorption measurements were conducted in a climate chamber with a constant temperature of 20°C and five steps of relative humidity: 25, 45, 65, 85 and 93%. The specimens were weighed when the mass was in equilibrium with the appropriate climate. According to

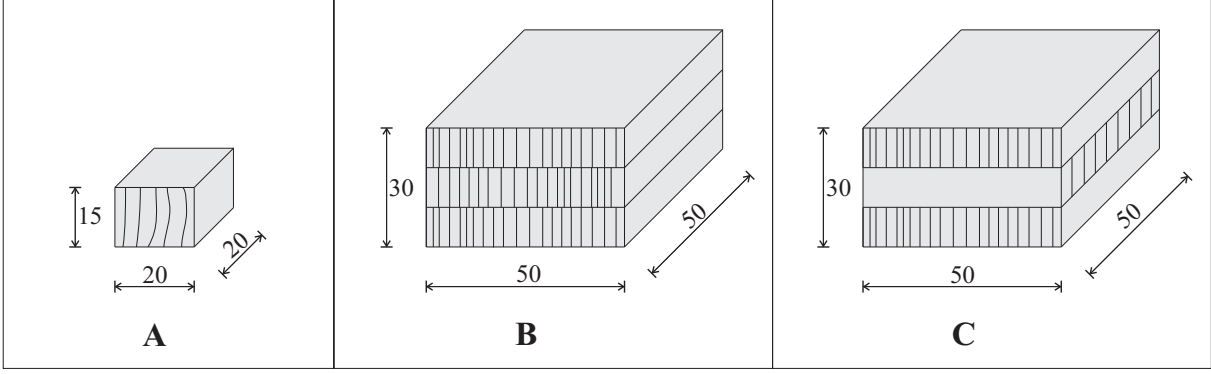


Figure 3.2 Types of specimen applied in the sorption tests, dimensions in mm

DIN 52183 [21] the moisture content is assumed to be in equilibrium with the ambient air, when the moisture contents of two measurements at intervals of 24 hours differ less than 0.1%. Finally, the moisture contents were determined by kiln drying (Eq. 2.1). The specimens were dried in an oven at 103°C. After reaching the equilibrium, the specimens were cooled down in an exsiccator with the presence of silica gel ($\varphi = 0\%$). Finally, the mass m_0 of the dried specimen was determined.

3.2.2 Results

Tab. 3.1 shows the results of the sorption measurements for specimens A, B and C. The differences between the sorption curves are not as significant as in Bader et al. [2] and Popper et al. [83] since the differences between the mean values are only small. A reason may be the differing specimen geometries. Bader et al. [2] and Popper et al. [83] tested cross-laminated solid wood panels with dimensions of 20mm $\times$ 200mm and different thicknesses. However, verified by statistics, the results for the equilibrium moisture content, ω_∞ , are:

- $\omega_{\infty,A} < \omega_{\infty,B} = \omega_{\infty,C}$ at 25, 45 and 65% RH,
- $\omega_{\infty,A} = \omega_{\infty,B} = \omega_{\infty,C}$ at 85% RH,
- $\omega_{\infty,A} > \omega_{\infty,B} > \omega_{\infty,C}$ at 93% RH,
- $\omega_{\infty,B/C} < \omega_{\infty,EXTB/C}$ at 25, 45, 65 and 85% RH.

The application of the adhesive and/or the pressing process (specimens B and C compared to A) influenced the equilibrium moisture content. Stresses due to the crosswise bonding

Table 3.1 Equilibrium moisture contents at different relative humidities φ of non-glued (A), glued (B, C) and external loaded (EXT) specimens from Norway spruce, mean and (standard deviation)

φ (%)	Equilibrium moisture content, ω_∞ (%)			
	A (n=100)	B (n=18)	C (n=27)	EXT (n=20)
25	6.88 (0.19)	7.19 (0.07)	7.26 (0.09)	7.11 (0.07)
45	9.31 (0.22)	9.59 (0.06)	9.62 (0.09)	9.47 (0.07)
65	12.31 (0.25)	12.53 (0.07)	12.49 (0.11)	12.40 (0.10)
85	17.49 (0.43)	17.57 (0.12)	17.47 (0.17)	17.37 (0.17)
93	21.46 (0.44)	21.27 (0.15)	21.04 (0.22)	21.04 (0.24)

(C compared to B) only influenced sorption at $\varphi = 93\%$. The external loadings affected the sorption curve. Proofed by statistical analysis, the moisture contents of external loaded samples were significantly lower at relative humidities less than 85% than samples without external loading. This may be due to the hindered swelling and, thus, compressive stresses. Differences between the loading directions could not be determined definitely. According to Skaar [98] compression leads to reduced equilibrium moisture content.

Since the equilibrium moisture contents of the specimen types A, B and C varied only a little and less significant as obtained before [2, 83], only the sorption curves of clear spruce samples (type A) are analysed in the following. There are several sorption theories applied to wood [see 94, 96, 99]. These theories provide an explanation of sorption in hygroscopic materials. The most well known applied to wood and textiles, which have sigmoid sorption isotherms, is the theory according to Hailwood and Horrobin [41]. However, according to Siau [94] and Simpson [97] a fifth degree polynomial function (six parameters) gives a better fit to the experimental data than any of the theoretical theories although it has no physical basis.

As mentioned, the concentration of moisture, c is used as the driving potential in the simulations of moisture flux in wood. The relation between concentration c , relative humidity φ and oven-dry density ρ_0 is stated in terms of the polynomial function

$$\begin{aligned}
 c(\varphi, \rho_0) = & f_0 + f_1\varphi + f_2\rho_0 + f_3\varphi\rho_0 + f_4\varphi^2 + f_5\varphi^2\rho_0 + f_6\varphi^3 \\
 & + f_7\varphi^3\rho_0 + f_8\varphi^4 + f_9\varphi^4\rho_0 + f_{10}\varphi^5 + f_{11}\varphi^5\rho_0 .
 \end{aligned} \tag{3.1}$$

The concentration is approached by a fifth degree polynomial function of relative humidity. The relation between concentration and oven-dry density was found to be linear (Fig. 3.3).

When the density of wood is increased, the ratio of cell walls to lumen is increased. Thus, denser wood is able to absorb more water; the concentration increases.

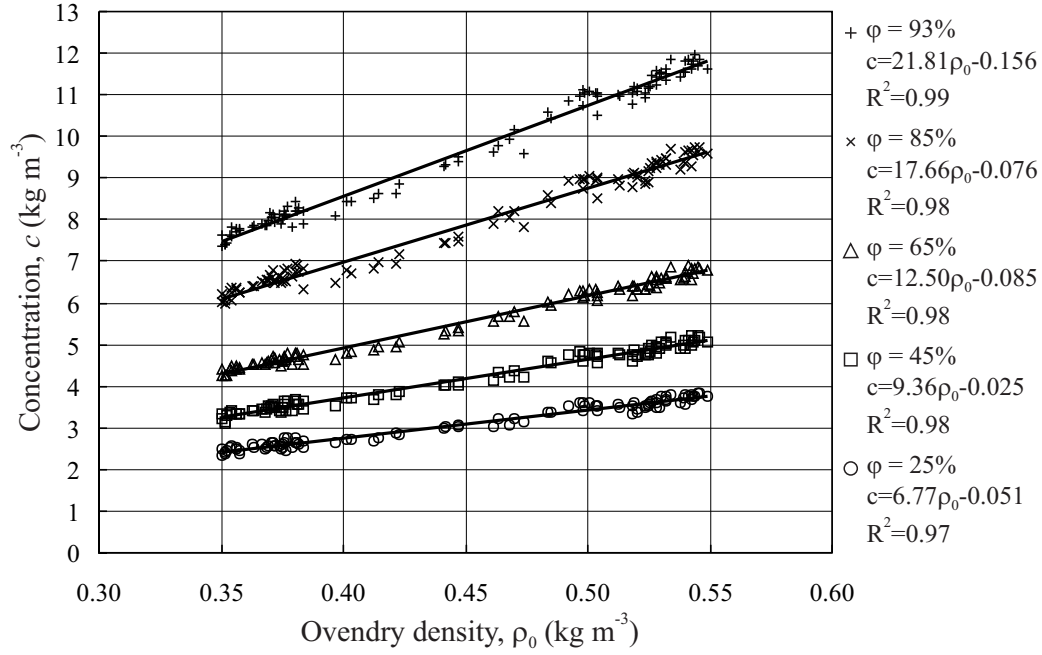


Figure 3.3 Linear correlations between concentration c and ovendry density ρ_0 in Norway spruce at different relative humidities φ , $T = 20^\circ C$

The shape factors $f_0 - f_{11}$ in Eq. 3.1, which fit the experimental data the best, were determined to

$$\begin{array}{ll}
 f_0 = -4.169 \cdot 10^{-4} & f_6 = 45.036 \\
 f_1 = 2.696 & f_7 = 0.715 \\
 f_2 = -1.501 \cdot 10^{-6} & f_8 = -47.823 \\
 f_3 = 6.714 \cdot 10^{-2} & f_9 = -0.797 \\
 f_4 = -18.438 & f_{10} = 18.515 \\
 f_5 = -0.294 & f_{11} = 0.338
 \end{array}$$

The shape factors inserted into Eq. 3.1 give the three-dimensional field as it is displayed in Fig. 3.4. The typical sigmoid shape of sorption isotherms of wood is clearly seen. With this knowledge it is now possible to calculate the concentration of water in spruce wood of a certain density at a given relative humidity. Note that Eq. 3.1 is only valid for ovendry densities between 350 and 550 kg m^{-3} , which represent Norway spruce in Central Europe. The sorption isotherm for the average density $\rho_0 = 450 \text{ kg m}^{-3}$ calculated by means of Eq. 3.1 and converted into moisture content by Eq. 2.2 is contrasted to the measured data in Fig. 3.5. An excellent fit to the data determined experimentally can be concluded.

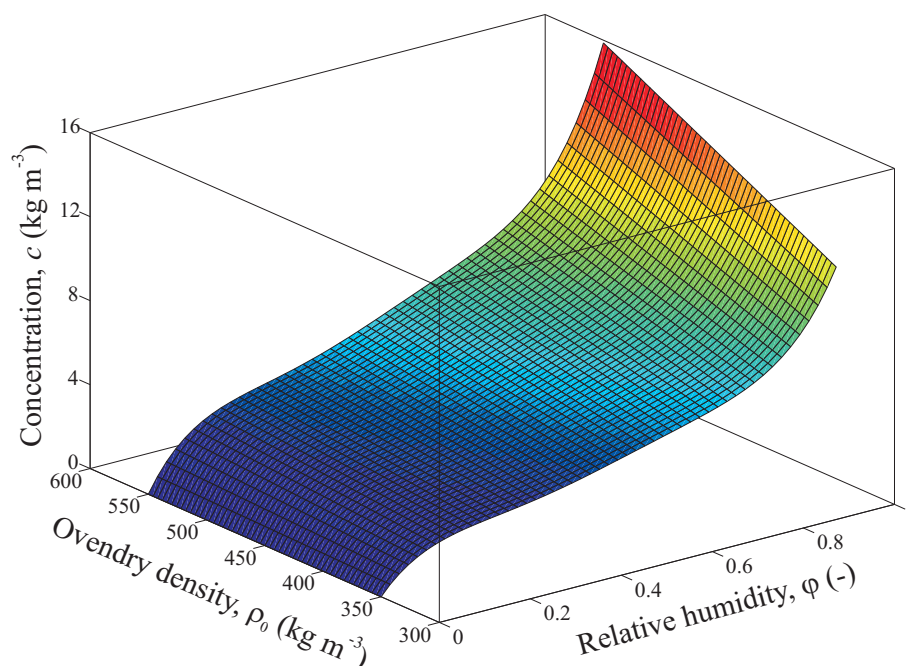


Figure 3.4 Concentration as a function of ovendry density and relative humidity in Norway spruce at 20°C

Despite all accuracy during the execution of the measurements of sorption isotherms, the results may be associated with errors. These errors may be due to a manipulation of the moisture content as a result of the timespan between removing the specimens out of the climate chamber or the oven and the weighing. Increasing standard deviations with increasing relative humidity may point to this systematic error (Tab. 3.1). Specimens with higher moisture contents are rather influenced by inaccurate measurements than specimens with low moisture contents.

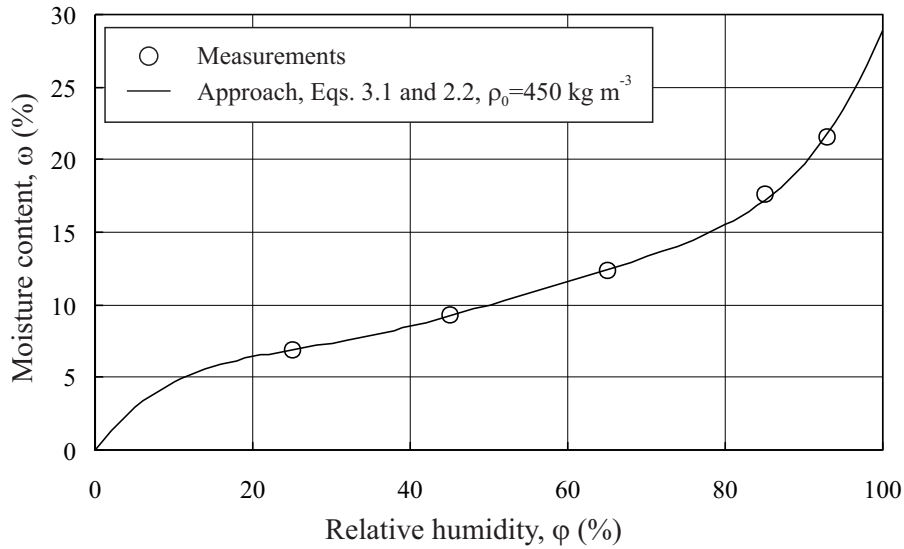


Figure 3.5 Sorption isotherm of Norway spruce at 20°C calibrated to the measurements

3.3 Moisture profiles of laminated solid wood

3.3.1 Material and method

The test set-up was made according to the standard DIN EN ISO 12572 [23]. Cylindrical specimens with vertical oriented annual rings (i.e. tangential diffusion direction) were used. Three layers, each 10mm thick, were assembled by applying 1K PUR adhesive. After turning, these specimens measured 30mm in thickness and 140mm in diameter. They were conditioned before and after gluing at $\phi = 65\%$ until moisture equilibrium was achieved. Then, the samples were fastened with rubber sleeves on cups, which were filled with distilled water. The sleeves prevented the specimens from moisture sorption at the edges. A climate gradient of 65/100% RH was applied by placing the cups in a climate room. After 14, 21, 28 and 170 days, two rectangular samples with a dimension of 40mm $\times$ 80mm were cut from the centre of each specimen (Fig. 3.6). Subsequently, they were split into layers of 5mm thickness (slices 1-6 in Fig. 3.6), whose moisture content was determined by drying. According to Teischinger and Vanek [109] two methods are suited to produce layers of that thickness: cutting with a band saw and splitting. With a thin saw blade, the cutting method provides more exact layer thicknesses. However, it may affect the moisture content by heating due to friction. Thus, the splitting method was chosen here.

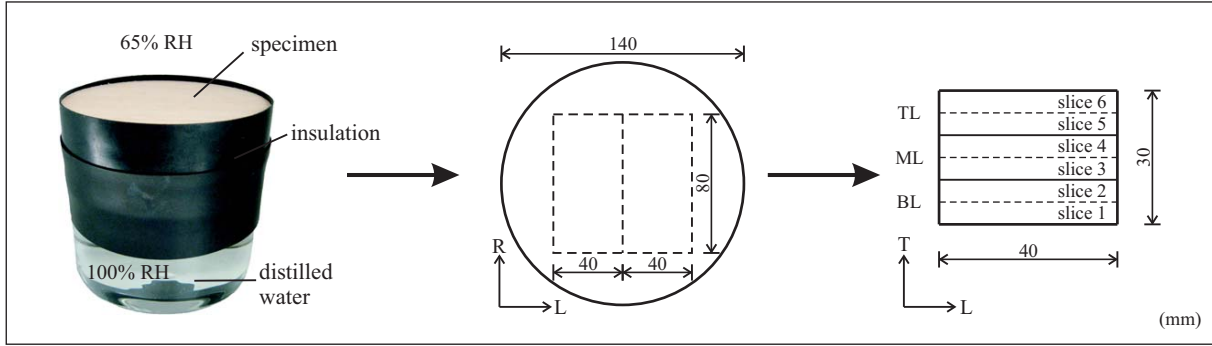


Figure 3.6 Procedure of the determination of moisture profiles, from left to right: test set-up, cutting scheme in the cylindrical sample, splitting into six layers

3.3.2 Results and discussion

The slices were checked optically after the measurements. Slices, where the layers were not separated accurately in the glue lines, were excluded. Altogether, the moisture contents of 72% of the slices were included in the analysis (Tab. 3.2). The induced climate gradient led to distinctive moisture profiles as it is shown in Tab. 3.2 and Fig. 3.7. The moisture content of the top layer (TL) sharply increased and did not change very much from day 14 to day 170. In contrast, the middle layer (ML) and the bottom layer (BL) showed a rapid increase in moisture during the first 14 days and also a significant increase in moisture from day 14 to 170. This is because the moisture penetrated into the samples from the bottom face.

Table 3.2 Moisture contents in each layer (in %), $\bar{w}$ - mean, STD - standard deviation, n - number of slices included in the analysis

Time (d)	Top layer (TL)			Middle layer (ML)			Bottom layer (BL)		
	$\bar{w}$	STD	n	$\bar{w}$	STD	n	$\bar{w}$	STD	n
14	13.2	0.30	10	15.8	0.72	7	25.3	1.38	9
21	13.8	0.31	9	16.6	0.58	5	26.6	0.99	8
28	13.9	0.39	10	17.1	1.13	7	27.2	0.64	9
170	14.3	0.37	10	19.2	0.54	9	29.0	0.68	11

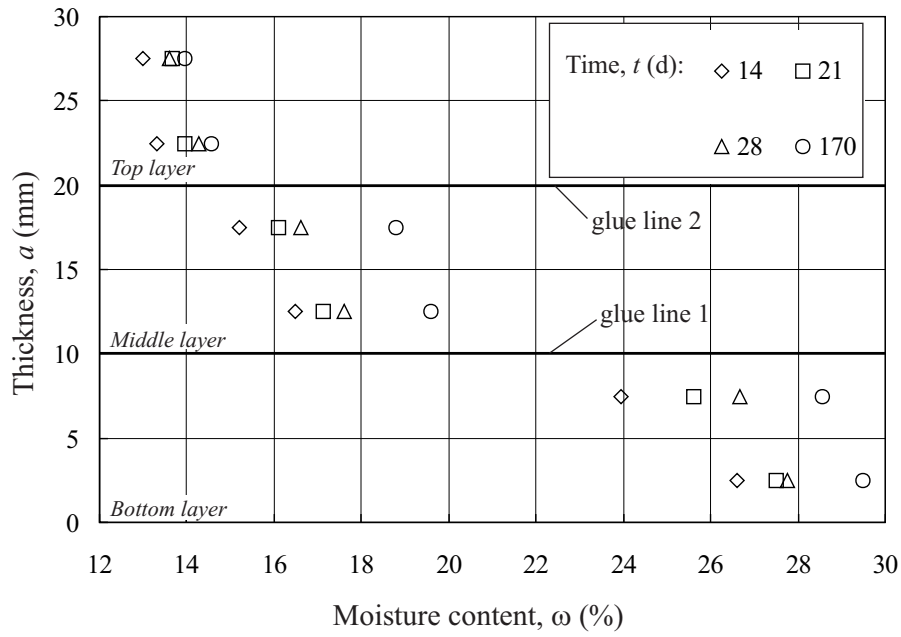


Figure 3.7 Measured wood moisture contents over panel thickness, climate conditions: $\varphi = 100\%$ at $a = 0\text{mm}$ and $\varphi = 65\%$ at $a = 30\text{mm}$, bold lines indicate the position of the glue lines

The influence of the glue lines is obvious and very significant. The largest gradient in moisture content could be detected between the bottom and the middle layer (Tab. 3.3). At day 170, the difference between the moisture contents of slices 2 (BL) and 3 (ML) was about 9% while it was 4% between slices 4 (ML) and 5 (TL). The moisture distribution is clearly not linear within the panel as it was assumed by Popper et al. [82].

Table 3.3 Absolute differences in moisture content (%) between slices 4 and 5 (ML/TL) and slices 2 and 3 (BL/ML)

Glue line	Time (d)			
	14	21	28	170
ML/TL	1.9	2.2	2.3	4.2
BL/ML	7.5	8.5	9.1	9.0

3.4 Modelling moisture transport

3.4.1 Material model

To model moisture transport in laminated wood, a five layer finite element model has been developed. Referring to the experiments, the model consists of three layers of spruce wood and two layers of adhesive. The dimensions are 10 and 0.1mm per layer, respectively. Modelling diffusion is based on the equations presented in Section 2.2, which were solved numerically using the finite element program ABAQUS. The material parameters necessary for modelling diffusion according to Fick's laws are the diffusion coefficients D_{ij} ($i = j = x, y, z$). The diffusion coefficients of Norway spruce, as it is applied in the present study, are well documented in the literature [5, 45, 94, 113, 116]. Fig. 3.8 summarizes the results obtained by some of these researchers. The diffusion coefficient of wood strongly depends on the moisture content. With increasing moisture content the resistance to diffusion decreases and, thus, D increases. Differences exist between wood species and anatomical direction. The diffusion transverse to the fibres is significantly smaller than diffusion in fibre direction. The diffusion coefficient of the adhesive layers is discussed in the next section.

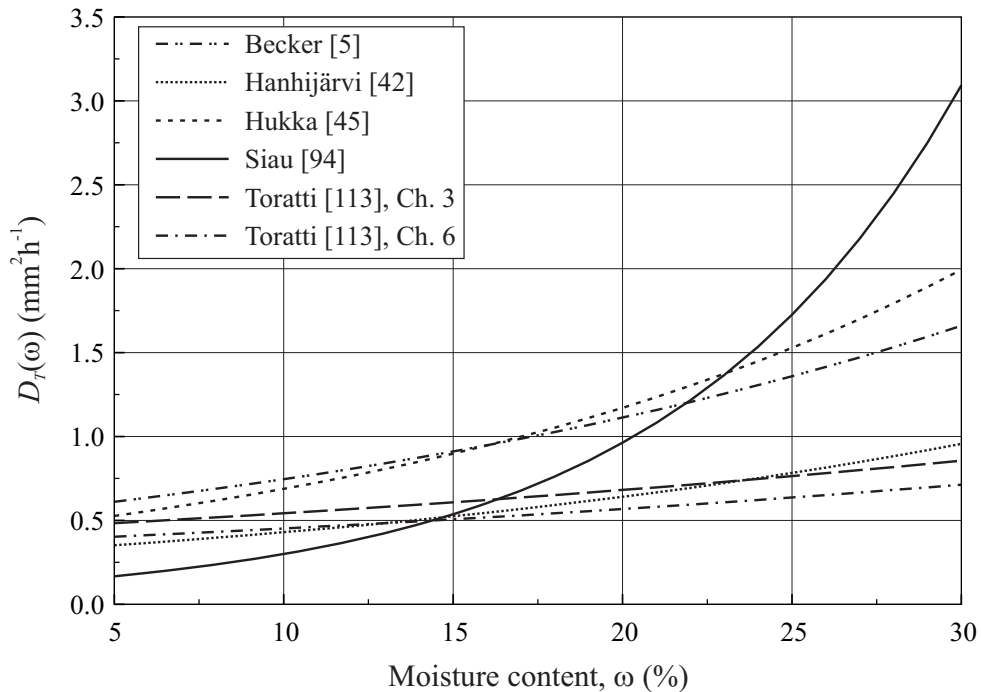


Figure 3.8 Diffusion coefficients of Norway spruce perpendicular to the grain vs. moisture content found by different authors

Due to the small ratio of radial aligned wood rays in Norway spruce, the diffusion coefficients are similar in radial and tangential direction, which is in opposition to hardwoods. They are commonly summarised in a transverse diffusion coefficient D_T , where the subscript T denotes the direction transverse to the fibres. The diffusion coefficient

$$D_T(\omega) = 0.288 e^{4\omega} \quad (3.2)$$

is assigned to the spruce layers (in $\text{mm}^2 \text{h}^{-1}$). It was found by Toratti [113], but the amount of dependency on ω was increased by Hanhijärvi [42] (compare curves in Fig. 3.8).

Due to significant Non-Fickian behaviour especially at high relative humidities (see Section 2.2), the diffusion coefficient was adapted to this phenomenon by

$$\begin{aligned} D_T(\omega) &= 0.288 e^{4\omega}, & t > t^* \\ D_T^*(\omega, t) &= D_T(\omega) \left[(1 - \kappa) \frac{t}{t^*} + \kappa \right], & t < t^* \end{aligned} \quad (3.3)$$

If $t < t^*$, the diffusion coefficient increases linearly from $\kappa D_T(\omega)$ to $D_T(\omega)$ (Fig. 3.9). This reduction of D_T in the beginning of the analysis refers to the effect that water diffuses into wood as water vapour and is then absorbed by the cell walls. During this process the diffusion slows down. Time-dependent sorption, i.e. the coupling between the two phases of water diffusion in wood, plays a significant role [29].

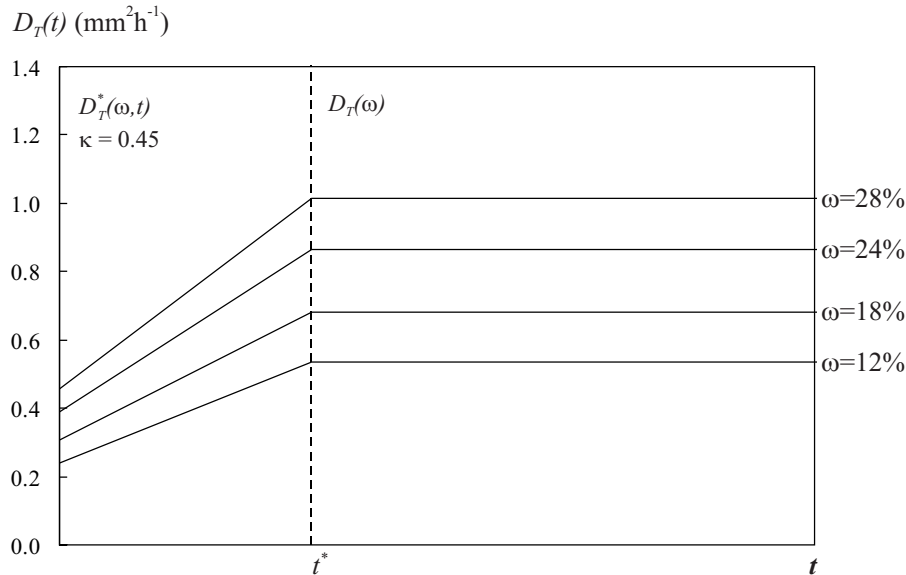


Figure 3.9 Time variation of the diffusion coefficient of wood for different moisture contents

During unsteady-state diffusion, D_T is assumed to depend not only on moisture content but also on time. The barrier between unsteady-state and steady-state diffusion is referred to as t^* . It has been determined from continuous weighing of the samples. A specific concentration $\hat{c}$ [kg m⁻³] was calculated by

$$\hat{c}(t) = \frac{m(t) - m_0}{V_0} , \quad (3.4)$$

where m_0 [kg] and V_0 [m³] are the weight and the volume of the sample at $t = 0$ (corresponds to the beginning of the experimental test, i.e. the beginning of the moisture gradient). The condition for steady-state diffusion is

$$\frac{\partial \hat{c}}{\partial t} = 0 . \quad (3.5)$$

The barrier time was determined to $t^* = 1000$ h, since concentration diversifies insignificantly afterwards (Fig. 3.10).

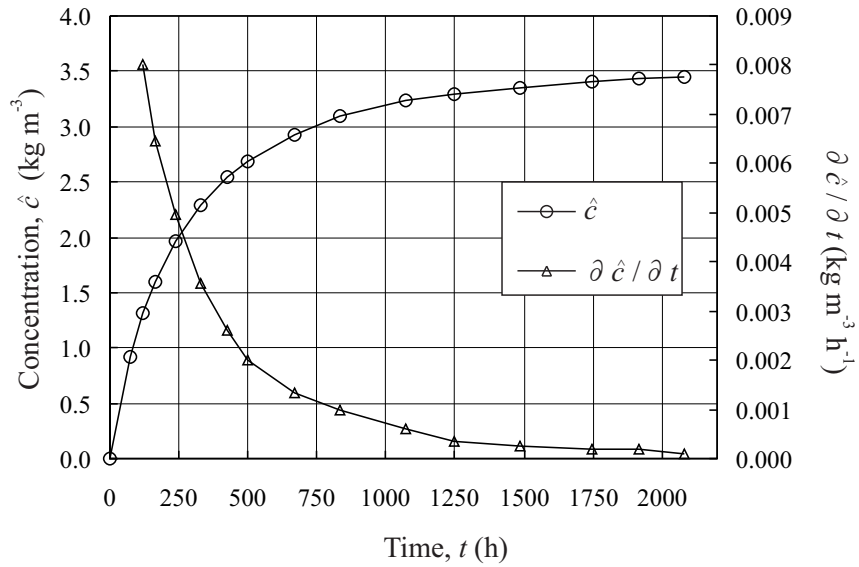


Figure 3.10 Concentration $\hat{c}$ and fractional concentration change $\partial \hat{c} / \partial t$

The air velocity was slow during the measurements, i.e. wood provided a surface resistance to sorption at the boundaries. Thus, Eq. 2.13 was applied at the boundary faces. The relative humidities of $\varphi = 65\%$ at the initial state and at the bottom face and of $\varphi = 100\%$ at the top face were converted into concentration by Eq. 3.1. Based on $\rho_{0,spruce} = 450$ kg m⁻³ the concentrations corresponding to the conditions in ambient air at the top and bottom surfaces, c_a^T and c_a^B , and the initial conditions, c_0 , are

$$\begin{aligned} c_0 = c_a^T &= 5.60 \text{ kg m}^{-3} \\ c_a^B &= 13.01 \text{ kg m}^{-3} . \end{aligned}$$

The surface-emission coefficient expressed on the basis of moisture content as the potential, h_m , is according to Hanhijärvi [42]

$$h_m(\omega) = 0.115 e^{4\omega} [\text{mm h}^{-1}] . \quad (3.6)$$

It increases in the same manner with moisture content as the diffusion coefficient of spruce (Eq. 3.2), i.e. with increasing moisture content the surface resistance to water vapour sorption decreases and, thus, the surface-emission coefficient increases. The surface-emission coefficient in terms of the concentration of moisture in wood, h_c , may be calculated from h_m as [94]

$$h_c = h_m \frac{\partial \omega}{\partial c} . \quad (3.7)$$

3.4.2 Diffusion coefficient of the adhesive

Introduction

As it was shown in the experimental tests (Section 3.3) the glue lines have a significant influence on the moisture distribution in laminated wood. It is hard to measure the diffusion coefficient of polyurethane directly, since the adhesive foams during the chemical reaction with water. Furthermore, the pure adhesive may have different characteristics than the adhesive-wood compound in a wooden panel. Forming pressure and the adhesive penetration into the wood structure are influences to account for. Thus, the diffusion coefficient was determined by the inverse method. Simulated moisture profiles were adjusted to the results obtained in the measurements (Fig. 3.7, Tab. 3.2).

Firstly, constant diffusion coefficients of the adhesive joints are tested. Secondly, it is chosen to depend linearly on the moisture content. In the end, the final formulation is presented.

Constant diffusion coefficient

In order to estimate the magnitude of the diffusion coefficient of the glue lines, two constant values are applied:

$$\begin{aligned} D_{adh,1} &= 10^{-3} \text{ mm}^2 \text{ h}^{-1} \\ D_{adh,2} &= 10^{-2} \text{ mm}^2 \text{ h}^{-1} \end{aligned}$$

These values are approximately 500 to 50 times smaller than the diffusion coefficient of spruce at $\omega = 12\%$ (Eq. 3.2).

Figs. 3.11 and 3.12 provide the results of material modelling. The history plots of the moisture contents in the three wood layers show specific characteristics (Fig. 3.11). In the top layer the curves, representing the two boundary diffusion coefficients of the adhesive, envelope the values obtained experimentally at all times. The two curves also represent the results determined at days 14, 21 and 28 in the middle and the bottom layer. However, the moisture contents at day 170 are overestimated in the middle layer and underestimated in the bottom layer by the simulations. The moisture contents calculated with $D_{adh,2}$ are higher than those calculated with $D_{adh,1}$ in the top and middle layer but lower in the bottom layer. The differences decrease with time.

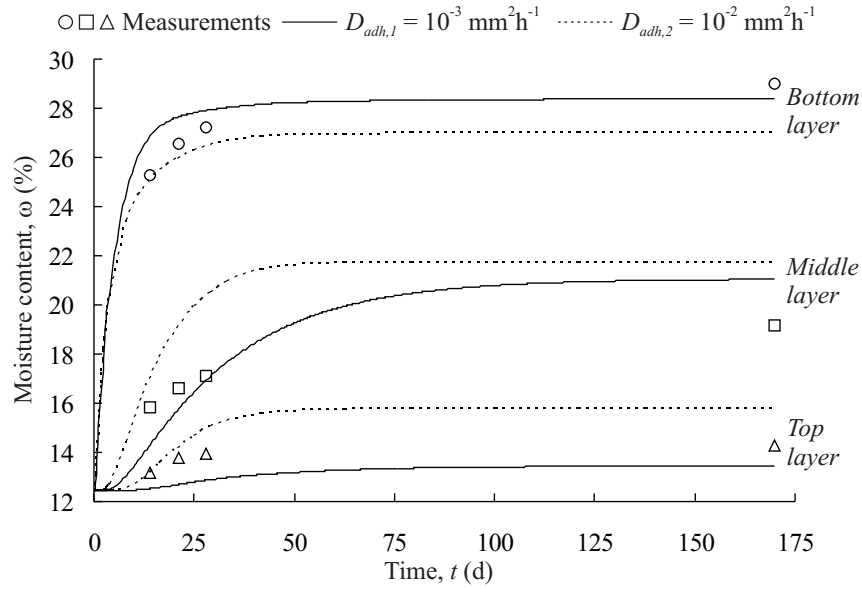


Figure 3.11 Moisture content development in the panel layers at varying constant diffusion coefficients of the glue lines compared to the measured data

The reasons are obvious in the profiles calculated at days 14 and 170 (Fig. 3.12). Increasing the diffusion coefficient of the glue lines decreases the moisture content of the bottom layer but increases the moisture content of the middle layer. This is due to the amount of water which is allowed to penetrate through the adhesive joints. The more water diffuses through the glue line (higher D_{adh}) the moister the middle layer and the drier the bottom layer. The differences between the bottom and the middle layer decrease, i.e. the water accumulation at the glue line is reduced. The influence of the upper glue line is less significant since the water diffuses from the bottom to the top.

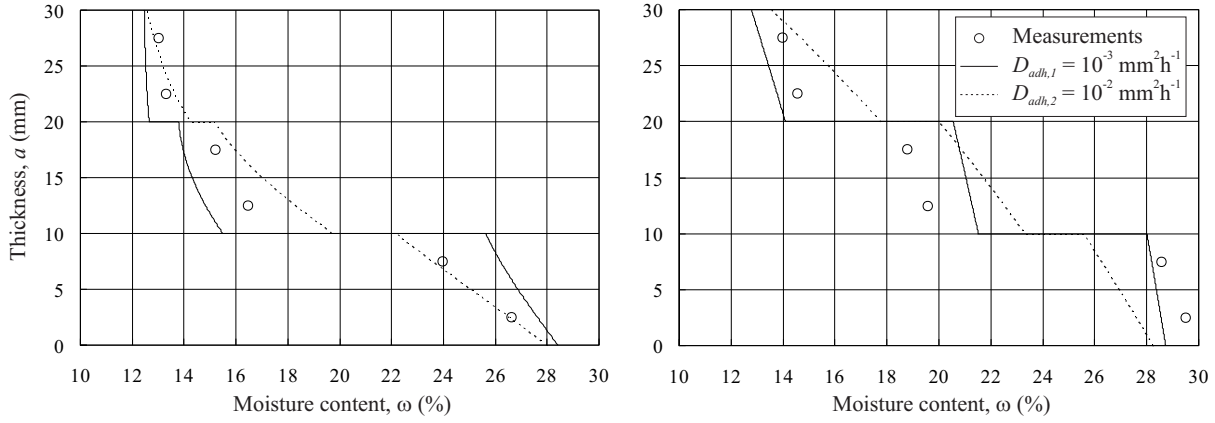


Figure 3.12 Moisture profiles at two constant D_{adh} compared to the measurements, left hand side: 14 days, right hand side: 170 days

In the bottom layer the profile simulated by means of $D_{adh,2}$ results in an excellent fit of the simulated to the measured data on day 14 but underestimates the moisture content on day 170 (Fig. 3.12). On the other hand, the profile simulated by means of $D_{adh,1}$ better correlates with the measured data on day 170, but the correlation to the experiments at day 14 is worse.

On day 14 the simulation on the one hand underestimates the experiments ($D_{adh,1}$) and on the other hand overestimates the experiments ($D_{adh,2}$). On day 170, however, the moisture content in the middle layer is overestimated by both simulations. Here the discrepancy between experiment and simulation is more conspicuous when $D_{adh,2}$ is applied.

In the top layer the fit to the data on day 14 is better when $D_{adh,2}$ is used. On day 170 both simulations do not accurately estimate the experiments.

Concentration dependent diffusion coefficient

It is assumed that water diffuses through the adhesive as water vapour. According to Siau [94] the diffusion coefficient of water-vapour diffusion in wood decreases with increased moisture content and, thus, concentration. This effect has also been observed in fibre- and particleboard, where water-vapour diffusion is assumed to be the governing process [31, 125]. The water-vapour diffusion coefficient of air in the lumens D_v [$\text{m}^2 \text{s}^{-1}$] was derived by Siau [94] to

$$D_v = \frac{0.018 D_a p_0}{G_\omega^\omega \rho_w R T} \frac{\partial \varphi}{\partial \omega} . \quad (3.8)$$

D_a [m^2s^{-1}] is the coefficient of interdiffusion of water vapour in bulk air where the subscript indicates concentration in air. p_0 [Pa] is the saturated vapour pressure, G_ω^w [-] is the specific gravity of the cell wall at moisture content ω , ρ_w is the normal density of water [1000 kg m^{-3}], R [$\text{Pa m}^3 \text{ mol}^{-1} \text{ K}^{-1}$] is the specific gas constant and T [K] is the temperature. $\partial\varphi/\partial\omega$ represents the inverse slope of the sorption isotherm. The effect that water-vapour diffusion slows down at higher moisture contents occurs beyond the first inflection point of the sorption curve. Here the ratio $\partial\omega/\partial\varphi$ increases rapidly with moisture content as it is evident in Fig. 3.5.

Two linear functions, F_1 and F_2 , of the diffusion coefficient dependence on concentration $D_{adh}(c)$ [$\text{mm}^2 \text{ h}^{-1}$] were tested:

$$\begin{aligned} F_1 : D_{adh}(c) &= 1.0 \cdot 10^{-1}c + 1.5 \cdot 10^{-2} \\ F_2 : D_{adh}(c) &= 5.0 \cdot 10^{-2}c + 8.0 \cdot 10^{-3} \end{aligned} \quad (3.9)$$

Both linear slopes are plotted in Fig. 3.13. They remain between the constant diffusion coefficients investigated in the previous section, but D_{adh} now sharply decreases with concentration.

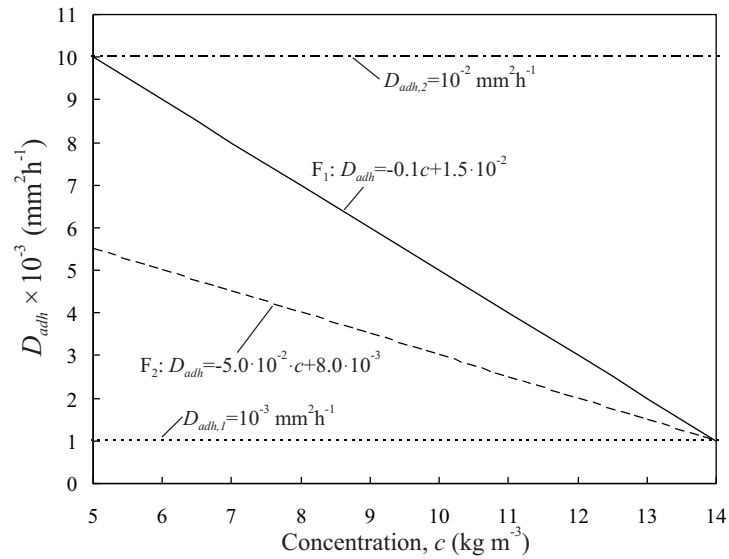


Figure 3.13 Functions F_1 and F_2 (Eq. 3.9) tested in the simulations

The simulated moisture content development with time is compared to the measurements in Fig. 3.14. Both parameter sets of the linear function between D_{adh} and concentration, F_1 and F_2 , conform with the measured moisture contents in the top layer, whereas F_2 is closer to the measurements than F_1 . In the middle layer, the simulated curves exceed the measured data above day 14. Again, parameter set F_2 better matches the measurements.

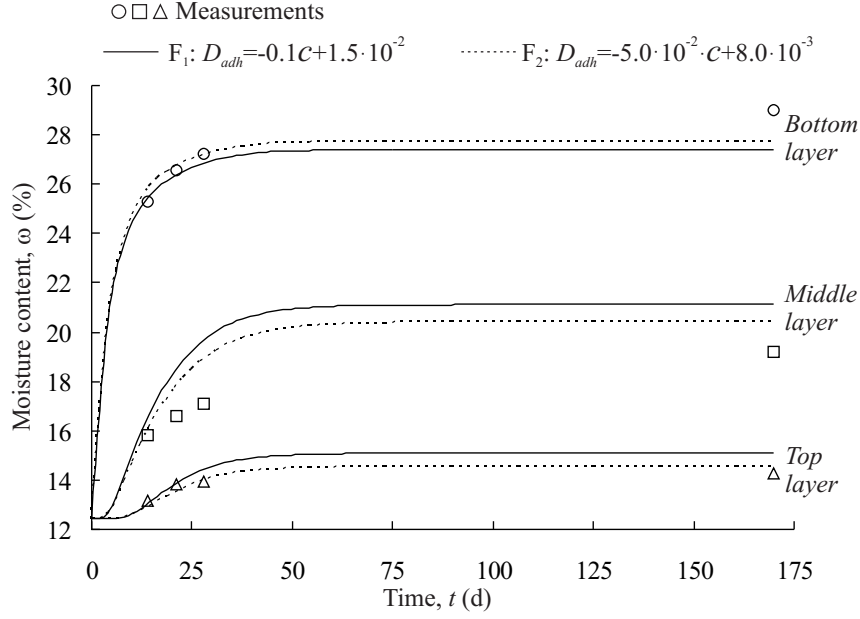


Figure 3.14 Moisture content development in the panel layers at varying linear functions of the diffusion coefficient of the adhesive compared to the measured data

In the bottom layer simulations excellently reflect the measured moisture contents at $14\text{d} < t < 28\text{d}$. However, the simulations underestimate the measurements at day 170.

A closer look at the moisture distributions within the layers reveals the excellent agreement between simulated and measured moisture profiles at day 14 (Fig. 3.15). As discussed before, higher D_{adh} (as it is the case in function F_1) lead to less significant offsets between the moisture contents on both sides of the glue line. This explains the differences between both parameter sets at the lower glue line. The simulated profiles at $t = 170\text{d}$ deviate from the results obtained at $t = 14\text{d}$. Here, the simulations clearly do not predict the measurements at least in the bottom and the middle layer. Altogether, the fitting of the simulation to the measured data needs more adaptations.

3.4.3 Results of material modelling

As a result of the previous findings, the formulation of the $D_{adh} - c$ relationship was modified to

$$D_{adh}(c) = \frac{A_1}{c^{A_2}} + A_3 . \quad (3.10)$$

This approach offers more possibilities on manipulating the shape of the function. In an optimisation process, the error between the real physical (experimental) data and the

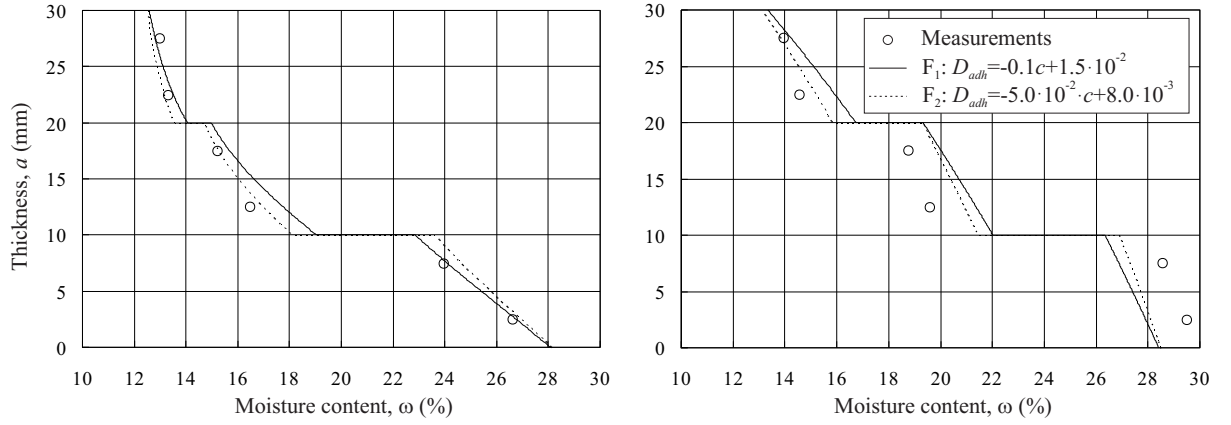


Figure 3.15 Moisture profiles simulated by means of two different linear $D_{adh} - c$ relationships compared to the measurements, left hand side: 14 days, right hand side: 170 days

numerical approximation was minimised. The shape factors were determined to (Fig. 3.16)

$$A_1 = 3.3 \cdot 10^{-3} \text{ mm}^2 \text{ h}^{-1}, \quad A_2 = 0.51 [-], \quad A_3 = -8.6 \cdot 10^{-3} \text{ mm}^2 \text{ h}^{-1}.$$

These parameters are an acceptable compromise, reflecting the measured moisture contents in all layers at all times considered.

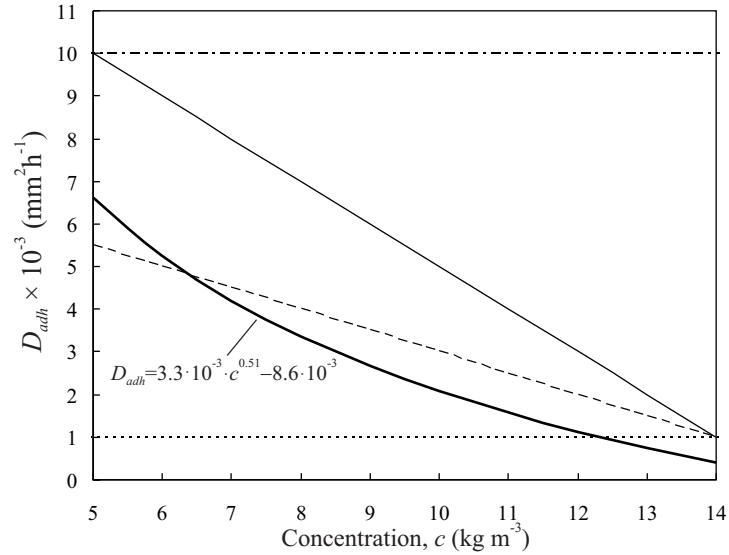


Figure 3.16 Final proposed dependency of D_{adh} on c

The results of material modelling are displayed in Figs. 3.17 - 3.19. The relative error between the experimental and the numerical data was minimised to values between -4%

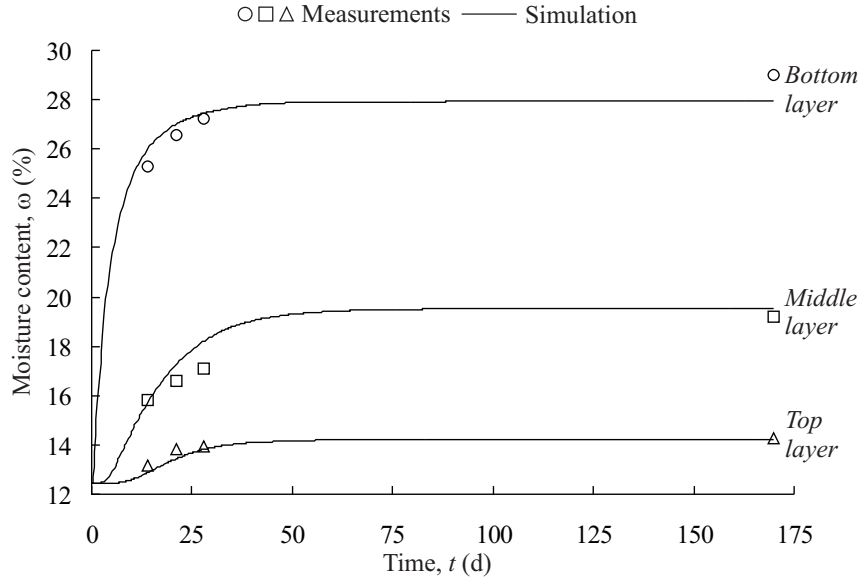


Figure 3.17 Results of material modelling applying Eq. 3.10

at $t = 170\text{d}$ in the bottom layer and $+6.2\%$ at $t = 28\text{d}$ in the middle layer. The difference is largest where the standard deviation of the measured data is largest (Tab. 3.2). This, despite all accuracy in execution of the experiments, may be due to errors in the measurements. Again, systematic errors due to a manipulation of the moisture content as a result of the timespan between splitting the samples or removing the specimens out of the oven and the weighing may be occurred. Increasing standard deviations with increasing moisture content point to this error (Tab. 3.2). Due to high moisture contents mainly the bottom layer may be affected by this error. Discrepancies between measured and calculated data may also be due to the applied moisture model. The swelling of wood, which is neglected in the mass balance, may affect the results. The assumption of a 0.1mm thick and planar layer of adhesive between two spruce layers is only a simplification of the reality.

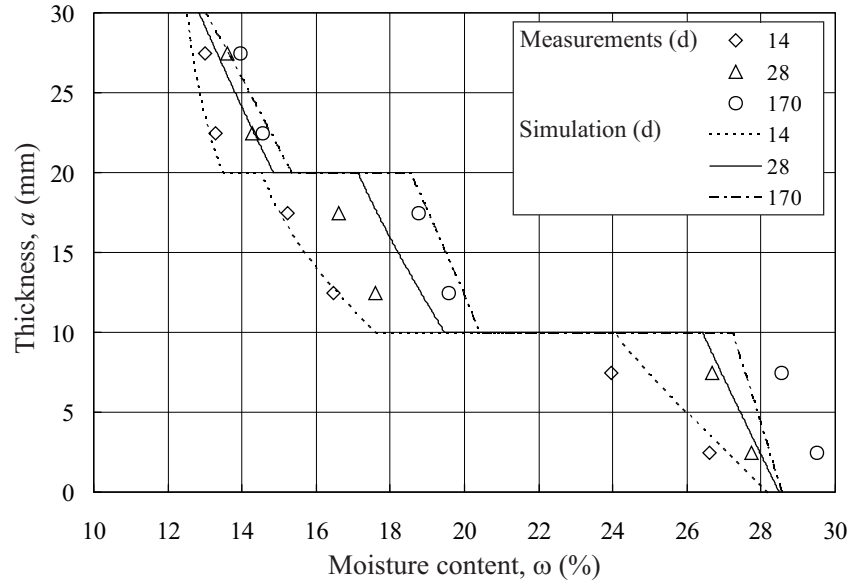


Figure 3.18 Simulated moisture profiles applying Eq. 3.10

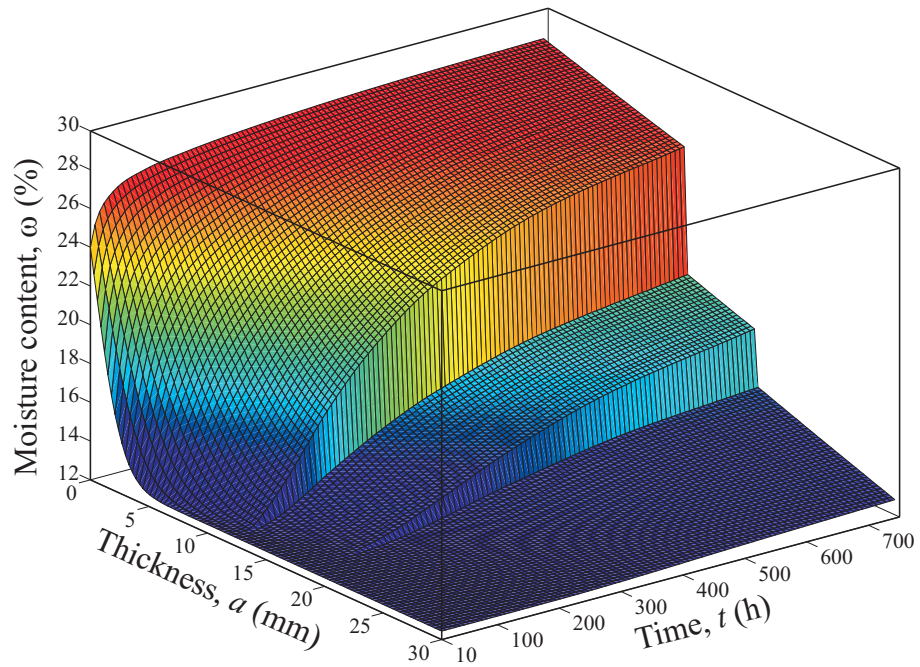


Figure 3.19 Simulated history development of moisture content within the panel, $a = 0\text{mm}$ refers to $\varphi = 100\%$ and $a = 30\text{mm}$ refers to $\varphi = 65\%$

Tab. 3.4 summarises the material modelling of moisture diffusion in three-layered laminated solid wood panels under the climatic conditions described in Section 3.3.

Table 3.4 Summary of the moisture model

Wood	$\rho_0 = 450 \text{ kg m}^{-3}$ $D_T(\omega) = 0.288 e^{4\omega}, t > t^* [\text{mm}^2 \text{ h}^{-1}] \quad (3.3)$ $D_T^*(\omega, t) = D_T(\omega) \left[(1 - \kappa) \frac{t}{t^*} + \kappa \right], t < t^* \quad (3.3)$ $\kappa = 0.45 [-]$ $t^* = 1000\text{h}$
Adhesive	$\rho_0 = 1000 \text{ kg m}^{-3}$ $D_{adh}(c) = \frac{A_1}{c^{A_2}} + A_3 \quad (3.10)$ $A_1 = 3.3 \cdot 10^{-3} \text{ mm}^2 \text{ h}^{-1}$ $A_2 = 0.51 [-]$ $A_3 = -8.6 \cdot 10^{-3} \text{ mm}^2 \text{ h}^{-1}$
Boundaries	initial: $c_0 = 5.60 \text{ kg m}^{-3}$ $nJ_m = h_c (c_a - c_s) \quad (2.13)$ $h_m(\omega) = 0.115 e^{4\omega} [\text{mm h}^{-1}] \quad (3.6)$ $h_c = h_m \frac{\partial \omega}{\partial c} \quad (3.7)$ top face: $c_a^T = 5.60 \text{ kg m}^{-3}$ bottom face: $c_a^B = 13.01 \text{ kg m}^{-3}$

3.5 Modelling spruce-MDF laminates

In order to prepare for the investigation of warping in spruce-MDF laminates (i.e. spruce in the outer layers assembled with an MDF middle layer), the moisture distribution is required. The five layer model was adapted to diffusion in composite laminates. The layers in particular are (from bottom to top):

- spruce ($a_1 = 10\text{mm}$),
- adhesive ($a_2 = 0.1\text{mm}$),
- MDF ($a_3 = 16\text{mm}$),

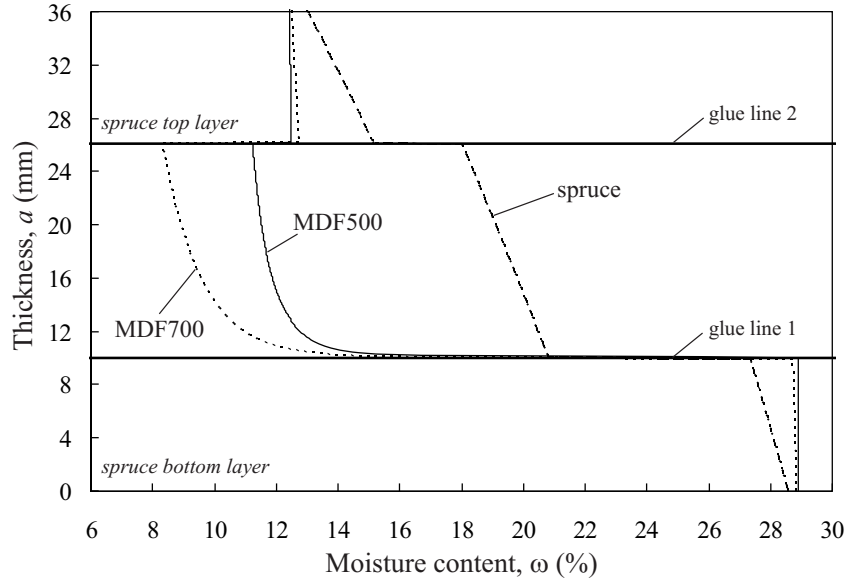


Figure 3.20 Simulated moisture profiles of two spruce-MDF laminates, $t = 170d$

- adhesive ($a_4 = 0.1\text{mm}$) and
- spruce ($a_5 = 10\text{mm}$),

which finally creates panels with a total thickness of

$$a_{tot} = \sum_{n=1}^5 a_n = 36.2\text{mm} .$$

The material parameters of both the wood and the adhesive layers were derived in the previous section and are listed in Tab. 3.4. The diffusion coefficients found by Ganey et al. [31] were assigned to the MDF middle layer (Eq. 2.35). Two types of MDF were tested: MDF500 with $\rho_0 = 500 \text{ kg m}^{-3}$ and MDF700 with $\rho_0 = 700 \text{ kg m}^{-3}$.

The simulated moisture profiles reveal differences between the two types of MDF middle layers as it is displayed in Fig. 3.20. The MDF500 middle layer is significantly moister than the MDF700 middle layer. On the one hand, this is caused by the differences in the diffusion velocity, which is quantified by the diffusion coefficients. They deviate between both materials, MDF500 and MDF700, as it is displayed in Fig. 3.21. On the other hand, the differences in the moisture content of the middle layers are caused by the differences in the equilibrium moisture content. These strongly depend on the density (Eq. 2.2), which differ between the two types of MDF. Furthermore, the sorption curve of MDF significantly differs from that of spruce. The reasons are the different densities of MDF and spruce, the application of an adhesive system in wood-based panels and the

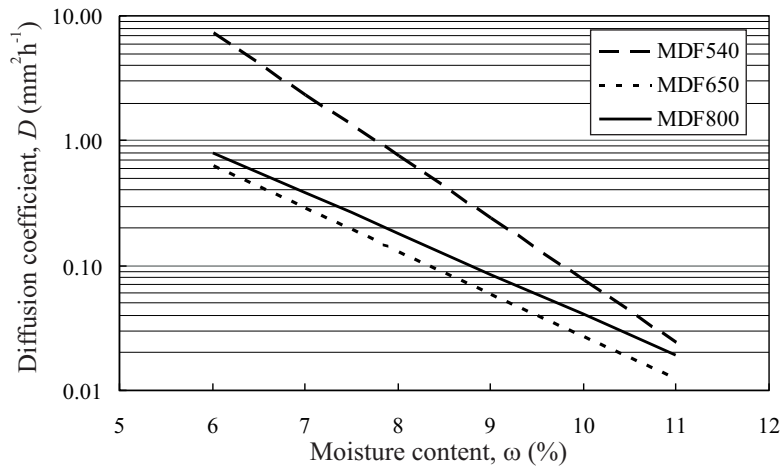


Figure 3.21 Dependency of the diffusion coefficient of MDF on moisture content and density (Eq. 2.35), Ganev et al. [31]

production process. The adhesive blocks sorption sites of the wood material, resulting in lower moisture content of the wood based panel compared to solid wood in the same climatic conditions. This results in significant moisture spruce middle layers compared to the MDF middle layers in laminated panels. The drying of the wood fibres during the production of MDF also results in lower equilibrium moisture contents similar to thermally treated wood.

The application of an MDF middle layer also significantly affects the spruce outer layers even when they have equal thicknesses (Fig. 3.22). Compared to panels consisting of three layers of spruce, the bottom layer is moister and the top layer remains drier in spruce-MDF laminates. This is mainly the result of the lower diffusion coefficient of MDF. They significantly differ between both materials, spruce and MDF (compare Eqs. 2.35 and 3.2). The diffusion in the MDF middle layers slowed down. This leads to more moisture in the bottom layer (water accumulation) and to less moisture in the top layer.

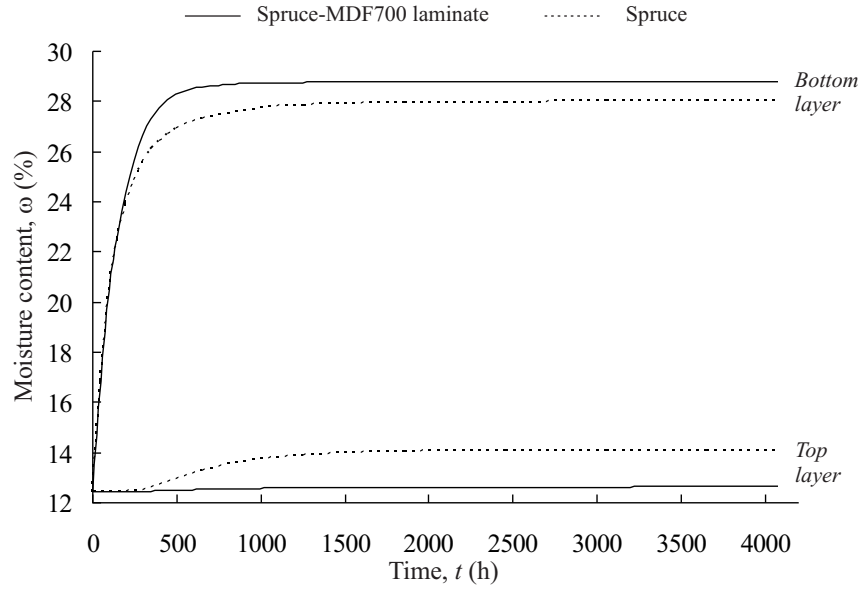


Figure 3.22 Moisture content development in the spruce top and bottom layers, comparison between solid spruce panel and spruce-MDF700 laminate, both with layer thicknesses of 10/16/10

3.6 Concluding remarks

This chapter describes the moisture transport through three-layered laminated wood panels by means of experimental tests and numerical simulations. Sorption measurements at $T = 20^{\circ}\text{C}$ of clear spruce specimens, laminated and cross-laminated spruce samples are provided. It was shown that only small differences between spruce and cross-laminated samples exist, which is in opposition to former investigations. Thus, the results of clear wood samples were analysed and influences on sorption, such as stresses, adhesive and production parameters, are not further considered. The stress and moisture fields are not coupled.

From measured moisture profiles of laminated wood samples, it can be concluded that the glue lines provide a high resistance to moisture diffusion. The coefficient of diffusion of 1K PUR glue lines was evaluated from the measurements. The diffusion through glue lines was found to depend on the concentration of moisture. The diffusion coefficient of the adhesive decreases with increased concentration. Thus, it was concluded that moisture transport through glue lines is a process governed by water-vapour diffusion.

Finally, moisture profiles of spruce-MDF laminates were calculated by means of the moisture transport model. The diffusion in MDF is governed by the density of the material.

It influences the diffusion coefficient and the equilibrium moisture content. Differences in the outer layer moisture contents were concluded to result from different middle layer materials.

Chapter 4

Hygroscopic warping

4.1 Introduction

Dimensional stability, i.e. the ability to resist warping, is of main interest for the application of wood-panels. The cross lamination of the layers is advantageous to warping, however, in large-scale panels warping was also observed. This, in practice reduces the serviceability. Thus, the warp deformation is investigated in the present study. Due to a climate gradient, considerable distortions (warp) in the form of cup and twist may occur. The types of warp are shown in Fig. 4.1. The cup deformation may occur in two directions, the xz - and the yz -plane. In boards, this is often termed cup and bow deformation.

Experimental tests are completed by numerical studies. Cross-laminated solid wood panels and composite laminates were exposed to a specific climate gradient. Influences such as the annual ring orientation in the individual layers, the layer ratio, pre-stressed outer layers and the middle layer structure are questioned in the experimental part. The material data for the mechanical model and results of mechanical material modelling are presented in this chapter. The findings on the moisture distribution and the history moisture variation as obtained in the previous chapter are applied as input to the mechanical model. The numerical simulations are validated against the experimental results and different constitutive laws are compared. The dependency of warp on adhesive properties such as element type, stiffness and diffusivity is studied by numerical simulations. Finally, simulations of spruce-MDF laminates are compared to experimental results.

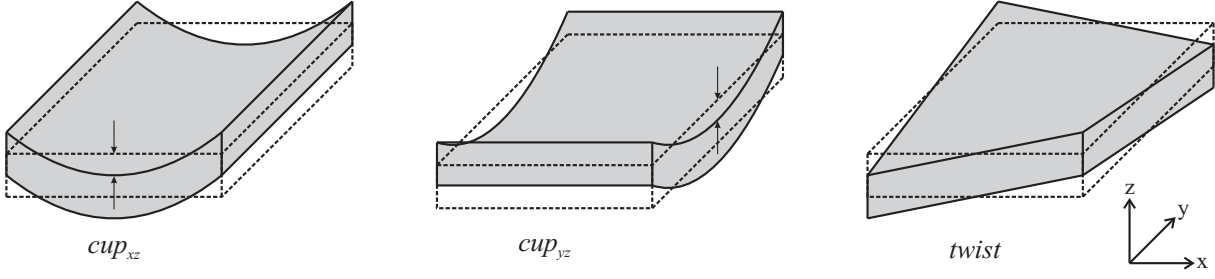


Figure 4.1 The types of warp deformation

4.2 Experimental tests

4.2.1 Material

In Tab. 4.1, the thicknesses and growth ring orientations of the panels and the individual layers are shown. The panels were made of Norway spruce and assembled either in the laboratory (indicated as type A in Tab. 4.1) or assembled by an industrial manufacturer (indicated as type B). In panels of production type A, firstly, lamellas measuring 100mm in width were sawn from spruce logs. Afterwards, the boards were glued at the edges. After planing to the final thickness, the layer-to-layer surfaces were glued using 1K PUR. Before and after the gluing the material was conditioned at 65% RH until equilibrium was reached. Type B panels were industrially produced using UF. The middle layers were sawn from wooden blocks that were glued together previously. The lamellas of both, the middle layer and the outer layers, measured 26 ± 1 mm in width. Altogether, the layer thicknesses ranged from 7mm to 10mm in the outer layers and 10mm to 24mm in the middle layer. The total thicknesses varied between 27mm and 38mm. Four groups of solid wood specimens were tested (Tab. 4.1).

The group of specimens indicated by AR refers to experiments where the influence of the annual ring orientation θ was studied. In opposition to the definition of the ring angle ψ_3 in Bodig and Jayne [8] (Fig. 2.6b), the angle θ is defined between the x_2 -axis of the lamella and the tangential axis of the orthotropic coordinate system, thus,

$$\theta = 90^\circ - \psi_3 . \quad (4.1)$$

In global panel coordinates (x, y, z) , θ is defined as the angle between the tangential axis and the horizontal coordinate of the geometric coordinate system (Fig. 4.2), in particular

$$\theta = \angle(x; T), \quad 0^\circ \leq \theta \leq 90^\circ$$

Table 4.1 Solid wood specimens (Norway spruce)

Group	ID	Prod. ¹	a_{OL} ²	a_{ML} ²	a_{tot} ²	θ ³	LR ⁴
			(mm)	(mm)	(mm)	(°)	(-)
Annual ring orientation	AR00	A	10	10	30	0	0.67
	AR45	A	10	10	30	45	0.67
	AR90	A	10	10	30	90	0.67
Layer ratio	LR37	B	7	24	38	60-90	0.37
	LR52	B	7	13	27	60-90	0.52
	LR57	A	10	15	35	60-90	0.57
Pre-stress	PS06	B	7	13	27	60-90	0.52
	PS12	B	7	13	27	60-90	0.52
	PS18	B	7	13	27	60-90	0.52
Slits	SWS	A	10	15	35	60-90	0.57

¹ Production: A - laboratory, 1K PUR; B - industry, UF

² a - thickness, OL - outer layers, ML - middle layer, tot - total

³ θ - angle between glue line and tangential direction of the growth ring, defined in Fig. 4.2

⁴ Layer ratio $LR = 2a_{OL}/a_{tot}$

in the middle layer and

$$\theta = \angle(y; T), \quad 0^\circ \leq \theta \leq 90^\circ$$

in the outer layers. Three orientations were tested: 0° , 45° and 90° . $\theta = 0^\circ$ describes horizontally oriented annual rings and $\theta = 90^\circ$ describes vertically oriented annual rings.

The group of specimens indicated by LR relates to testing where the influence of the layer ratio LR on warping was studied. The layer ratio is defined as

$$LR = \frac{2a_{OL}}{a_{tot}}, \quad (4.2)$$

where a_{OL} is the thickness of one outer layer (OL) and $a_{tot} = 2a_{OL} + a_{ML}$ is the total panel thickness. The layer ratio varied between 0.37 and 0.67.

A possibility in reducing hygroscopic deformations and stresses in three-layered cross-laminated panels may be the application of pre-stressed outer layers. This influence was

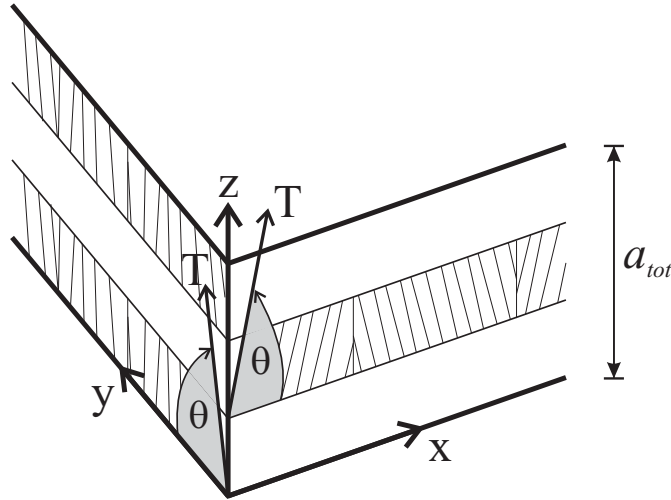


Figure 4.2 Definition of the angle θ

investigated on panels indicated by PS. Three levels of pre-stress were tested: 0.6 N mm^{-2} (PS06), 1.2 N mm^{-2} (PS12) and 1.8 N mm^{-2} (PS18). The panels were industrially manufactured. Pre-stresses were applied by means of mechanical compression perpendicular to the grain during gluing. These previously induced compressive stresses (termed pre-stress) may reduce tensile stresses due to shrinkage and, thus, also reduce the likelihood of cracks developing in the surface layers. In terms of hygroscopic warping, pre-stresses may lead to reduced warping. The compressive stresses may counteract the tensile stresses that are perpendicular to the grain and that occur in the bottom layer during warping.

In panels indicated by SWS, slits in the middle layer were tested on their influence on hygroscopic warping. Slits measuring $2.7\text{mm} \times 11\text{mm}$ were sawn into the middle layer as shown in Fig. 4.3. They were aligned in the fibre direction and alternated from the top to the bottom side of the layer. The distance between two slits of the same edge was 20mm.

Wood-based panels in the middle layer were tested on their influence on hygroscopic warping. The application of wooden composites in the core allows for reduced production costs. Wood-based panels are characterised by a more homogeneous property distribution in the in-plane directions than wood, where the differences between properties parallel and perpendicular to the fibre alignment are very large. Spruce layers with a thickness of 10mm were assembled with three different middle layer materials: MDF, OSB and particleboard (PB) as indicated in Tab. 4.2. Two types of MDF were applied. They varied in their density: $\rho = 530 \text{ kg m}^{-3}$ (CMDF5) and $\rho = 710 \text{ kg m}^{-3}$ (CMDF7). Both remaining materials, OSB and PB, had a density in between the two types of MDF: $\rho = 620 \text{ kg}$

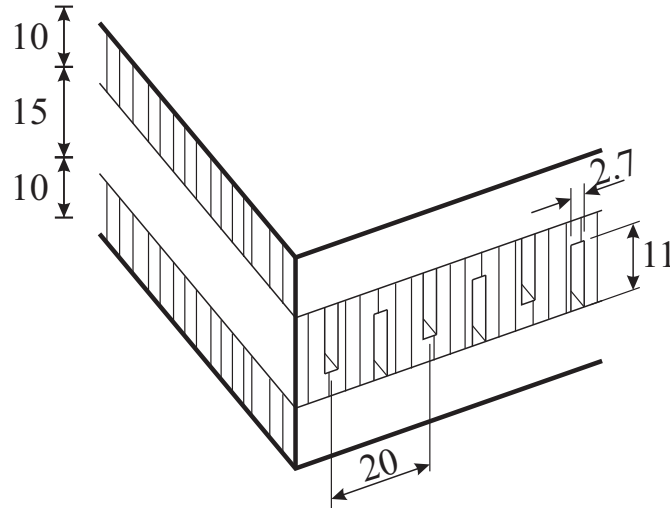


Figure 4.3 Characteristics of panels termed SWS (dimensions in mm)

m^{-3} (COSB) and $\rho = 650 \text{ kg m}^{-3}$ (CPB). The middle layers measured 15mm and 16mm in thickness, which resulted in total thicknesses of 35mm and 36mm, respectively. The individual layers were assembled in the laboratory using 1K PUR.

Table 4.2 Panel specifications of composite laminates (10mm spruce outer layers in combination with a wood-based material in the middle layer)

ID	Middle layer			a_{tot} (mm)	LR (-)
	Material	a_{ML} (mm)	ρ (kg m ⁻³)		
CMDF5	MDF	15	530	35	0.57
CMDF7	MDF	16	710	36	0.56
COSB	OSB	16	620	36	0.56
CPB	PB	15	650	35	0.57

4.2.2 Method

The hygroscopic warping of three-layered cross-laminated wood panels was determined on panels measuring 300mm $\times$ 300mm and that were initially conditioned at 65% RH. The small faces of the samples were sealed with lacquer to enforce the moisture flow within the panels in the thickness direction. The specimens were then placed in a box on three supports as shown in Fig. 4.4a and b. The contact area between the panel edges and the box top cover was insulated by a rubber joint (Fig. 4.4c). The climate difference of 65% RH and 100% RH between the upper and lower surfaces was induced. These two levels of relative humidity were obtained by means of water in the bottom of the box and a constantly conditioned climate room, in which the test set-up was stored. Relative displacements in the z-direction of the measuring points, indicated in Fig. 4.5a, were recorded by means of dial gauges placed in a steel plate (Fig. 4.4d). Stop positions guaranteed identical placement of the steel plate in every measurement. The displacement measurements were carried out on days 1, 2, 4, 7, 10, 14, 21 and 31.

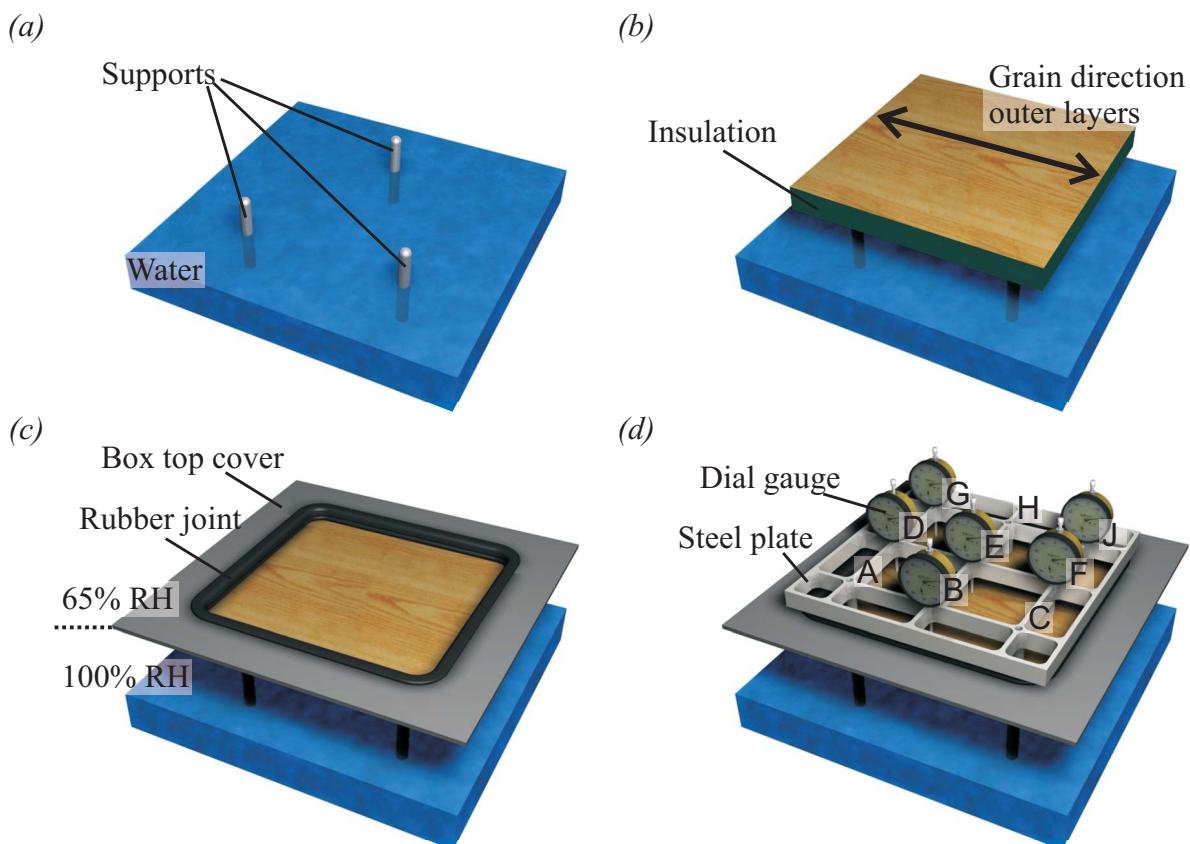


Figure 4.4 Test set-up of the hygroscopic warping experiments (box not displayed)

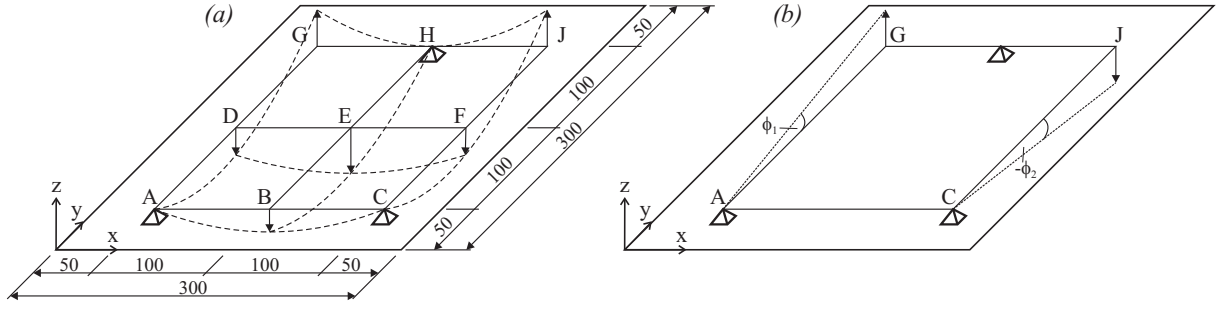


Figure 4.5 a) Measuring points

b) Definition of angles ϕ_1 and ϕ_2 for the calculation of twist deformation

The cup deformation in the xz -plane was determined as

$$cup_{xz} = \frac{1}{2} \left(\bar{u}_z^{A,D,G} + \bar{u}_z^{C,F,J} \right) - \bar{u}_z^{B,E,H} . \quad (4.3)$$

and the cup deformation in the yz -plane as

$$cup_{yz} = \frac{1}{2} \left(\bar{u}_z^{A,B,C} + \bar{u}_z^{G,H,J} \right) - \bar{u}_z^{D,E,F} \quad (4.4)$$

The notation $\bar{u}_z^{i,j,k}$ indicates the mean value of the displacements in the measuring points i , j and k as shown in Fig. 4.5a. The twist deformation was calculated by

$$\begin{aligned} twist &= |\phi_1 - \phi_2| \\ twist &= \left| \arctan \left(u_z^G / 200 \right) - \arctan \left(u_z^J / 200 \right) \right| . \end{aligned} \quad (4.5)$$

The angles ϕ_1 and ϕ_2 are shown in Fig. 4.5b.

4.2.3 Results and discussion

The results of the warping tests are listed for $t = 31d$ in Tab. 4.3. The greatest amount of warping was found in the yz -plane, since the perpendicular to the grain direction of the outer layers was in the y -direction. This resulted in a dominant swelling of the bottom layer in this direction. Due to the hindrance caused by the middle layer, the panel warped in the yz -plane. The supports may also play a significant role in the formation of warp. Due to the triangular arrangement in points A, C and H, the development of warp may be forced to occur in the yz -plane. Numerical studies, as indicated in Section 4.3, will resolve this influence. The groups of test samples significantly differed in their warp deformations. The maximum warp was obtained in composite laminates. The differences between the groups of panels are discussed below.

Table 4.3 Results of the warping experiments after 31 days as arithmetic mean (standard deviation), ID according to Tabs. 4.1 and 4.2

ID	n	cup_{xz} (mm)	cup_{yz} (mm)	$twist$ ($^{\circ}$)
AR00	6	0.06 (0.05)	0.42 (0.05)	0.11 (0.11)
AR45	6	0.06 (0.06)	0.22 (0.08)	0.11 (0.09)
AR90	6	0.10 (0.15)	0.27 (0.08)	0.06 (0.03)
LR37	6	0.11 (0.19)	0.08 (0.02)	0.04 (0.04)
LR52	6	0.07 (0.09)	0.14 (0.04)	0.05 (0.03)
LR57	3	0.04 (0.27)	0.12 (0.08)	0.06 (0.06)
SWS	3	0.17 (0.08)	0.08 (0.06)	0.11 (0.10)
PS06 ¹	4	0.14 (0.05)	0.02 (0.03)	0.06 (0.02)
PS12 ¹	4	0.05 (0.12)	0.03 (0.04)	0.05 (0.02)
PS18 ¹	4	0.02 (0.05)	0.06 (0.04)	0.04 (0.01)
CMDF5	3	0.08 (0.06)	1.15 (0.25)	0.10 (0.03)
CMDF7	3	0.15 (0.10)	0.80 (0.25)	0.04 (0.01)
COSB	3	0.19 (0.03)	0.44 (0.03)	0.03 (0.01)
CPB	3	0.13 (0.09)	1.29 (0.30)	0.03 (0.01)
¹ mean extrapolated from nonlinear regression, standard deviation calculated from results at $t = 21d$				

Compared to beech panels, as investigated in Gereke et al. [35], spruce panels showed a good dimensional stability. The maximum cup_{yz} in three-layered cross-laminated beech panels with layer thicknesses of 10mm and $LR = 0.67$ was detected to 1.14mm. The cup in spruce panels was measured to be 70% smaller (AR90).

Spruce samples reached their maximum cup deformation cup_{yz} between day 3 and day 12 of the observation period (Tab. 4.4). Series AR45, LR37 and PS18 showed no significant maximum. For AR45, cup_{yz} became constant from day 5, whereas for LR37 and PS18 a slight but constant increase of cup_{yz} was recorded during the whole testing period. The decrease in cup was governed by the differences in moisture content between the layers. As shown in Chapter 3, firstly, the moisture content of the bottom layer was found to increase rapidly. Hence, the expansion of the bottom layer, which was mainly in the y-direction, led to a strong increase in cup_{yz} . The decrease in cup_{yz} after the maximum was reached was caused by an increase in moisture content of the middle layer and later

Table 4.4 Time to maximum warp deformation t_{max} , maximum warp x_{max} and ratio of final warp (x_{31}) to maximum warp

ID	t_{max} (d)		x_{max} (mm)		x_{31}/x_{max} (-)	
	cup_{xz}	cup_{yz}	cup_{xz}	cup_{yz}	cup_{xz}	cup_{yz}
AR00	3	3	0.07	0.47	0.84	0.88
AR45	31	6	0.22	0.22	1.00	0.98
AR90	31	4	0.10	0.36	1.00	0.78
LR37	31	31	0.11	0.08	1.00	1.00
LR52	31	4	0.07	0.15	1.00	0.86
LR57	4	10	0.07	0.15	0.48	0.88
SWS	31	12	0.17	0.10	1.00	0.77
PS06	31	6	0.14	0.07	1.00	0.24
PS12	5	3	0.09	0.04	0.58	0.86
PS18	5	31	0.07	0.06	0.35	1.00
CMDF5	31	31	0.08	1.15	1.00	1.00
CMDF7	31	31	0.15	0.80	1.00	1.00
COSB	31	31	0.19	0.44	1.00	1.00
CPB	31	31	0.13	1.29	1.00	1.00

of the top layer. The moisture increase and, therefore, swelling of these two layers acted in opposition to the deformation of the bottom layer. Thus, cup_{yz} decreased and cup_{xz} increased.

In all composite laminates, warping continued to increase from the initial state to day 31. This was most probably due to the moisture profile as it was calculated exemplary for MDF in the previous section (Fig. 3.20). The wood-based panel in the middle layer acted as a moisture barrier. The moisture content of the bottom layer was significantly higher in composite laminates than in spruce panels. Thus, the swelling of the bottom layer was higher, which resulted in a large cup_{yz} . The mechanical properties of the middle layers may also play a significant role. This is discussed below.

The cup deformation, cup_{xz} , as well as the twist deformation mostly showed no maximum. They increased from the initial state to the final measurement. cup_{xz} in panels AR00, LR57, PS12 and PS18 developed a maximum between day 3 and day 5 and significantly

decreased afterwards. However, the variation between different samples of one series was high (Tab. 4.3), which does not allow for a closer examination.

The maximum cup_{xz} was obtained in a panel of series LR37 (Fig. 4.6). This was most probably due to the little layer ratio. The lower the layer ratio, the lower was cup_{yz} and the higher was cup_{xz} . cup_{yz} was found to be maximum in a panel of type CPB. The different behaviours of panels with both, dominant cup in the xz-plane and dominant cup in the yz-plane, are clearly seen in Fig. 4.6. The largest *twist* was found in a panel of type AR00. The displacement of point G is aligned upwards while the displacement of point J is aligned downwards, which led to a high *twist*. This may be due to the cylindrical shape and the arrangement of the annual rings in the 100mm wide lamellas.

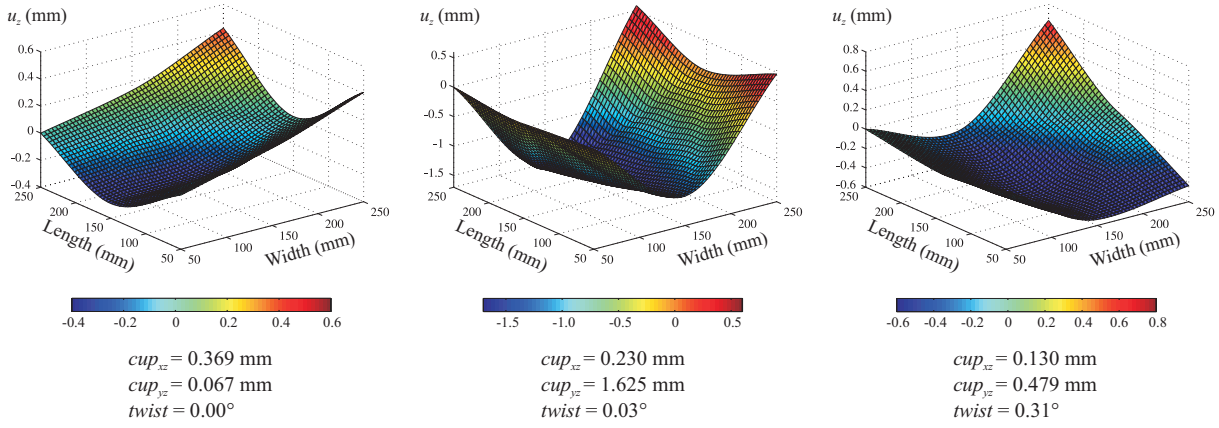


Figure 4.6 Hygroscopic warping of different panels after 31 days in the climate difference 65/100% RH

Left: maximum cup_{xz} in a panel of type LR37

Centre: maximum cup_{yz} in a panel of type CPB

Right: maximum *twist* in a panel of type AR00

Influence of the annual ring orientation

Fig. 4.7 shows the history plots of cup_{xz} , cup_{yz} and $twist$ for panels with different annual ring orientations in the lamellas: AR00 ($\theta = 0^\circ$), AR45 ($\theta = 45^\circ$) and AR90 ($\theta = 90^\circ$). The influence of the annual ring orientation, θ , on the warp deformation was conspicuous. It was governed by the difference in swelling, stiffness properties and material orientation. The largest cup_{yz} was found for panels with horizontal annual rings (AR00) reflecting the large swelling in the bottom layer. Tangential swelling is about two times higher than radial swelling as it occurred in series AR90. The smallest cupping, cup_{yz} , was recorded for the panels with $\theta = 45^\circ$, which most probably is due to the very small stiffness of softwood in that direction. It is about one half of the tangential stiffness and about one quarter of the radial stiffness (see Section 2.3). Significant differences in cup_{xz} and $twist$ could not be detected for different material orientations.

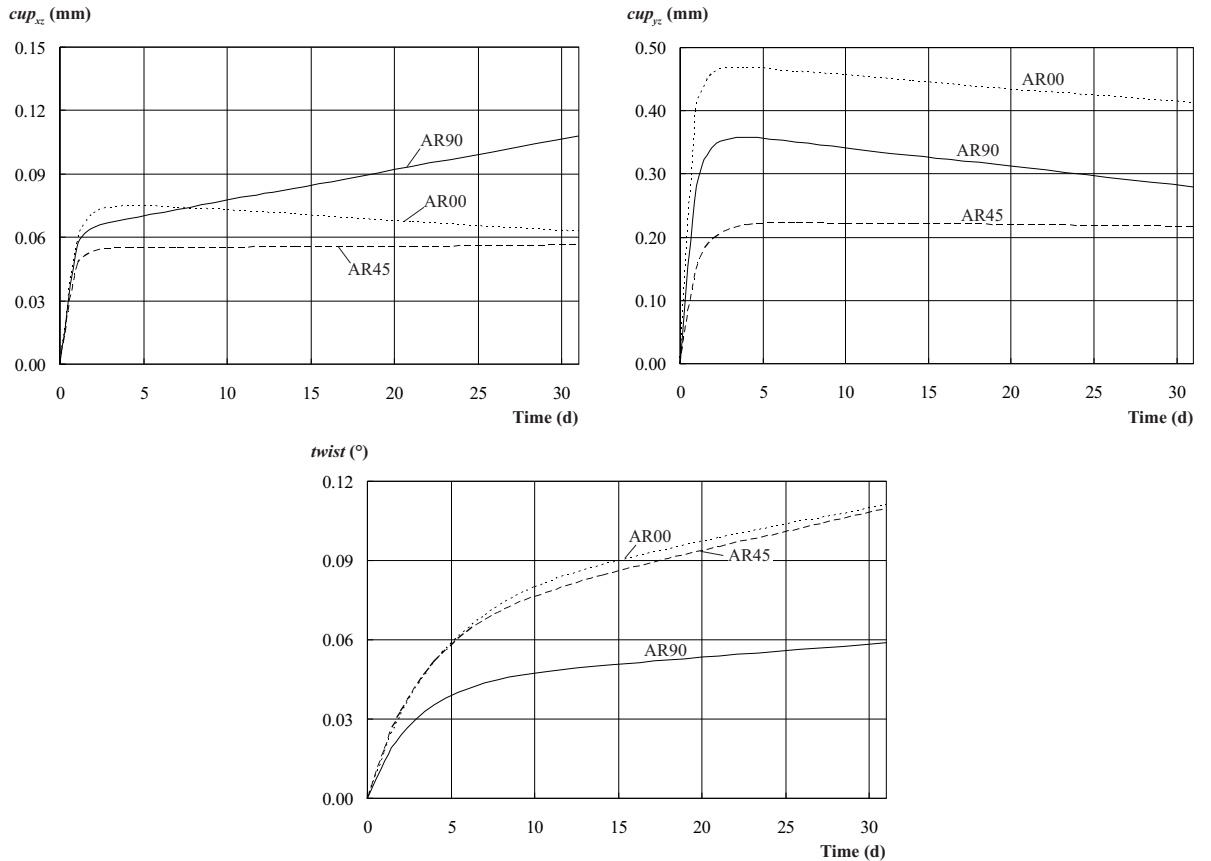


Figure 4.7 History plots of cup and twist deformations, influence of annual ring orientation, curves showing regression functions $x(t) = \delta_1(1 - e^{-\delta_2 t}) + \delta_3 t$ of average values

Influence of the layer ratio

Fig. 4.8 shows the history plots of cup_{xz} , cup_{yz} and $twist$ for panels with different layer ratios: LR37 ($LR = 0.37$), LR52 ($LR = 0.52$), LR57 ($LR = 0.57$) and AR90 ($LR = 0.67$). The tests of different layer ratios, LR , show increased cupping, cup_{yz} , with increased LR . An increased layer ratio gave an increased ratio of outer layers, i.e. a larger ratio of transverse oriented fibres in the y-direction. This effect yielded a larger cup_{yz} at higher layer ratios. However, the results may be superimposed by the different panel thicknesses. The influence of LR on warping is further investigated by means of numerical simulations in Chapter 6.

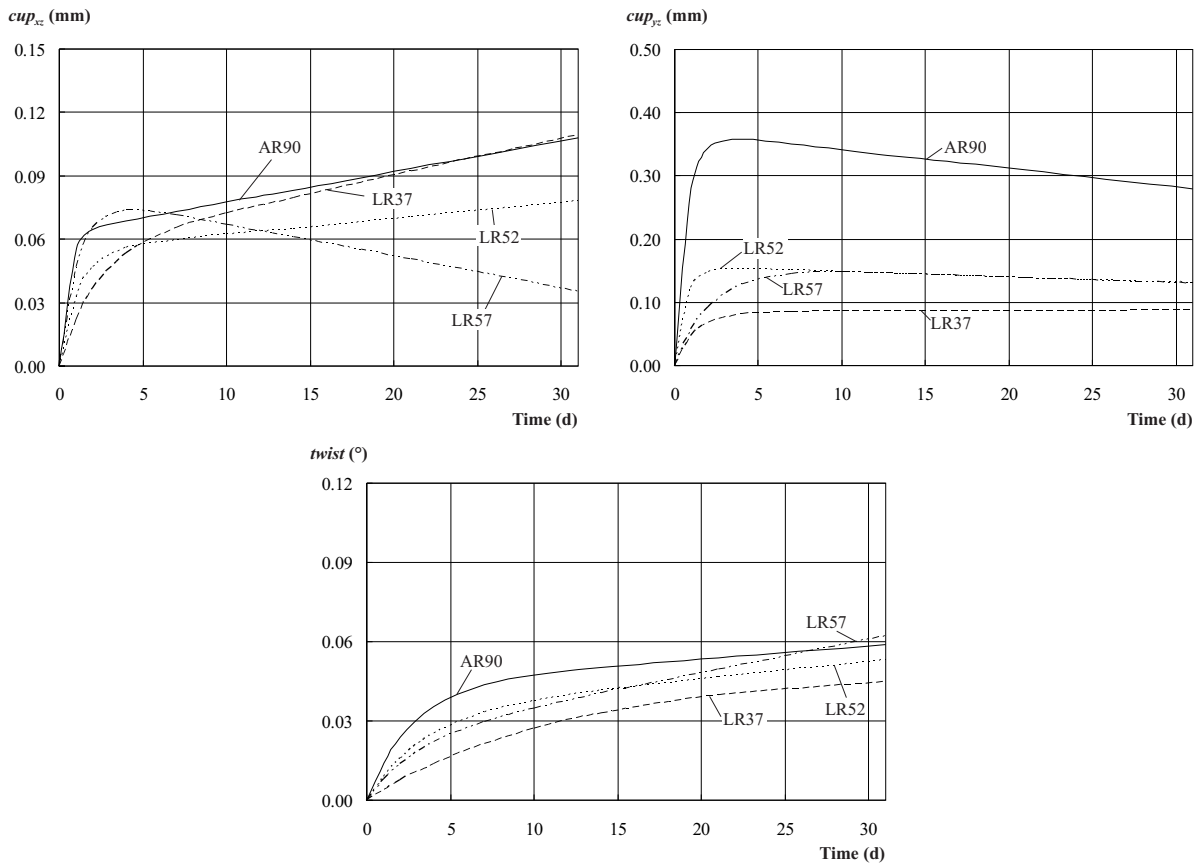


Figure 4.8 History plots of cup and twist deformations, influence of layer ratio, curves showing regression functions $x(t) = \delta_1(1 - e^{-\delta_2 t}) + \delta_3 t$ of average values

Influence of pre-stressed outer layers

Fig. 4.9 shows the history plots of cup_{xz} , cup_{yz} and $twist$ for panels with different magnitudes of pre-stress in the top layers: LR52 ($\sigma_0 = 0 \text{ N mm}^{-2}$), PS06 ($\sigma_0 = 0.6 \text{ N mm}^{-2}$), PS12 ($\sigma_0 = 1.2 \text{ N mm}^{-2}$) and PS18 ($\sigma_0 = 1.8 \text{ N mm}^{-2}$). The results show a significant influence of pre-stressed outer layers. cup_{yz} was lower in series PS compared to LR52, where the layer ratios and the total panel thicknesses were identical. However, the investigated magnitudes of pre-stress had no substantial influence on warp deformations, both cup and twist. The maximum cup deformations in the xz-plane were higher compared to those in the yz-plane in pre-stressed panels. This was in contrast to the unstressed panels LR52. The pre-stressing yielded initial compressive stresses. These, acting in opposition to tensile stresses that developed in the outer layers, reduced the cupping in the yz-plane.

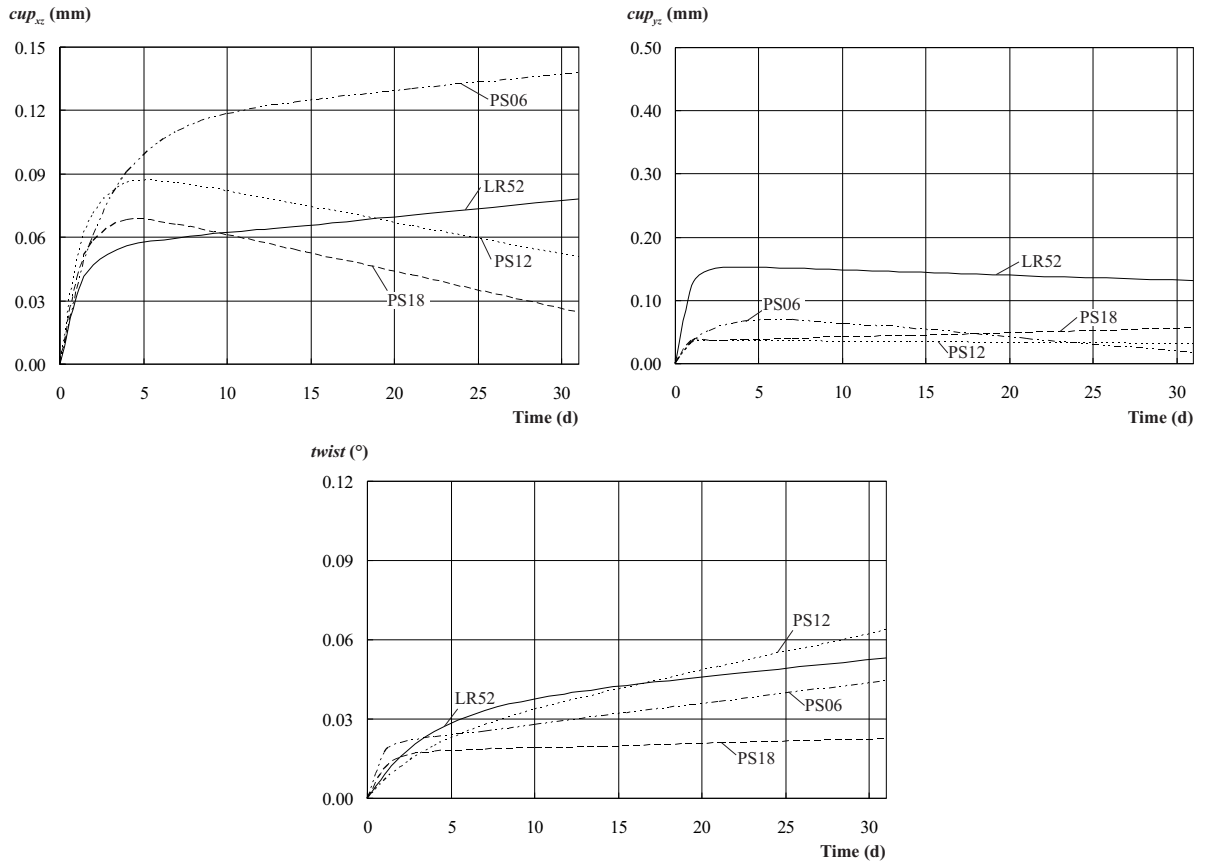


Figure 4.9 History plots of cup and twist deformations, influence of pre-stressed outer layers, curves showing regression functions $x(t) = \delta_1(1 - e^{-\delta_2 t}) + \delta_3 t$ of average values

Influence of the middle layer structure

Fig. 4.10 shows the history plots of cup_{xz} , cup_{yz} and $twist$ for panels with different middle layers but similar layer ratios: LR57 (middle layer: spruce), SWS (spruce, slits), CMDF5 (MDF500), CMDF7 (MDF700), COSB (OSB) and CPB (PB). The results indicate that the different middle layers differed considerably in the influence that they had on warp deformation. The slitted middle layer (SWS) had no significant influence on warp deformation. The composite laminates, on the other hand, showed substantially higher warp deformations than the spruce panels LR57 and SWS. The difference was most significant in cup deformation cup_{yz} . The largest cup deformations, cup_{yz} , were obtained for composite laminates containing particleboard and MDF500, while it was lower in MDF700- and OSB-composite laminates. This strongly correlates with the in-plane modulus of elasticity of the wood-based panels as indicated in Tab. 4.5. Not all composite laminates reached an equilibrium state, i.e. constant warp deformation. Rather, their warp deformations increased during the data recording.

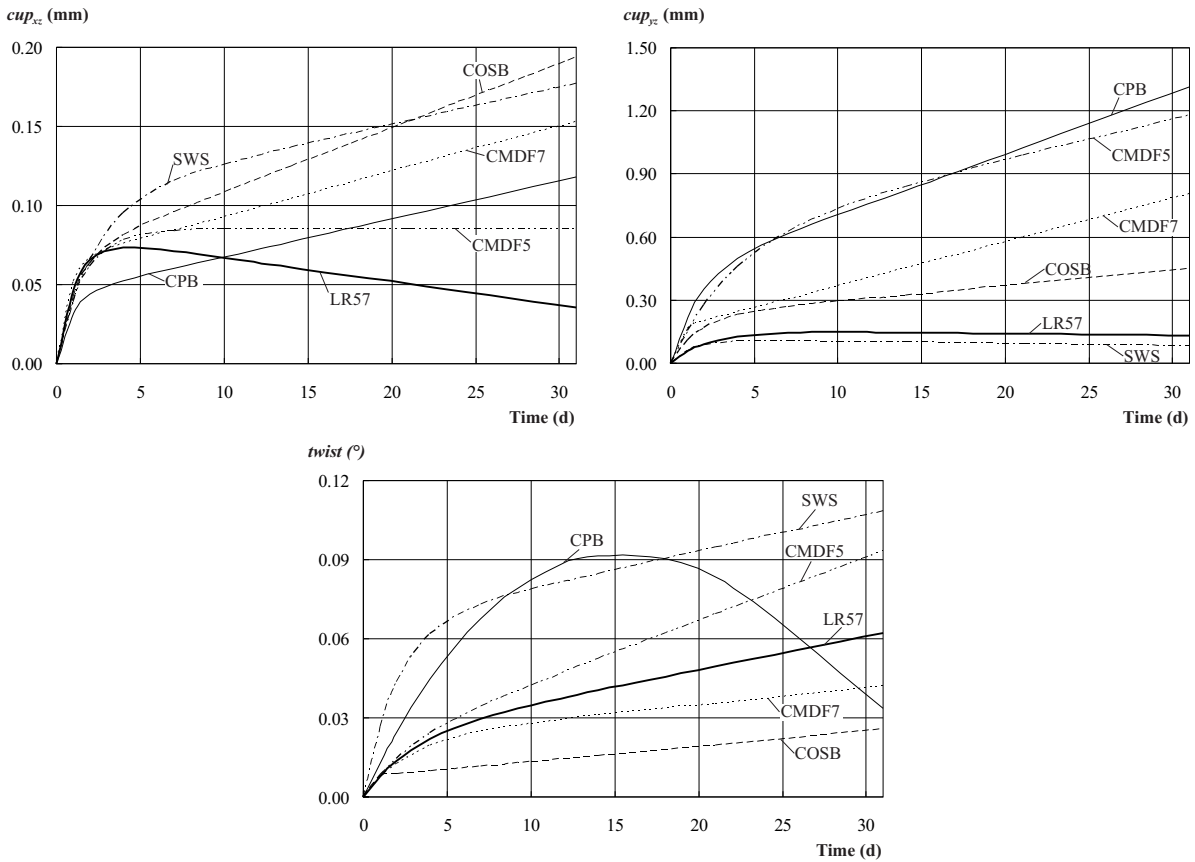


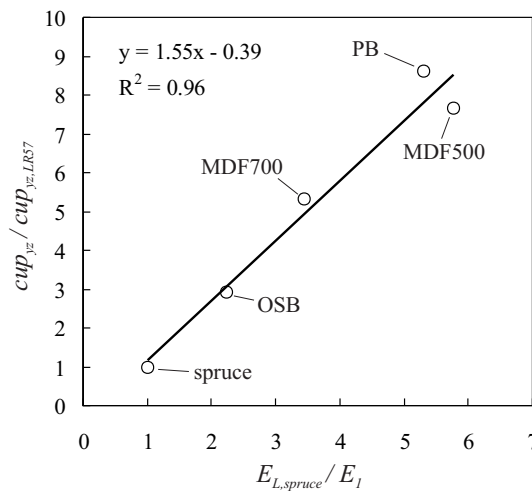
Figure 4.10 History plots of cup and twist deformations, influence of middle layer structure, curves showing regression functions $x(t) = \delta_1(1 - e^{-\delta_2 t}) + \delta_3 t$ of average values

Table 4.5 Moduli of elasticity of the different middle layer materials and maximum cup_{yz}

ID	Material ML	E_1 (N mm ⁻²)	E_2 (N mm ⁻²)	cup_{yz} (mm)
LR57	spruce		11000 ¹	0.15
CMDF5	MDF500	1907 ²	1550 ²	1.15
CMDF7	MDF700	3189 ²	3095 ²	0.80
COSB	OSB	4904 ²	2795 ²	0.44
CPB	PB	2076 ²	1597 ²	1.29

¹ Sell [92], E_L ² Schreiber et al. [90], tensile test, thickness 15mm (MDF500, OSB) and 16mm (MDF700, PB)

Due to the deformation of the bottom layer, the whole panel is exposed to a bending deformation towards the minor axis (perpendicular to the fibre direction of the outer layers). Since the top layer has only a small modulus of elasticity in that direction, it is only able to minimally constrain warp deformation of the panel. Thus, the ability to resist warp depends, to a great extent, on the ability of the middle layer to constrain the deformation of the bottom layer. Therefore, the modulus of elasticity of the middle layer is of great importance. The comparison of the moduli of elasticity of the different materials along the minor axis, as indicated in Tab. 4.5, confirms these considerations. cup_{yz} is inversely proportional to the modulus of elasticity of the middle layer, i.e. with increasing E , cup_{yz} decreases. Thus, the ratio $cup_{yz}/cup_{yz,LR57}$ strongly correlates with the ratio $E_{L,spruce}/E_1$ as indicated in Fig. 4.11.

**Figure 4.11** Relation between modulus of elasticity of the middle layer material in production direction, E_1 , and cup deformation cup_{yz}

Cup deformations in the minor axis show similar characteristics when compared to results obtained by Tobisch [110] (Tab. 4.6, see also Fig. 2.14). Discrepancies between both investigations are a result of the different moisture gradients, different panel thicknesses, different layer ratios, different panel dimensions and different positioning on the supports. However, the comparison of panels with similar layer ratios in spruce panels and composite laminates reveals larger cup deformations in composite laminates compared to the spruce panels in both investigations (Tab. 4.6).

Table 4.6 Comparison of cup deformations in the minor axis to results obtained by Tobisch [110] for panels with different middle layer materials, $t = 48\text{h}$

Source	Material ML	a_{tot} (mm)	LR (-)	Climate difference (%)	Moisture gradient (% mm ⁻¹)	cup (%)
Tobisch [110]	spruce	27	0.67	35/85	1.85	100
	OSB	27	0.63	35/85	1.85	134
	PB	27	0.63	35/85	1.85	229
Pre- sent thesis	spruce	35	0.57	65/100	1.00	100
	OSB	36	0.56	65/100	0.97	176
	PB	35	0.57	65/100	1.00	381

4.3 Numerical analysis

4.3.1 Material model and boundary conditions

Material modelling is based on the model that was established and validated by Ormarsson [79] (Section 2.6) on drying wood sticks. In the present study, the total strain rate is assumed to be the sum of the elastic, the moisture-induced and the mechano-sorptive strain rate. Creep is omitted from the material model since it is assumed that, in the investigated time period, mechano-sorptive creep has a much greater effect on panel warping than creep.

The following boundary conditions were applied (Fig. 4.12):

$$\begin{aligned}
 u_z &= 0 & \text{at } (50, 50, 0) \\
 u_x = u_y = u_z &= 0 & \text{at } (150, 250, 0) \\
 u_x &= 0 & \text{at } (150, y, z) .
 \end{aligned} \tag{4.6}$$

The first two boundary conditions at the bottom face of the panel ($z=0$) referred to the supports that were applied in the experiments. In point A' ($x = 50\text{mm}$, $y = 50\text{mm}$, see Fig. 4.5 and 4.12) displacements were restricted in the z -direction and in point H' ($x = 150\text{mm}$, $y = 250\text{mm}$) displacements were restricted in all directions. The last boundary condition referred to the effect that the panel structure and the supports were symmetrical. This allowed simulating only one half of the panel. The symmetry plane was at $x = 150\text{mm}$. Thus, panels dimensioned $150\text{mm} \times 300\text{mm}$ were simulated. This saved computational time and gave accurate results.

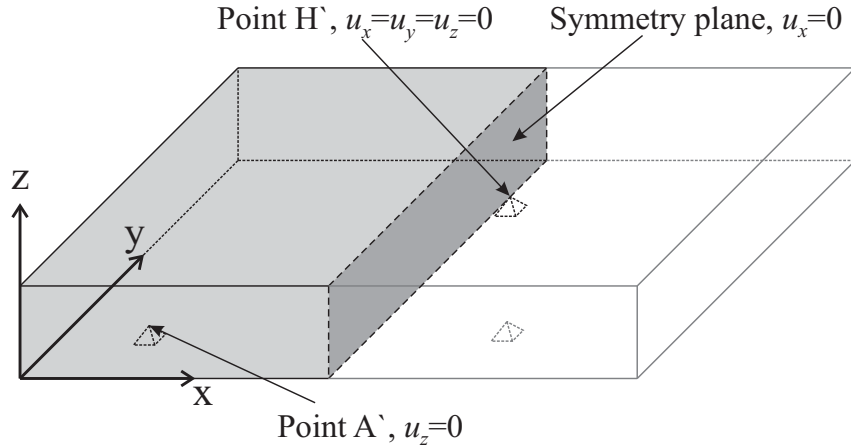


Figure 4.12 Boundary conditions and global coordinate system of the mechanical model to simulate hygroscopic warping

4.3.2 Material data

Elastic coefficients

As indicated in Section 2.3, the elastic coefficients describing the compliance matrix $\mathbf{S}$ (Eq. 2.18) depend on the moisture content. Neuhaus [74] determined these dependencies for Norway spruce. The relationship between compliance and moisture content was already displayed in Fig. 2.7. From the findings of Neuhaus [74] (summarised in the Appendix) and Eqs. 2.19, 2.20 and 2.22 the following dependencies of the moduli and Poisson's ratios on the moisture content can be found:

$$\begin{aligned}
 E_{ii} &= E_{ii,0} + E_{ii,1} \cdot \omega + E_{ii,2} \cdot \omega^2 + E_{ii,3} \cdot \omega^3 \\
 G_{ij} &= G_{ij,0} + G_{ij,1} \cdot \omega + G_{ij,2} \cdot \omega^2 + G_{ij,3} \cdot \omega^3, \quad i \neq j \\
 \nu_{ij} &= \nu_{ij,0} + \nu_{ij,1} \cdot \omega + \nu_{ij,2} \cdot \omega^2 + \nu_{ij,3} \cdot \omega^3, \quad i \neq j
 \end{aligned} \tag{4.7}$$

The stiffness parameters also depend on the temperature. Since the temperature was kept constant during the measurements, this relation factor is neglected. Due to these dependencies on moisture content, the rate of the compliance matrix $\dot{\mathbf{S}}$ in Eq. 2.25 is

$$\dot{\mathbf{S}} = \begin{bmatrix} -\frac{\dot{E}_L}{E_L^2} & \frac{\dot{E}_R \nu_{LR}}{E_R^2} - \frac{\dot{\nu}_{LR}}{E_R} & \frac{\dot{E}_T \nu_{LT}}{E_T^2} - \frac{\dot{\nu}_{LT}}{E_T} & 0 & 0 & 0 \\ \frac{\dot{E}_L \nu_{RL}}{E_L^2} - \frac{\dot{\nu}_{RL}}{E_L} & -\frac{\dot{E}_R}{E_R^2} & \frac{\dot{E}_T \nu_{RT}}{E_T^2} - \frac{\dot{\nu}_{RT}}{E_T} & 0 & 0 & 0 \\ \frac{\dot{E}_L \nu_{TL}}{E_L^2} - \frac{\dot{\nu}_{TL}}{E_T} & \frac{\dot{E}_R \nu_{TR}}{E_R^2} - \frac{\dot{\nu}_{TR}}{E_R} & -\frac{\dot{E}_T}{E_T^2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\dot{G}_{LR}}{G_{LR}^2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\dot{G}_{LT}}{G_{LT}^2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\dot{G}_{RT}}{G_{RT}^2} \end{bmatrix} \quad (4.8)$$

In Eqs. 4.7 the moisture content ω is assumed to be a function of time. With $\dot{\omega}$ as the rate of change of moisture content below the fibre saturation point, the rate of change of the stiffness parameters is:

$$\begin{aligned} \dot{E}_{ii} &= E_{ii,1} \cdot \dot{\omega} + E_{ii,2} \cdot (2\omega\dot{\omega} + \dot{\omega}^2) + E_{ii,3} \cdot (3\omega^2\dot{\omega} + 3\omega\dot{\omega}^2 + \dot{\omega}^3) \\ \dot{G}_{ij} &= G_{ij,1} \cdot \dot{\omega} + G_{ij,2} \cdot (2\omega\dot{\omega} + \dot{\omega}^2) + G_{ij,3} \cdot (3\omega^2\dot{\omega} + 3\omega\dot{\omega}^2 + \dot{\omega}^3), \quad i \neq j \\ \dot{\nu}_{ij} &= \nu_{ij,1} \cdot \dot{\omega} + \nu_{ij,2} \cdot (2\omega\dot{\omega} + \dot{\omega}^2) + \nu_{ij,3} \cdot (3\omega^2\dot{\omega} + 3\omega\dot{\omega}^2 + \dot{\omega}^3), \quad i \neq j \end{aligned} \quad (4.9)$$

The parameters $E_{L,k}$, $E_{R,k}$, $E_{T,k}$, $G_{LR,k}$, $G_{LT,k}$, $G_{RT,k}$, $\nu_{LR,k}$, $\nu_{RL,k}$, $\nu_{LT,k}$, $\nu_{TL,k}$, $\nu_{RT,k}$ and $\nu_{TR,k}$ ($k = 0 \dots 3$) describe the rate of change of the stiffness parameters in Eqs. 4.7 and 4.9. They are displayed in Tabs. 4.7 and 4.8.

The influence of moisture content on the moduli of elasticity and the shear moduli is displayed in Figs. 4.13 and 4.14. The figures show that E_{ii} and G_{ij} decrease with moisture content. In the range of $0\% \leq \omega \leq 6\%$ (chemical adsorption of water) the moduli slightly increase. According to Kollmann [57], this is possibly due to a decrease of the inner friction. Thus, the elasticity may be increased. The moduli decrease with moisture content during the physical adsorption of water ($6\% \leq \omega \leq 15\%$) and capillary condensation ($15\% < \omega < FSP$ ³). E_T and E_R decrease within the hygroscopic range to ca. 50% of their value at the oven-dry state, while E_L only decreases to ca. 85%. The shear moduli

³FSP - fibre saturation point, $\omega \approx 28 \dots 32\%$

Table 4.7 Parameters $E_{L,k}$, $E_{R,k}$, $E_{T,k}$, $G_{LR,k}$, $G_{LT,k}$ and $G_{RT,k}$ ($k = 0 \dots 3$) of the moisture dependency of the moduli of elasticity and the shear moduli (Eqs. 4.7 and 4.9), according to Neuhaus [74], $0 \leq \omega \leq 0.28$

Modulus (N mm ⁻²)	$k = 0$ ($\times 10^2$)	$k = 1$ ($\times 10^2$)	$k = 2$ ($\times 10^4$)	$k = 3$ ($\times 10^4$)
$E_{L,k}$	127.92	15.22	-9.01	18.85
$E_{R,k}$	10.00	3.61	-2.09	4.67
$E_{T,k}$	5.06	5.00	-1.35	2.97
$G_{LR,k}$	7.63	5.93	-1.99	4.77
$G_{LT,k}$	8.81	1.39	-1.39	2.77
$G_{RT,k}$	0.61	-1.07	-0.06	0.17

Table 4.8 Parameters $\nu_{LR,k}$, $\nu_{RL,k}$, $\nu_{LT,k}$, $\nu_{TL,k}$, $\nu_{RT,k}$ and $\nu_{TR,k}$ ($k = 0 \dots 3$) of the moisture dependency of the Poisson's ratios (Eqs. 4.7 and 4.9), according to Neuhaus [74], $0 \leq \omega \leq 0.28$

Poisson's ratio (-)	$k = 0$	$k = 1$	$k = 2$	$k = 3$
$\nu_{LR,k}$	0.046	0.14	-0.46	-0.21
$\nu_{RL,k}$	0.232	-0.86	28.78	-78.62
$\nu_{LT,k}$	0.021	0.26	-1.43	2.19
$\nu_{TL,k}$	0.286	-4.10	71.91	164.23
$\nu_{RT,k}$	0.153	1.08	3.98	-19.12
$\nu_{TR,k}$	0.335	0.13	26.40	-82.19

also decrease with moisture content: G_{LR} and G_{LT} decrease to ca. 50% and G_{RT} decreases to ca. 30% from the oven-dry state to the fibre saturation point.

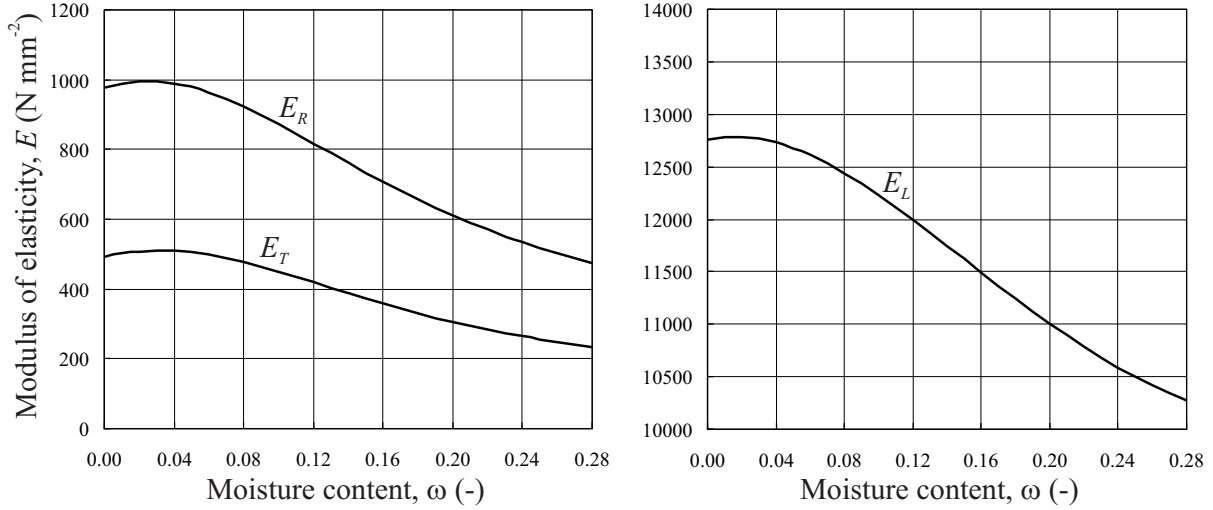


Figure 4.13 Moduli of elasticity E_L , E_R and E_T of Norway spruce according to Neuhaus [74] (Eq. 4.7 and Tab. 4.7)

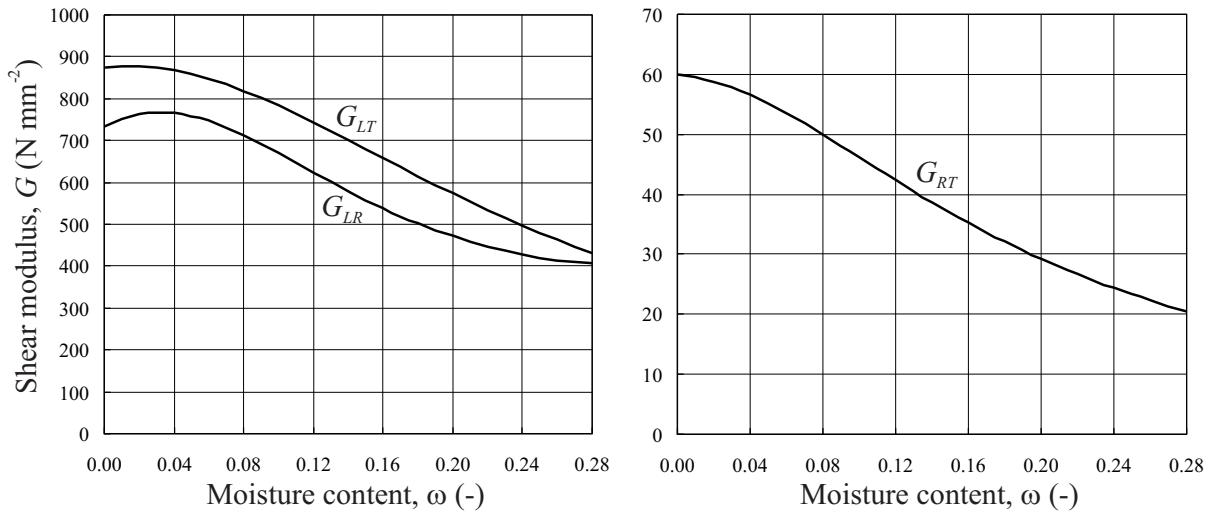


Figure 4.14 Shear moduli G_{LR} , G_{LT} and G_{RT} of Norway spruce according to Neuhaus [74] (Eq. 4.7 and Tab. 4.7)

Fig. 4.15 shows a different behaviour for the Poisson's ratios. ν_{LR} and ν_{LT} remain nearly constant within the hygroscopic range ($\nu_{LR} > \nu_{LT}$) while ν_{TL} , ν_{TR} , ν_{RL} and ν_{RT} increase when the moisture content is increased.

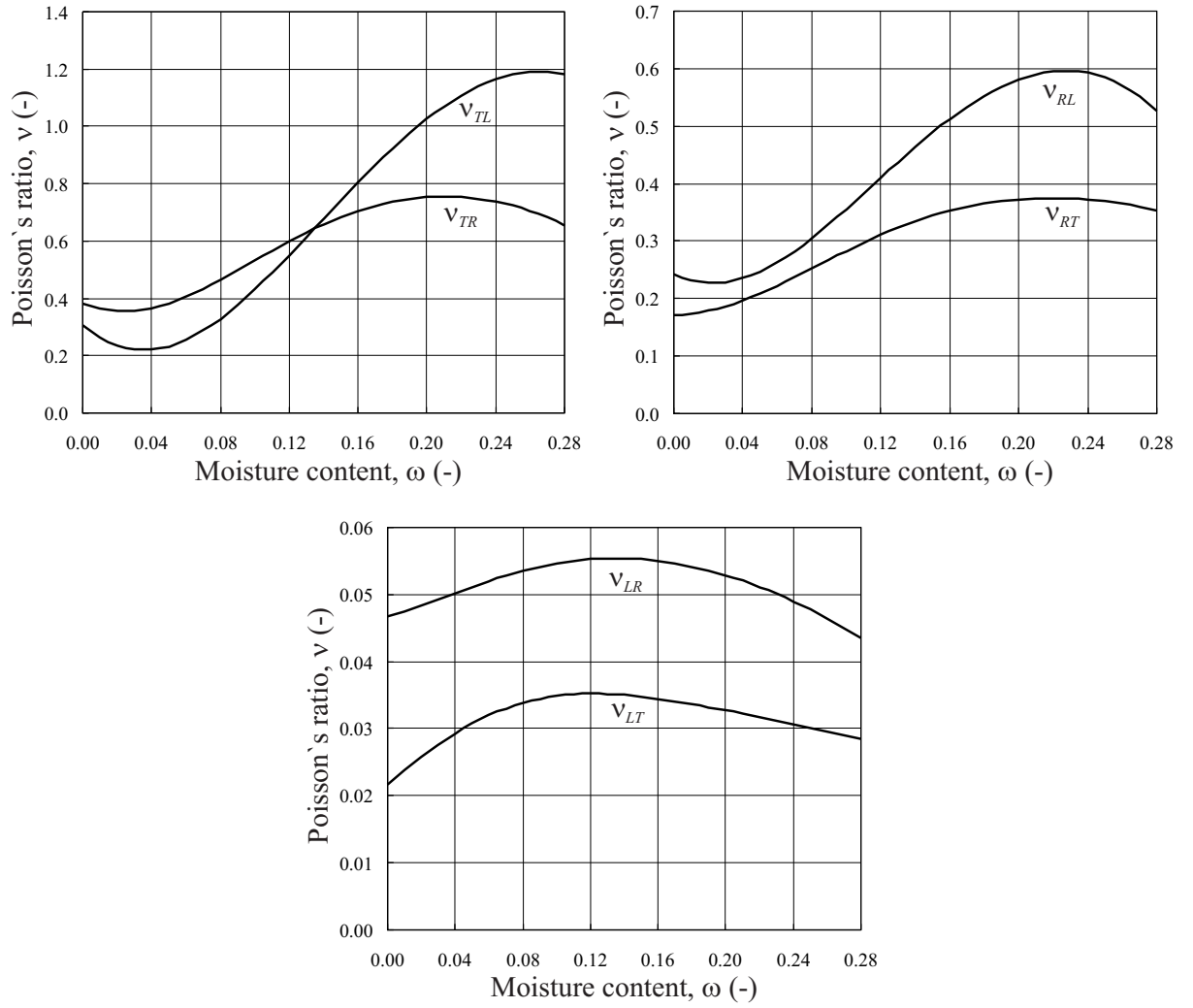


Figure 4.15 Poisson's ratios ν_{LR} , ν_{RL} , ν_{LT} , ν_{TL} , ν_{RT} and ν_{TR} of Norway spruce according to Neuhaus [74] (Eq. 4.7 and Tab. 4.8)

Coefficients of hygroexpansion

The coefficients of hygroexpansion, characterising the swelling in different orthotropic directions (Eq. 2.27), were chosen according to Neuhaus [74] and Sell [92] as

$$\alpha_R = 1.7 \cdot 10^{-3}$$

$$\alpha_T = 3.3 \cdot 10^{-3} .$$

Spruce wood swells twice as much in the tangential than in the radial direction. The longitudinal swelling is negligible compared to swelling in the transverse directions. It has been implemented into the model with

$$\alpha_L = 5.0 \cdot 10^{-5} ,$$

which was found by Dahlblom et al. [19], see also Ormarsson [79]. The swelling parameters are set to constant values, but they may be dependent on moisture content.

Mechano-sorptive parameters

The mechano-sorptive material parameters are based on results obtained by Santaoja et al. [88] and Mårtensson [71] and are set to be independent of the moisture content. The mechano-sorption coefficients in the orthotropic directions and planes that describe the mechano-sorption material matrix $\mathbf{m}$ (Eq. 2.29) are, according to Ormarsson [79] and Santaoja et al. [88]:

$$\begin{aligned} m_L &= 1.0 \cdot 10^{-4} & m_R &= 0.15 & m_T &= 0.20 \\ m_{LR} &= 0.008 & m_{LT} &= 0.008 & m_{RT} &= 0.8 \quad (\text{all mm}^2 \text{ N}^{-1}). \end{aligned}$$

The coupling terms μ_{ij} are

$$\mu_{LR} = 0 \quad \mu_{LT} = 0 \quad \mu_{RT} = 1 \quad (\text{all dimensionless}).$$

4.3.3 Model validation

In the preceding sections the material model, the boundary conditions and the material data were presented. In this section the numerical simulation is compared to the experimental results as shown in Section 4.2.3. Therefore, the spruce panel of series AR90 was modelled in the finite element software ABAQUS. Firstly, the moisture distribution within the panel was calculated numerically by means of the moisture model as presented in Section 3.4 and summarised in Tab. 3.4. The climatic conditions were

- initial climate: $\varphi = 65\%$,
- test climate: $\varphi = 65\%/100\%$.

The results of the simulation of moisture distribution were shown in Figs. 3.17 - 3.19.

Fig. 4.16 contrasts the results of the simulations against the experimental results. The simulated curves show good agreement to the experiments. However, the simulation overestimates the experimentally obtained mean values of cup_{yz} between $100\text{h} < t < 400\text{h}$. The calculated curve of cup deformation, cup_{xz} , overestimates the mean values of the experiments. However, the variation in cup_{xz} in different panels was very high. Thus, the calculated curve of cup_{xz} is a good fit to the measured mean values.

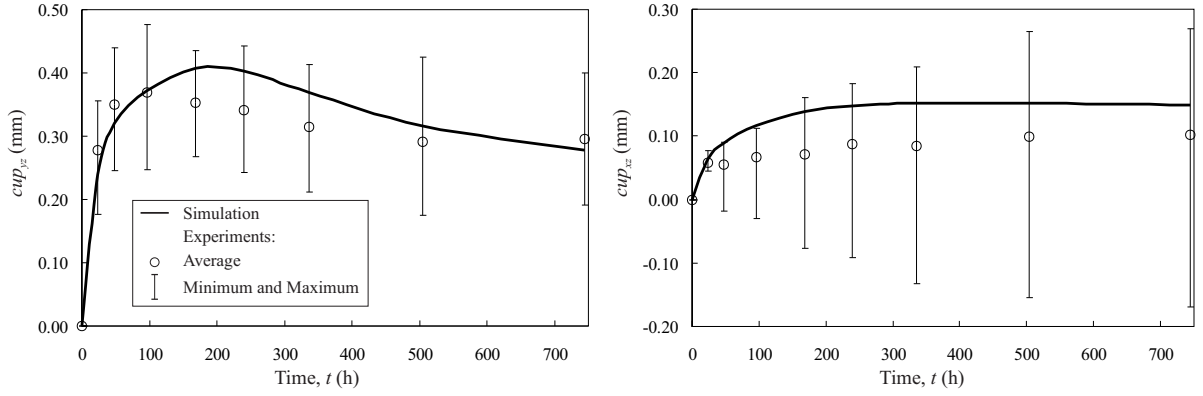


Figure 4.16 Comparison of simulated cup deformations and the measured data, series AR90

4.3.4 Basic studies

Element mesh density

In this section, the fineness of the element mesh necessary to obtain accurate results is determined. Therefore, a panel of type AR90 was studied with different element mesh densities. The material data is the same as the data presented in Section 4.3.2. The moisture profile is calculated according to the moisture model in Chapter 3 for $t = 744\text{h}$. The mesh density

$$NOE = N_x \cdot N_y \cdot (3N_{z,wood} + 2N_{z,adh}) \quad (4.10)$$

is illustrated in Fig. 4.17. NOE is the number of elements used in the simulations. N_x and N_y are the number of elements along the x- and y-axis, respectively. $N_{z,wood}$ and $N_{z,adh}$ are the number of elements in the z-direction in the three wood layers and the two adhesive layers, respectively. The adhesive layers were modelled by three elements in thickness ($N_{z,adh} = 3$). Two element mesh densities in the x- and y-directions were tested: $N_x \times N_y = 60 \times 120$ and $N_x \times N_y = 75 \times 150$, which refers to element lengths of 2.5mm and 2.0mm in both directions. Convergence was studied for different number of elements along the thickness. In each case, an eight-node volume element (C3D8) was used.

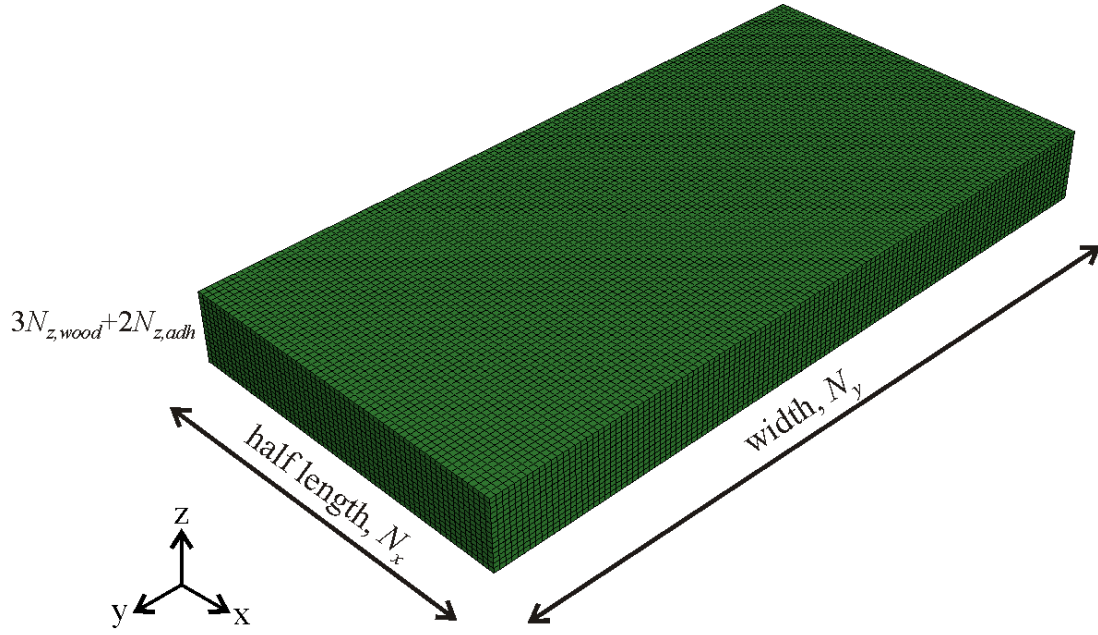
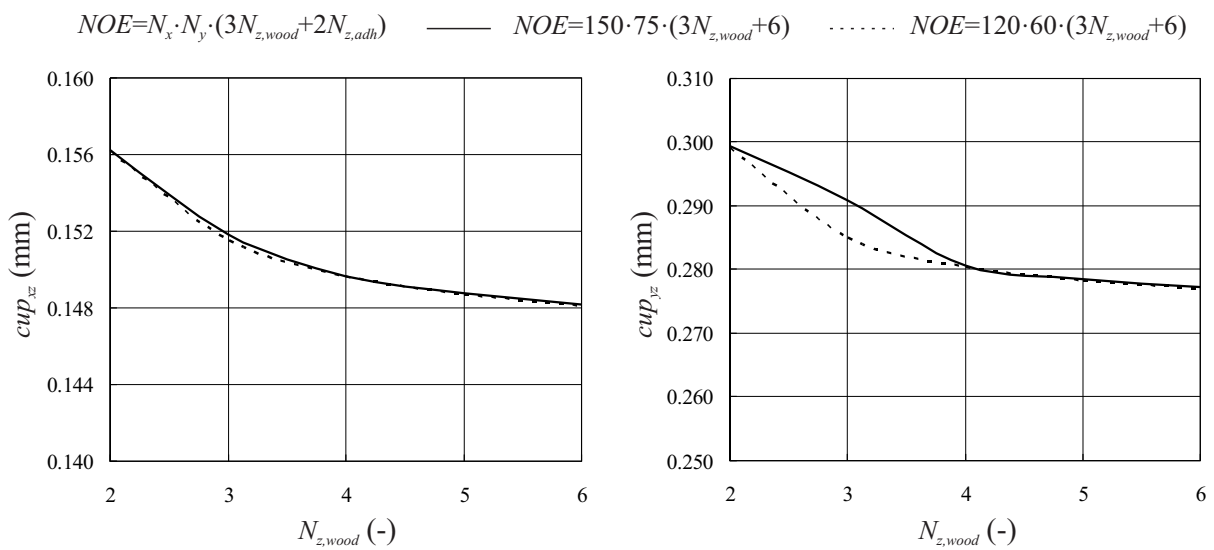


Figure 4.17 Element mesh

The results, as shown in Fig. 4.18, show decreasing cup deformations with an increasing number of elements along the thickness. The results for the different number of elements in the in-plane directions show only small differences. The cup deformations remain nearly constant for $N_{z,wood} = 5$ and $N_{z,wood} = 6$. Thus, if not otherwise mentioned, an element mesh density of $N_x = 60$, $N_y = 120$, $N_{z,wood} = 5$ and $N_{z,adh} = 3$ was chosen, which results in $NOE = 151200$.

Figure 4.18 Influence of element mesh density NOE on the hygroscopic warping of panel AR90

Influence of supports

In this section, it is numerically investigated if the arrangement of the panel on the triangular oriented supports plays a significant role in the formation of warp. As indicated in Fig. 4.19, two types of arrangements were tested. Type A refers to the supports applied in the experimental tests (see Figs. 4.5a and 4.12). The fibre direction of the outer layers (x-axis) is aligned parallel to the axis that is described by the points D, E and F (see Fig. 4.5a). Type B refers to investigations where the panel is aligned perpendicular to the orientation of type A, i.e. the fibres of the outer layers are aligned perpendicular to the axis DEF. Again panel type AR90 was studied with the moisture profile calculated according to Chapter 3 ($t = 744\text{h}$).

The calculated cup deformations are plotted in Fig. 4.20. The results show that cupping is only slightly influenced by the orientation of the supports. The dominant cup deformation was found in the yz-plane. Due to the triangular, i.e. asymmetric, orientation of the supports, small differences could be detected.

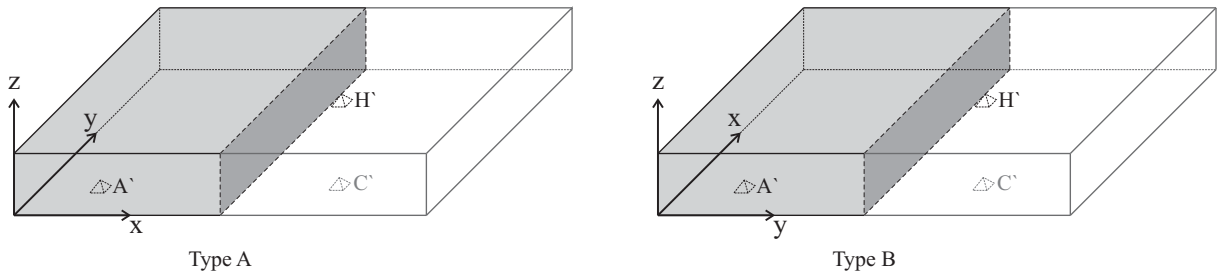


Figure 4.19 Types of panel orientation on the supports

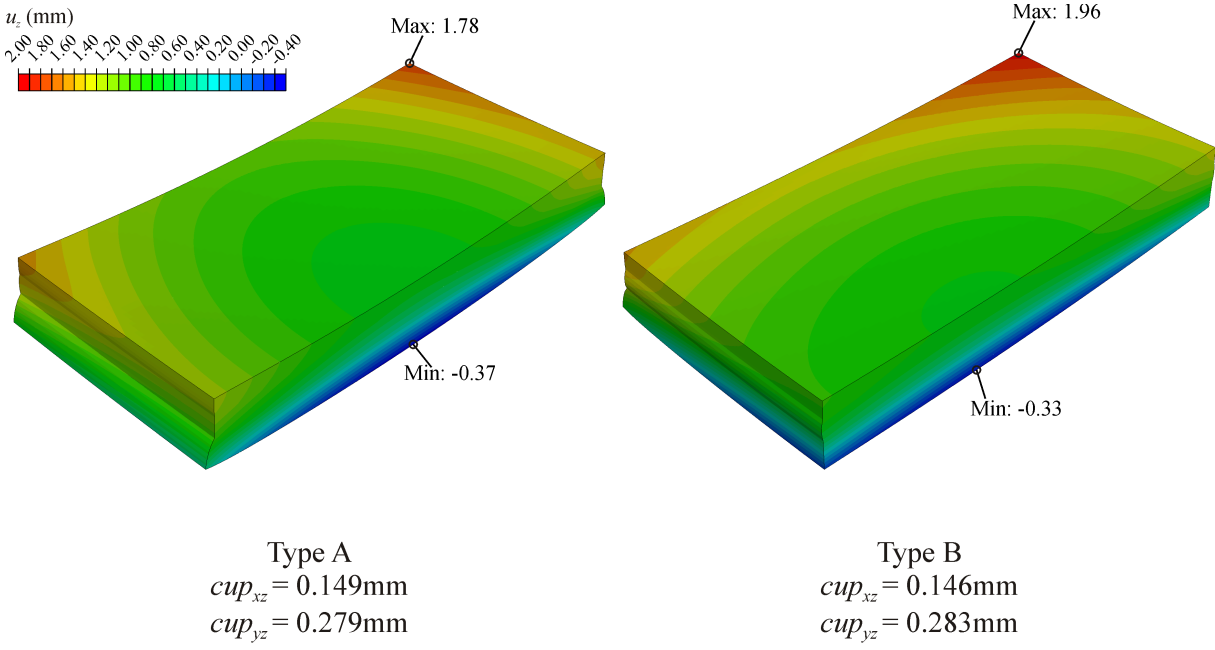


Figure 4.20 Displacement u_z [mm] for different arrangements of the panel on the supports, displacement scale factor: $\times 10$

Influence of the constitutive law

All earlier simulations were based on the material model for wood described in Section 2.6. The total strain rate $\dot{\epsilon}$ is assumed to be the sum of the elastic strain rate $\dot{\epsilon}_{el}$, the moisture induced strain rate $\dot{\epsilon}_{\omega}$ and the mechano-sorptive strain rate $\dot{\epsilon}_{\omega\sigma}$. In this section, the influence of alternative material models on deformations and stresses are examined. Three alternatives to the reference model were tested:

$$\dot{\epsilon} = \dot{\epsilon}_{el} + \dot{\epsilon}_{\omega} + \dot{\epsilon}_{\omega\sigma}$$

$$\text{Model 1 } \dot{\epsilon} = S\dot{\sigma} + \dot{S}\sigma + \alpha\dot{\omega} + m\sigma|\dot{\omega}|$$

$$\text{Model 2 } \dot{\epsilon} = S\dot{\sigma} + \alpha\dot{\omega} + m\sigma|\dot{\omega}|$$

$$\text{Model 3 } \dot{\epsilon} = S\dot{\sigma} + \dot{S}\sigma + \alpha\dot{\omega} + 0$$

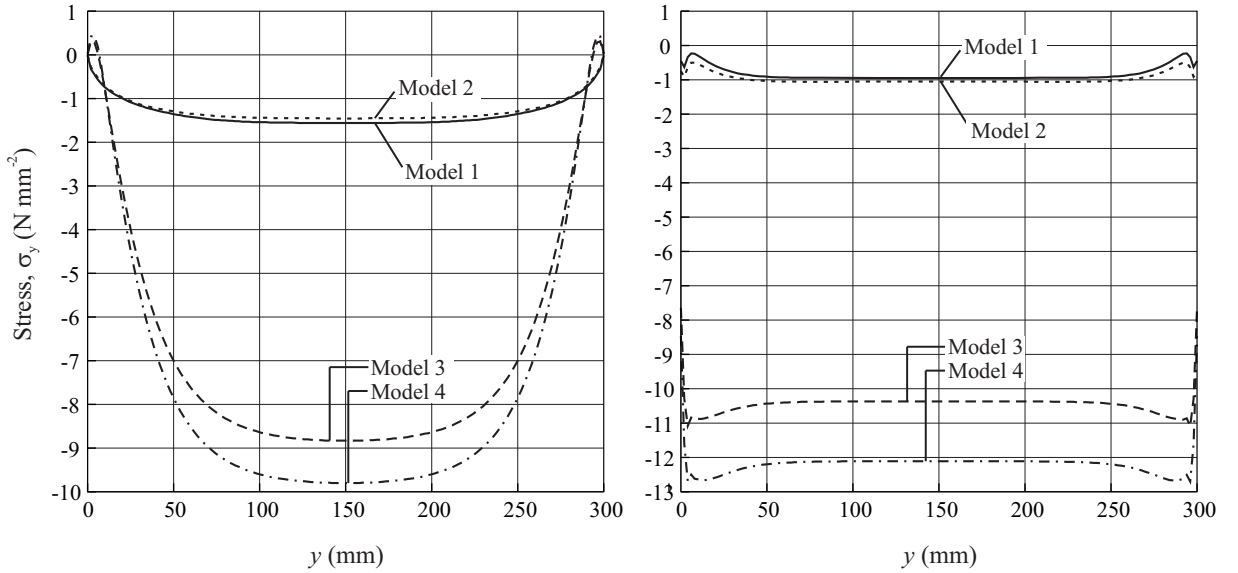
$$\text{Model 4 } \dot{\epsilon} = S\dot{\sigma} + \alpha\dot{\omega} + 0$$

Material model 1 is the reference as it is applied in most of the simulations in this study. In the second model, the changes in the material stiffness are not considered. In models 3 and 4 the elastic strain rate is the same as in models 1 and 2 but the mechano-sorptive strain rate is neglected. In Tab. 4.9, the cup deformations for panel AR90 are plotted for each of these four models.

Table 4.9 Cup deformations in panel AR90 for different material assumptions, % ref is the percentaged difference to the reference cup

Model	cup_{xz}		cup_{yz}	
	(mm)	% ref	(mm)	% ref
Model 1 (ref)	0.149	-	0.278	-
Model 2	0.146	-1.7	0.268	-3.8
Model 3	0.200	34.3	2.763	892.5
Model 4	0.204	37.2	3.095	1011.8

The results show a considerable influence of the mechano-sorptive strain rate on deformations (Tab. 4.9). Neglecting the mechano-sorptive effect yields deformations about 10 times greater. The choice of formulation of the elastic strain rate has very little influence on warping deformation. In terms of drying deformations, Ormarsson [79] found the same effect. Whether mechano-sorption is considered or not has a strong influence on radial stresses in the outer layers (Fig. 4.21). Mechano-sorption reduces stresses. These become lower than those in the elastic formulation. The choice of elastic formulation has little influence when mechano-sorption is considered, but has a strong influence when it is neglected.

**Figure 4.21** Radial stress component along the y -axis for different constitutive models, $x = 150\text{mm}$, $z = 0\text{mm}$ (lefthand side) and $z = 10\text{mm}$ (righthand side)

Influence of the type of elements in the glue lines

In the preceding sections, solid elements were applied for both the wood and the adhesive layers. The adhesive layers are very small in thickness, which may cause convergence problems when solid elements are used. Thus, two types of elements were tested in the glue lines:

- solid (continuum) elements and
- cohesive elements.

Rectangular elements with a linear basis function, i.e. eight-node elements, were applied. In ABAQUS they are denoted as C3D8 (solid) and COH3D8 (cohesive). The connectivity of cohesive elements is like that of continuum elements, except that cohesive elements are composed of two faces separated by a thickness. The relative change in position of the top and the bottom face along the thickness direction quantifies opening and closing of the interface. The relative displacement of the two faces measured in the plane represents the transverse shear behaviour of the cohesive element. The results indicate that the type of element has minimal influence on warp (Tab. 4.10) and stresses.

Table 4.10 Cup deformations for different types of elements in the glue lines

Elements		cup_{xz}		cup_{yz}	
type	$N_{z,adh}$	(mm)	% ref	(mm)	% ref
solid (ref)	3	0.149	-	0.279	-
solid	1	0.149	0.2	0.285	2.2
cohesive	1	0.149	0.2	0.286	2.5

Influence of the stiffness of the glue lines

The properties of adhesives can vary over a very large spectrum. A lot of different types of adhesives are applied in the wood industry. Few are licensed for the application of load bearing timber constructions. In order to determine the elastic properties of polymer films, Konnerth et al. [62] investigated polymer films of seven different adhesives by means of tensile tests. Furthermore, polymer films and bond lines were tested by means of nanointendation by Konnerth et al. [61]. The tensile tests showed a wide range of elastic properties from $E_{adh} = 470 \text{ N mm}^{-2}$ in 1K PUR to $E_{adh} = 6300 \text{ N mm}^{-2}$ in MUF in tensile tests. The tests performed by means of nanointendation generally yielded higher elastic moduli than the tensile tests.

In the present study, the modulus of elasticity of the bond lines (thickness 0.1mm) varied between $E_{adh} = 470 \text{ N mm}^{-2}$ and $E_{adh} = 10000 \text{ N mm}^{-2}$ in order to analyse the influence of the bond line stiffness on stresses and deformations. The results shown in Tab. 4.11 indicate that cup_{xz} is not influenced by the glue line stiffness. The cup deformation cup_{yz} is, however, significantly influenced by the glue line stiffness. cup_{yz} decreases linearly with E_{adh} as displayed in Fig. 4.22.

Table 4.11 Cup deformations for different moduli of elasticity of the glue lines

E_{adh} (N mm ⁻²)	cup_{xz}		cup_{yz}	
	(mm)	% ref	(mm)	% ref
470 (ref)	0.149	-	0.278	-
1000	0.149	0.0	0.278	-0.3
5000	0.149	0.1	0.271	-2.5
10000	0.149	0.3	0.264	-5.1

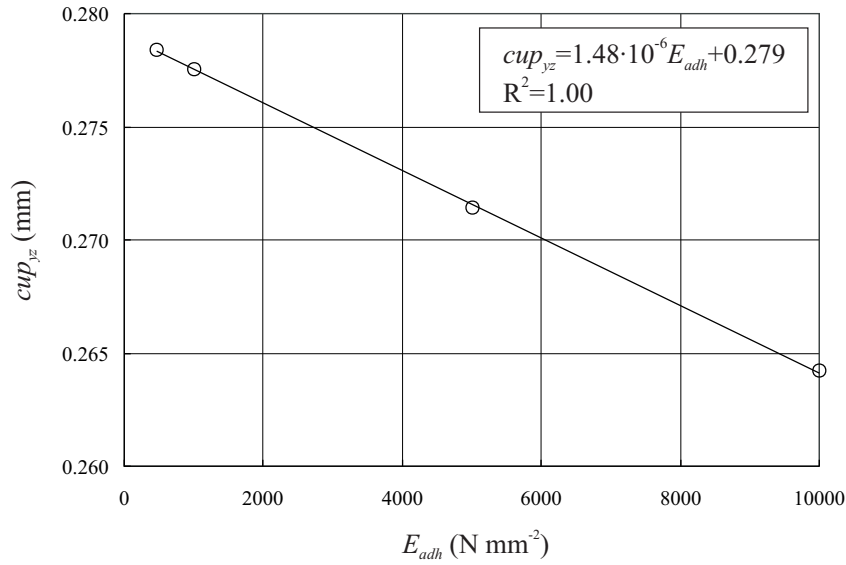


Figure 4.22 Dependency of the cup deformation cup_{yz} on modulus of elasticity of the glue lines E_{adh}

Influence of the diffusion coefficient of the glue lines

The influence of the moisture profile on warping is questioned in this section. The moisture profile may be influenced by the parameters that describe the moisture dependency of the diffusion coefficient of the glue lines, A_1 , A_2 and A_3 , as

$$D_{adh}(c) = \frac{A_1}{c^{A_2}} + A_3 \quad (4.11)$$

The reference value $A_1 = 3.3 \cdot 10^{-3} \text{ mm}^2 \text{ h}^{-1}$ (see Section 3.4.2) was both decreased and increased with two steps of 5%. The resulting relationships between concentration and diffusion coefficient, D_{adh} , are displayed in Fig. 4.23. Furthermore, waterproof glue lines ($D_{adh} = 0 \text{ mm}^2 \text{ h}^{-1}$) and glue lines where $D_{adh} = D_{wood}$ were applied.

The results are shown in Figs. 4.24 and 4.25. Differing A_1 yielded different moisture profiles. The middle layer was significantly influenced by a change in the water diffusivity of the bond lines. The influence on the moisture contents of the top and the bottom layer was less significant. At the end of the investigated time span, which was $t = 744\text{h}$, the moisture contents calculated by both extremal A_1 varied by 0.69% in the bottom layer, 2.12% in the middle layer and 0.80% in the top layer. A larger A_1 resulted in a larger D_{adh} and, thus, in a larger amount of water that was allowed to penetrate through the adhesive. Consequentially, the moisture content in the top and the middle layer increased when A_1 increased. On the other hand, the moisture content of the bottom layer decreased. When

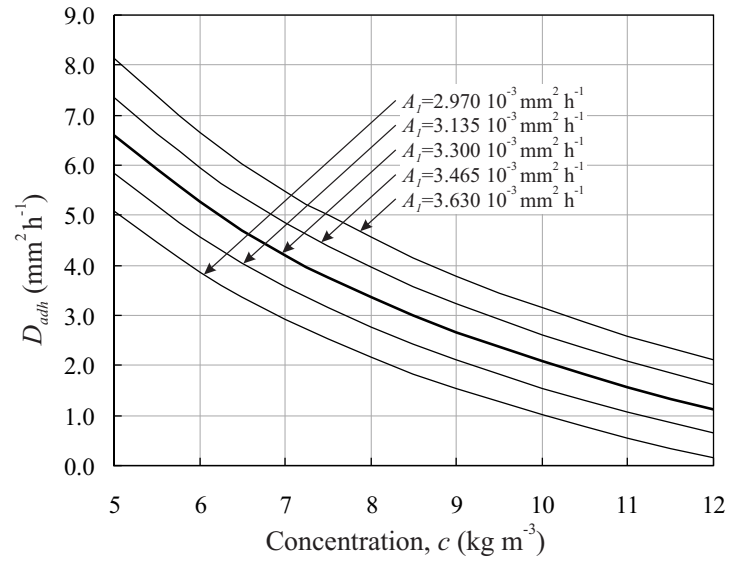


Figure 4.23 Relationship between diffusion coefficient of the bond lines and concentration for different A_l , Eq. 3.10

$D_{adh} = 0 \text{ mm}^2 \text{ h}^{-1}$, only the bottom layer was affected by the induced moisture gradient. Since the panels were assumed to be insulated at the ends and moisture flow was assumed to be one-dimensional, the moisture content of the top and the middle layer was constant and the moisture content of the bottom layer increased rapidly. This increase was larger than the increase, when $D_{adh} > 0$. Glue lines with the same diffusivity as wood yielded a linear moisture distribution in the panel (Fig. 4.24).

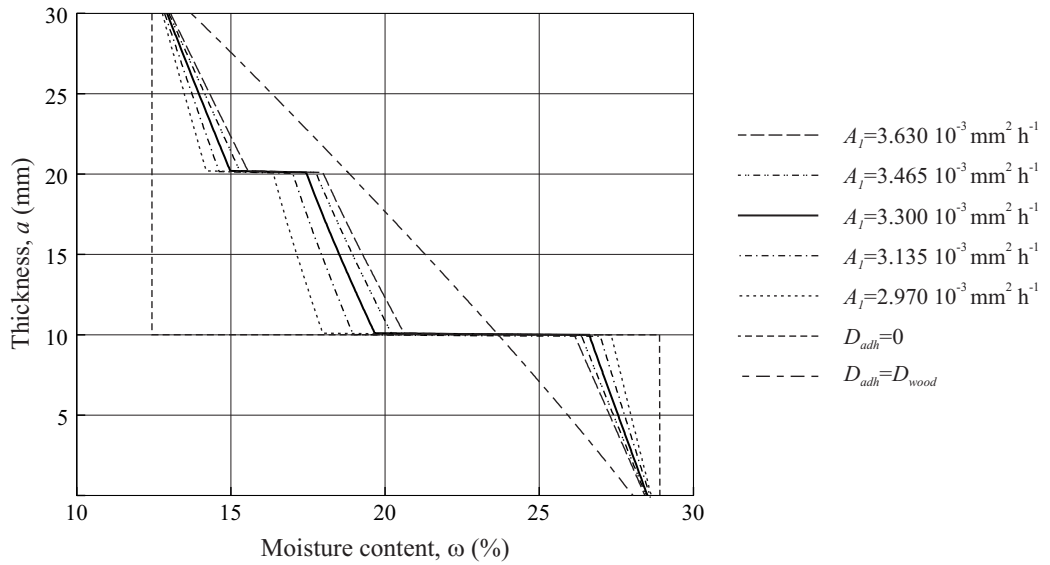


Figure 4.24 Moisture profiles due to different D_{adh} (Fig. 4.23), $t = 744\text{h}$

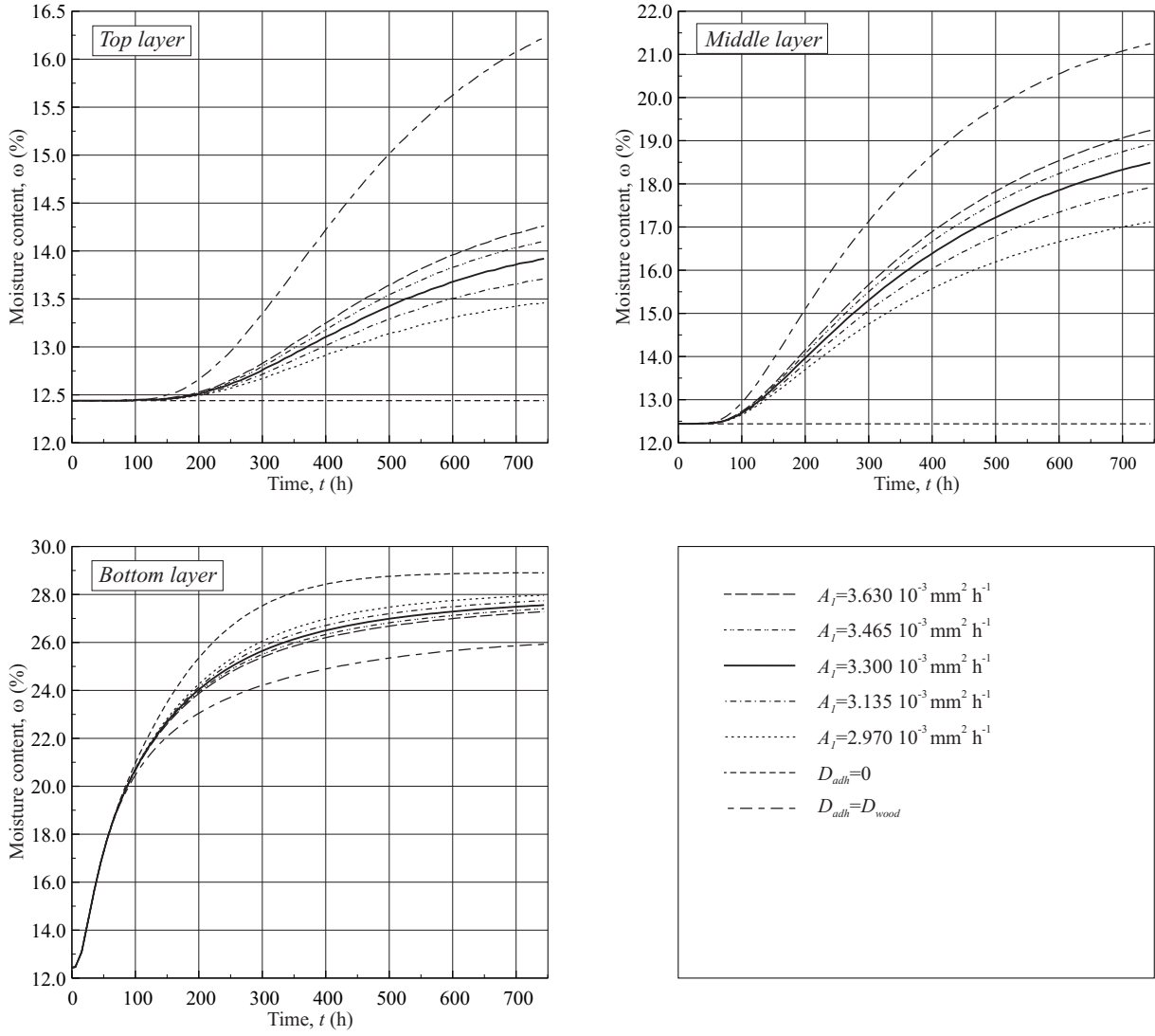


Figure 4.25 History plots of moisture content in different layers

The influence of different D_{adh} on warping was tested in panels of the type AR90. The moisture profiles determined at $t = 744\text{h}$ were applied in the simulations (Fig. 4.24). Tab. 4.12 summarises the results. The tests on different moisture distributions indicate that both cup_{xz} and cup_{yz} were influenced. $D_{adh} = 0$ yielded considerably larger cup deformations than the reference cuppings, since the bottom layer expansion perpendicular to the grain was very large. On the other hand, a balanced (linear) moisture distribution ($D_{adh} = D_{wood}$) led to smaller cup deformations (-17.8% compared to the reference cup_{xz} and -24.7% compared to the reference cup_{yz}).

As expected, the cup deformations decreased with increased A_1 . This was a result of the drier bottom layer and the moister middle and top layers (compared to the reference). The influence on cup_{yz} was more significant than on cup_{xz} . The increase and the decrease

of A_1 , each of 10%, resulted in a 17.6% increased cup_{yz} and a 5.8% decreased cup_{yz} . On the other hand, cup_{xz} was influenced only with 8.1% and -3.4%.

Table 4.12 Cup deformations for different moisture profiles (Fig. 4.24)

Moisture profile	cup_{xz}		cup_{yz}	
	(mm)	% to ref	(mm)	% to ref
$D_{adh} = 0$	0.189	27.0	0.380	36.6
$A_1 = 2.970 \cdot 10^{-3} \text{ (-10.0)}^1$	0.161	8.1	0.327	17.6
$A_1 = 3.135 \cdot 10^{-3} \text{ (-5.0)}^1$	0.152	2.4	0.290	4.2
$A_1 = 3.300 \cdot 10^{-3} \text{ (ref)}^1$	0.149	-	0.279	-
$A_1 = 3.465 \cdot 10^{-3} \text{ (5.0)}^1$	0.146	-1.9	0.270	-3.1
$A_1 = 3.630 \cdot 10^{-3} \text{ (10.0)}^1$	0.144	-3.4	0.262	-5.8
$D_{adh} = D_{wood}$	0.122	-17.8	0.210	-24.7
¹ in $\text{mm}^2 \text{ h}^{-1}$ (% to reference)				

4.3.5 Warping of spruce-MDF laminates

The simulation of spruce-MDF laminates, i.e. panels with a substituted middle layer, requires changes in the material modelling. The five layer model was adapted to the simulation of spruce-MDF laminates by the layer structure

- spruce, 10mm
- adhesive, 0.1mm
- MDF500 or MDF700, 16mm
- adhesive, 0.1mm
- spruce, 10mm

The orthotropic material model

$$\dot{\epsilon} = \dot{\epsilon}_{el} + \dot{\epsilon}_{\omega} + \dot{\epsilon}_{\omega\sigma} \quad (4.12)$$

was assumed to act in the spruce layers. The adhesive layers were assumed to act as a linear elastic material. As discussed earlier (Section 2.8), plane isotropic properties were assumed for the MDF middle layer. The calculation of the moisture distribution in composite laminates was shown in Section 3.5. The material data found by Ganey et al. [32; 34] was applied to the warp calculation (see Section 2.8). For the simulation of CMDF5 the properties of MDF540 were used, and for the calculation of CMDF7 the properties of MDF800 were used (Tabs. 2.1 and 2.2).

The results of material modelling are contrasted against the measured cup deformations in Tab. 4.13. The simulated cuppings show good agreement to the measured data in both series, CMDF5 and CMDF7. To illustrate the differences in warping between panels with MDF middle layers of different densities, the deformations in thickness direction, u_z , of these panels are displayed in Fig. 4.26. The results clearly indicate that CMDF7 is much more shape stable than CMDF5. However, both composite laminates are less dimensionally stable towards moisture-induced deformations than spruce cross-laminates.

Table 4.13 Cup deformations in spruce-MDF-laminates at $t = 744\text{h}$

ID	cup_{xz} (mm)		cup_{yz} (mm)	
	Experiment ¹	Simulation	Experiment ¹	Simulation
CMDF5	0.08 (0.06)	0.13	1.15 (0.25)	1.29
CMDF7	0.15 (0.10)	0.15	0.80 (0.25)	0.79
¹ mean value (standard deviation)				

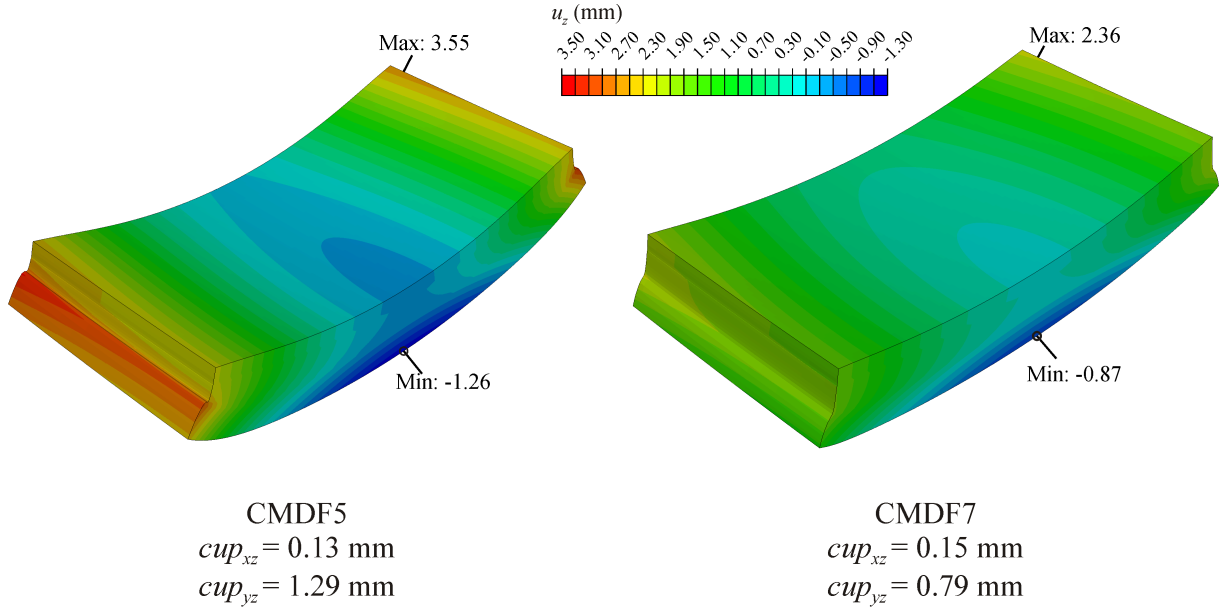


Figure 4.26 Displacement u_z [mm] of spruce-MDF laminates, $t = 744\text{h}$, displacement scale factor: $\times 10$

4.4 Concluding remarks

This chapter investigated hygroscopic warping, which was induced by a moisture gradient of 65/100% RH. Due to the experimental set-up and the panel characteristics (orientation of the layers), the main form of warping was found to be cup_{yz} , which is a cup deformation along the minor axis. The test results show significant influences of the annual ring orientation, the layer ratio, pre-stressed outer layers and the middle layer structure.

Annual ring orientation

The annual ring orientation, θ , is a highly important parameter that has a strong influence on warping. Cup deformations are governed by swelling properties, stiffness and annual ring orientation. Cupping was found to be lowest when $\theta = 45^\circ$ and highest when $\theta = 0^\circ$.

Layer ratio

The layer ratio of a panel is an important parameter for the warp deformation. The higher the layer ratio, the higher the cup deformation in the minor direction, cup_{yz} .

Pre-stressed outer layers

Pre-stressing the outer layers perpendicular to the fibres gives significantly lower cup deformation than the reference panel. An influence of the magnitude of pre-stress could not be detected.

Middle layer structure

The cupping of boards with different middle layer materials is highly dependent on the modulus of elasticity of the material used. Lower moduli of elasticity cause larger cupping. Thus, the lowest cup deformations were found for panels consisting of OSB cores. It is, therefore, recommended to apply OSB in the middle layer if the substitution of the spruce layer is required (reduction of costs). However, the warping of spruce laminates is significantly lower than the warping of composite laminates, which is due to the differences in stiffness. Slitted middle layers show no significant effect on warping. They can, therefore, be applied in industrial production if a larger thermal insulation (as found by Bader et al. [2]) is required.

The three-dimensional material model for the calculation of deformations and stresses in wood was successfully validated to spruce cross-laminates and spruce-MDF laminates. The most important part of the material model is the mechano-sorptive deformation. Neglecting the mechano-sorptive effect yields ten times higher deformations. The glue line stiffness affects warping only a little. Thus, the influence of the different adhesives applied in the experiments, is negligible. The coefficient of diffusion of the glue lines has a strong influence on warp deformations, both cup_{xz} and cup_{yz} . With increased diffusivity of the adhesive, warping is reduced. Thus, adhesives with low moisture resistances should be applied to avoid panel warping. In Section 6.2 further numerical studies on hygroscopic warping are given.

Chapter 5

Internal stresses

5.1 Introduction

Stresses in wood perpendicular to the grain are of great interest. In this direction, wood has a minimal strength and stiffness compared to the longitudinal direction. Thus, wood breaks along the longitudinal axis when dried. These drying cracks weaken the wood while causing contact surfaces for water to penetrate into the material, and the visual surface quality of the wood is diminished.

By changing the temperature and humidity of the ambient air, stresses occur in the layers due to moisture uptake and cross-lamination. These stresses would lead to an expansion or a contraction if the panel were to be separated into the individual layers. Thus, stresses could be measured by a simple but effective method. The deformations that result from the stresses could be released by separating the layers, the change in length could be measured and stresses could then be calculated from the released strain and the modulus of elasticity. This method was applied in the present study.

Internal stresses due to moistening were determined. The advantage to drying is that the wood does not fail in terms of cracking, which leads to stress relaxation. The results are nevertheless transferable to drying stresses. Limitations may be the small differences in the shrinkage and swelling parameters. Furthermore, compressive stresses, as they occur when cross-laminated wood panels are moistened, may yield plastification of the material. This has an influence on the measurable stresses.

Different types of panels were tested on their influence on internal stresses. The material orientation of the individual lamellas in the RT plane was varied in steps of 0° , 45° and 90° . Furthermore, a large number of samples were tested in order to study the influence of different middle layer materials. Another group of panels was applied to test for the influence of pre-stressed outer layers.

In Section 5.2 the experimental investigations will be presented. Firstly, the strains on the lateral surfaces of two panels with vertically and horizontally aligned growth rings were evaluated by digital image correlation. The knowledge about unhindered shrinkage at the boundaries aided in the preparation for the tests of swelling properties. Besides free swelling, internal stresses due to moistening were measured in a single climate step and the results were compared against each other. The tests were completed by a visual evaluation of the surfaces of dried panels. In Section 5.3 numerical simulations will be presented and the element mesh and the boundary conditions will be shown. The model will be validated to the experimental results and the stress distributions will be displayed.

5.2 Experimental tests

5.2.1 Material

The material, as indicated in Section 4.2.1, was also applied in the study of internal stresses. The groups of panels that were studied in the present chapter and the types of investigations are summarised in Tab. 5.1. Panels of series AR relate to investigations of the annual ring angle, θ , as defined in Fig. 4.2. Again, three values of θ were tested: 0° , 45° and 90° . Another group of panels were used to investigate the middle layer structure. Two middle layers from Norway spruce were tested: 15mm (LR57) and 15mm with slits (SWS, Fig. 4.3). These panels were contrasted against panels where the spruce layer was substituted by a wood based panel: MDF500 (CMDF5), MDF700 (CMDF7), particleboard (CPB) and OSB (COSB). Pre-stressed outer layers were tested for their influence on swelling and internal stresses. Again, three magnitudes of pre-stress were applied: 0.6 N mm^{-2} , 1.2 N mm^{-2} and 1.8 N mm^{-2} .

The spruce layers of panels AR00, AR45, AR90, LR57, SWS, CMDF5, CMDF7 and COSB were produced in the laboratory. Using preconditioned lamellas, layers were assembled by gluing the lateral surfaces. The lamella width of the spruce layers was 100mm in

AR00, AR45 and AR90 and 70mm in LR57, SWS, CMDF5, CMDF7 and COSB. After a period of adhesive curing and after planing, the individual layers were bonded to three-layered cross-laminated panels. The adhesive applied was again 1K PUR. The spruce outer layers of the composite laminates (CMDF5, CMDF7, COSB, CPB) that were applied in the drying test were provided by an industrial manufacturer. In contrast to the layers produced in our laboratory, these layers contained knots and had a lamella width of ca. 70mm. The industrially produced panels (LR52, PS06, PS12, PS18) were assembled with UF. The individual layers of series LR52 were provided by an industrial manufacturer but assembled in the laboratory to three-layered panels after conditioning in different climates.

The influence of the adhesive on the internal stress state was investigated by means of panel types SWPUR and SWPUF. Two types of adhesives were studied: 1K PUR (PURBOND® HB 110, Purbond AG, Switzerland) and UF (Placol 4500, Geistlich Ligamenta AG, Switzerland). By means of tensile tests, Konnerth et al. [62] measured moduli of elasticity of 1K PUR (PURBOND® HB 110) and MUF (Dynamel L-435 with hardener H469, Dynea Austria GmbH). The results show significant differences between both adhesives: $E_{PUR} = 470 \text{ N mm}^2$ and $E_{MUF} = 6300 \text{ N mm}^2$. The two adhesives applied in this thesis were investigated by means of shear-tension tests according to DIN EN 205 [22] by Allenspach [1] and Schnider [89]⁴. The results show similar characteristics as the tests of Konnerth et al. [62]. The tensile-shear strengths were $\tau_{PUR} = 11.6 \text{ N mm}^{-2}$ (wood fracture percentage 4%) [89] and $\tau_{UF} = 15.1 \text{ N mm}^{-2}$ (85%) [1]. Both investigations reveal that UF has a larger modulus of elasticity and it applies a higher shear-tensile strength than 1K PUR. UF is much more brittle than 1K PUR. From the wood fracture percentages it can be concluded that the strength of 1K PUR is lower than that of the tested wood and the strength of UF is higher than that of wood. This thesis investigates whether application of these adhesives affects the internal stress state of cross-laminated panels.

⁴Gluing and testing at 20°C/65% RH, wood species: European Beech (*Fagus sylvatica* L.)

Table 5.1 Investigated panels related to the performed tests, ID according to Tabs. 4.1 and 4.2

Group	ID	Digital image correlation	Internal stresses	Swelling properties	Visual surface evaluation
		<i>drying</i> ¹	<i>moistening</i> ¹	<i>moistening</i> ¹	<i>drying</i> ¹
Annual ring orientation	AR00	✓	✓	✓	
	AR45		✓	✓	
	AR90	✓	✓	✓	
Middle layer	LR57		✓	✓	✓
	SWS		✓	✓	
	CMDF5		✓	✓	✓
	CMDF7		✓	✓	✓
	COSB		✓	✓	✓
	CPB				✓
Pre-stressed outer layers	PS06		✓	✓	✓
	PS12		✓	✓	✓
	PS18		✓	✓	✓
Adhesive type	SWPUR ²		✓		
	SWUF ³		✓		
Initial RH	LR52				✓

¹ Test conditions² SWPUR - solid wood, 1K PUR, $a_{OL} = a_{ML} = 10\text{mm}$, $LR = 0.67$ ³ SWUF - solid wood, urea resin, $a_{OL} = a_{ML} = 10\text{mm}$, $LR = 0.67$

5.2.2 Methods

Digital image correlation

Digital image correlation (DIC) is a suitable method to measure strains on a surface [54]. It is a non-contact optical method and, thus, it does not influence the specimen during the test. The panels that were used in the tests measured $200\text{mm} \times 200\text{mm}$ in width and length. On the lateral face of the samples, a random dot texture termed speckle pattern was sprayed on the specimen. An airbrush gun and finely pigmented acrylic paint was used in order to obtain a high-resolution pattern. Firstly, a white background and then black speckles were applied. This range of grey values is needed for the interpretation of displacements on the specimen's surface by the DIC software.

The samples first were placed in a climate room with constant conditions of $\varphi = 85\%$ until a consistent weight was achieved. The measurement configuration was set up in a dry climate with average conditions of $T = 20^\circ\text{C}$ and $\varphi \approx 40\%$ RH. The speckle fields of two parallel aligned specimens were filmed by a CCD camera with a distortion-free objective lens. The principal axis of the camera was aligned perpendicular to the panel surface as shown in Fig. 5.1. Images were recorded by the camera with a frequency of 12h^{-1} . A source of cold light provided even illumination of the speckle field. Reflections that were too intense from the surface of the specimens were avoided in order to provide accurate data for the analysis of deformations on the wood surface by the software.

Finally, the recorded images were run through a strain mapping software (VIC2D, Correlated Solutions). The strain was determined from the surface displacements that were calculated on the basis of a cross-correlation algorithm. Therefore, the grey values in between a defined area (subset) around a central pixel are used to locate the best matching pattern after deformation by maximising the cross-correlation between two subsets.

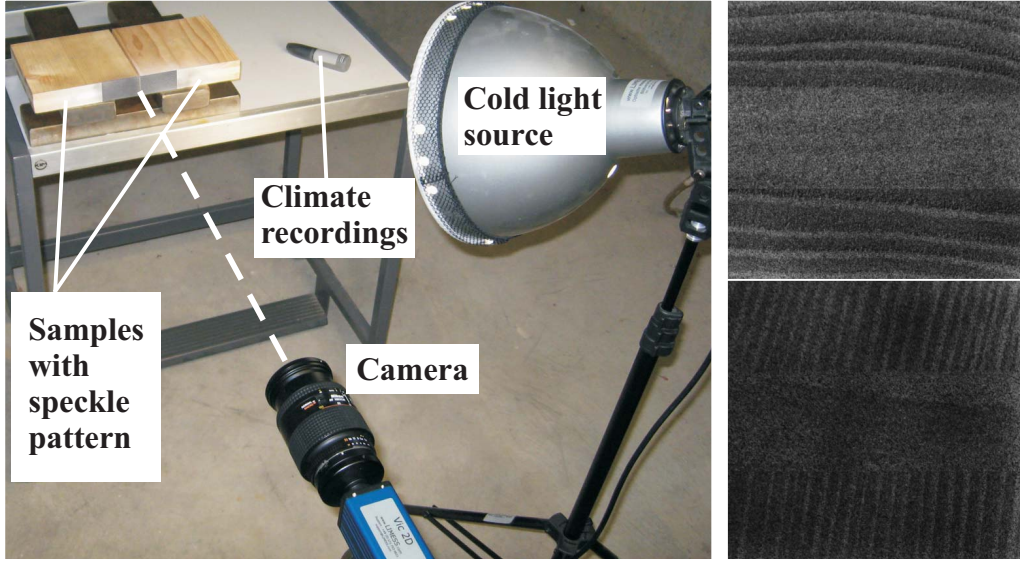


Figure 5.1 Test set-up for measuring shrinkage strains on the lateral panel surfaces (lefthand side) and samples of type AR00 (upper) and AR90 (lower) with speckle patterns (righthand side)

Swelling properties

The swelling in the three panel directions x , y and z was investigated with the equipment displayed in Fig. 5.2. The specimens were conditioned at $\varphi = 35\%$ until moisture balance and were then placed in a climate room with constant conditions of $\varphi = 85\%$. To exclude friction influences during swelling, the samples were positioned on point-supports which additionally allowed contact of all surfaces to the air. Inductive displacement transducers measured the changes in length due to swelling as shown in Fig. 5.2. Different free swelling in the individual layers occurred at the boundaries. This is discussed in more detail in Section 5.2.3. The in-plane displacements were determined using screws ($\varnothing 5\text{mm}$) that were turned in the samples over the whole panel thickness with a distance of 20mm to the edges. The displacements were gripped at the screw heads while the thickness swelling was measured directly at the wood surface. The linear swelling q_k [%] in the panel axes k was calculated by

$$q_k = \frac{a_{k,wet} - a_{k,dry}}{a_{k,dry}} \times 100, \quad k = x, y, z, \quad (5.1)$$

where $a_{k,dry}$ [m] is the appropriate dimension in the initial (dry) climate and $a_{k,wet}$ [m] is the dimension in the test (wet) climate. A swelling coefficient α_k [% %⁻¹] was calculated from the linear swelling and the change in moisture content $\Delta\omega$ [%] to

$$\alpha_k = q_k \Delta\omega^{-1}, \quad k = x, y, z. \quad (5.2)$$

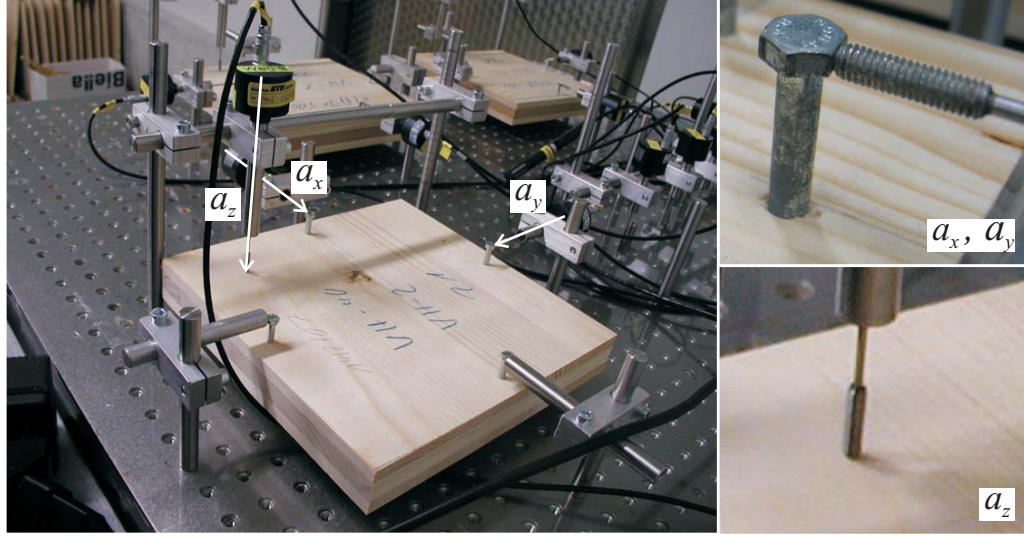


Figure 5.2 Measurement equipment for the continuous recording of swelling displacements a_k in the panel axes $k = x, y, z$

Internal stresses

The procedure is illustrated in Fig. 5.3. Firstly, the specimens were conditioned at $\varphi = 35\%$ until mass equilibrium. Afterwards, they were transferred into a constantly conditioned climate room with $\varphi = 85\%$. At several sampling points after conditioning 25mm of all edges were removed (Fig. 5.3a) and then the specimens were divided into $200\text{mm} \times 20\text{mm}$ wide strips by means of a circular saw (Fig. 5.3b). Removing the edges prevented influences of unhindered swelling of the boundaries. The length was quantified immediately before (l_0) and after (l_1) slicing into the individual layers (Fig. 5.3c). Released deformations were measured by a linear gage (Mitutoyo) with an accuracy of 10^{-2}mm . The contact of the gage head onto the small surface was at the same place in both measurements. Compressive stress leads to an extension and tensile stress to a contraction of the slices. A potential bending of the sliced specimen due to the annual ring curvature could be manually adjusted with this technique. The mean stress σ_m [N mm^{-2}] in each slice is

$$\sigma_m = E \varepsilon_m . \quad (5.3)$$

The mean strain ε_m [-] was calculated from the initial length l_0 [m] and the final length l_1 [m] to be

$$\varepsilon_m = \frac{l_0 - l_1}{l_0} . \quad (5.4)$$

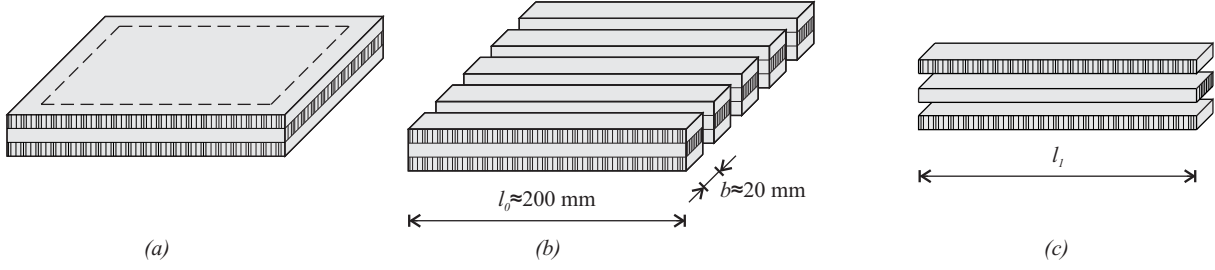


Figure 5.3 Procedure to determine the internal stress state of laminated wood panels: a) removal of the edges, b) cutting into strips, c) release of strain by sawing in the glue lines

This is not the maximum stress in the slice, since stress distribution throughout the slice is not uniform, and the stress at the ends of the slice must be zero. For glulam samples perpendicular to the grain, Jönsson and Svensson [52] calculated the maximum stress to be 30% higher than the mean stress.

In each individual slice the modulus of elasticity was determined by means of ultrasonic waves. This dynamic method was chosen because preliminary investigations by means of bending tests partly lead to failure of the small specimens due to a high variation in the densities of early- and latewood. This problem occurred especially when the fibre direction was aligned perpendicular to the specimens longitudinal axis. The sound velocity measurements were carried out with an ultrasonic tester (BP 5, Steinkamp, $f = 50 \text{ kHz}$). A transmitter and a receiver were installed on each end of the slices and the sound running time of a constant pulse was measured over the sample length. The modulus of elasticity is approximated by

$$E_{dyn} = c^2 \rho , \quad (5.5)$$

where $c \text{ [m s}^{-1}\text{]}$ is the sound velocity and $\rho \text{ [kg m}^{-3}\text{]}$ is the raw density [9, 10]. This is valid for bars only when their height and width are small compared to the wavelength. The smallest sound velocities measured in this study corresponded to a wavelength of around 35mm (AR45), which was larger than the dimensions of the cross-section of the slices. Even in an isotropic system, the dynamic modulus of elasticity is higher than the statically determined value. However, since the comparison between the variants was of main interest, the dynamic elastic modulus was not transformed into a static one. The application of sound velocities for the determination of elastic parameters is well accepted and has frequently been applied [4, 10, 13, 60, 65].

In the transverse direction, the sound waves penetrated through a varying number of glue

lines. Since a glue line is very thin compared to the total length of the specimen, its influence is negligible. Parameters such as contact pressure or placement of the transducers were an important factor. Possible errors in the determination of the internal stress state in cross-laminated wood panels were:

- inaccuracy in the strain measurements especially parallel to the grain due to the very small change in length,
- errors in measuring the dynamic modulus of elasticity.

Visual surface evaluation

Studies of the surface behaviour were performed in order to test the internal stress state during drying. Tensile stresses develop in the outer layers due to shrinkage occurring perpendicular to the grain and constraining by the middle layer. When the tensile strength is exceeded, gaps open up transverse to the fibres and these cracks then expand in the fibre direction. Another scenario may be the delamination in the glue line between two lamellas of the surface layers. Due to the straight orientation of the gaps along the fibres, it becomes possible to measure their length.

The tested panels measured 500mm $\times$ 500mm in width and length. The crack development was observed during 21 days and the crack lengths at the two large panel faces were measured by means of a sliding calliper with an accuracy of 10^{-2} mm. Once opened, the ends of cracks were marked and further investigated. Additionally, the middle layer was visually observed at the lateral faces.

The initial and the test conditions are summarised in Tab. 5.2. The group of samples with different middle layer materials, i.e. panels LR57, CMDF5, CMDF7, COSB and CPB, were initially conditioned at 20°C and 85% RH and then transferred to the test climate room with constant conditions of 20°C and 35% RH. Panels with pre-stressed outer layers were also preconditioned at 20°C and 85% RH, but the samples were tested in two different climates. The test climate of 20°C and 35% RH was generated in a climate room and the test climate of 25°C and 30% RH was generated in a climate chamber with constant air circulation. The layers of series LR52 were initially conditioned at 20°C and 45% RH, 65% RH and 85% RH, respectively, until moisture equilibrium was achieved. The layers were then assembled according to the scheme that is drawn in Tab. 5.2. Immediately after pressing, the panels were stored in the test climate, which was 20°C and 45% RH.

The test of different preconditioned layers related to the investigation where the influence of a moister middle layer was studied. Varying climate differences between the middle and the surface layers were induced. These panels were compared against panels with identical preconditioned layers.

Table 5.2 Initial and test climates of the drying tests in °C/% RH

ID	Initial climate	Test climate
LR57	20/85	20/35
CMDF5	20/85	20/35
CMDF7	20/85	20/35
COSB	20/85	20/35
CPB	20/85	20/35
PS06	20/85	25/30
	20/85	20/35
PS12	20/85	25/30
	20/85	20/35
PS18	20/85	25/30
	20/85	20/35
TL/ML/BL (20°C/% RH)		
LR52	85/85/85	20/45
	65/65/65	20/45
	65/85/65	20/45
	45/85/45	20/45
	45/65/45	20/45

5.2.3 Results and discussion

Digital image correlation

The surface displacement due to shrinkage of samples that were previously in a moist climate ($\varphi = 85\%$), were recorded over 160 hours in a dry climate ($\varphi \approx 40\%$). Strains were calculated from these displacements. The yz-plane was investigated at the unhindered boundaries. Two samples were tested, one with vertically aligned growth rings (AR90)

and one with horizontally aligned growth rings (AR00). Fig. 5.4 presents the strains in the y-direction (i.e. strains perpendicular to the fibres in the outer layers and strains in the fibre direction of the middle layer) for different distances to the unhindered edge for both boards, AR00 and AR90. In Fig. 5.5 the calculated strains are illustrated over the measured area.

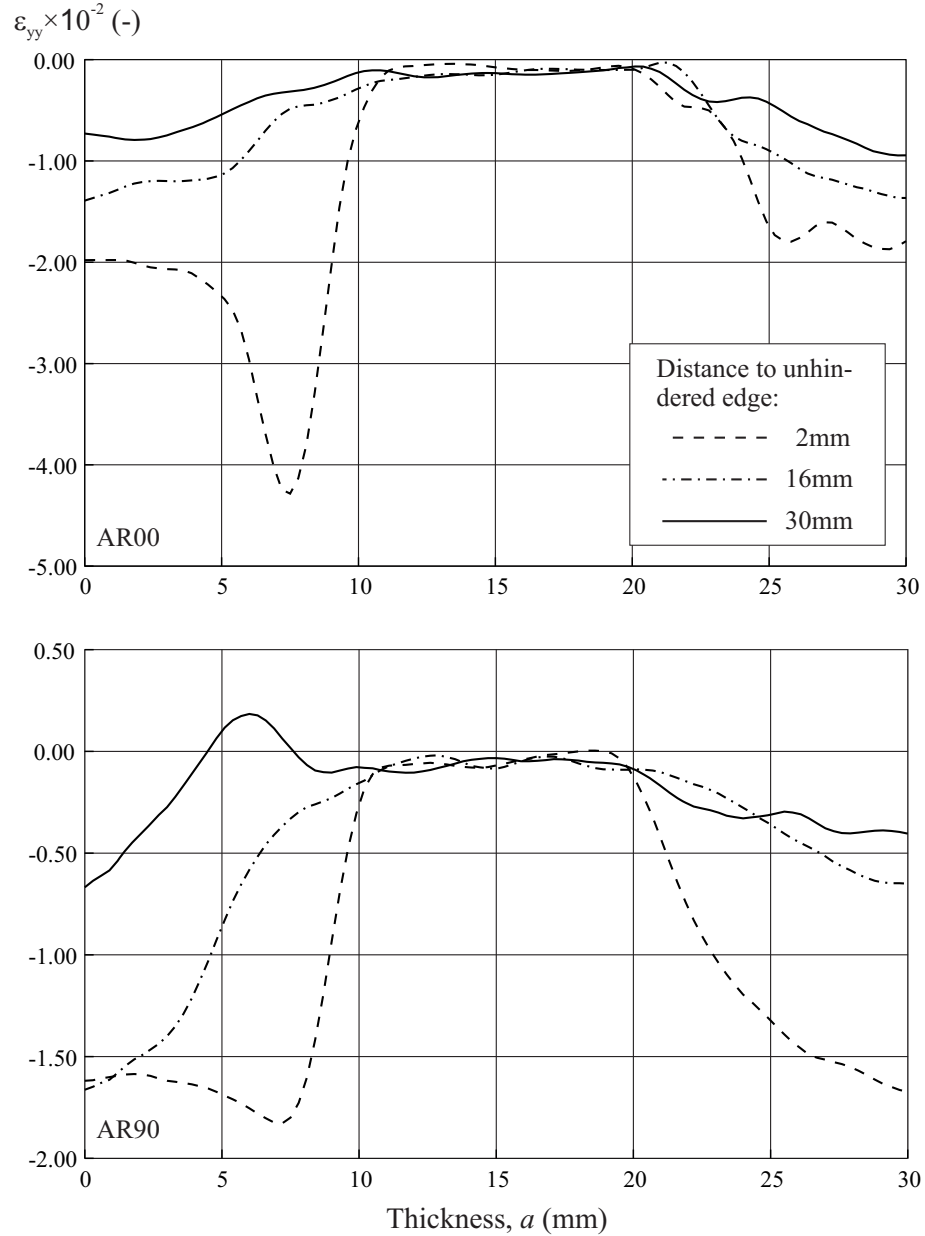


Figure 5.4 Distribution of the horizontal strain fragment, ε_{yy} , over panel thickness in samples AR00 (upper) and AR90 (lower) measured by digital image correlation, $t = 160\text{h}$

The differences between the outer layers and the middle layer are obvious and very significant. The middle layer contracted minimally along the y-axis, since longitudinal shrinkage

of wood is very small. In the outer layers, shrinkage strains perpendicular to the grain could be detected. The results indicate that these strains decreased from the edge towards the centre of the samples. As shown in Fig. 5.5, the strain distribution was nearly balanced over the panel thickness at a distance of 30mm to the unhindered edge. Significant differences between both growth ring orientations were found. As expected, due to higher shrinkage in the tangential direction, the calculated strains were larger in panel AR00. Figs. 5.4 and 5.5 show a gap opening in the top layer of sample AR00. This was expressed by a local increase in the shrinkage strains. This crack was found to develop between the borders of early- and latewood, which is due to the large density differences.

Swelling properties

The results shown in Tab. 5.3 indicate that the swelling in the thickness direction was the dominant swelling direction. Swelling in the board planes was significantly lower, i.e.

$$\alpha_z \gg \alpha_y > \alpha_x .$$

This sequence was a result of the orthotropic swelling characteristics of wood. Due to the crosswise bonding of the layers, swelling in the x- and y-directions was restrained and, thus, swelling in the z-direction became dominant. The difference between α_x and α_y was governed by the orientation of the layers. The larger the ratio of transversely oriented fibres, the larger the swelling in that direction.

The annual ring orientation affected the swelling properties. The difference between radial and tangential swelling yielded differences in the deformation of cross-laminated wood panels. The largest swelling (α_y) in AR00 was a result of the large tangential swelling and the lowest swelling of AR90 was a result of the lower radial swelling.

The middle layer structure affected the swelling of laminated panels. The swelling coefficient in the major direction, α_x , was larger in panels where the spruce middle layer was substituted by wood composites. The swelling along the minor axis, α_y , in spruce-composite laminates was governed by the stiffness of the middle layer. With increased modulus of elasticity, the deformation decreased. The swelling in the thickness direction, α_z , was significantly larger in panels with a wood composite in the middle layer. This resulted from the larger swelling of wooden composites in that direction.

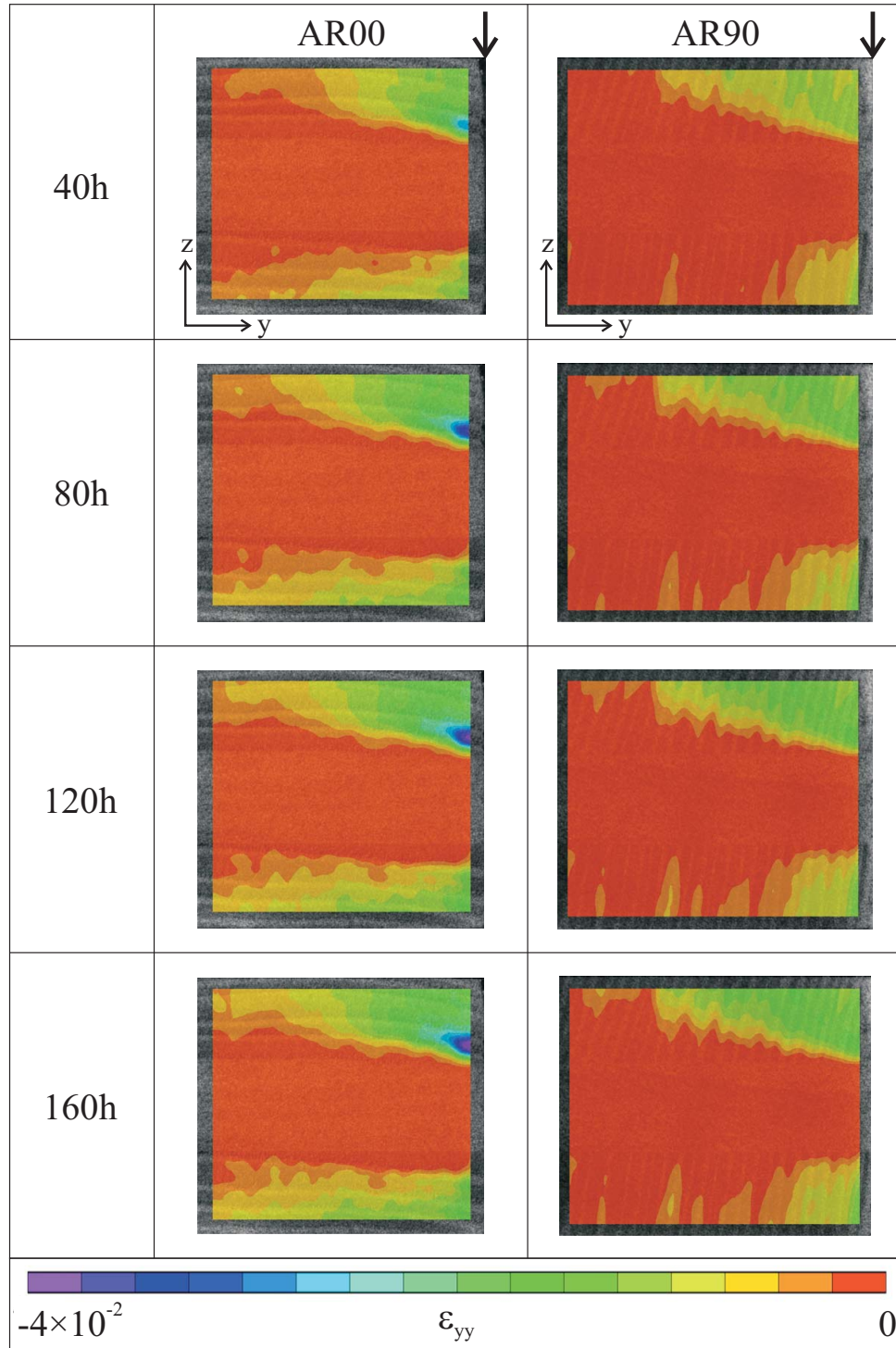


Figure 5.5 History development of the horizontal strain fragment, ε_{yy} , on the lateral surface, measured by digital image correlation, arrows indicate unhindered boundaries

Table 5.3 Swelling coefficients α_k ($k = x, y, z$) after 14 days wetting from 35% RH to 85% RH

ID	n	α_x (% % ⁻¹)	α_y (% % ⁻¹)	α_z (% % ⁻¹)	α_x/α_y (-)	α_x/α_z (-)
AR00	4		0.039			
AR45	4		0.021			
AR90	4		0.019			
LR57	3	0.007	0.048	0.345	7.3	52.9
SWS ¹	3	0.007	0.032	0.379	4.4	52.4
CMD5	3	0.013	0.065	0.518	4.9	39.5
CMD7	3	0.010	0.032	0.423	3.2	42.8
COSB	3	0.014	0.022	0.518	1.6	37.6
PS06	3	0.028	0.065	0.762	2.3	27.3
PS12	3	0.027	0.025	0.690	0.9	25.6
PS18	3	0.029	0.047	0.634	1.6	21.6

¹ Assumption: $\omega_{ML} = \omega_{OL}$

Internal stresses

Influence of the annual ring orientation

Moisture-induced stresses and deformations due to a single climate step from $\varphi_1 = 35\%$ to $\varphi_2 = 85\%$ were studied. However, during the tests a climate change was induced in the otherwise constantly conditioned climate room due to a defect. Thus, the relative humidity decreased from day 23 onwards as displayed in Fig. 5.6.

Fig. 5.7 illustrates the measured moisture content in the outer and middle layers. The moisture content of different panel types did not significantly differ and, thus, the samples of the three types of θ were summarised in one boxplot. The moisture content increased from the initial state to day 14. The increase was from 10.6% to 15.3% in the outer layers and from 11.1% to 14.4% in the middle layers (Fig. 5.7).

The panels were investigated perpendicular to the grain for the outer layers. The measured dynamic moduli of elasticity differed significantly in different material directions (Fig. 5.8). The modulus of elasticity was largest in radial direction ($\theta = 90^\circ$) and smallest for $\theta = 45^\circ$. In average the ratio was 1.0 : 2.2 : 3.7 (45° : 0° : 90°). The considerably small shear modulus, G_{RT} , of spruce wood caused the very small modulus of elasticity

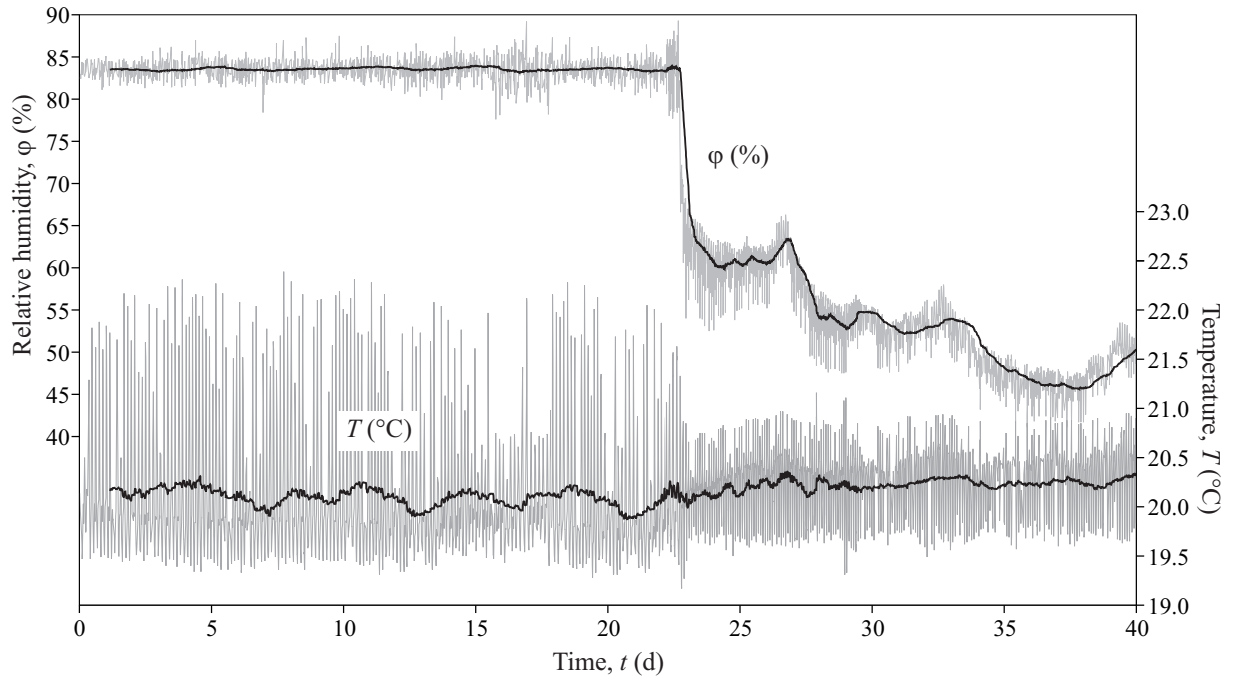


Figure 5.6 Test climate conditions

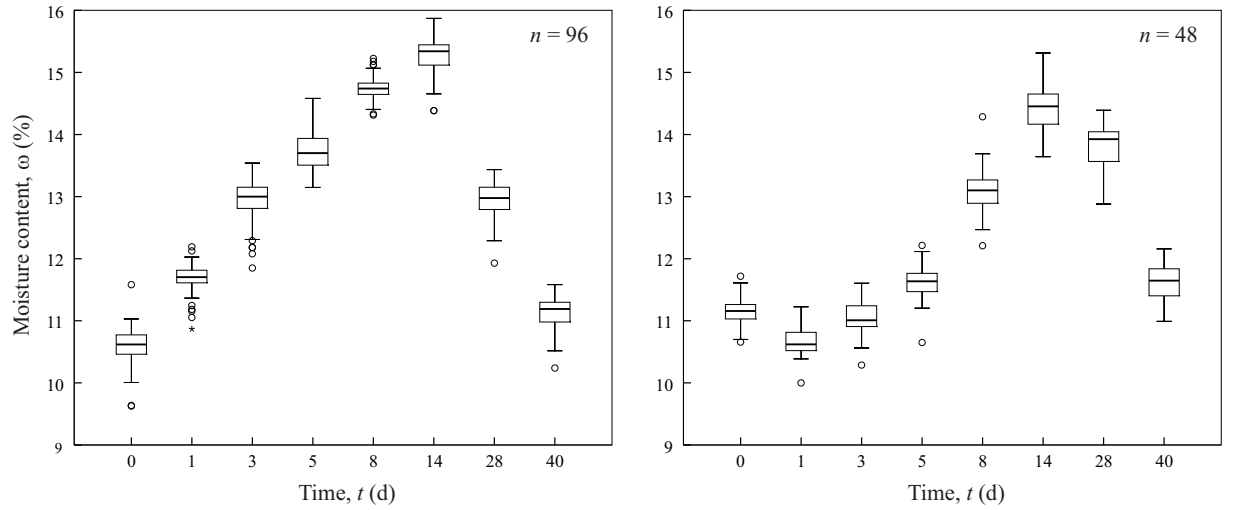


Figure 5.7 Moisture content in the outer layers (lefthand side) and in the middle layers (right-hand side) of panels of series AR (time axes not scaled), initial climate: 20°/35% RH, test climate: Fig. 5.6

under 45° orientation (Section 2.3). The relationship between modulus of elasticity and moisture content was not significant in the investigated samples. The highest variation was detected in radially aligned fibres ($\theta = 90^\circ$). Due to the cylindrical shape of the log, considerable variations in the annual ring curvature over the board width and thickness occurred in the laboratory lamellas with a width of 100mm. Thus, the ring angle θ was not exactly 0° , 45° or 90° in the boards. $E(\theta = 90^\circ)$ was particularly affected by an anomaly in the ring curvature, since the reduction of the modulus of elasticity is substantial when the annual rings deviate from the exact alignment of 90° .

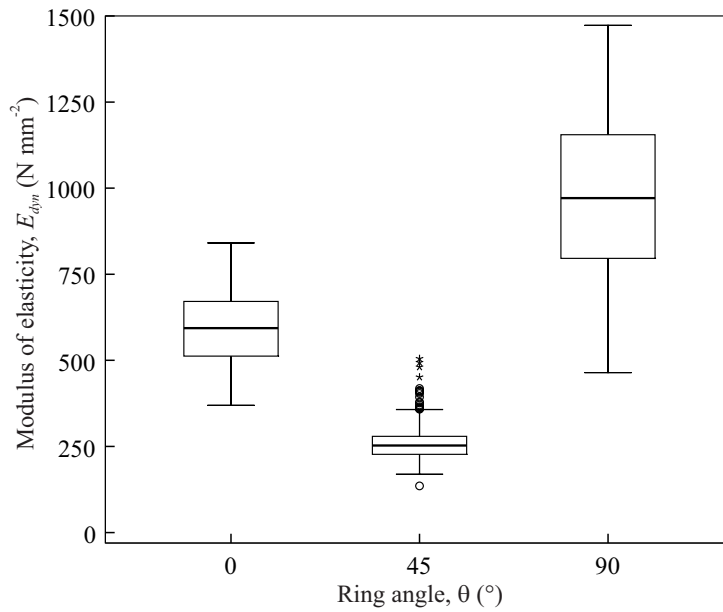


Figure 5.8 Moduli of elasticity measured by means of ultrasonic waves in the outer layers perpendicular to the grain for different ring angles θ , 0° - tangential, 90° - radial, $n = 192$

Compared to the moduli of elasticity found by Neuhaus [74] (see Tab. 4.7), which were determined in static tension tests, the present results are around 17% larger in the radial direction and around 46% larger in the tangential direction. This is a known phenomenon, which is due to the determination of elastic constants by means of ultrasound measurements. The calculation of the elastic modulus from the measured sound velocities, according to Eq. 5.5, does not consider any transverse contraction. The difference between dynamically and statically determined moduli of elasticity depends on the Poisson's ratio ν . For an isotropic material, Keunecke et al. [55] calculated differences of 10% when $\nu = 0.2$ and of 50% when $\nu = 0.5$. For spruce wood, Keunecke et al. [55] found differences of 34% to the values of Neuhaus [74] in the radial and tangential directions.

The moistening of the panels resulted in an expansion of the sliced strips. The formation of strains was governed by different swelling in different material orientations (Fig. 5.9). The sequence followed the swelling coefficient: 1.0 : 1.7 : 1.9 (90° : 45° : 0°). The stress plot (Fig. 5.10) is based on calculation of the product of strain and dynamic modulus of elasticity. Thus, the development of stresses was governed by different swelling and different modulus of elasticity in different material orientations: 1.0 : 2.1 : 2.5 (45° : 90° : 0°). The expansion and, thus, compressive stresses were maximum between days 5 and 8 depending on the panel.

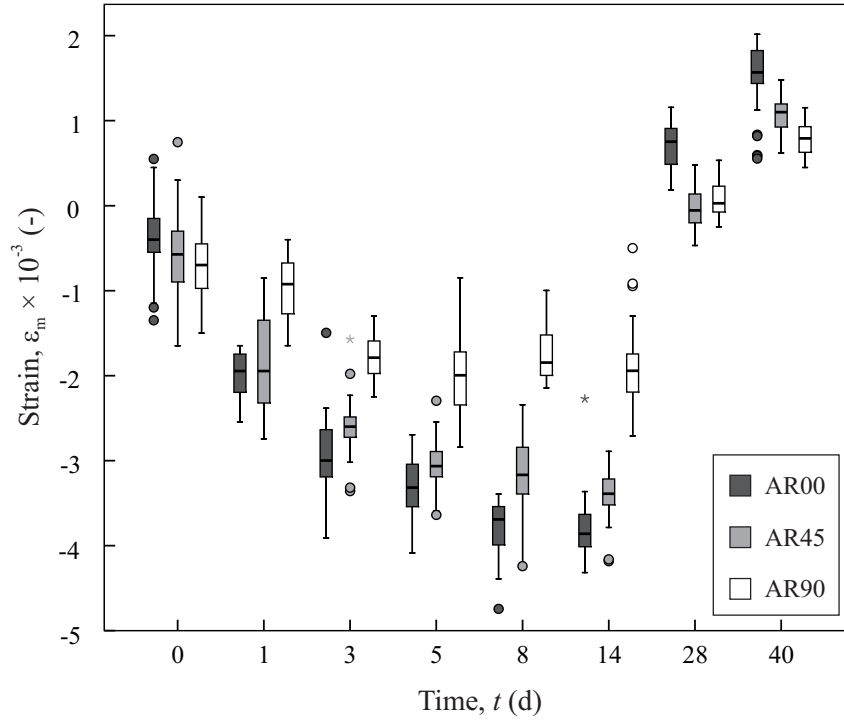


Figure 5.9 Mean strains perpendicular to the grain in the outer layers in samples with different material orientations θ , initial climate: $20^\circ/35\%$ RH, test climate: Fig. 5.6, time axis not scaled, $n = 32$

Eight strips were sawn from each sample as is illustrated in Fig. 5.11. The consequent numbering of the strips allowed for the visualisation of the moisture and stress distribution from the edges towards the centre of the samples. Fig. 5.12 shows the development of moisture content and stresses perpendicular to the grain across the panel length. Initially, the moisture contents differed between the edges and the centre of the panels and equalised afterwards. After the decrease in relative humidity ($t > 23\text{d}$) the moisture content was again unevenly distributed, i.e. the edges were drier than the centre of the boards. This was due to the large moisture conductivity parallel to the grain, which, despite previously

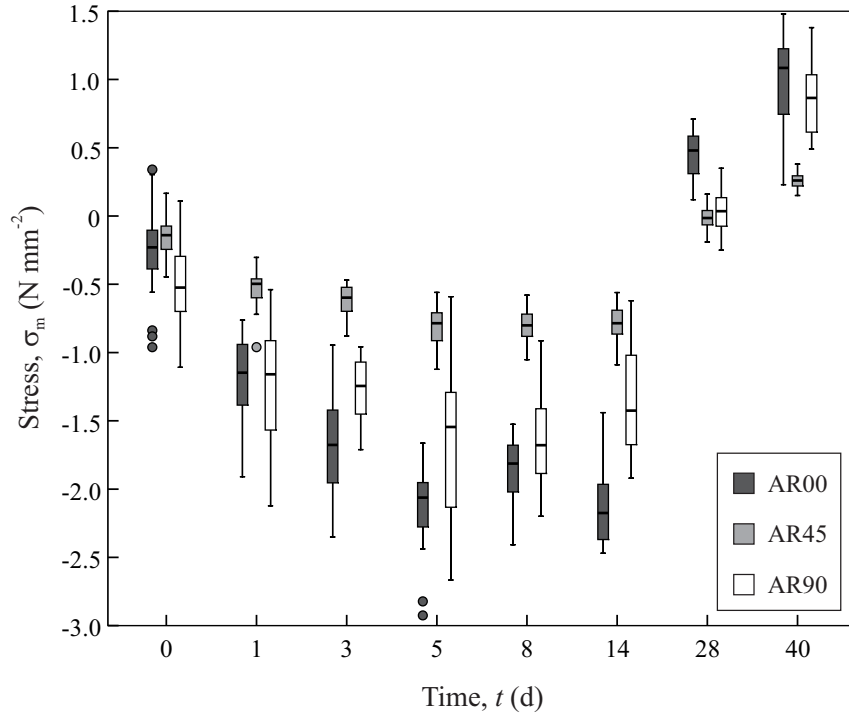


Figure 5.10 Mean stress perpendicular to the grain in the outer layers in samples with different material orientations θ , initial climate: $20^\circ/35\%$ RH, test climate: Fig. 5.6, time axis not scaled, $n = 32$

removing 25mm of both boundaries, affected the border areas (most noticeably in strips 1 and 8). The stress plots reveal that the compressive stress was initially higher at the edges but was then evenly distributed without a noteworthy alteration. The tensile stresses due to drying ($t = 28\text{d}$ and $t = 40\text{d}$) were larger at the edges than in the centre in series AR00 and AR90.

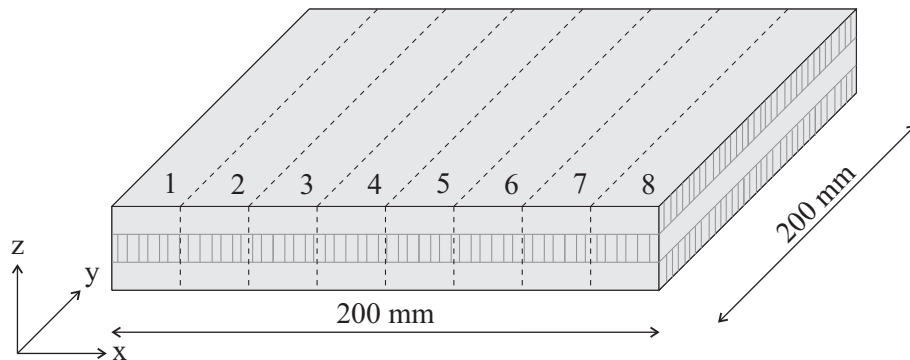


Figure 5.11 Cutting scheme and strip numbering

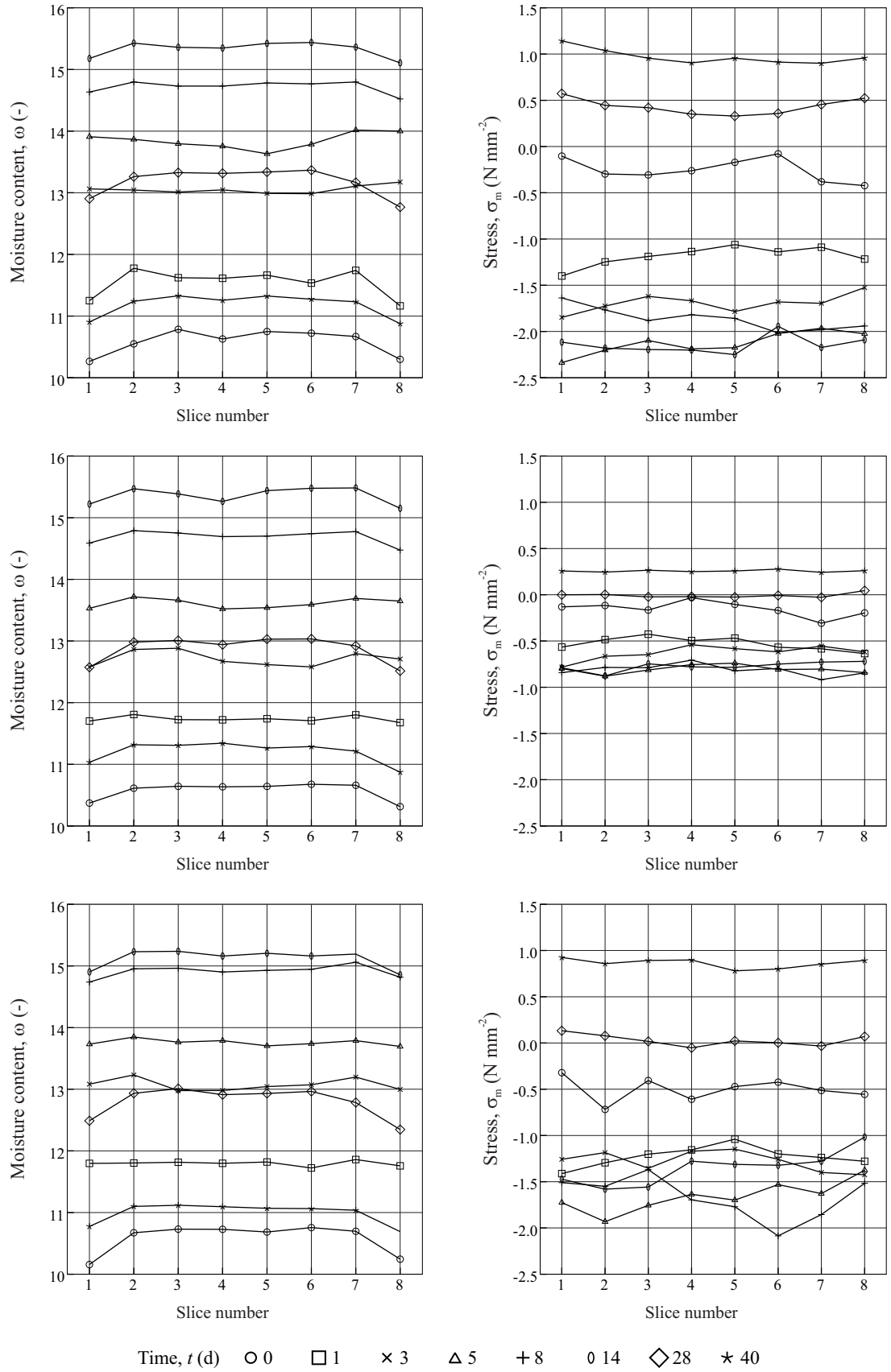


Figure 5.12 Stress and moisture content distributions in the outer layers within panels of type AR00 (upper), AR45 (middle) and AR90 (lower)

Influence of the middle layer structure

Eigenstresses in laminated panels with varying middle layer structures were investigated over 21 days. The following middle layers were applied: spruce (LR57), spruce with slits (SWS, Fig. 4.3), MDF500 (CMDF5), MDF700 (CMDF7) and OSB (COSB). The thickness of the outer layers was consistently 10mm. The thickness of the middle layers was between 15mm and 16mm. All panel types were initially conditioned at 35% RH and then tested at 65% RH and 85% RH, respectively.

In Tab. 5.4, the measurements are evaluated for $t = 21\text{d}$. In this process, outer and middle layers are regarded separately. The following characteristics are shown by the table:

- For panel type SWS, only the outer layers were tested, since an analysis of the middle layer would fail due to the slits.
- The measured strains were very small in the composite middle layers compared to the strains perpendicular to the grain in solid wood. The standard deviations are significantly higher than the arithmetic mean, which made the analysis difficult. A contraction of the strips and, thus, tensile stresses were measured in some of the composite middle layers. No further analysis will be made on the middle layers due to the small number of tests and the large variations between them.
- The annual rings were aligned randomly in the lamellas. However, the small moduli of elasticity that were measured in wood perpendicular to the grains point to a large ratio of annual rings where $\theta \approx 45^\circ$.

The history stress developments for both test climates are plotted in Fig. 5.13. The plots show stresses in the outer layers that were measured perpendicular to the grain. Again, marginal tensile stresses were observed in the initial state ($t = 0\text{d}$). Due to wetting, compressive stresses developed until the maximum, which was between day 5 and 7 in most of the panels. Afterwards, the compressive stresses remained constant or decreased until day 21. A distinct maximum was observed for CMDF7 tested at 85% RH ($t = 5\text{d}$).

Between panels with different middle layer structures, significant differences in the stresses (perpendicular to the grain in the outer layers) were detected. Fig. 5.14 displays the mean values and the 95% confidence intervals of the measured moduli of elasticity, strains and stresses in different panel types in the two test climates, 65% RH and 85% RH (summarised

Table 5.4 Moduli of elasticity, E , and stresses, σ_m , perpendicular to the grain and moisture content change $\Delta\omega_{21} = \omega(t = 21\text{d}) - \omega(t = 0\text{d})$ after 21 days in the test climate of 85% RH

ID	Layer	n	E (N mm ⁻²)		σ_m (N mm ⁻²)		$\Delta\omega_{21}$ (%)
			$\bar{X}$	STD	$\bar{X}$	STD	$\bar{X}$
LR57	OL	32	347.0	54.1	-0.75	0.17	5.8
	ML	8	275.0	7.2	-1.12	0.10	5.5
SWS	OL	48	334.2	85.0	-0.94	0.17	6.1
CMDF5	OL	32	316.4	44.9	-0.72	0.14	6.5
	ML	8	2153.3	30.7	-0.12	0.36	4.3
CMDF7	OL	30	301.4	78.1	-0.84	0.15	6.2
	ML	16	4171.4	89.4	0.17	0.32	1.7
COSB	OL	32	277.9	81.3	-0.65	0.13	5.9
	ML	16	5451.6	633.2	0.39	0.85	3.3

for all $t > 0\text{d}$). The plots have to be interpreted as follows: if the confidence intervals of two samples do not coincide, the mean values of both samples differ significantly from each other. The stress plots show the following characteristics:

- CMDF5 differed significantly from all other panel types when tested in 65% RH and from CMDF7 and SWS when tested in 85% RH.
- The variation of CMDF7 was significant to all other panels except COSB in the 65% test climate.
- The compressive stresses were significantly higher in COSB than in all other panels except CMDF7 when tested in 65% RH. In the 85% test climate, COSB differed significantly only from CMDF7 and LR57.
- SWS differed significantly from all panels except LR57 (65% RH) and COSB (85% RH).
- As a consequence, the difference of stresses in LR57 was significant to all other panels except SWS (65% RH) and CMDF5 (85% RH).

The strong differences between CMDF7 and the remaining panel types in the test climate 85% RH were a result of the significantly higher moduli of elasticity of CMDF7. These

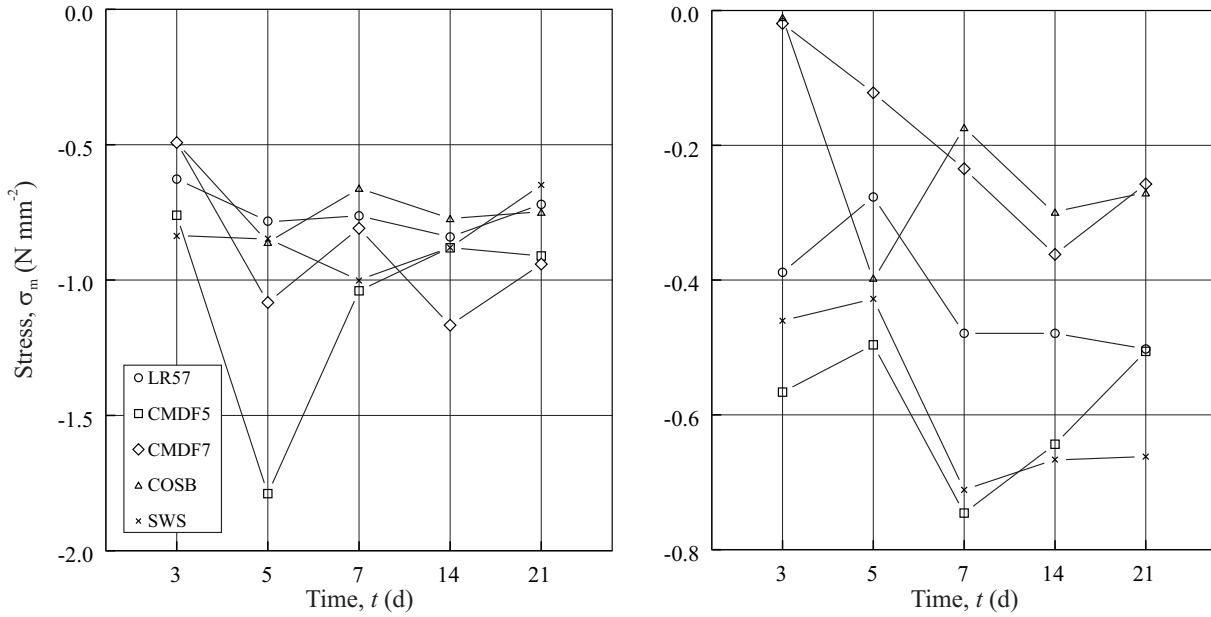


Figure 5.13 Average stresses in the outer layers perpendicular to the grain in panels with different middle layers, lefthand side: test climate $\varphi = 85\%$, righthand side: test climate $\varphi = 65\%$, time axis not scaled

were detected at $t = 5$ d. These wood influences also explain the significant stress maximum at this time (Fig. 5.13). The results obtained in both test climates differ from each other. Under normal conditions (65% RH) strains and stresses in composite laminates (CMD5, CMD7, COSB) were found to be smaller than in solid wood laminates (SWS, LR57). Under wet conditions (85% RH) strains and stresses were found to be smallest in CMD7, COSB and SWS.

The moisture and stress distributions within panels of type LR57 that were tested in 85% RH are illustrated in Fig. 5.15. The plots indicate that moisture content and moisture history have a strong influence on the stress distribution. In the outer layers, the moisture content increased rapidly until day 3, which was followed by a slight increase until day 14. Then constant moisture content was maintained. The high diffusion resistance of the glue lines caused a delay in the increase of the moisture content of the middle layer. Here, a strong increase was observed until day 14 and a slight increase until day 21. An influence of diffusion through end-grain areas of the lateral faces of the panels is evident, since the moisture content was higher in the outer zones than in the core of the middle layer.

The stress plots show a strong increase in compressive stresses until day 5 in the outer layers, afterwards these stresses decreased. In the middle layer, however, the compressive stresses increased until day 14 to a maximum of 1.74 N mm^{-2} , similarly, afterwards

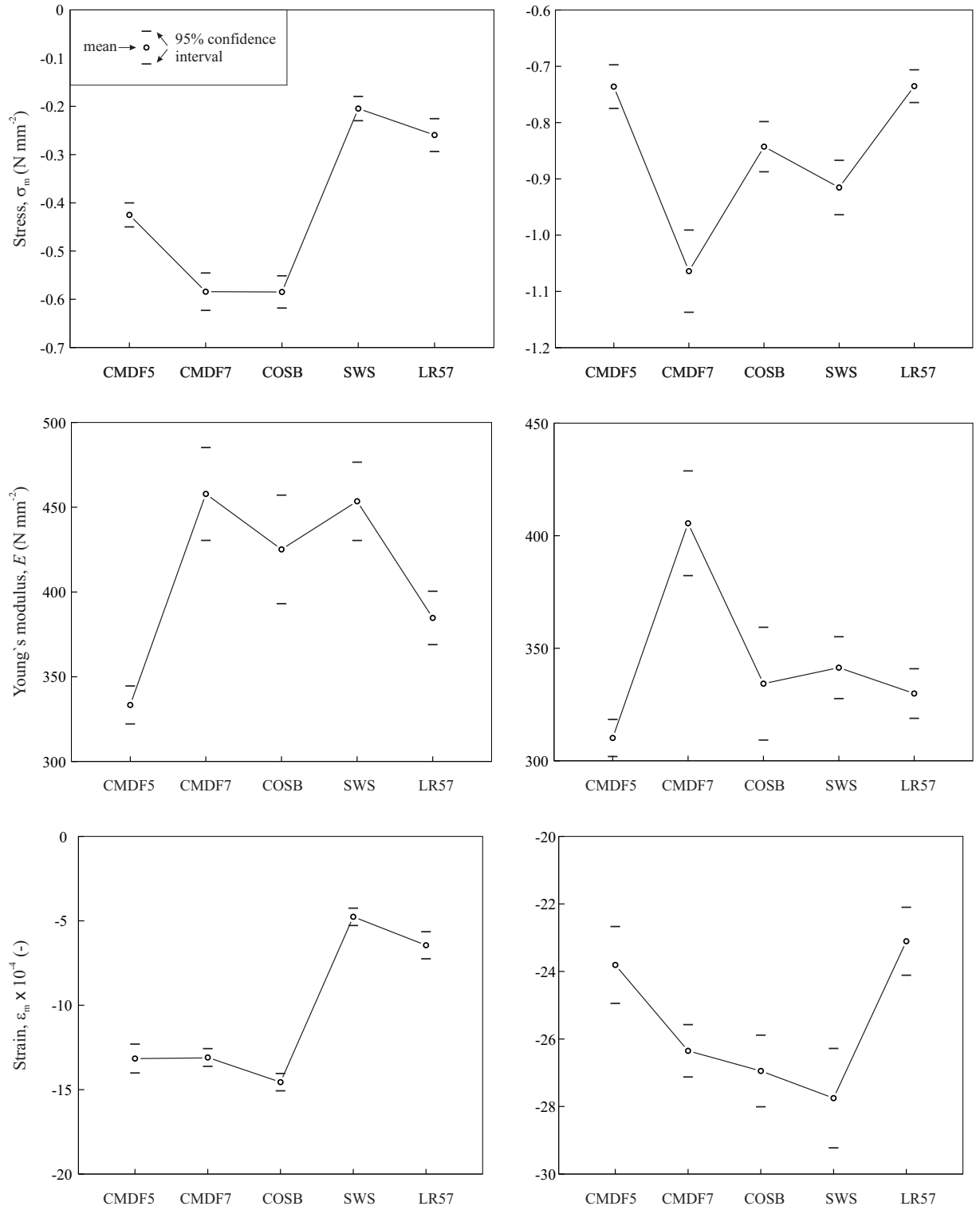


Figure 5.14 Mean values and 95% confidence intervals of stresses, strains and Young's moduli in the outer layers perpendicular to the grain in panels with different middle layers, lefthand side: test climate $\varphi = 65\%$, righthand side: test climate $\varphi = 85\%$ (summarised for all $t > 0\text{d}$)

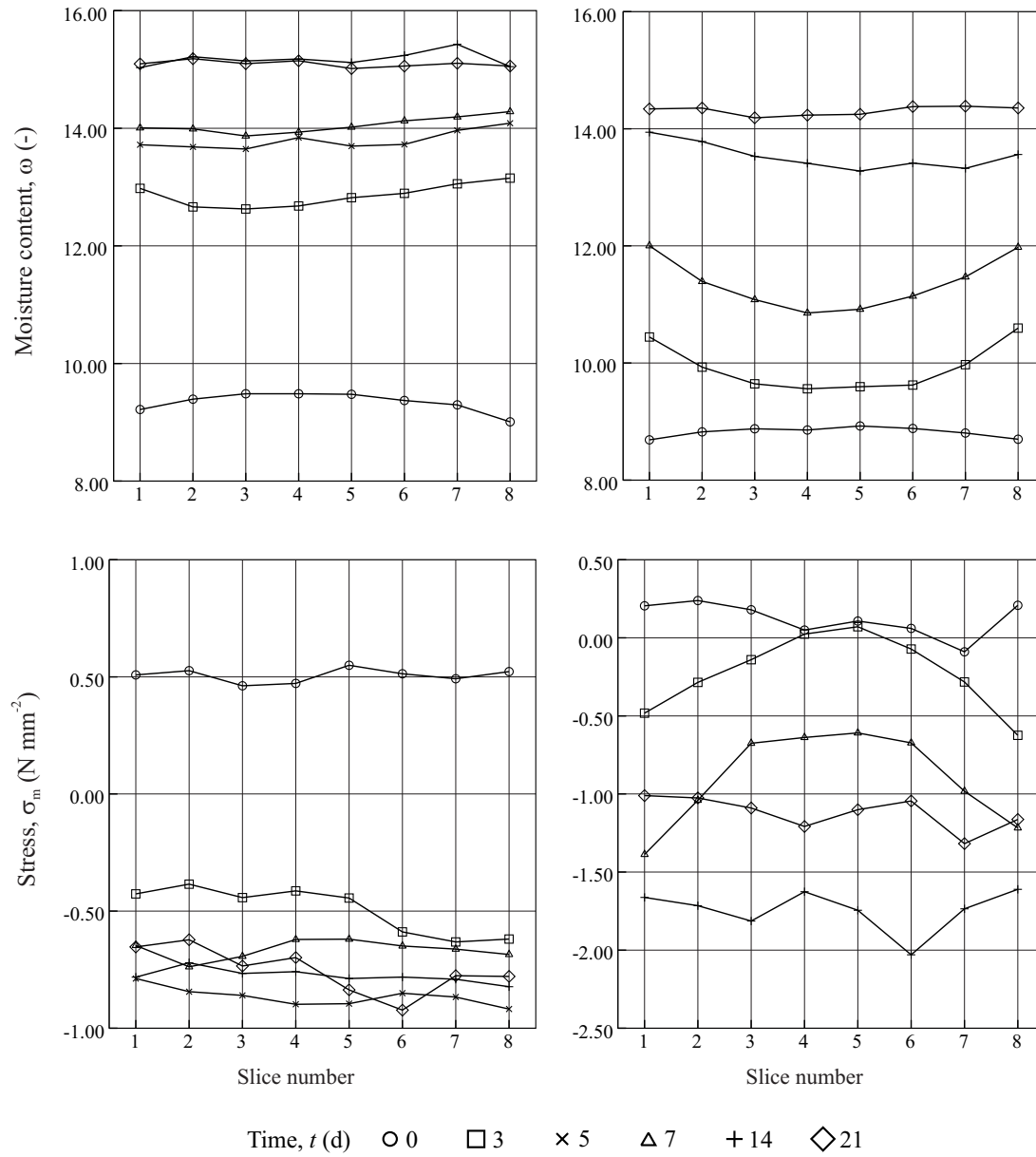


Figure 5.15 Stress and moisture content distributions within the panel type LR57 in the outer layers (lefthand side) and the middle layer (righthand side), test climate $\varphi = 85\%$ (stresses measured perpendicular to the grain)

the compressive stresses decreased. The increase in compressive stresses was delayed compared to the outer layers, which was due to the moisture development. The maximum stress in the middle layer was higher than in the outer layers, since the middle layer was hindered from both faces while the outer layers were able to swell at the outer surface.

The comparison to stresses measured in panels of series AR indicates similar maximum compressive stresses and larger minimum compressive stresses (Fig. 5.16). The maximum compressive stresses in composite laminates, thus, corresponded to stresses measured in AR00. The smallest compressive stresses were detected in panel AR45. Stresses in series AR were governed by the annual ring orientation. Compared to composite laminates, the layer ratio differs between both series. This influence is investigated by means of numerical studies in Section 6.3.

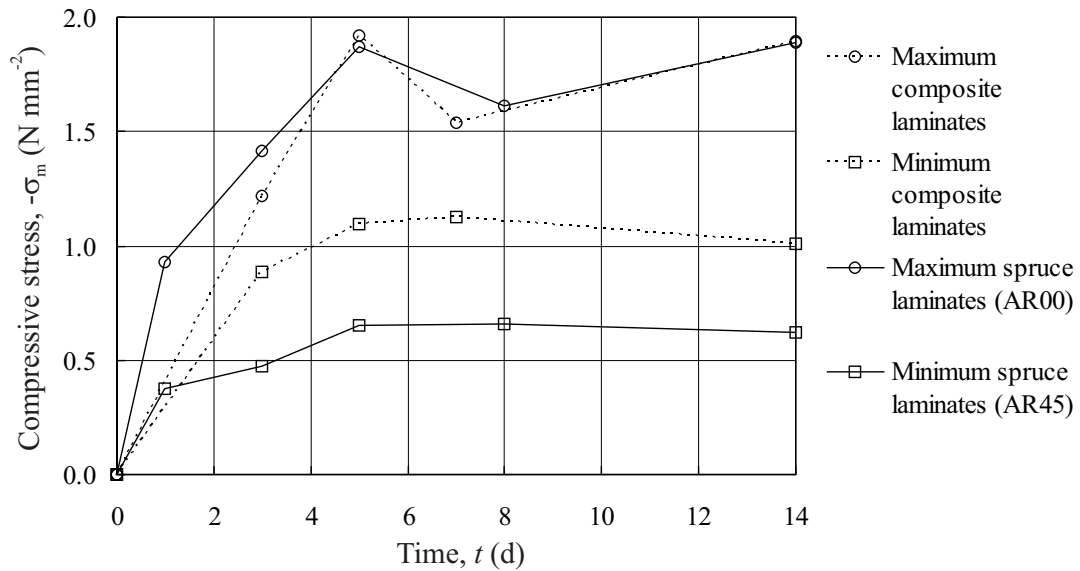


Figure 5.16 Maximum and minimum compressive stresses perpendicular to the grain in the outer layers in spruce cross-laminates (series AR) and spruce-composite laminates, values referred to $\sigma(t = 0)$

Influence of pre-stressed outer layers

Fig. 5.17 shows the history plots of moisture content, modulus of elasticity, strain and stresses of series PS for the outer layers. The panels were preconditioned in 35% RH and then transferred to 85% RH. The plots summarise different pre-stresses, since significant differences were only detected for $t = 21$ d in the outer layers. With increased moisture content, the modulus of elasticity decreased. Strains and stresses decreased until day 3 and then continued to remain nearly constant.

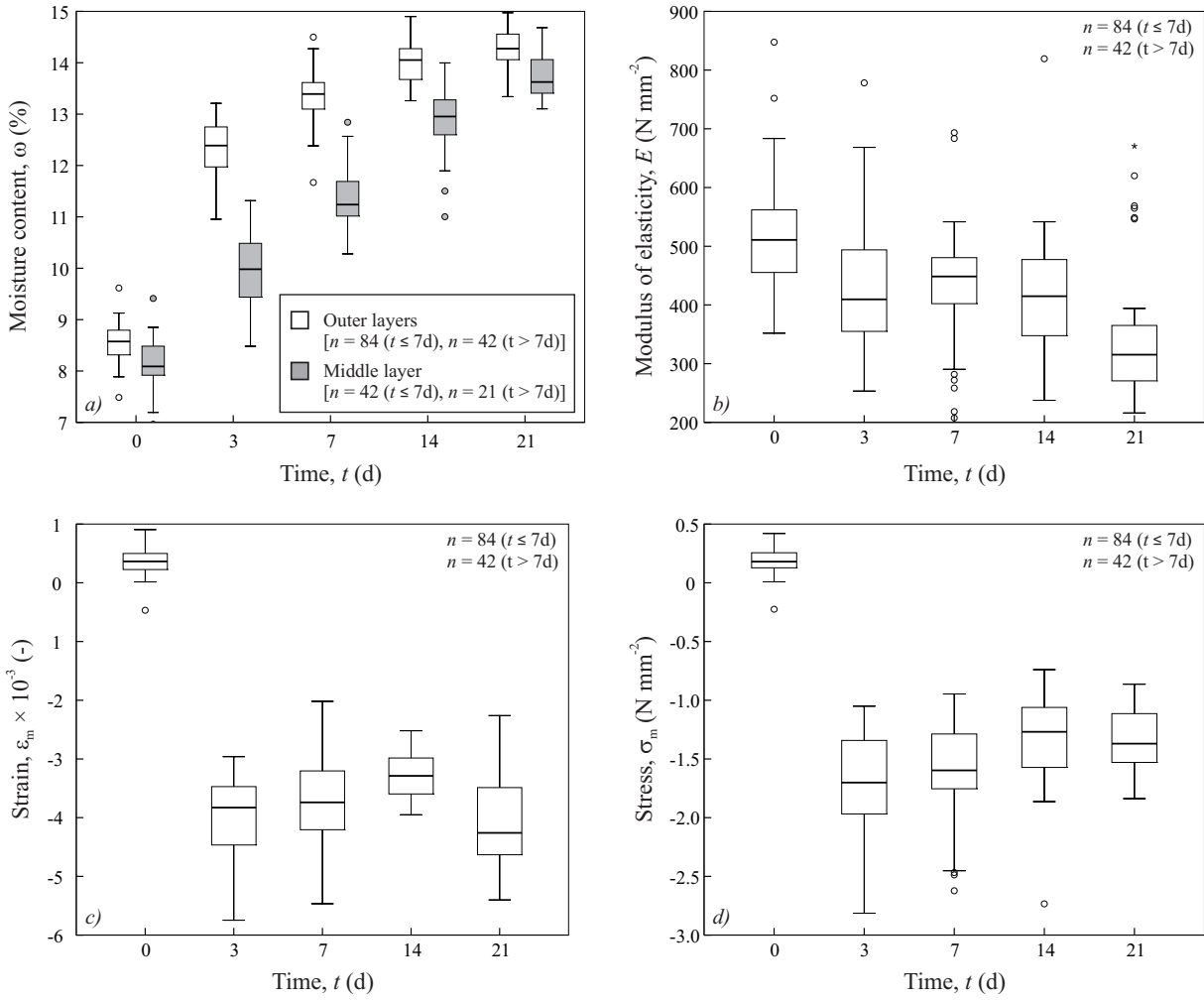


Figure 5.17 History plots of (a) moisture content, (b) modulus of elasticity, (c) strain and (d) stress perpendicular to the grain of the outer layers, series PS, moistening from 35% RH to 85% RH

At day 21, compressive stresses were minimal in panel PS12 ($\sigma_m = -1.02$ N mm⁻²). Stresses in PS06 ($\sigma_m = -1.52$ N mm⁻²) and PS18 ($\sigma_m = -1.48$ N mm⁻²) were found to be similar. This minimal compressive stress in PS12 was found to be the result of high moisture content. This, accompanied with a lower modulus of elasticity, resulted in lower

stresses. The reason for the higher moisture content is incomprehensible. Significant differences compared to AR90 were not found, since the previous drying generated tensile stresses perpendicular to the grain in the outer layers, which reduced the pre-stresses (relaxation).

Influence of the type of adhesive

The history development of moisture content and stresses in the outer and middle layers of panels SWPUR and SWUF is illustrated in Fig. 5.18. These panels were previously conditioned in 35% RH and then tested in 85% RH. The panels differed in the adhesive that was used for assembling lamellas and layers. Both 1K PUR (SWPUR) and UF (SWUF) were applied.

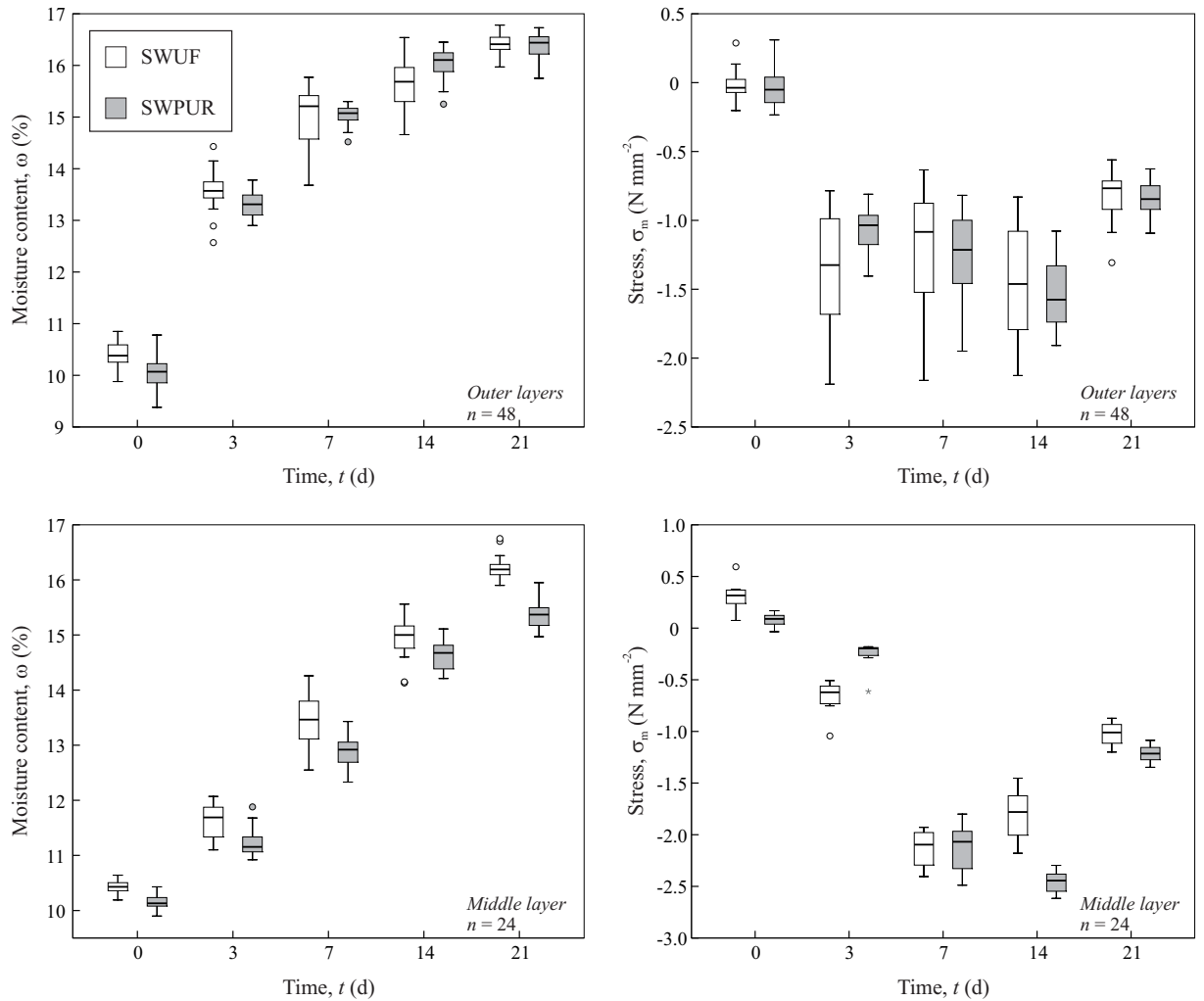


Figure 5.18 History plots of moisture content and stresses perpendicular to the grain of series SWUF and SWPUR

The moisture plots show significant lower moisture content in the middle layer of type SWPUR. This may be due to different diffusion coefficients of both adhesives. The results point to a lower diffusion coefficient of the 1K PUR adhesive joints. The detected differences in moisture of both series led to different strain and stress distributions.

In the outer layers, no significant divergences between both adhesive systems were detected. The compressive stresses increased within the first 14 days. On day 21 a reduction was observed. In the middle layer, the compressive stresses increased slower but to a larger value. The compressive stresses were significantly larger in panel SWPUR on day 14 and 21.

Both findings, lower moisture content and larger compressive stresses in the middle layer, in panel type SWPUR are inconsistent with one another. Lower moisture content would lead to lower compressive stresses, since the expansion of the middle layer is reduced. The contrary behaviour may be due to wood influences. Thus, the influence of different moisture contents of the middle layer on moisture-induced swelling and stresses is drawn and discussed in Section 5.3.4 by means of numerical analysis.

Visual surface evaluation

As mentioned previously, surfaces of panels that were initially exposed to a moist climate and then tested in a drier climate were optically investigated. The crack lengths on the surfaces were measured and added for each panel. Tabs. 5.5 - 5.7 show the mean values of these crack length sums after $t = 21$ d. However, cracks had already appeared after 24 hours. They were mainly observed between two lamellas but also within one lamella (Fig. 5.19). The differences in the ring angle between two adjacent lamellas yield differences in the shrinkage properties. Since the tangential shrinkage is twice the radial shrinkage, stresses developed between the two lamellas. These stresses acted together with stresses between the outer and the middle layers. Consequently, areas of maximum stress developed in the glue lines between two lamellas, which were released by cracks. Inside the lamellas, cracks were aligned mainly in the radial direction due to the lowest stiffness in the tangential direction.

The surface evaluation of different preconditioned layers yielded significant differences in the crack lengths. Tab. 5.5 shows the results after 21 days in the test climate 20°C/45% RH. Before gluing, the layers were conditioned at 45%, 65% and 85% RH, respectively.



Figure 5.19 Types of cracks on the panel surfaces: cracks between two lamellas (►) and cracks within one lamella (—→)

Just after assembling three layers to one panel, the panels were conditioned in the test climate. Panels with the same moisture in all layers showed the most cracks and the highest crack length. Panels that were previously conditioned at 85% RH showed more cracks than panels that were previously conditioned at 65% RH. This was due to the larger moisture difference between initial and test climate. The combination of a moist middle layer with drier outer layers (relative humidities of the individual layers: 65/85/65) yielded significantly less cracks, even when compared to panels with a lower moisture content (65/65/65). This was due to the effects of pre-stressed outer layers as a result of the moister middle layer. This leads to compressive stresses perpendicular to the grain in the outer layers and, thus, a reduced risk of cracking. Nearly no cracks were observed for panels where the outer layers were initially conditioned at 45% RH, which was in agreement with the test climate.

Table 5.5 Crack lengths on the surfaces after $t = 21$ d, panel type LR52, test climate $\varphi = 45\%$, temperature constant 20°C

Initial climate		Σ crack length	
OL (% RH)	ML (% RH)	n	(mm)
85	85	3	3841.7
65	65	3	2375.2
65	85	3	1433.2
45	85	3	123.2
45	65	3	143.1

Tab. 5.6 shows the results for panels with different middle layer materials. All panels were previously conditioned at $20^\circ\text{C}/85\%$ RH and then tested at $20^\circ\text{C}/35\%$ RH. Compared to the solid wood variant (LR57), CPB and COSB showed larger crack lengths. In panels with MDF in the middle layer, cracks were observed at the lateral faces in the MDF layer. They were aligned in the x -direction (Fig. 5.20). These cracks were due to transverse tension and the very low strength of MDF in the thickness direction. Depending on the manufacturing, MDF has a distinctive density profile with a higher density in the outer parts and a low density in the core (Fig. 5.20). The cracks were observed at the borders of low and high density. Thus, shear stresses due to different shrinkage may be the reason for the cracks in the middle layer. This release of stresses in the middle layer led to fewer cracks on the large faces.

Table 5.6 Crack lengths on the surfaces after $t = 21$ d in the test climate, panels with different middle layer materials

ID	Initial climate (°C/% RH)	Test climate (°C/% RH)	$\sum$ crack length n (mm)
LR57	20/85	20/35	3 96.7
CMDF5	20/85	20/35	3 68.2
CMDF7	20/85	20/35	3 67.0
COSB	20/85	20/35	3 267.5
CPB	20/85	20/35	3 177.4

Tab. 5.7 shows the crack lengths in pre-stressed panels that were exposed to different climates for 21 days. The results show the fact that more cracks developed on the surfaces of panels that were tested in the climate chamber (test climate 25°C/30% RH). This was due to the higher temperature, the lower relative humidity and the higher air velocity. Thus, the panels dried faster and more cracks developed. The results also revealed that the least amount of cracks arose in panels of type PS12 in both test climates. The most cracks were found on PS06. The previous moistening of the panels yielded compressive stresses perpendicular to the grain in the outer layers. These stresses acted together with the pre-stresses. This led to plastic deformations mainly in panel type PS18 due to the largest compressive stresses. These stiff areas hindered the shrinkage when the panels were dried, and more cracks developed at the surfaces. This is the reason why PS18 showed more cracks than PS12. In type PS06, no or few plastic areas developed. Here the crack development was governed by the amount of pre-stress. The largest crack length

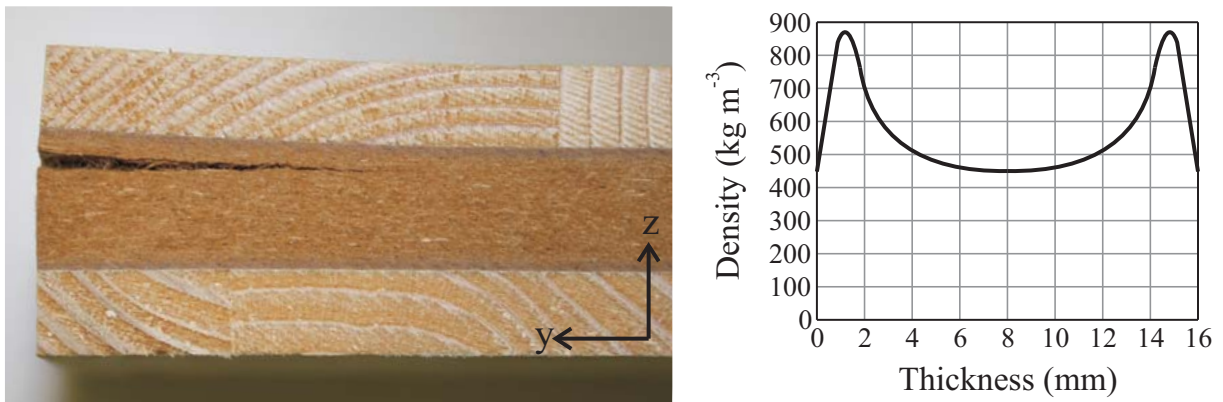
**Figure 5.20** Lefthand side: cracks in the middle layer of a CMDF7 panel, righthand side: typical density profile of MDF

Table 5.7 Crack lengths on the surfaces after $t = 21$ d in the test climate, panels with pre-stressed outer layers

ID	Initial climate (°C/% RH)	Test climate (°C/% RH)	n	Σ crack length (mm)
PS06	20/85	25/30	2	1270.4
	20/85	20/35	2	598.5
PS12	20/85	25/30	2	639.4
	20/85	20/35	2	241.5
PS18	20/85	25/30	2	1043.4
	20/85	20/35	2	528.4

sum in panels of series PS06 was due to the lowest pre-stress and, thus, the largest drying tensile stresses. Compared to the panel with the same layer ratio but without pre-stressed outer layers (Tab. 5.5, initial climate 85% RH), pre-stressed panels showed significantly lower cracking on their surfaces.

5.3 Numerical analysis

5.3.1 Material model and boundary conditions

The material model that was described in Section 2.6 will now be employed in the simulation of deformations and internal stresses due to a single climate step. The total strain rate is the sum of the elastic strain rate, the moisture-induced strain rate and the mechano-sorptive strain rate. In Section 4.3.2, the input data used in the simulations was given. In Fig. 5.21, the element mesh, the geometry and the coordinate system are given. The local coordinate system that is employed in the individual layers is a rectangular system. This indicates a large distance to the pith. As defined in Fig. 1.2, the longitudinal axis of these local coordinate systems is aligned parallel to the x-axis in the outer layers and parallel to the y-axis in the middle layer. Due to double symmetry, it is possible to model one quarter of the panel. Thus, by introducing two planes of symmetry and applying appropriate boundary conditions, the simulated dimensions are 200mm $\times$

200mm. The following boundary conditions were applied:

$$\begin{aligned}
 u_x &= 0 && \text{at } (100, y, z) \\
 u_y &= 0 && \text{at } (x, 100, z) \\
 u_x = u_y = u_z &= 0 && \text{at } (100, 100, 0) .
 \end{aligned} \tag{5.6}$$

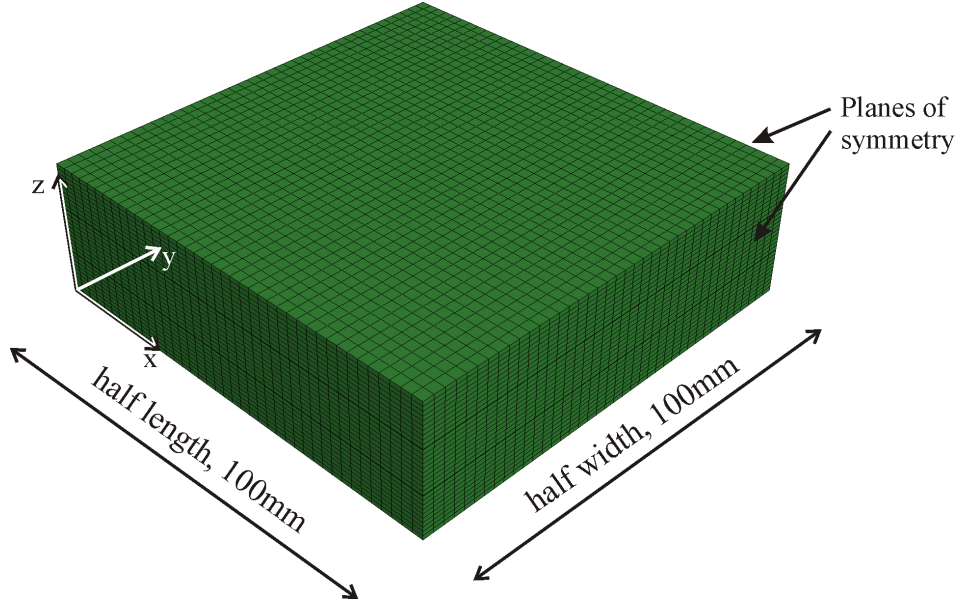


Figure 5.21 Element mesh applied in the present section

5.3.2 Model validation

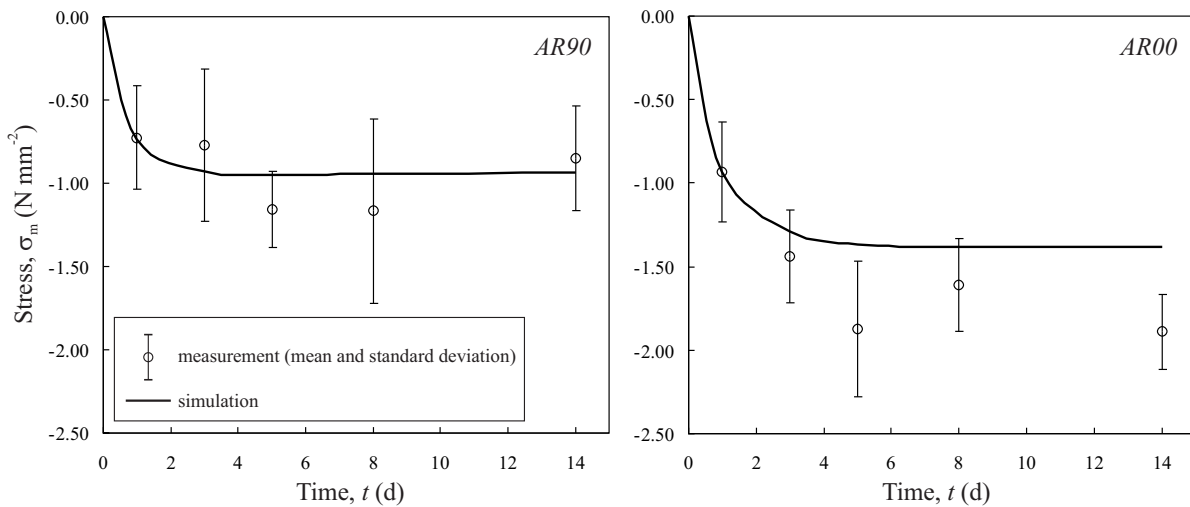
In this section, numerical simulations are compared to results obtained in experimental tests. Panels AR00 and AR90 were conditioned from 35% RH to 85% RH over a 14 day period. The measured moisture contents, moduli of elasticity, released strains and stresses were shown in Section 5.2.3. The mean moisture content of the individual layers, summarised in Tab. 5.8 are applied to the mechanical model. The variation of moisture within the layers is neglected. Different moisture contents in different layers at the initial state are omitted in the simulations. The moisture content at $t = 0$ d was set to $\omega = 10.5\%$. From the measured stress values, the initial stress state (measured at $t = 0$) was subtracted. In order that, $\sigma(t = 0) = 0$ in both, the simulations and the measurements.

Two paths along the y-axis are analysed, both at $x = 100$ mm, so that, the paths are located in the centre of the panel. Two positions in the thickness direction are evaluated: $z = 0$ mm and $z = 10$ mm. The first path is, thus, situated at the bottom of the

Table 5.8 Moisture content (in %) in the outer and middle layers applied in the simulations

Layer	Time, t (d)					
	0	1	3	5	8	14
OL	10.5	11.7	12.9	13.7	14.7	15.3
ML	10.5	10.7	11.1	11.6	13.1	14.4

panel and the second path is situated directly at the glue line in the bottom layer. The stresses calculated along both paths are summarised to a mean stress in the bottom layer and then compared to the mean stresses measured previously. The results of material modelling and experimental tests shown in Fig. 5.22 indicate a good agreement between simulations and measurements. However, the compressive stresses in the outer layers are underestimated by the material modelling, especially in panel AR00. This is caused by an overestimation of the dynamically determined moduli of elasticity. The difference was 46% in the tangential direction and 17% in the radial direction compared to the data of Neuhaus [74], which was applied in the simulations.

**Figure 5.22** Measured mean stresses perpendicular to the grain in the outer layers compared to simulations

5.3.3 Maximum stress

The stresses shown in Fig. 5.22 were calculated on the basis of the average released strain (experiments) or averaged over the layer (simulation). Now the stress distribution within

the layer and, thus, the maximum stress is regarded. As indicated before, the stress is evaluated along two paths in the bottom layer. These paths are situated at $z = 0\text{mm}$ and $z = 10\text{mm}$ along the y -axis at $x = 100\text{mm}$. The stress distributions along both paths are displayed in Figs. 5.23 and 5.24.

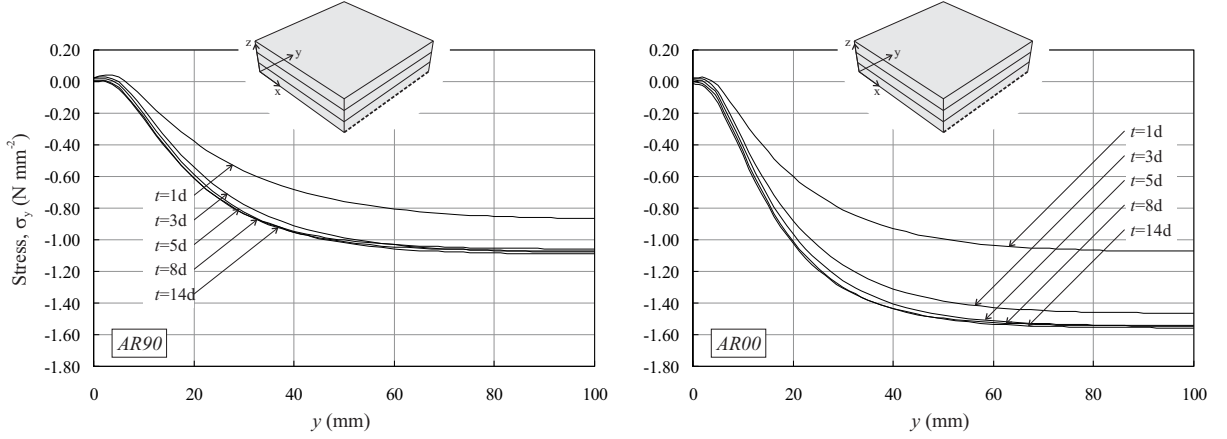


Figure 5.23 Stresses σ_y at $z = 0\text{mm}$, moisture content according to Tab. 5.8

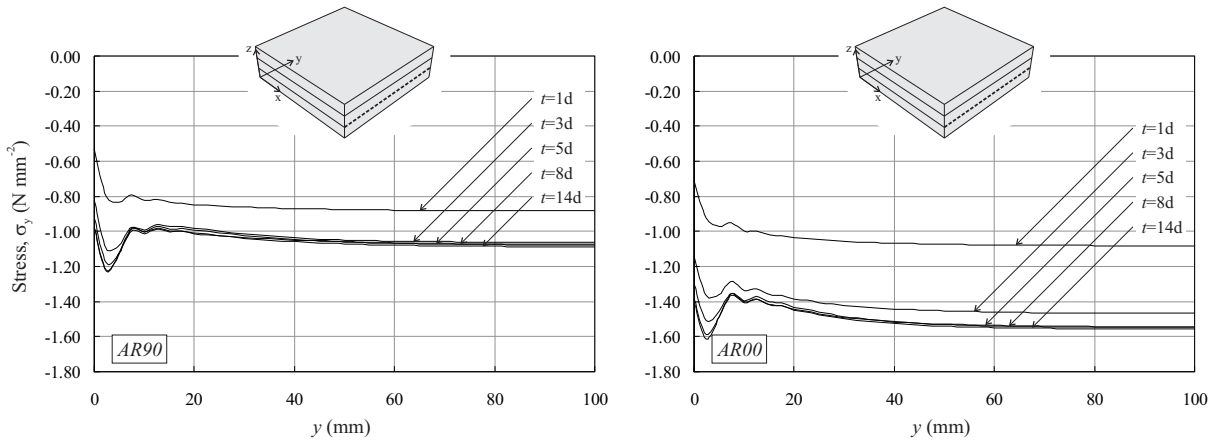


Figure 5.24 Stresses σ_y at $z = 10\text{mm}$, moisture content according to Tab. 5.8

Both figures show that compressive stresses develop in the outer layers. These compressive stresses are zero at the ends ($y = 0\text{mm}$, $z = 0\text{mm}$) and maximum at $y = 100\text{mm}$, which is the centre (symmetry plane) of the panel. The stresses at the bottom face decrease sharply from $y = 0\text{mm}$ to $y \approx 40\text{mm}$, then decrease slightly to $y \approx 70\text{mm}$ and remain constant until $y = 100\text{mm}$ (Fig. 5.23). Compared to the mean stresses, the maximum compressive stresses are 12% to 17% higher (Tab. 5.9). Jönsson and Svensson [52] determined a larger stress difference in glulam, specifically 33%.

Table 5.9 Relative differences between mean and maximum stress in the outer layers (in %)

Panel ID	Time, t (d)				
	1	3	5	8	14
AR00	14.7	13.7	13.1	12.5	12.0
AR90	17.0	16.2	15.0	13.7	13.3

5.3.4 Basic studies

Comparison between spruce cross-laminates and spruce-MDF laminates

In the present section, four types of panels are compared. In Tab. 5.10, the layer characteristics of the boards studied are presented. The investigation tests different layer ratios of spruce panels (S1 and S2) and of different middle layer materials (S2 - S4). The annual ring angles of the spruce layers are constant in all boards tested: $\theta = 90^\circ$. The study focuses on the modelling of deformation and stress development. Therefore, the moisture distribution within the different panels is first calculated.

In the present simulations, the panels are assumed to be insulated at the ends and moisture transport is assumed to be one-dimensional in the cross-sectional direction. The moisture model is as expressed in Section 3.4. The relative humidity of the ambient air is increased from $\varphi_1 = 35\%$ to $\varphi_2 = 85\%$, which, according to Eq. 3.1, relates to $c_1 = 3.56 \text{ kg m}^{-3}$ and $c_2 = 7.74 \text{ kg m}^{-3}$. Fig. 5.25 illustrates the history plots of moisture in the outer and middle layers. Each curve represents the moisture content of a node in the centre of

Table 5.10 Characteristics of boards S1 - S4

ID	Outer layers		Middle layer		LR (-)
	Material	a_{OL} (mm)	Material	a_{ML} (mm)	
S1	spruce ¹	10	spruce ¹	10	0.67
S2	spruce ¹	10	spruce ¹	15	0.57
S3	spruce ¹	10	MDF540 ²	15	0.57
S4	spruce ¹	10	MDF800 ²	15	0.57

¹ Material data according to Section 4.3.2

² Material data according to Section 2.8

the layer. Strong differences in the moisture content of the middle layer are governed by thickness and material. The thinner the panel, the faster the middle layer saturates with moisture. Thus, the moisture content of board S1 is initially larger than that of board S2, and the moisture in the S2 middle layer adapts to the moisture in board S1 with time. Moisture content is also governed by material density and diffusivity in different materials. Thus, the moisture contents of the MDF middle layers of boards S3 and S4 are significantly lower than the spruce layers of boards S1 and S2.

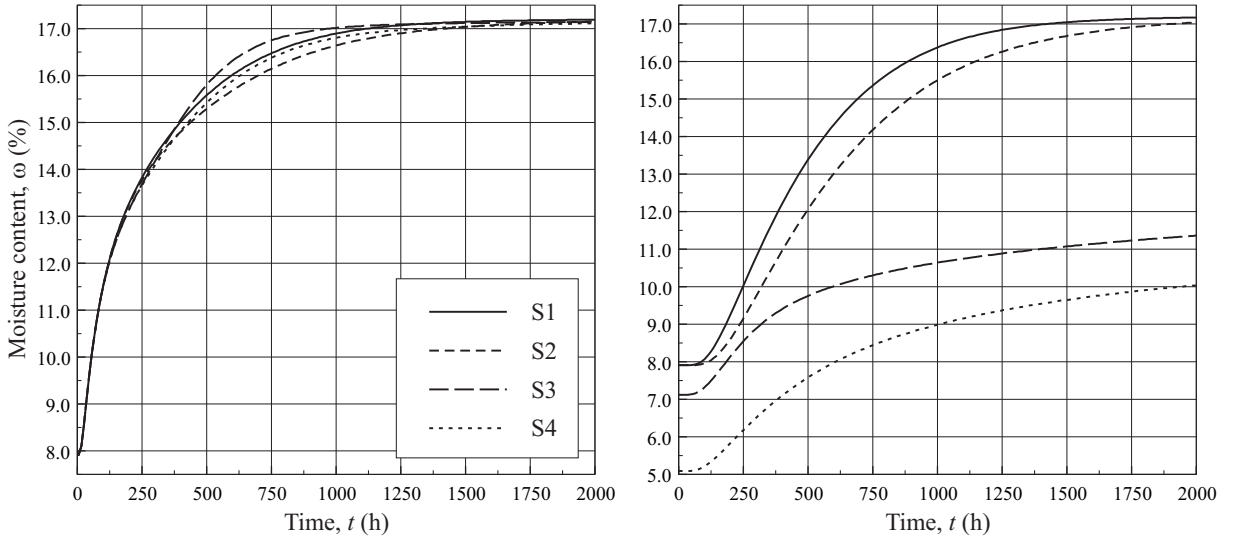


Figure 5.25 Moisture contents in the outer (lefthand side) and middle layers (righthand side) in boards S1-S4, moistening from 35% RH to 85% RH

Fig. 5.26 shows the history developments of the displacements u_y for a node at the unhindered edge ($x = 100\text{mm}$, $y = z = 0\text{mm}$) and the average radial stresses σ_y in the outer layer. The figure shows that deformation is larger in spruce-MDF laminates than in spruce cross-laminates. This is due to the lower stiffness and the larger swelling of MDF compared to spruce in the longitudinal direction. Outer layers are hindered to a greater extent by a spruce middle layer than by a MDF middle layer. The swelling of panels S3 and S4 are governed by the density and, thus, stiffness and swelling coefficients of the MDF middle layer. Deformations in board S3 (MDF540) are larger than those of S4 (MDF800). The stress plot confirms these findings. Compressive stresses perpendicular to the grain in the outer layers are lower in composite laminates than in spruce cross-laminates. This is a result of the stronger constraint by the spruce core. Fig. 5.27 illustrates the plots of the displacements in the minor axis, u_y , and radial stresses in the top layer, σ_y , in panels S1 - S4 after 2000 hours of wetting from 35% RH to 85% RH. The displacements increase from $u_y = 0\text{mm}$ at the centre of the panels ($y = 100\text{mm}$) to the maximum value at the

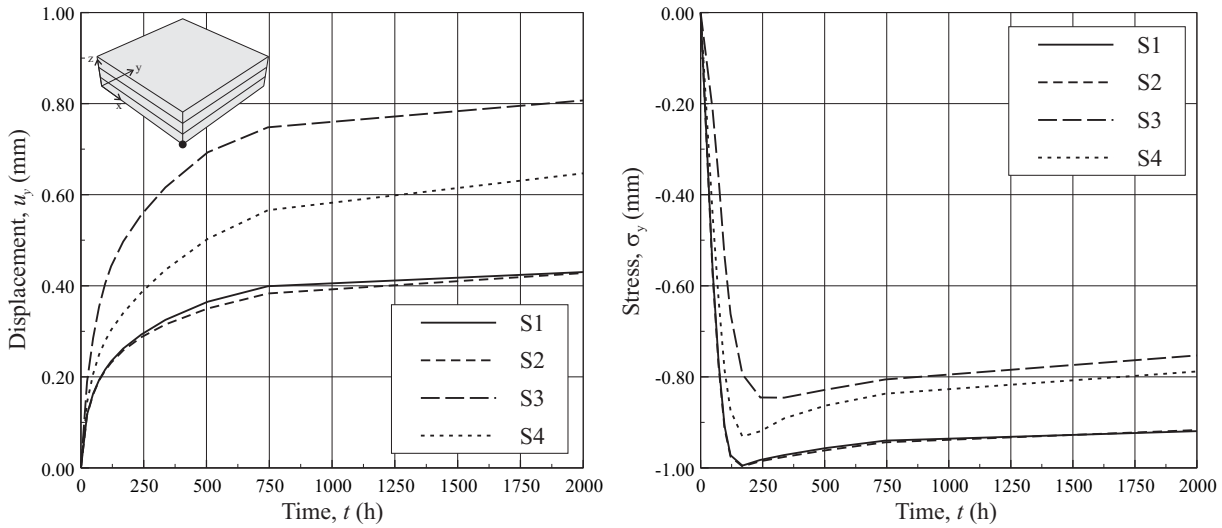


Figure 5.26 Simulated displacements in radial direction (lefthand side) and average radial stresses (righthand side), both in the outer layer and determined for panels S1 - S4, wetting from 35% RH to 85% RH

unhindered edge ($y = 0\text{mm}$). The figure also shows the compressive stresses increasing from the unhindered boundary towards the centre of the panel.

The plots demonstrate the strong differences between the outer and the middle layers and the differences between spruce cross-laminates and spruce-MDF laminates. When MDF is employed in the middle layer, the deformation of the panel in the direction of the panel axes x and y increases. This leads to a reduction of the compressive stresses in the outer layers. This behaviour depends on the density of the applied MDF material. The larger the density, the smaller the displacements and the larger the compressive stresses. As shown in Fig. 5.26, the displacement, u_y , is about 88% higher in panel S3 (MDF540) than in panel S2 (spruce). However, the compressive stress is only reduced by 20%. Thus, a gain in stress reduction occurs with an increase in swelling displacements, which is disadvantageous.

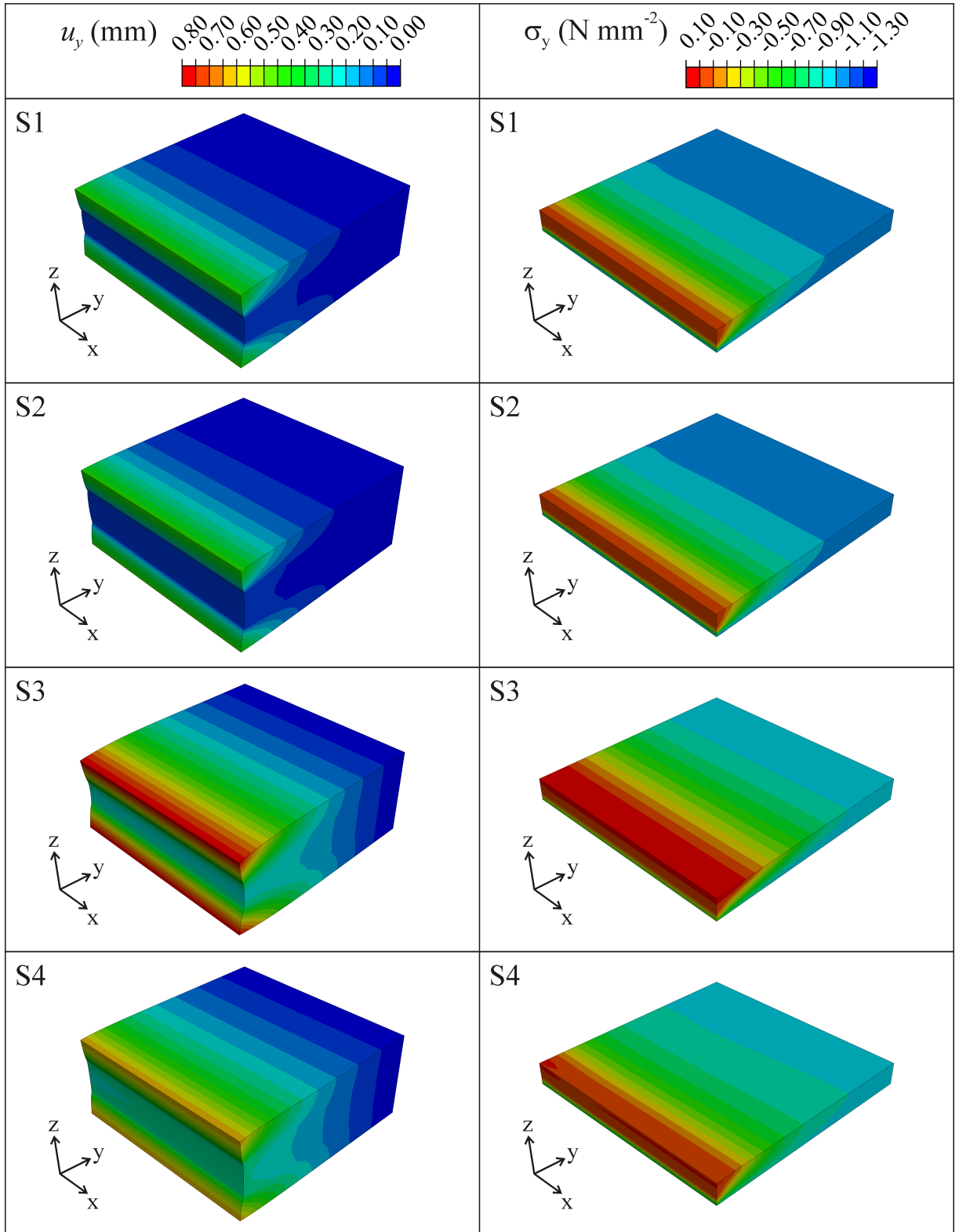


Figure 5.27 Displacement u_y (lefthand side) and radial stresses in the top layer (righthand side) in panels S1-S4, wetting from 35% RH to 85% RH, $t = 2000\text{h}$, displacement scale factor: $\times 5$

Influence of the diffusivity of the glue lines on internal stresses

In order to investigate the influence of the moisture profile on the internal stress state, different moisture contents of the middle layer were tested in a panel of type S1 (Tab. 5.10). The initial moisture content of the panel was 8%. The moisture is increased to 13% in the outer layers and is varied in five steps in the middle layer: 8.0%, 8.5%, 9.0%, 9.5% and 10.0%. Different moisture contents in the middle layer may be due to different diffusion resistances of the glue lines or due to wood influences.

Fig. 5.28 illustrates the displacement u_y along the z -axis. The figure includes two subplots showing u_y at different distances to the unhindered edge, $y = 0\text{mm}$ and $y = 20\text{mm}$. With increased moisture content of the middle layer, the elongation of this layer increases. This also affects the deformation of the outer layers. Here the expansion also increases since the opposite constraint is reduced. This is also shown by the stress plots (Fig. 5.29). Different moisture contents of the middle layer imply only small differences in stresses occurring in the outer layers perpendicular to the grain (Fig. 5).

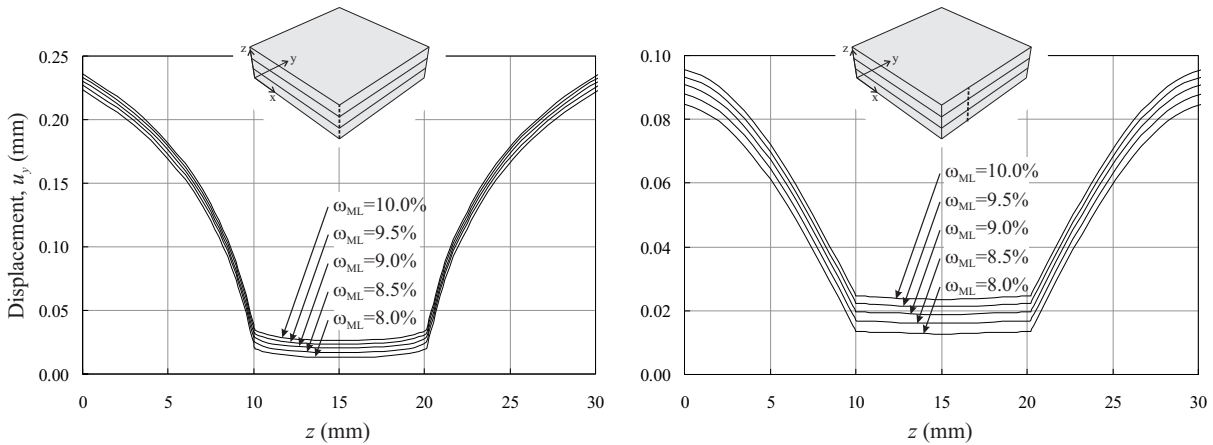


Figure 5.28 Displacements u_y along the thickness, lefthand side: $y = 0\text{mm}$, righthand side: $y = 20\text{mm}$, influence of different moisture profiles, $\omega_0 = 8\%$, $\omega_{OL} = 13\%$

However, significant differences of the radial stresses in the middle layer are shown by Fig. 5.30. When $\omega_{ML} = 8\%$, tensile stresses occurred perpendicular to the grain in the middle layer. The moisture content of this layer was constant compared to the initial state and only the outer layer moisture content increased. The expansion of the outer layers yielded tensile stresses. When the moisture content of the middle layer was also increased, compressive stresses developed in the middle layer. The expansion of the middle layer occurring perpendicular to the grain was hindered by the cross-lamination. A moister middle layer yielded increased expansion and, thus, increased compressive stresses.

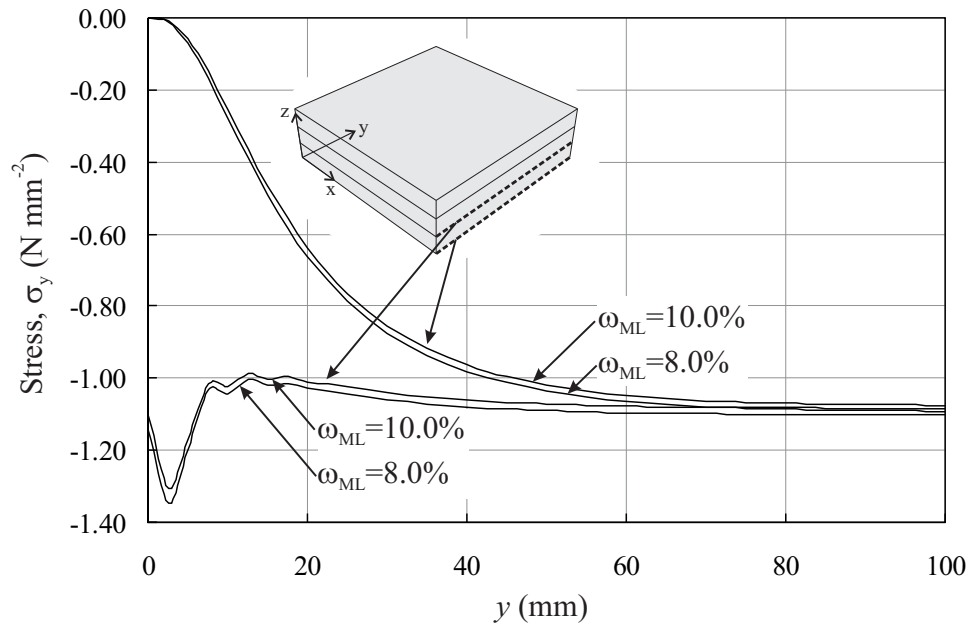


Figure 5.29 Stresses σ_y (radial stresses) along the y-axis, influence of different moisture profiles, $\omega_0 = 8\%$, $\omega_{OL} = 13\%$

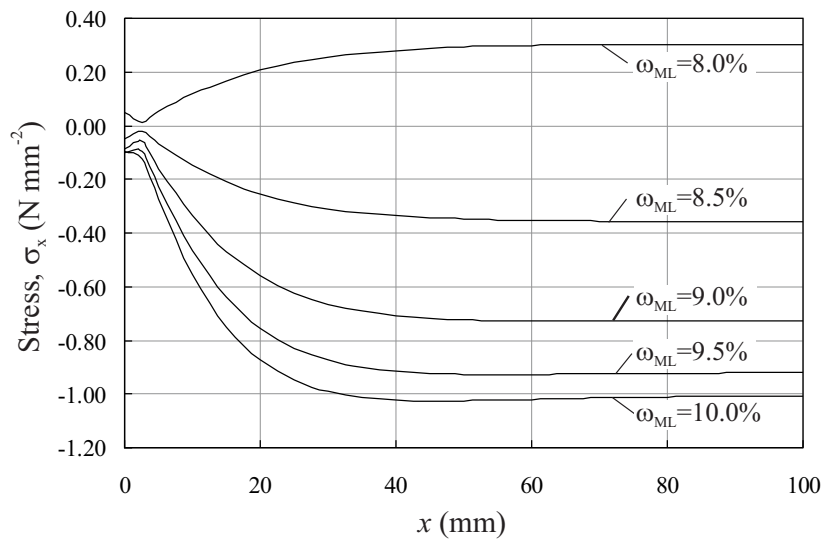


Figure 5.30 Stresses σ_x (radial stresses) along the x-axis in the middle layer, influence of different moisture profiles, $\omega_0 = 8\%$, $\omega_{OL} = 13\%$

5.4 Concluding remarks

In this chapter, test results of measurements and computations of moisture-induced stresses due to a single climate step were presented.

From the analysis of strains at the lateral faces by means of digital image correlation, it was concluded that it is preferable not to measure swelling deformations at the boundaries. Instead, measurements of swelling properties were performed by turning screws, with a specific distance to the edge, through all three layers. This enabled the measurements of displacements of the entire panel and not only of the individual layers. Thus, effects of different swelling at the unhindered boundaries were excluded from the swelling measurements.

It could be observed that, in the investigated climate step between 35% RH and 85% RH, the compressive stresses reached maximum between days 5 and 8. The parameters that influence the stresses are:

- the climate step (as shown by the drying test of pre-stressed panels),
- the annual ring angle θ ,
- the middle layer material,
- pre-stressed outer layers, and
- the induced moisture profile, which may be influenced by the pre-conditioning of the layers and the diffusivity of the glue lines.

The annual ring angle was found to have a significant influence on the development of internal stresses in cross-laminated solid wood panels. The formation of stresses was governed by stiffness, swelling coefficient and annual ring orientation. The swelling properties, which were measured on the entire panel, and the strains, which were released by sawing the panels into slices, showed similar characteristics: the deformations were determined to be lowest when $\theta = 90^\circ$ and largest when $\theta = 0^\circ$. The compressive stresses, however, were lowest when $\theta = 45^\circ$, which is due to the low stiffness of spruce in that direction. In regard to load bearing constructions, the low stiffness of panels has disadvantages, namely, larger cross-sections or lower span lengths when calculating building statics. However, the application as decorative wall cladding without static function is recommended. The risk of stress cracks, and, as indicated previously, the amount of warp deformation are low.

The largest stiffness and the lowest swelling in the radial direction make a panel with vertically oriented annual rings more preferable. Here, medium stresses and warping were determined. It is recommended to avoid horizontal annual rings ($\theta = 0^\circ$), since the large tangential swelling and the low tangential stiffness of spruce increase stresses and the low tangential strength increases the risk of surface cracking.

Wood composite middle layers may be applied in industrial processes. They yield advantages compared to solid wood: reduced stresses and reduced costs. Disadvantages are the cracking of the middle layer during drying processes (MDF) and the large expansion and contraction when the ambient climate changes. This may lead to problems in areas where two panels are connected, such as gap opening or failure due to compression. Composite laminates should, therefore, be investigated in large-sized sample tests.

The application of different adhesives provides parameters with which one can influence the internal stress state, namely the moisture diffusivity and the stiffness of the composition of adhesive and wood, i.e. of the glue line. While the stiffness has a very little influence, the diffusivity was found to strongly influence stresses. Different moisture diffusivities of the glue lines yield different moisture contents, mainly in the middle layer. In numerical analysis, it was found that a rise in the moisture content of the middle layer, with the outer layer moisture kept constant, resulted in increased compressive stresses perpendicular to the fibres in the middle layer. Therefore, a larger diffusivity of the glue lines would increase the compressive stresses perpendicular to the fibres in the middle layer when the panel is moistened. In contrast, tensile stresses would develop when the panel would be dried. The stress development would lead to plastified areas (compression) or failure due to cracking (tension). It is, therefore, recommended to apply a waterproof adhesive or an additional waterproof layer to reduce stresses in the middle layer. The results showed that the outer layers were only slightly influenced by the diffusivity of the glue lines. However, the influence of moisture accumulation in the outer layers was neglected from the study. A lower diffusivity (or even zero diffusivity) would lead to moisture accumulation and, thus, a higher moisture content of the outer layers. This would give larger stresses and an increased risk of cracking on the surfaces. The tests of different adhesive diffusivities in Chapter 4 indicated an increase in cup deformations, both cup_{xz} and cup_{yz} , when the diffusivity was reduced. Since the prevention of warping and of stresses in the outer layers is more important than the formation of stresses in the middle layer, it is recommended to apply an adhesive that has a high permeability for moisture. Alternatively, one could apply another connection system, which gives $D_{adh} = D_{wood}$, but

ensures the effect of the crosswise orientation of the layers.

In drying tests, the influence of different parameters on the formation of cracks on the surface was tested. The pre-stress in the outer layers showed a significant influence. Pre-stress can be applied mechanically (as in panel series PS) or by manipulating the initial moisture contents of the middle and outer layers. In such a way, pre-stressed panels generated less cracks when dried. However, a high value of pre-stress, as in panel PS18, yields plastified areas as a result of the initial compressive pre-stress and the previous wetting. These increased the crack lengths. However, no compressive pre-stresses were detected in panels that were previously dried. The shrinkage of the outer layers reduced the compressive stresses and generated tensile stresses. As a conclusion, pre-stressing is a valid method to reduce cracks and warping (as shown in Chapter 4), but the production costs and the relaxation of pre-stresses when the panel is dried are critical factors.

Chapter 6

Numerical studies

6.1 Introduction

The simulation of moisture-induced stresses and deformations of wood is complex. It is based on a large number of material parameters, which need to be determined through experimental investigations. This chapter presents results of a parameter study where different values for the input parameters are examined. The objective is to obtain information on how warping and internal stresses are influenced by modifying the parameters.

In Section 6.2, parametric studies of the hygroscopic warping will be shown. The mechanical material model was introduced in Section 4.3. The influences of the elastic, hygroscopic and mechano-sorptive input parameters are examined. In experimental tests, the layer ratio was found to influence the amount of warp deformations. This interrelationship is studied further with numerical calculations. In all previous calculations, the lamella layout of the layers was not considered. In this chapter, it is examined in numerical calculations how the geometry and the annual ring orientation of the individual lamellas influence warping. The density affects the elastic parameters of wood [57, 75]. This relationship is analysed by means of investigations by Keunecke et al. [55], who measured moduli of elasticity and shear moduli of Norway spruce by means of ultrasonic waves under normal conditions. These dynamic moduli were transferred into static values and fitted to the moduli-moisture relationship of Neuhaus [74]. Afterwards, calculations of different density configurations are presented. The studies are completed with an analysis of the warping of spruce-MDF laminates, where the layer ratio and the modulus of

elasticity of MDF are varied.

In Section 6.3, parametric studies of the internal stress state due to a single climate step are shown. The model is as shown in Section 5.3. The influences of the input parameters and the layer ratio are discussed.

6.2 Parametric study of hygroscopic warping

6.2.1 Influence of the material parameters

To investigate the influence of different material parameters on simulated warping deformations, calculations using different input parameter values were performed and were compared with the reference calculation. In this section, the influences of a variation of the moduli of elasticity E , the shear moduli G , the Poisson's ratios ν , the coefficients of hygroexpansion α and the mechano-sorptive material parameters m are studied. In every calculation, one parameter was varied either in the complete panel or in the individual layers, while the other parameters were kept the same as in the reference calculation. Panel AR90 ($a_{tot} = 30\text{mm}$, $LR = 0.67$, $\theta = 90^\circ$) was used.

No influences could be detected by varying the shear moduli and the Poisson's ratios. Panel type AR90 is also very insensitive to modifications of E_T .

Fig. 6.1 shows the influence of a 30% increase and a 30% decrease of E_L , E_R , α_L and α_R in the individual layers and in the complete panel. The cupping, cup_{yz} , is significantly influenced by a change in the longitudinal parameters E_L and α_L , when changed in the middle layer, and of the radial parameters E_R and α_R , when changed in the outer layers. The cup deformation, cup_{xz} , is strongly influenced by a change of α_L when changed in the outer layers. With an increase in the coefficients of hygroexpansion the deformations increase and vice versa. A 30% variation of α_L in the outer layers influences cup_{xz} by about 30%. In contrast, the variation of E_L , E_R and α_R has nearly no influence on cup_{xz} . The modification of the coefficient of hygroexpansion causes a larger expansion of the layers in the appropriate direction. The longitudinal hygroexpansion influences the major direction. This results in a larger cup_{xz} but only slightly changes cup_{yz} . The panel shows the opposite behaviour when α_R is modified, since the minor direction is mainly affected.

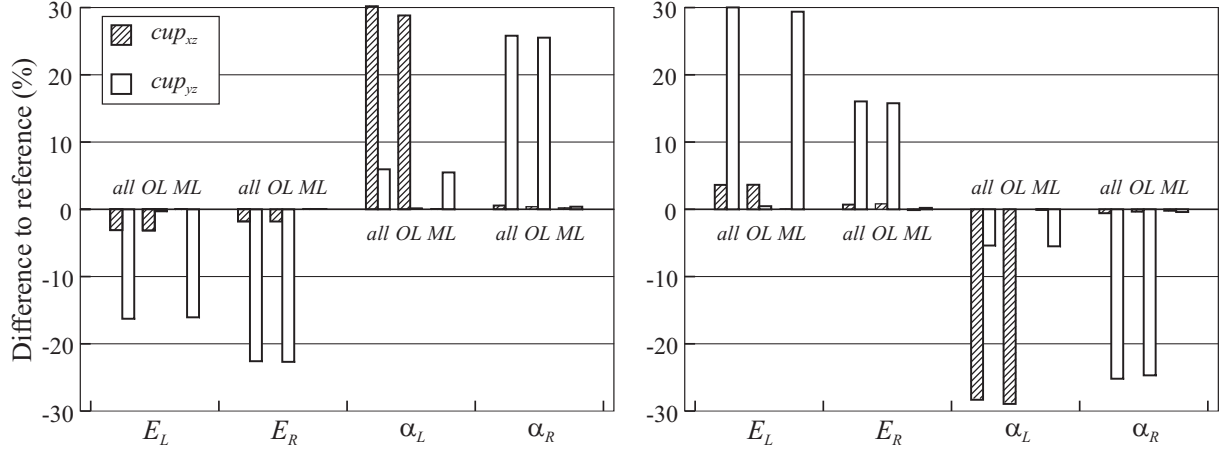


Figure 6.1 Relative change of the cup of panel type AR90 caused by a 30% increase (lefthand side) and a 30% decrease (righthand side) of material parameter values for the complete panel (all) and the individual layers (OL - outer layers, ML - middle layer)

The mechano-sorptive strain was found to be very important for the cup deformation and for the results obtained in the parameter study. While a change of the mechano-sorptive coefficient, m_L , causes a small change in cup deformations, m_R strongly affects cup_{yz} and also significantly influences cup_{xz} (Fig. 6.2). This impact was detected to occur mainly in the outer layers. The mechano-sorptive strain is also important for the influence of E_R and α_R on cupping. An increase (or decrease) of E_R gives a smaller (or larger) cup_{yz} if mechano-sorption is considered and larger (or lower) cup_{yz} if not considered. A variation of α_R causes a larger influence on cup deformations when mechano-sorption is neglected.

Fig. 6.3 presents the influence on cup_{xz} and cup_{yz} caused by a reduction of the material parameter values with the strongest impact on warping, namely E_L , E_R , α_L , α_R and m_R . The relationship between the change in cup deformation and the relative decrease of the material parameter value is generally linear. The variation of m_R leads to an exponential increase in cup_{yz} .

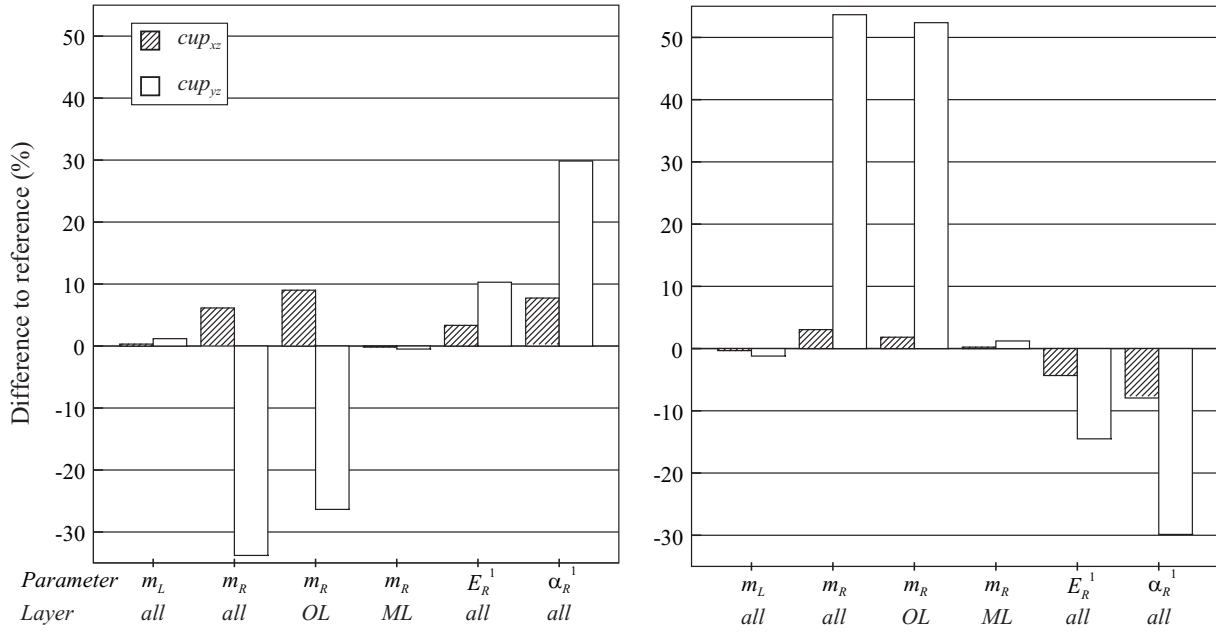


Figure 6.2 Relative change of the cup of panel type AR90 caused by a 30% increase (lefthand side) and a 30% decrease (righthand side) of material parameter values for the complete panel (all) and the individual layers (OL - outer layers, ML - middle layer), ¹ the mechano-sorptive fraction is omitted in both, the parameter study and the reference calculation

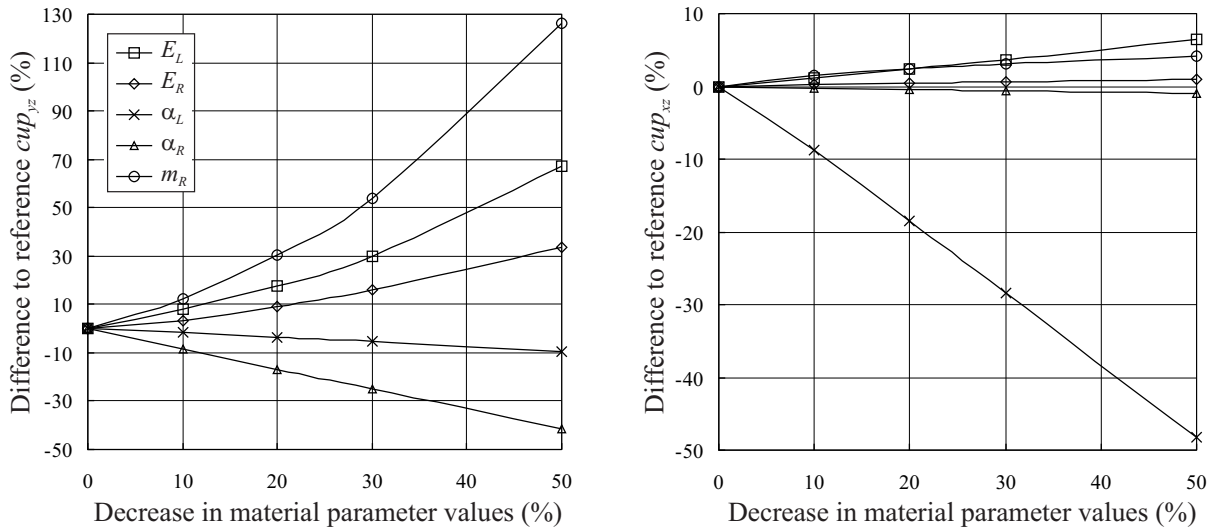


Figure 6.3 Differences to the reference cupplings cup_{yz} (lefthand side) and cup_{xz} (righthand side) due to a decrease in the material parameter values

The distribution of radial stresses along the y-axis is shown in Fig. 6.4. The figure shows the influence of a 30% decrease in the input parameters. Stresses are evaluated for two paths, $z = 0\text{mm}$ (bottom face) and $z = 10\text{mm}$ (at the glue line). The figure shows a strong influence of E_R and α_R on the stresses at the bottom face. It indicates a strong influence made by m_R and also a significant influence made by E_R and α_R on the stresses at the glue line. The longitudinal parameters E_L and α_L only slightly effect radial stresses in the bottom layer. The decrease of E_R and α_R yields an increase in stresses at the bottom face. The decrease of α_R increases the stresses at the glue line while the decrease of E_R and the mechano-sorptive coefficient m_R decreases these stresses.

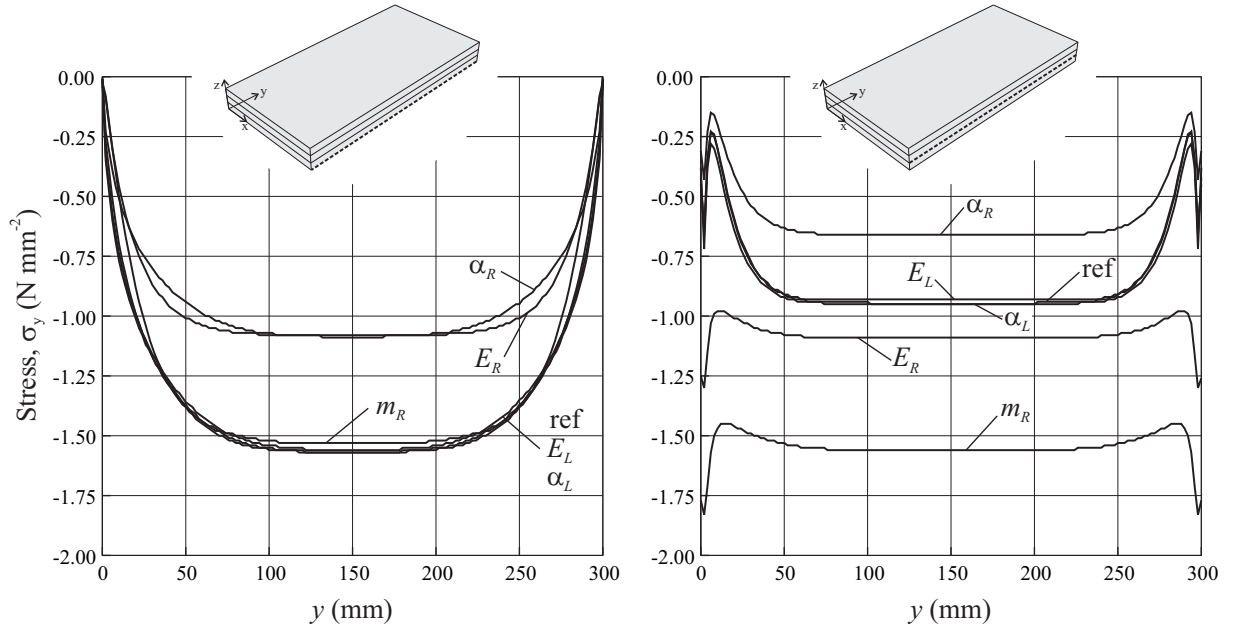


Figure 6.4 Stress distributions, σ_y , along the y-axis in the bottom layer in panel AR90 after 744h of climate difference 65/100% RH, 30% decrease of the material parameter values, lefthand side: $z = 0\text{mm}$, righthand side: $z = 10\text{mm}$

6.2.2 Influence of the layer ratio

In this section, the influence of the layer ratio on hygroscopic warping is examined. As was noted in Section 4.2.1, the layer ratio, LR , is the relative thickness of the outer layers to the panel thickness. It was found in the experimental tests (Section 4.2.3) that the layer ratio significantly influences warping. To investigate this influence, various calculations were performed with two different panel thicknesses, 20mm and 30mm. The layer ratio was varied between 0.4 and 0.8, thus, the minimum layer thickness was 4mm (Tab. 6.1).

Table 6.1 Investigated panel configurations

LR (-)	$a_{tot} = 20\text{mm}$		$a_{tot} = 30\text{mm}$	
	a_{OL} (mm)	a_{ML} (mm)	a_{OL} (mm)	a_{ML} (mm)
0.40	4.0	12.0	6.0	18.0
0.50	5.0	10.0	7.5	15.0
0.67	6.7	6.6	10.0	10.0
0.80	8.0	4.0	12.0	6.0

The moisture distribution was calculated by means of the model that was presented in Section 3.4. The induced climate difference, 65/100% RH (top face/bottom face), resulted in different moisture gradients due to different panel thicknesses. The moisture gradients were:

- 1.75 % mm^{-1} when $a_{tot} = 20\text{mm}$ and
- 1.17 % mm^{-1} when $a_{tot} = 30\text{mm}$.

The results of moisture modelling are shown in Fig. 6.5. The moisture data was collected for material points that were located in the centre of the examined layer. The calculated period was $t = 744\text{h}$. The moisture content highly depends on panel thickness and layer ratio. A larger panel thickness gives lower moisture content in all layers. However, these differing moisture contents are balanced after a specific time period. In the bottom layer, the moisture content is smaller, the greater the layer ratio. Due to the larger volume of the bottom layer, caused by the greater layer ratio, the concentration of moisture increases. The middle layer, however, provides a smaller volume, the larger the layer ratio. Thus, the moisture content is greater than in panels with lower LR . The top layer is influenced by

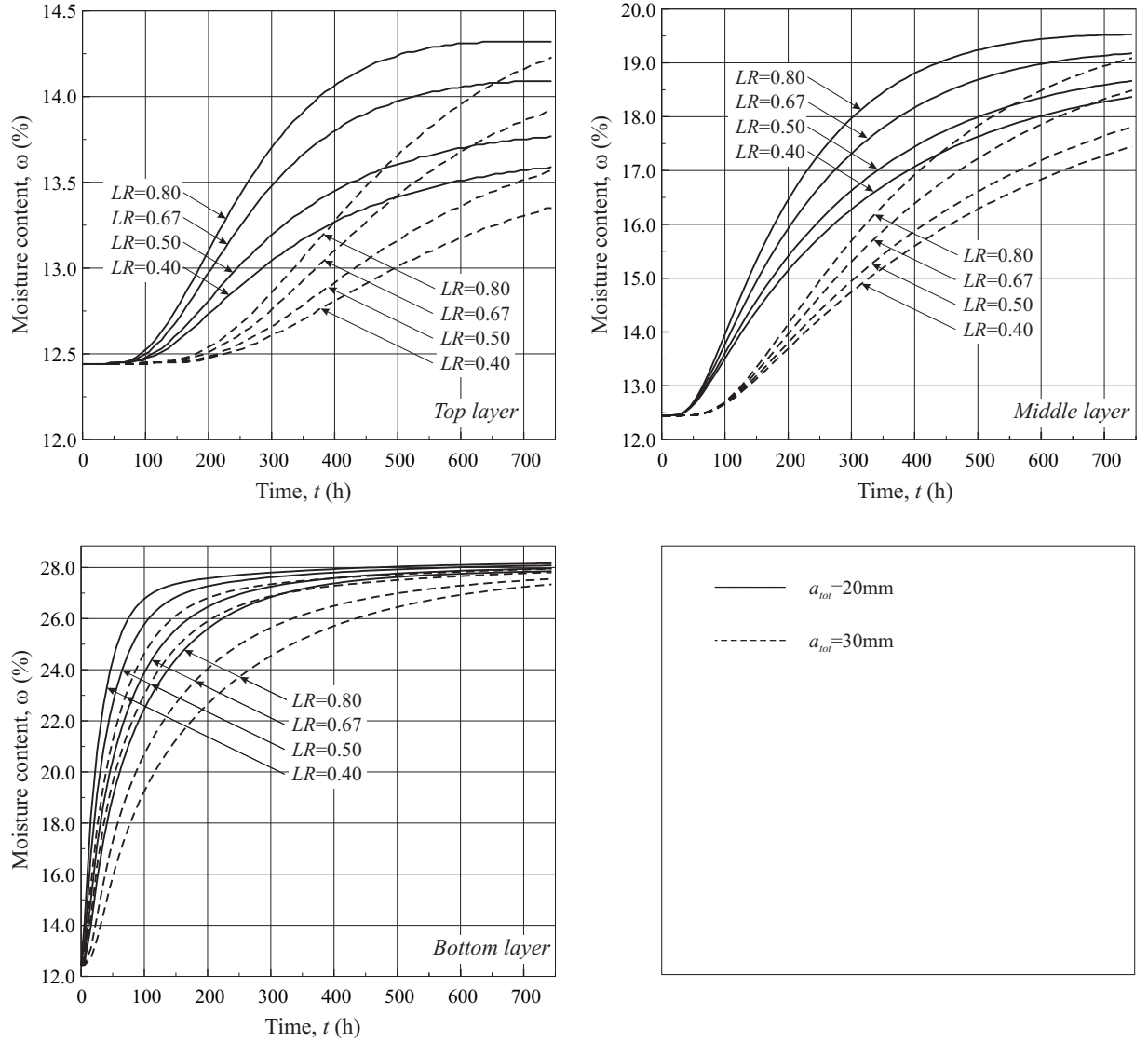


Figure 6.5 Moisture content in the individual layers, influence of panel thickness and layer ratio

the middle layer: the greater the core moisture content, the greater the top layer moisture content, since more water reaches the glue line.

Fig. 6.6 shows the cup deformations, cup_{xz} and cup_{yz} , as a function of the layer ratio. The results indicate that LR has a very strong influence on cup_{yz} . Cup deformation in the yz -plane increases exponentially with increased layer ratio. Initially, panels with a thickness of 30mm have a greater cup_{yz} , but when $LR > 0.55$, the influence of LR is stronger on thinner panels. A layer ratio of approximately 0.5 yields $cup_{yz} = 0\text{mm}$ for panels of both thicknesses. This is caused by the ratio of middle and outer layers, which is $2a_{OL}/a_{ML} = 1$. The displacements u_z , which are the basis of the calculation of cup

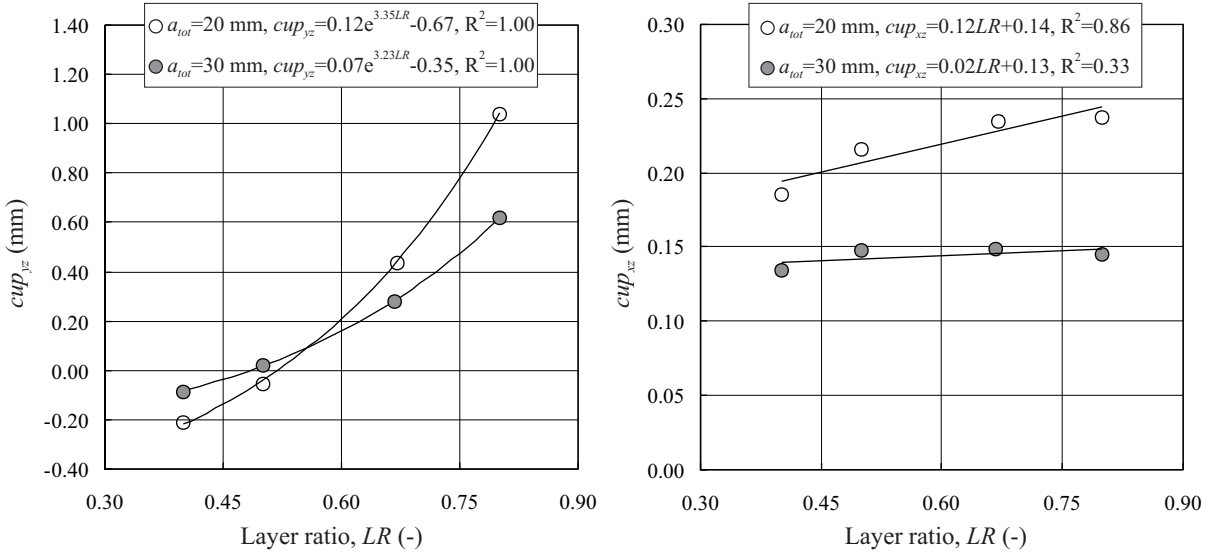


Figure 6.6 Relationship between cup deformations and layer ratio, $t = 744h$

deformations, are displayed for such a configuration in Fig. 6.7.

Cupping in the xz -plane is barely influenced by a variation in LR . Again cup_{xz} increases with decreasing a_{tot} . The influence of LR on cup_{xz} increases with reduced panel thickness, which is indicated by the larger slope.

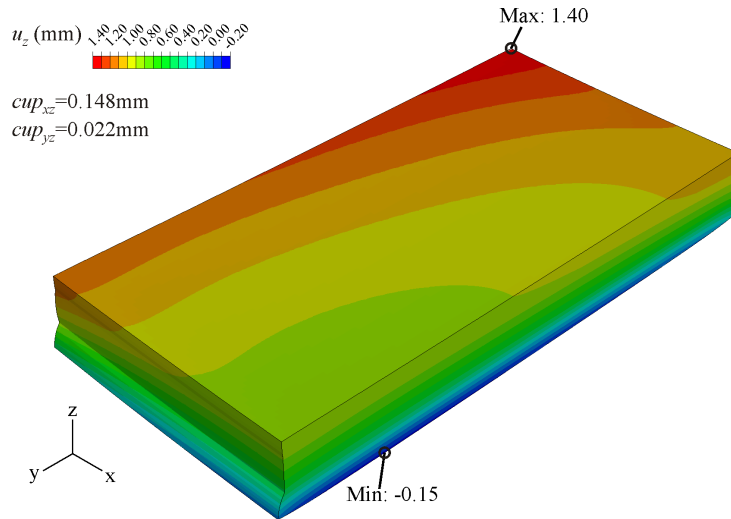


Figure 6.7 Displacement u_z for a panel with $LR = 0.5$, $a_{tot} = 30$ mm, $t = 744h$, displacement scale factor: $\times 10$

6.2.3 Influence of the lamella characteristics

In which way the geometry and the annual ring orientation of individual lamellas affect warping, is studied in the present section. The parameters that describe the geometry and the position in the timber log are displayed in Fig. 6.8. The geometry is expressed by three parameters: the length l , the width b and the thickness a . The length is kept constant at $l = 300\text{mm}$ and the thickness is kept constant at $a = 10\text{mm}$. A variation of the lamella width on warping is investigated. The position in the log is indicated by two parameters: e_v and e_h . They describe the vertical distance of point A to the pith and the horizontal distance of point B to the pith, respectively. Point A is located at the lefthand side and point B is located at the bottom side of the panel face in the $x_2 - x_3$ plane as shown in Fig. 6.8. In Fig. 6.9 two lamellas are defined: lamella I and lamella II. They are arranged alternatively in the top and bottom layer. It is possible to model one half of the panel by means of symmetry boundary conditions (Section 4.3.1).

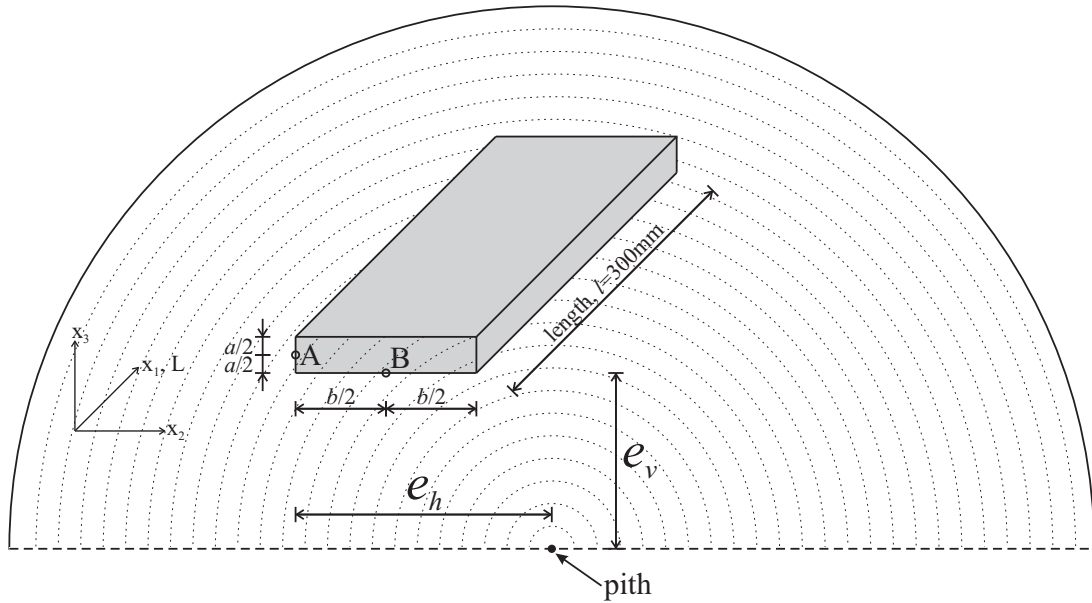
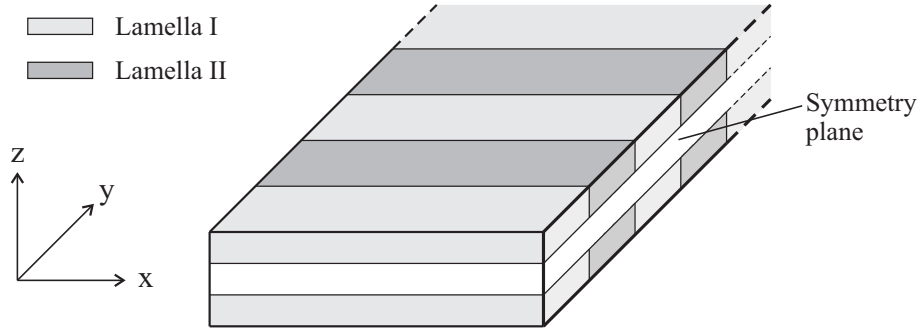


Figure 6.8 Lamella parameters

The positions of the lamellas have been chosen so as to include different distances from the pith and different growth ring orientations (vertical and horizontal). Tab. 6.2 defines the investigated configurations A, B, ... H. In configurations A, B, C and D the vertical parameter e_v is kept constant to zero and e_h is varied. This indicates vertical orientation of the annual rings the larger the distance is to the pith. In configurations E, F, G and H the horizontal parameter e_h is kept constant at zero and e_v is varied. This indicates horizontal annual rings the larger the distance is to the pith. The choice of e_v and e_h in lamellas I

**Figure 6.9** Definition of lamellas I and II

and II, respectively, employs identical or contrary ring angles in both lamella types. The lamella width, b , is varied in two steps: $b_1 = 30\text{mm}$ and $b_2 = 100\text{mm}$. The moisture difference of 65/100% is applied. The glue lines between two lamellas are neglected in the material modelling, since their influence on warping is very small. The mechanical model and the material data are as shown in Section 4.3.

Table 6.2 Alternative lamella configurations

Configuration	b (mm)	e_v (mm)	e_h (mm)	I/II ¹
A	100	0	var.	$e_{h,I} = e_{h,II}$
B	100	0	var.	$e_{h,I} = -e_{h,II}$
C	30	0	var.	$e_{h,I} = e_{h,II}$
D	30	0	var.	$e_{h,I} = -e_{h,II}$
E	100	var.	0	$e_{v,I} = e_{v,II}$
F	100	var.	0	$e_{v,I} = -e_{v,II}$
G	30	var.	0	$e_{v,I} = e_{v,II}$
H	30	var.	0	$e_{v,I} = -e_{v,II}$

¹ Lamellas I and II according to Fig. 6.9

Since there is a dominant influence perpendicular to the grain and causing the outer layers to dominate warping, cup_{yz} is strongly affected and cup_{xz} is minimally affected by a variation of the lamella parameters. The influence of different lamella characteristics on cup deformation cup_{yz} is shown in Figs. 6.10 and 6.11. The figures present cup_{yz} as a function of e_h and e_v , respectively.

The variation of e_h shown in Fig. 6.10 ($e_v = 0\text{mm}$) yields significant differences between different lamella widths, b_1 (configurations C and D) and b_2 (A and B). The cupping cup_{yz} is smaller when b_1 is applied. Equal cupping in all four configurations is found at $e_h \geq 75\text{mm}$, since the annual ring orientation is vertically with that distance to the pith. No influence of alternating growth ring orientations could be detected.

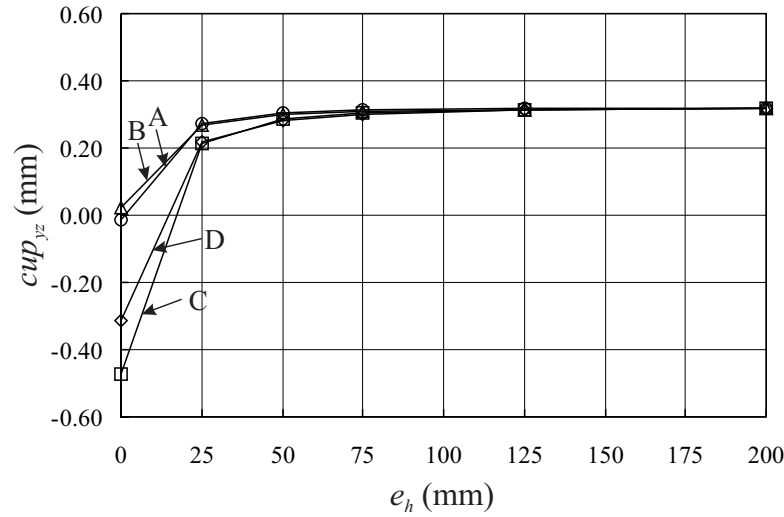


Figure 6.10 cup_{yz} as a function of e_h , influence of lamella configurations (Tab. 6.2)

Fig. 6.11 indicates a strong influence of lamella width and alternating ring orientations on cup_{yz} as a function of e_v ($e_h = 0\text{mm}$). The cup deformation increases with an increase in the vertical distance to the pith e_v . When $e_v \geq 50\text{mm}$, the cup deformation is larger for 100mm lamellas. Alternating ring orientations increase cup_{yz} when b is 100mm ($F > E$) and decrease cup_{yz} when b is 30mm ($H < G$). These differences between both lamella widths may result from the ratio of lamellas I and II in the layers. When b_2 is used, the number of both lamella types is unequal while it is equal when b_1 is used. When the ring angle alternates between lamellas I and II, cup_{yz} is found to be smaller when b is 30mm (configuration H compared to F). The small lamella width in combination with alternating annual rings yields smallest cup_{yz} .

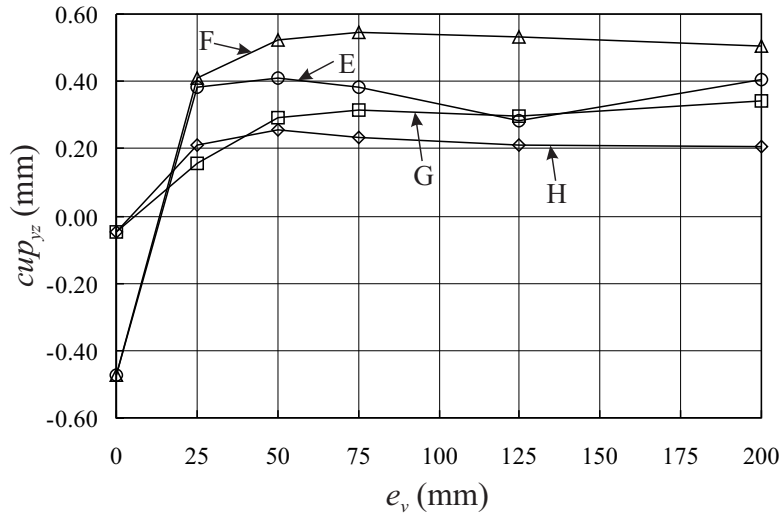


Figure 6.11 cup_{yz} as a function of e_v , influence of lamella configurations (Tab. 6.2)

6.2.4 Influence of density

Introduction

The growth of a tree is influenced by its conditions regarding location, such as climate, soil, forest stand, etc. Since these conditions strongly vary between different trees and stands, the mechanical properties of wood (modulus of elasticity, shear modulus, etc.) are liable to considerable variations. The elastic parameters of spruce wood, therefore, not only depend on the moisture content but also on the density of the sample. A strong correlation between density and elastic parameters was reported by many researchers: e.g., Eberhardsteiner [28], Glos [36], Kollmann [57], Niemz [75] and Spengler [101]. It is of interest to analyse how density affects the moisture-induced deformations of cross-laminated solid wood panels. Firstly, the correlation between density and the elastic parameters was derived from investigations by Keunecke et al. [55] and then the influence of varying the density on the hygroscopic warping was tested. The possible dependency of other parameters on density is neglected.

Data analysis

Keunecke et al. [55] determined Young's and shear moduli by means of ultrasonic measurements. Cubic specimens of clear wood with an edge length of 10mm were tested. The moduli were determined by means of two wave types: longitudinal waves and transverse

waves. The shear moduli can be calculated directly from the sound velocity of transverse waves c_{ij} [m s⁻¹], where i is the direction of wave propagation and j the direction of oscillation, as

$$G_{ij} = c_{ij}^2 \rho . \quad (6.1)$$

The modulus of elasticity is approached by applying the sound velocity of longitudinal waves c_{ii} by the same relationship

$$E_{ii} = c_{ii}^2 \rho . \quad (6.2)$$

Tab. 6.3 gives an overview of the elastic constants calculated from ultrasonic measurements by Keunecke et al. [55] and compares them with static measurements from Neuhaus [74]. The comparison between the Young's moduli measured by Keunecke et al. [55] and values from Neuhaus [74] reveals higher values determined by the dynamic method. The relative increase is higher for E_R and E_T than for E_L . This may be explained by the microscopic structure of spruce. The softwood structure is a system of closed tubes. Tracheids providing high acoustical consistency in the longitudinal direction with their length-to-diameter aspect ratio of 30:1 to 100:1. Ray cells, which are oriented in the radial direction, are shorter than tracheids but still able to transmit acoustical waves. In the tangential direction an acoustical conducting structure is completely missing.

The data of Keunecke et al. [55] was analysed in order to provide data for material modelling. The measured densities of the small specimens ranged between 320...400...530 kg m⁻³ (minimum...mean...maximum). The relationship between density and the moduli were found to be linear with different strong correlations ($0.26 \leq R^2 \leq 0.81$, see Appendix). Since these moduli were measured at 20°C/65% RH, Keunecke et al. [55] reported a moisture content of 12%. For this reason and due to similar mean densities,

Table 6.3 Elastic parameters of Norway spruce in N mm⁻² as measured by means of ultrasonic waves [55] and static measurements [74], $\omega = 12\%$

Source	ρ (kg m ⁻³)	E_L	E_R	E_T	G_{LR}	G_{LT}	G_{RT}
[55] $\bar{X}$	400	13800	1800	1170	617	587	53
V(%)	9.7	20.0	10.6	21.1	12.1	10.2	10.9
[74] $\bar{X}$	415	11996	817	420	624	743	42
$\xi = \frac{C_{dyn}}{C_{stat}}$		1.2	2.2	2.8	1.0	0.8	1.3

the values can be compared to the static determined moduli of Neuhaus [74] as was previously done for the mean values in Tab. 6.3. With the data of Keunecke et al. [55], the relationship between moduli and density is available. The data of Neuhaus [74] provides the relationship between the moduli and moisture content. The comparison of dynamically determined moduli C_{dyn} [55] and moduli determined by means of static methods C_{stat} [74] yields coefficients ξ [-] with

$$\xi = \frac{C_{dyn}(\rho)}{C_{stat}(\omega)} . \quad (6.3)$$

In several steps, the data of Keunecke et al. [55] and Neuhaus [74] were compared for $\rho = 415 \text{ kg m}^{-3}$ and ω_i :

$$\xi = \frac{C_{dyn}(415)}{C_{stat}(\omega_i)} . \quad (6.4)$$

It is assumed that the relationship between moduli and density is valid for all moisture contents between $0.00 \leq \omega_i \leq 0.28$. By means of Eq. 6.4, the new dependency of the moduli on density $\hat{C}(\rho)$ could be calculated with

$$\hat{C}(\rho) = \xi \cdot C_{dyn}(\rho) . \quad (6.5)$$

The values presented in Tab. 6.4 result for $\omega_i = 12\%$. $\hat{C}(\rho)$ calculated for $0.00 \leq \omega_i \leq 0.28$ yields $\hat{C}(\omega)$. This is valid for $320 \leq \rho \leq 530 \text{ kg m}^{-3}$.

Table 6.4 Calculated density dependency of the moduli for $\omega = 12\%$, $\hat{C}(\rho) = r_1\rho + r_2$ (ρ in kg m^{-3})

Modulus $\hat{C}$	r_1 (N mm ⁻²)	r_2 (N mm ⁻²)
E_L	33.12	-1748.98
E_R	1.74	94.91
E_T	1.13	-48.86
G_{LR}	1.59	83.23
G_{LT}	1.38	52.88
G_{RT}	0.10	-0.46

Hygroscopic warping

In all previous numerical simulations, a homogenous density distribution throughout the whole panel was used. In Section 6.2.3, lamellas were introduced and different ring angles in the lamellas were tested. In this section, the density of the lamellas is varied. The material parameters used in the previous sections affiliate to the mean density $\rho = 415 \text{ kg m}^{-3}$ [74]. Two extremum densities are applied additionally, $\rho = 330 \text{ kg m}^{-3}$ and $\rho = 500 \text{ kg m}^{-3}$. The aforementioned procedure provides a suitable relationship between moduli and moisture content for the three densities. Fig. 6.12 provides the curves of the $E_L(\omega)$ relation. The tested combinations of densities in the lamellas I and II (outer layers) and the middle layer are summarised in Tab. 6.5. Configurations A1, A2, ..., A8 relate to testing of different densities of the outer and middle layers. Panels with vertical and horizontal annual rings are tested. Configurations B1, B2, ..., B8 relate to testing where the influence of different densities in both lamellas is investigated. The density of the middle layer is kept constant at $\rho = 415 \text{ kg m}^{-3}$. It is tested how the lamella width and the growth ring orientation affect the warping.

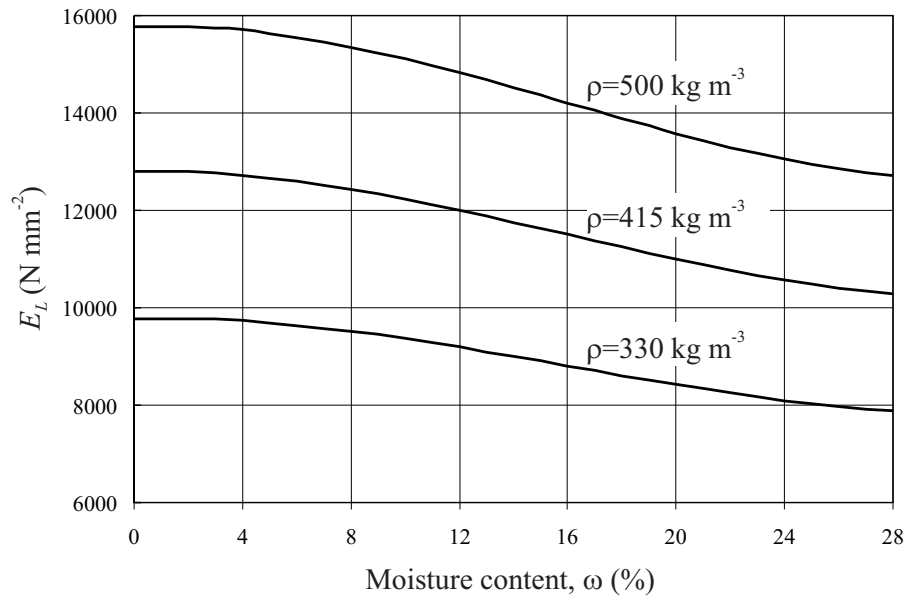


Figure 6.12 Relationship between E_L and moisture content ω for different densities ρ (Norway spruce), data according to Keunecke et al. [55] and Neuhaus [74]

Fig. 6.13 shows the cup deformations of series A. These are compared to the reference calculations where the density was constant 415 kg m^{-3} . The results indicate that cup_{yz} is smallest in panels where the annual rings are aligned vertically ($\theta = 90^\circ$). The density has

Table 6.5 Panel configurations, $a_{tot} = 30\text{mm}$, $LR = 0.67$, $e_{v,I} = -e_{v,II}$, $e_{h,I} = -e_{h,II}$

Configu- ration	$e_{v,I}$ (mm)	$e_{h,I}$ (mm)	Lamella width (mm)	Density, ρ_0 (kg m ⁻³)		
				Lamella I	Lamella II	ML
A1	0	250	100	330	330	330
A2	0	250	100	500	500	500
A3	0	250	100	330	330	500
A4	0	250	100	500	500	330
A5	205	0	100	330	330	330
A6	205	0	100	500	500	500
A7	205	0	100	330	330	500
A8	205	0	100	500	500	330
B1	0	250	100	330	500	415
B2	0	250	100	500	330	415
B3	0	250	30	330	500	415
B4	0	250	30	500	330	415
B5	205	0	100	330	500	415
B6	205	0	100	500	330	415
B7	205	0	30	330	500	415
B8	205	0	30	500	330	415

a significant influence on panel warping. The smallest cupping is detected when $\rho = 500$ kg m⁻³ in all layers (configurations A2 and A6). It is highest when $\rho = 330$ kg m⁻³ in the complete panel (A1 and A5). Different densities in the outer and middle layers imply different warping characteristics in panels with vertically and horizontally oriented growth rings. When $\theta = 90^\circ$, varying dense layers (A3 and A4) yield cuppings between both maximum densities (A1 and A2). Both configurations result in smaller cuppings than the reference series. cup_{yz} is smaller when the outer layer density is 330 kg m⁻³ and the middle layer density is 500 kg m⁻³ than vice versa. When $\theta = 0^\circ$, different densities in the layers yield greatly deviating results. Panel A7, where $\rho_{OL} = 330$ kg m⁻³ and $\rho_{ML} = 500$ kg m⁻³, shows small cupping, which is near the minimum (A6). The converse density distribution (A8), however, gives the highest cup_{yz} . Fig. 6.14 shows the cup deformations of series B and the reference configurations, where the density is constant at 415 kg m⁻³. A vertical orientation of the growth rings gives a smaller cup deformation than a horizontal orientation. Both annual ring orientations show opposite characteristics. When $\theta = 90^\circ$, different densities of lamellas I and II yield smaller cup_{yz} than the reference configurations. cup_{yz} is larger than the reference, when $\theta = 0^\circ$. Compared to 100mm wide

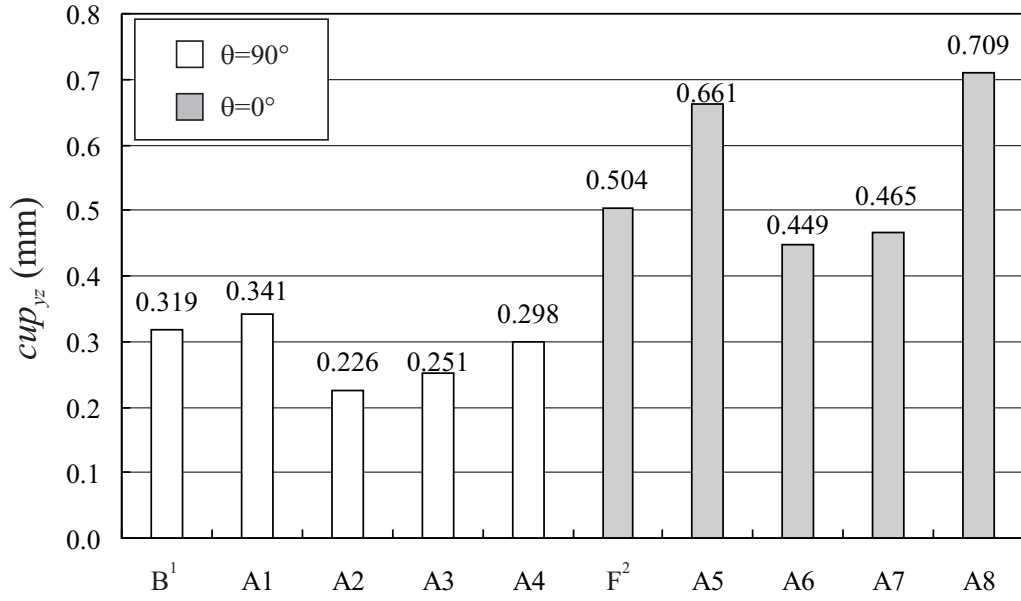


Figure 6.13 Cup deformations cup_{yz} for different densities in the middle and outer layers, configurations A1-A8 according to Tab. 6.5

^{1,2} configurations according to Tab. 6.2 ($\rho = 415 \text{ kg m}^{-3}$)

¹ $e_{v,I} = 0\text{mm}$, $e_{h,I} = -e_{h,II} = 250\text{mm}$

² $e_{h,I} = 0\text{mm}$, $e_{v,I} = -e_{v,II} = 205\text{mm}$

lamellas, a small lamella width ($b = 30\text{mm}$) gives larger cupping, when the annual ring orientation is vertical (B3 and B4 compared to B1 and B2), and considerably smaller cupping, when the annual ring orientation is horizontal (B7 and B8 compared to B5 and B6). Different densities in lamellas I and II, lamella width and annual ring orientation set constant, influence cup_{yz} only when $b = 100\text{mm}$ (B1 and B2, B5 and B6). The asymmetric number of lamellas I and II account for it.

Fig. 6.15 illustrates the displacements in thickness direction, u_z , of panels B1, B3, B5 and B7. The figure shows that the centre of each panel moves downwards (minimum displacement) and that the edges move upward (maximum displacement). The swelling of small lamellas ($b = 30\text{mm}$) and the cylindrical shape of the local material coordinate system lead to bowing at the bottom surfaces of panels B3 and B7.

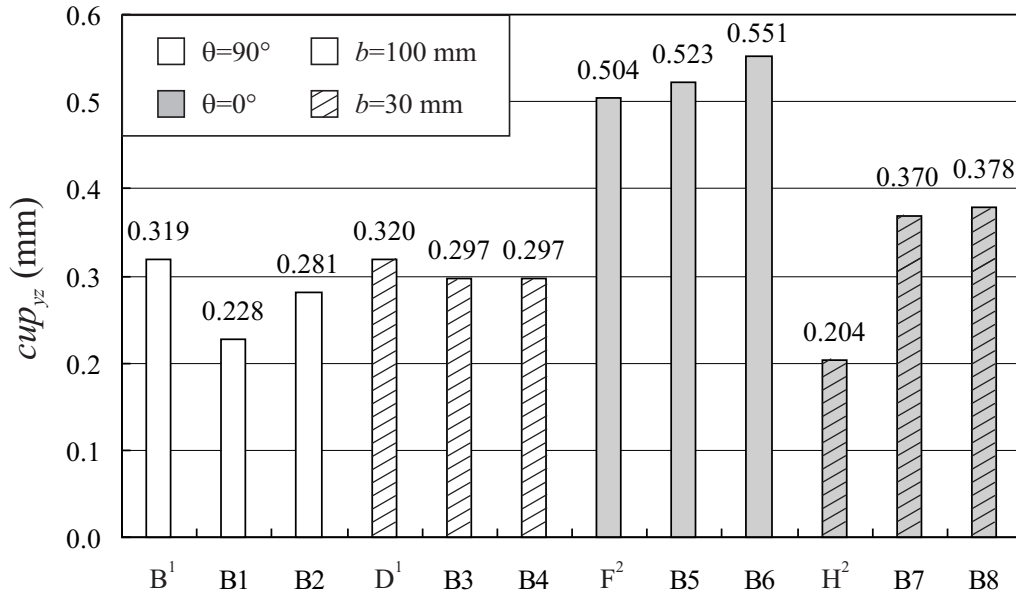


Figure 6.14 Cup deformations cup_{yz} for different densities in lamellas I and II, configurations B1-B8 according to Tab. 6.5

^{1,2} configurations according to Tab. 6.2 ($\rho = 415 \text{ kg m}^{-3}$)

¹ $e_{v,I} = 0\text{mm}$, $e_{h,I} = -e_{h,II} = 250\text{mm}$

² $e_{h,I} = 0\text{mm}$, $e_{v,I} = -e_{v,II} = 205\text{mm}$

6.2.5 Spruce-MDF laminates

Influence of layer ratio

In this section, the influence of the layer ratio on cup deformations in spruce-MDF laminates is tested. MDF700 is employed as the middle layer. The moisture profiles are calculated according to the model and input data that were presented in Section 3.5. The mechanical material model is shown in Section 4.3.5. The total thickness is kept constant at 36.2mm. The thicknesses of the spruce outer layers vary between 7mm and 12mm. Therefore, the middle layer thickness varies between 12mm and 22mm. This gives layer ratios between 0.39 and 0.67.

The cup deformations, cup_{xz} and cup_{yz} , are plotted as a function of LR in Fig. 6.16. The numerical test of different layer ratios, with the total thickness kept constant, shows a linearly increasing cup_{yz} with increased LR . The cupping, cup_{xz} , is not greatly influenced when LR is varied, since the outer layers are the dominant layers for warping. With increased LR the relative thickness of the outer layers increases and, thus, cup_{yz} increases.

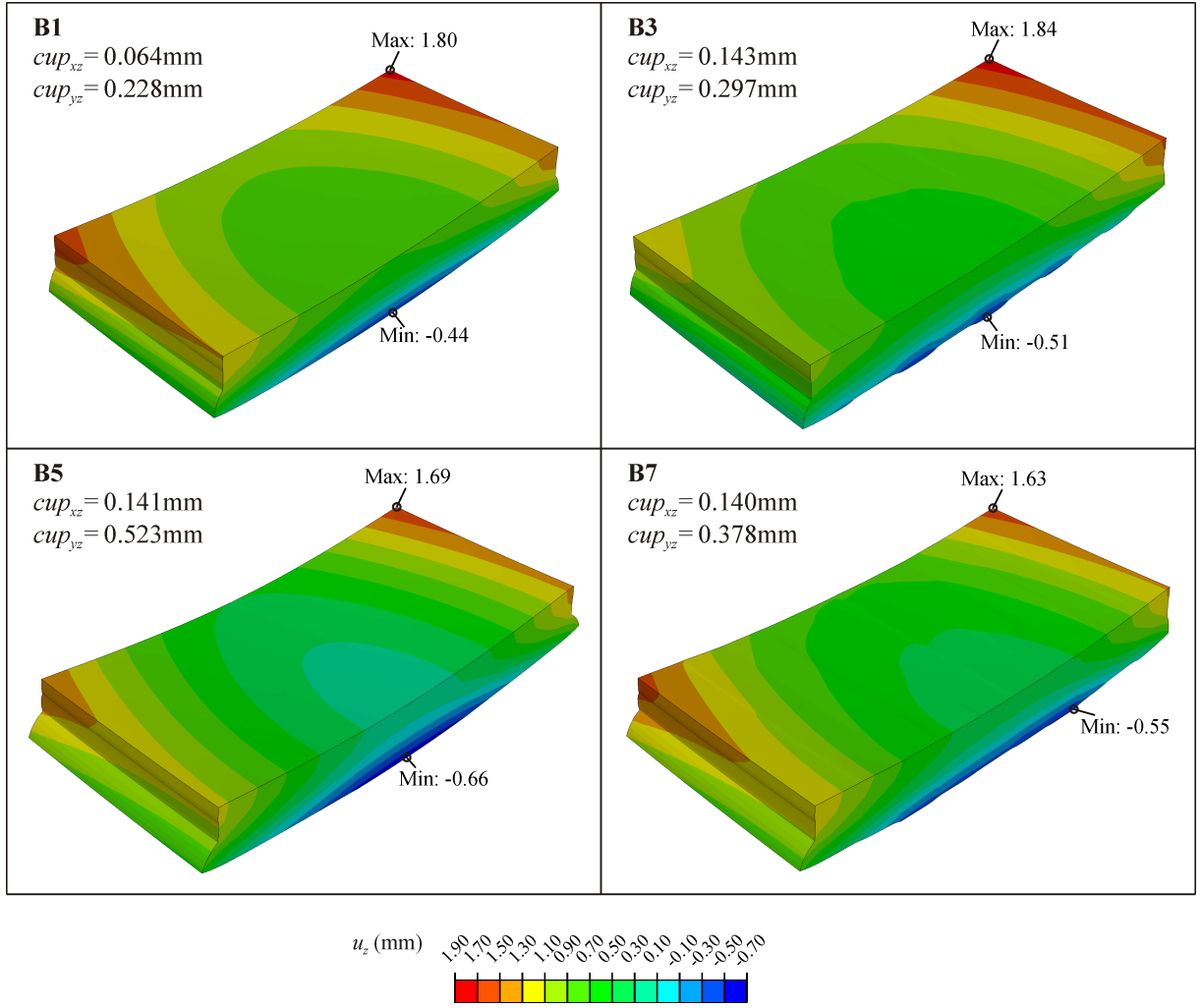


Figure 6.15 Displacement u_z in boards B1, B3, B5 and B7 (Tab. 6.5), displacement scale factor: $\times 10$

Influence of modulus of elasticity of MDF

The influence of a variation of the modulus of elasticity of the MDF middle layer on warping is examined in this section. The reference modulus is indicated in Tab. 2.1 ($\rho_0 = 800 \text{ kg m}^{-3}$) to depend on the moisture content. The reference configuration is varied over six steps between $0.2 \times E_{1,ref}$ and $1.8 \times E_{1,ref}$. The thicknesses of the individual layers are 10mm and 16mm for the spruce outer layers and the MDF middle layer, respectively. This gives a total panel thickness of 36.2mm and a layer ratio of 0.56.

In Fig. 6.17, the cup deformation cup_{yz} is plotted as a function of $E_1/E_{1,ref}$. The results indicate that cup_{yz} is greatly influenced when E_1 is varied. The decrease of the cup deformation is clearly nonlinear. The cup deformation in the minor axis, cup_{xz} , is not

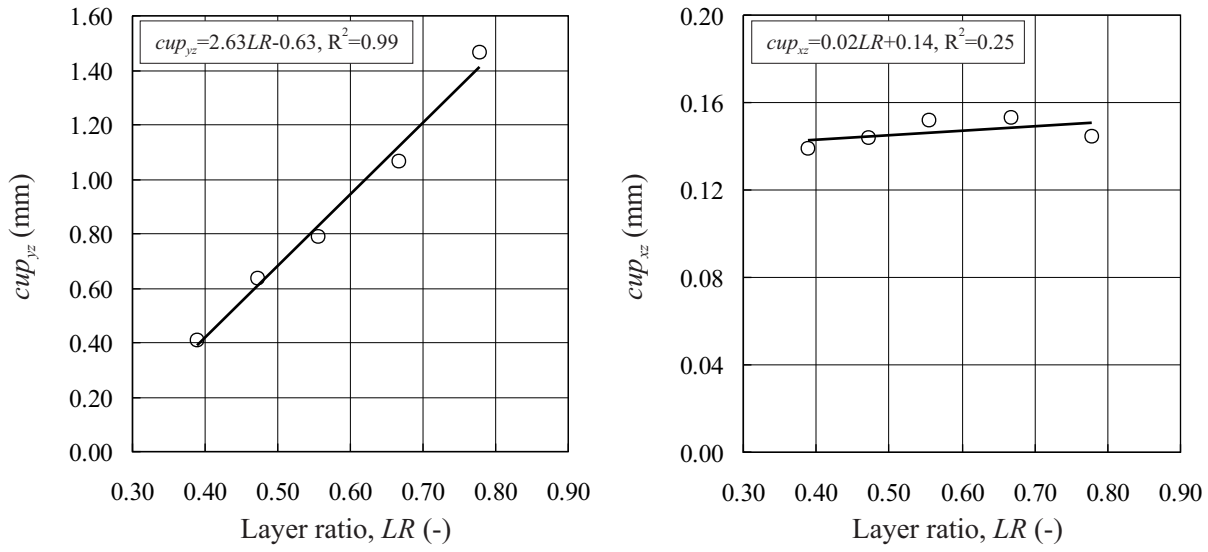


Figure 6.16 Relationship between cup_{yz} and layer ratio LR in spruce-MDF700 laminates, $a_{tot} = 36.2\text{mm}$, $t = 744\text{h}$

influenced by a variation of the modulus of elasticity of MDF even though the moduli of elasticity in the 1- and the 2- directions ($E_1 = E_2$) were varied.

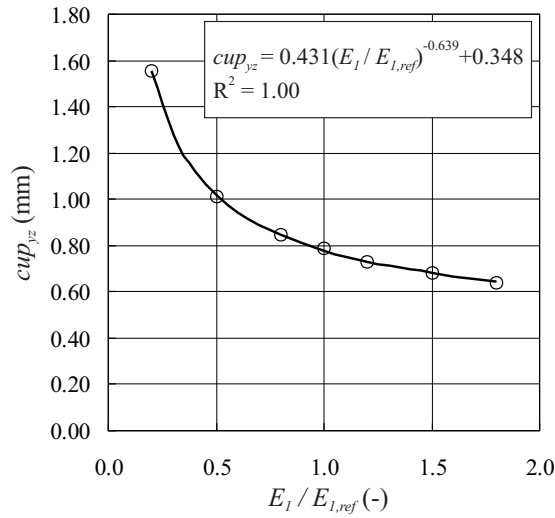


Figure 6.17 Relationship between cup_{yz} and modulus of elasticity of the MDF middle layer $E_1 = E_2$, $E_{1,ref}$: Tab. 2.1 ($\rho_0 = 800 \text{ kg m}^{-3}$), $a_{tot} = 36.2\text{mm}$, $t = 744\text{h}$

In Fig. 6.18, radial stresses are presented as a function of the width b . The figure includes two subfigures showing the stress results for paths lying at $z = 0\text{mm}$ and $z = 10\text{mm}$. At $z = 0\text{mm}$, the stresses in the bottom layer decrease with increased modulus of elasticity of

the MDF middle layer, and at $z = 10\text{mm}$ the stresses increase. The relationship between the modulus of elasticity and stresses is nonlinear.

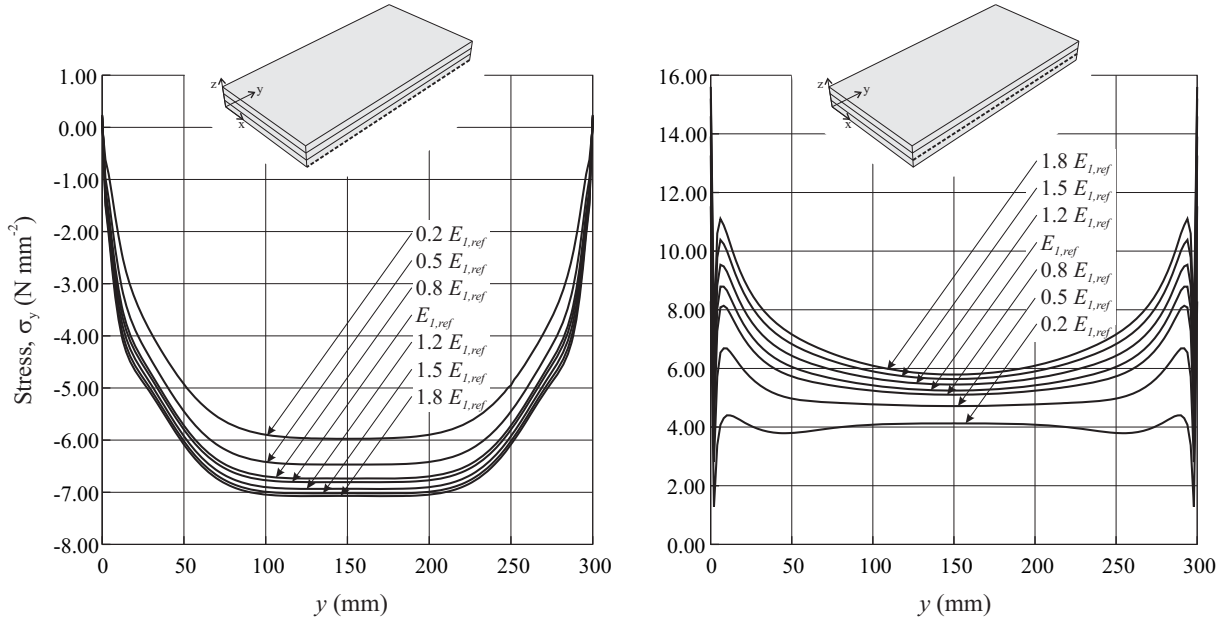


Figure 6.18 Stress distributions, σ_y , along the y -axis in the bottom layer in panel CMD7 after 744h of climate difference 65/100% RH, lefthand side: $z = 0\text{mm}$, righthand side: $z = 10\text{mm}$

6.3 Parametric study of internal stresses

6.3.1 Influence of the material parameters

In order to find out what influence a variation of the input parameters has on the internal stress state, a parametric study was performed. The following parameters were tested in a panel of type AR90: E_L , E_R , α_L , α_R and m_R . The moisture content was set to 10% in the initial state and then it was increased to 13% in the outer layers. The panel studied here is displayed in Fig. 5.21. The mechanical boundary conditions and the element mesh density are presented in Section 5.3.1.

The influence of a 30% decrease of the material parameters on the displacement, u_y , is displayed in Fig. 6.19. The figure contains two subplots: the results for different paths along the z -axis at $y = 0\text{mm}$ and $y = 20\text{mm}$. This gives the displacements in the radial direction in the outer layers and in the longitudinal direction in the middle layer. The

results indicate a strong influence by α_R and m_R and also a significant influence by E_L and E_R . The variation of α_L , however, does not affect the displacement u_y . Compared to the reference configuration, u_y decreases when α_R and E_R are decreased and u_y increases when E_L and m_R are decreased. The changes of α_R and E_R effect the outer layers in the radial direction (y-axis). As a result, u_y is influenced in the same way as both parameters. The influence of α_R is more significant than E_R . When E_L is decreased, the weakening of the middle layer increases the displacement in the outer layers. A smaller m_R increases the compressive stresses in the radial direction in the outer layers and, thus, decreases the displacement.

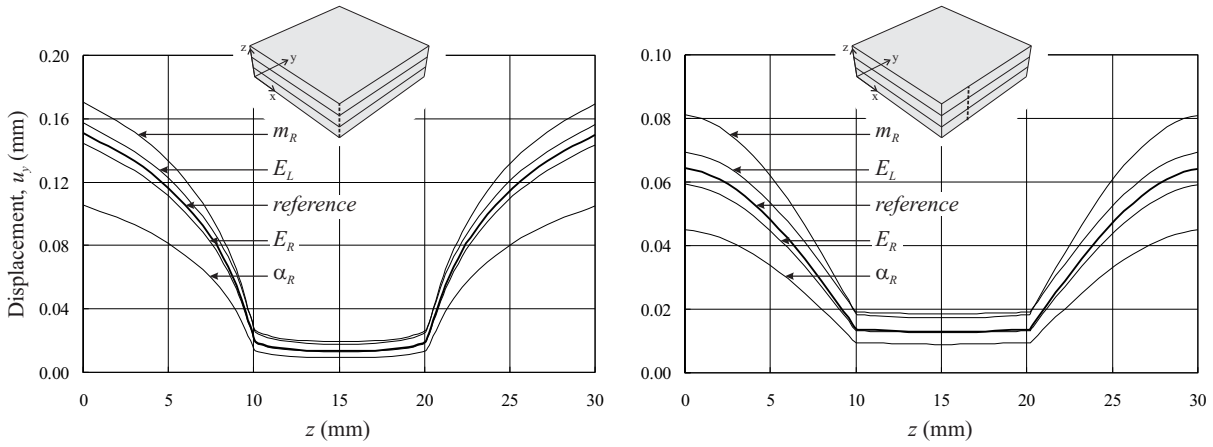


Figure 6.19 Displacements u_y along the thickness, lefthand side: $y = 0\text{mm}$, righthand side: $y = 20\text{mm}$, change of the material parameters of -30%

Figs. 6.20 and 6.21 display the radial stresses σ_y along the y-axis corresponding to a 30% decrease and a 30% increase of the material parameter values. Each figure contains two subplots, showing the stresses at $z = 0\text{mm}$ and $z = 10\text{mm}$. A 30% variation of E_L , E_R and α_L has no or little influence on stresses. The compressive stresses in the outer layers significantly decrease (or increase) when α_R is decreased (or increased). This is obvious since a less (or greater) swelling decreases (or increases) the deformation of the outer layers. A decrease (or increase) of m_R results in larger (or lower) compressive stresses.

In Fig. 6.22 the influences of a 30% decrease and a 30% increase of the material parameters on stresses σ_x along the x-axis in the middle layer are displayed. Tensile stresses perpendicular to the grain were detected. These stresses decrease when E_R , α_L and α_R are decreased and increase when E_L and m_R are decreased (and vice versa). The changes are strongest when the stiffness parameters E_L and E_R are varied.

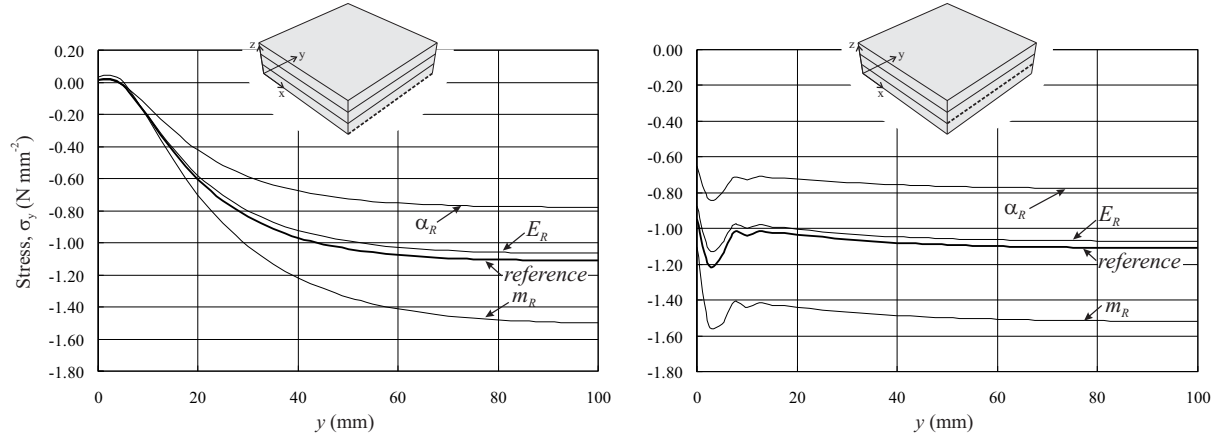


Figure 6.20 Stresses σ_y (radial stresses) along the y-axis, lefthand side: $z = 0\text{mm}$, righthand side: $z = 10\text{mm}$, change of the material parameters of -30%

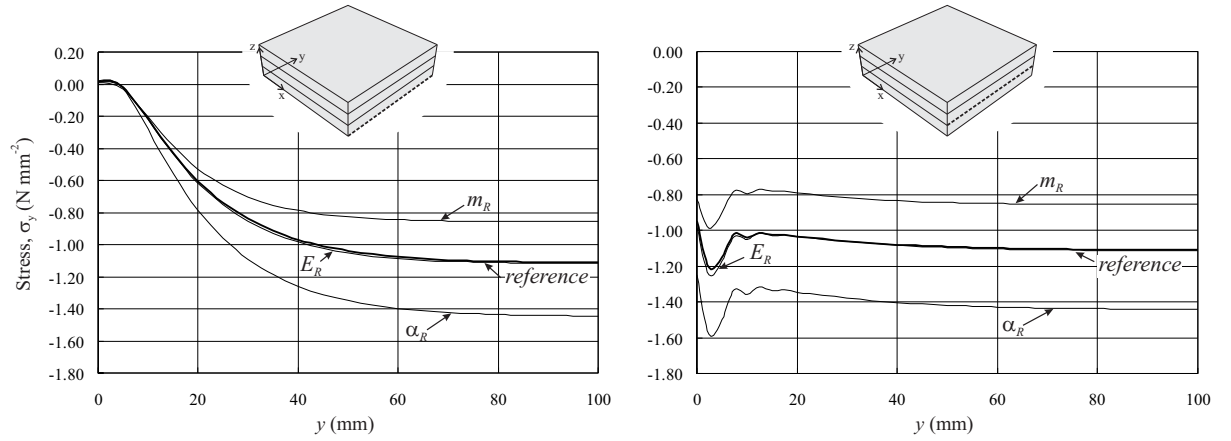


Figure 6.21 Stresses σ_y (radial stresses) along the y-axis, lefthand side: $z = 0\text{mm}$, righthand side: $z = 10\text{mm}$, change of the material parameters of +30%

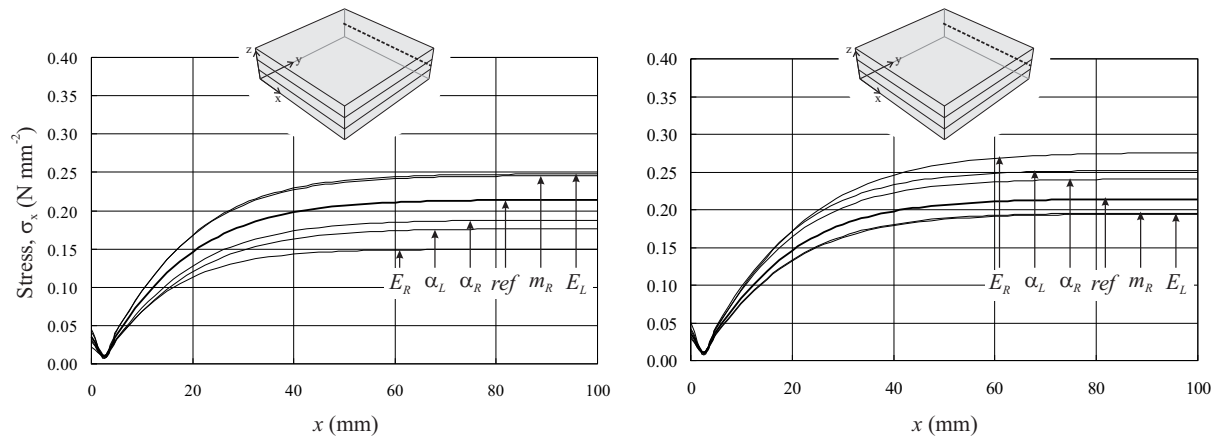


Figure 6.22 Stresses σ_x (radial stresses) along the x-axis in the middle layer ($z = 15.1\text{mm}$), 30% decrease (lefthand side) and 30% increase (righthand side) of material parameter values

6.3.2 Influence of the layer ratio

In this section, the influence of the layer ratio and the panel thickness on the internal stress state is evaluated. Three panel thicknesses are tested: 20mm, 30mm and 40mm. The layer ratio is varied between 0.3 and 0.9. The moisture content was increased from 10% in the initial state to 13% in the outer layers. This moisture content was applied in all panel configurations investigated. However, different structural parameters, such as total panel thickness or outer layer thickness, effect the moisture distribution within the panel and, thus, the stress development. The influence of different moisture distributions is neglected in order to determine the pure influence of the panel structure on stresses and deformations.

In Fig. 6.23 one can easily see that swelling highly depends on the layer ratio and the panel thickness. The displacement, u_y , at the investigated boundary clearly increases with increased layer ratio and panel thickness. Both give thicker outer layers and, thus, a larger distance to the middle layer, which hinders the outer layer in swelling.

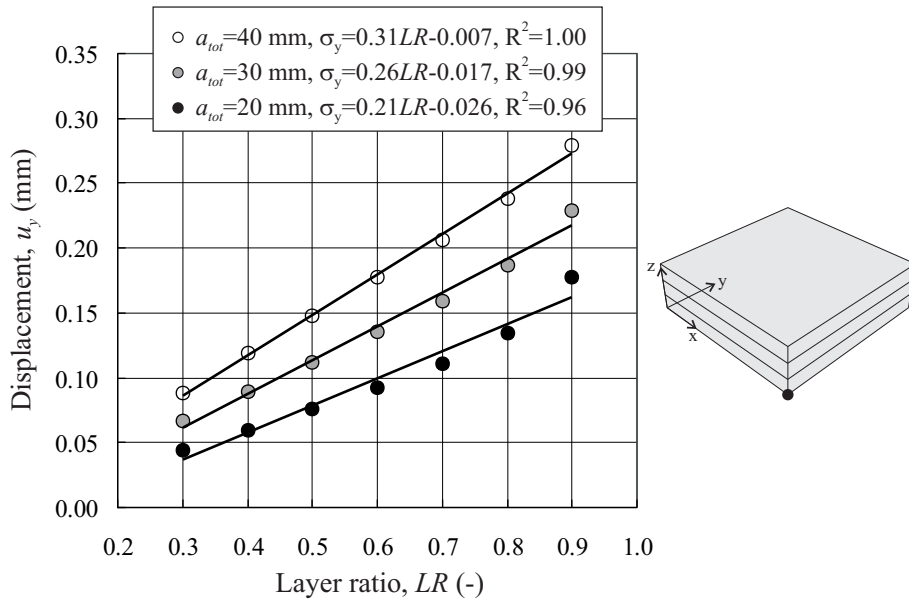


Figure 6.23 Influence of layer ratio and panel thickness on displacement u_y at $x = 100$ mm, $y = 0$ mm, $z = 0$ mm (labeled in the sketch)

The results shown in Figs. 6.24 and 6.25 indicate a very strong influence of layer ratio and panel thickness on stresses. The stresses increase from -1.1 N mm^{-2} for $LR = 0.3$ and $a_{tot} = 20$ mm to -0.26 N mm^{-2} for $LR = 0.9$ and $a_{tot} = 40$ mm ($y = 20$ mm, Fig. 6.24). The compressive stresses in the outer layer increase with increased panel thickness and

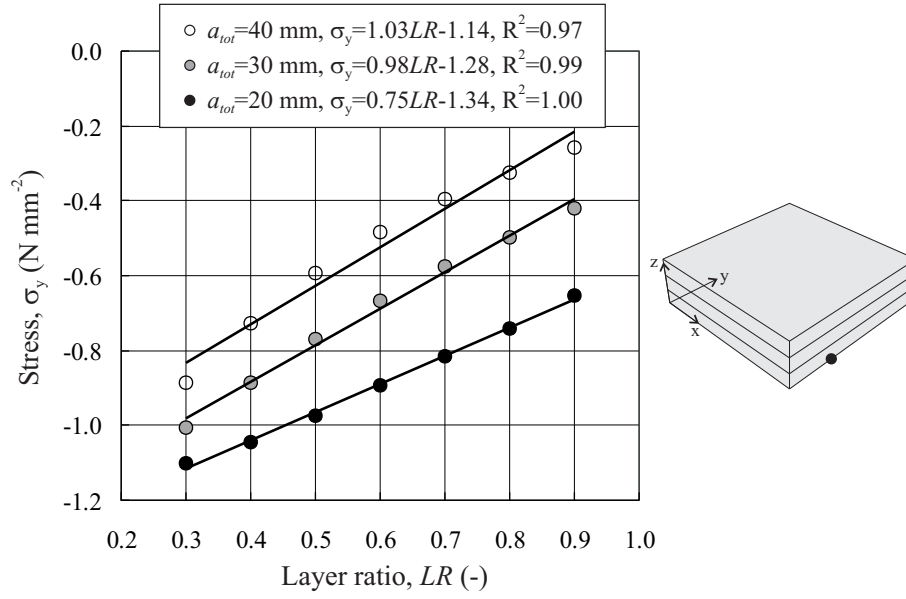


Figure 6.24 Influence of layer ratio and panel thickness on radial stress at $x = 100\text{mm}$, $y = 20\text{mm}$, $z = 0\text{mm}$ (labeled in the sketch)

layer ratio. Fig. 6.25 demonstrates small differences between different configurations at the boundaries but a large variation in between. When the layer ratio is increased and the panel thickness is kept constant, the outer layer thickness increases. Consequentially, forces are absorbed by a larger volume and the stress decreases.

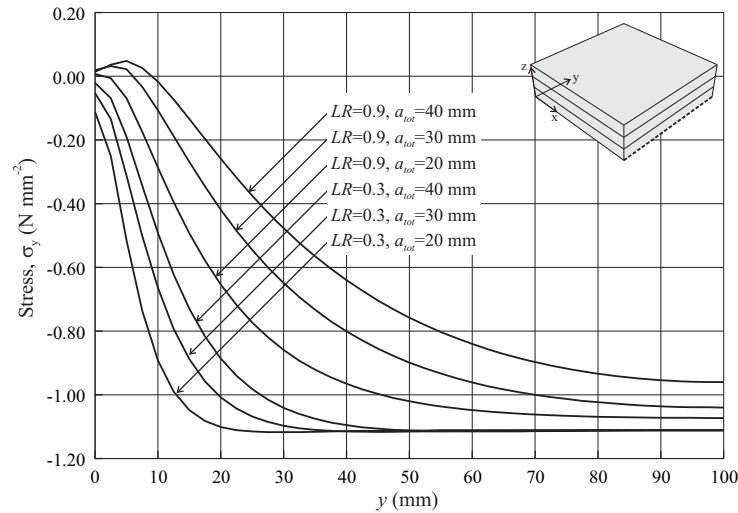


Figure 6.25 Influence of layer ratio and panel thickness on radial stresses in the bottom layer ($x = 100\text{mm}$, $y = 0\text{mm}$)

6.4 Concluding remarks

In the present chapter, parametric studies were presented in order to investigate the influence of different parameters on moisture-induced deformations and stresses due to a moisture gradient and a single climate step. The simulations provided information about the influence of material parameter values, the layer ratio and lamella characteristics in spruce cross-laminates and spruce-MDF laminates.

The hygroscopic warping and the internal stress state depend, to a high degree, on the values for stiffness and moisture swelling in the longitudinal and transverse directions to the fibres, which are oriented in the panel plane. An accurate experimental determination of these parameter values is, therefore, highly important for finite element calculations of cross-laminated wood panels. A relevant theoretical description and accurate material parameter values for mechano-sorption were also found to be very important for the simulation of moisture-induced stresses and deformations in cross-laminated solid wood panels.

Stresses and deformations induced by both, a moisture gradient and a climate change, depend to a high degree on the layer ratio of the panel. With increased relative thickness of the outer layers, the cup deformation, cup_{yz} , induced by a moisture gradient, increased however internal compressive stresses due to a climate change decreased. Thus, with the correct choice of panel thickness and layer ratio with respect to the static requirements and the expected climatic conditions, warping and stresses could be minimised.

The lamella parameters such as the distance to the pith (i.e. the annual ring orientation) or the density distribution influenced warping. The results showed that the lowest warping developed when the annual rings were oriented vertically in the lamellas, and when the density of the panel was large. Thus, the choice of the lamella characteristics and their arrangement in the panel could minimise warping and stresses.

In spruce-MDF laminates, two parameters were tested for their influence on warping: the layer ratio and stiffness of the MDF middle layer. The cup deformation increased when the layer ratio was increased. With increased modulus of elasticity of the MDF middle layer, cup_{yz} was reduced. Again, an accurate choice of layer ratio minimises warping.

Chapter 7

Summary and Outlook

7.1 Summary of the thesis

Moisture-induced deformations and stresses play an important role in the application of cross-laminated wood panels. In the present study, these deformations and stresses were investigated by means of experimental tests and numerical simulations. Three moisture scenarios, which cause problems in practice, were examined:

1. moisture difference between both large panel faces,
2. moistening and
3. drying in a single climate step.

Experimental tests on sorption and moisture distribution were completed by numerical estimation of the glue line diffusivity. The hygroscopic warping induced by a moisture gradient of 65/100% RH was studied and the internal stress state due to a single moistening step was evaluated.

Sorption measurements indicated a statistically significant but very small difference between the moisture adsorption in clear spruce samples and the moisture adsorption in cross-laminated spruce samples. The influence of the stress state due to the cross-lamination or the production process (cold pressing) was neglected from the analysis of sorption and stress and moisture fields were not coupled. From measured moisture profiles of laminated wood samples, it was concluded that the glue lines provide a high resistance to moisture diffusion. The coefficient of diffusion of 1K PUR glue lines was

evaluated from the measurements and found to depend on the concentration of moisture. The diffusion coefficient of the adhesive decreases with increased concentration. Thus, it was concluded that moisture transport through glue lines is a process governed by water-vapour diffusion.

Stresses and deformations caused by a change in the humidity of ambient air depend to a high degree on such structural panel properties as

- the annual ring angle θ ,
- the layer ratio LR ,
- pre-stresses,
- the middle layer material and
- the stiffness and coefficients of hygroexpansion of the raw material.

The properties of cross-laminated wood panels can be adjusted in a wide range.

Lowest stresses and warping were found in panels where the ring angle was 45° , which is due to the lowest stiffness of spruce in that direction. As a result of largest stiffness and lowest swelling in radial direction, medium warp and stresses and lowest expansion were detected in panels where the ring angle was 90° . It was, therefore, concluded to apply vertical annual rings ($\theta = 90^\circ$) and to avoid horizontal annual rings ($\theta = 0^\circ$).

Stresses and deformations induced by both, a moisture gradient and a climate change, depend to a high degree on the layer ratio of the panel. With increased relative thickness of the outer layers, the cup deformation, cup_{yz} , induced by a moisture gradient, increased, however internal compressive stresses due to a climate change decreased. Thus, with the correct choice of panel thickness and layer ratio with respect to the static requirements and the expected climatic conditions, warping and stresses could be minimised.

The pre-stress in the outer layers showed a significant influence on warping and stresses. Pre-stress can be applied mechanically (as in panel series PS) or by manipulating the initial moisture contents of the middle and outer layers. In such a way, pre-stressed panels generated less cracks when dried. Mechanically pre-stressed panels showed a significant smaller cupping than the reference panel when exposed to a moisture difference. However, a high value of pre-stress yielded plastified areas as a result of the initial compressive pre-stress and the previous wetting. These, e.g., increased the measured crack lengths when

dried. It was concluded that pre-stressing is a valid method to reduce cracks and warping, but the production costs and the disappearance of pre-stresses when the panel is dried are critical factors.

Wood composite middle layers may be applied in industrial processes. They yield advantages compared to solid wood: reduced stresses and reduced costs. Disadvantages are the cracking of the middle layer during drying processes (MDF) and the large warping and expansion/contraction. The warping of boards with different middle layer materials is highly dependent on the modulus of elasticity of the material used. Lower moduli of elasticity cause larger cupping. Thus, the lowest cup deformations were found for panels consisting of OSB cores. It is, therefore, recommended to apply OSB in the middle layer if the substitution of the spruce layer is required (reduction of costs, increase in thermal insulation). However, the warping of spruce laminates is significantly lower than the warping of composite laminates, which is due to the differences in stiffness. Slitted middle layers show no significant effect on warping and stresses. They can, therefore, be applied in industrial production if a larger thermal insulation is required.

The application of different adhesives provides parameters with which one can influence the internal stress state, namely the moisture diffusivity and the stiffness of the glue line. While the stiffness has a very little influence on warping and stresses, the diffusivity was found to strongly influence both. The tests of different adhesive diffusivities indicated a decrease in cup deformations, both cup_{xz} and cup_{yz} , when the coefficient of diffusion of the glue lines was increased to the value of wood (Section 4.3.4). However, stresses in the middle layer were increased (Section 5.3.4). Since the prevention of warping and of stresses in the outer layers is more important than the formation of stresses in the middle layer, it was recommended to apply an adhesive that has a high permeability for moisture. Alternatively, one could apply another connection system, which gives $D_{adh} = D_{wood}$, but ensures the effect of the crosswise orientation of the layers.

The three-dimensional theory for numerical simulation of the wood behaviour during moisture variations was successfully validated to spruce cross-laminates and spruce-MDF laminates. The most important part of the material model is the mechano-sorptive deformation. Neglecting the mechano-sorptive effect yields ten times higher deformations. A relevant theoretical description and accurate material parameter values for mechano-sorption were found to be very important for the simulation of moisture-induced stresses and deformations in cross-laminated solid wood panels. The hygroscopic warping and the internal stress state depend, to a high degree, on the values for stiffness and moisture

swelling in the longitudinal and transverse directions to the fibres, which are oriented in the panel plane. An accurate experimental determination of these parameter values is, therefore, highly important for finite element calculations of cross-laminated wood panels and wood in general.

7.2 Ideas on future work

Since the focus of this thesis was on moisture-induced stresses, influences of temperature on equilibrium moisture content, diffusion and mechanical parameters were neglected. However, temperature affects these variables and should, therefore, be investigated. A coupled heat and moisture transport model would be of great interest here.

The behaviour under static and dynamic loads was excluded from the investigations of the present thesis. These investigations are, nevertheless, important for the application of cross-laminated wood panels. Experimental tests of large-sized samples are required to extend the knowledge about the physical behaviour of wooden panels.

The diffusion of moisture through glue lines was regarded from a macroscopic view. In future investigations it would be interesting to study in which way the penetration depth of the adhesive into the wood material, influenced by such adhesive properties as the cross linking density and adhesive viscosity, influences the diffusion coefficient of the glue line. Furthermore, tests of supplementary adhesives should be examined in order to obtain more material data. The tests of diffusion properties of adhesive films should be considered. The examination of the diffusion process in wood on a mesoscopic or microscopic level would be of interest. On both structural levels, wood consists of elements with varying properties. On the mesoscopic scale, early- and latewood provide different properties for diffusion. On the microscopic level differences consist between bound water diffusion in the cell walls and water vapour diffusion in the lumen. Experimental data on proper material data is limited and it is, thus, essential to perform experimental and numerical investigations.

To reflect longterm behaviour of wooden panels the material model needs to be extended so as to include creep phenomena. Furthermore, the modelling of cracks could provide relevant information under which conditions cracks develop or how to avoid cracking. The modelling of plastification would allow to draw conclusions on how cross-laminated panels

act on large stresses.

The simulation models tested, both the moisture model and the mechanical model, provide an excellent basis for further investigations. The delamination of wood adhesive joints or the moisture transport through wooden wall elements with a multi-layer structure are possible fields of application.

7.3 Publications

Parts of the present thesis have been reported elsewhere. These publications and other papers concerning the topic of this thesis are listed below.

Papers in reviewed journals

- Gereke T., Gustafsson P.J., Persson K., Niemz P.: Experimental and numerical determination of the hygroscopic warping in cross-laminated solid wood panels. *Holzforschung*, 63:340-347, 2009.
- Gereke T., Schnider T., Hurst A., Niemz P.: Identification of moisture-induced stresses in cross-laminated wood panels from beech wood (*Fagus sylvatica* L.), *Wood Science and Technology*, 43:301-315, 2009.
- Gereke T., Hass, P., Niemz, P.: Moisture-induced stresses and distortions in spruce cross-laminates and composite laminates. *Holzforschung*, *in press*, 2009.
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- Gereke T., Schnider T., Hass P., Niemz P.: Moisture induced stresses in laminated wood panels - A comparison of beech, thermally treated beech and spruce wood. *The 3rd Conference on Hardwood Research and Utilisation in Europe*, Sopron/Hungary, pp. 20-26, September 3-4, 2007.
- Gereke T., Gustafsson P.J., Persson K., Niemz P.: Modelling the hygroscopic warping of laminated wood panels. *Cost Action E44 - Modelling the wood chain: forestry - wood industry - wood product markets*, Helsinki, Finland, pp. 39-48, September 17-19, 2007.

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- Gereke T., Gustafsson P.J., Persson K., Niemz P.: Modelling of cross-laminated solid wood panels. *8th World Congress on Computational Mechanics (WCCM8); 5th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS 2008)*, Venice, Italy, June 30 - July 5, 2008.

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Appendix

Elastic coefficients according to Neuhaus

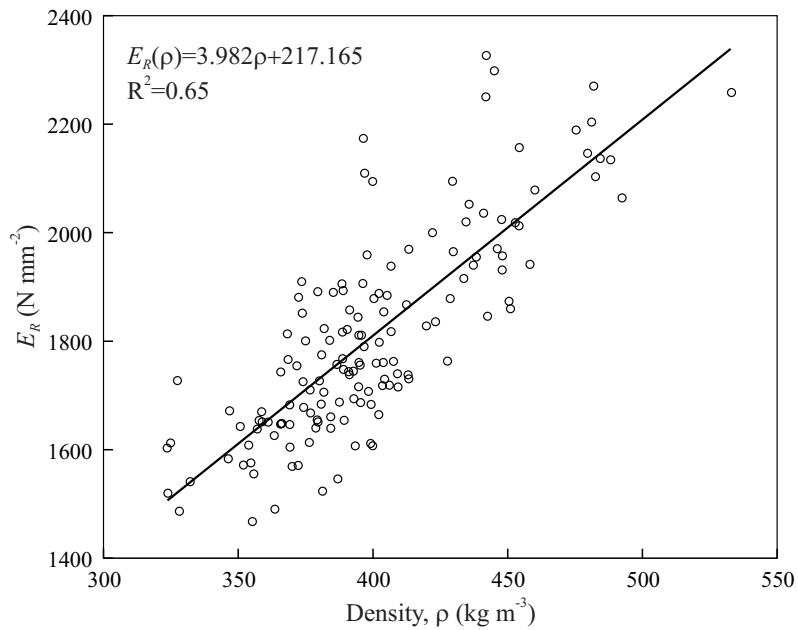
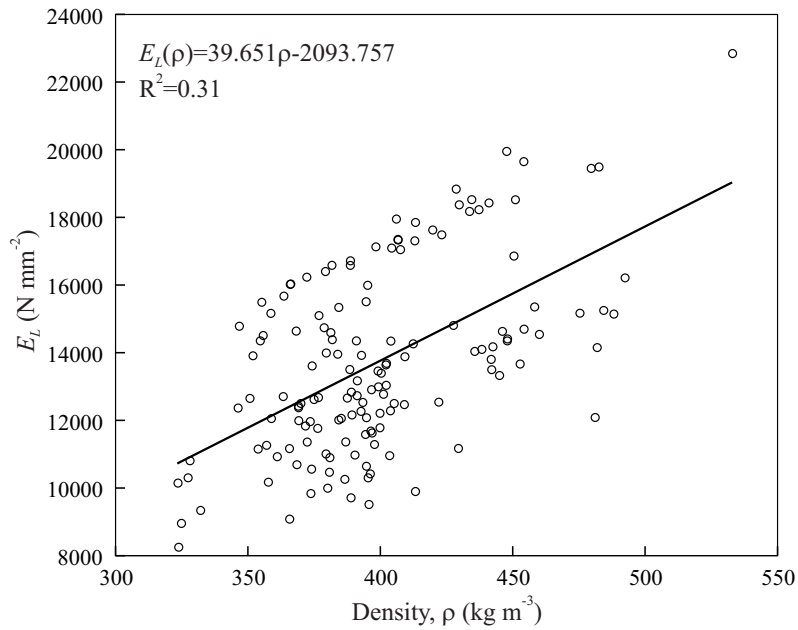
$$s_{ij}(\omega) = a_0 + a_1 \omega + a_2 \omega^2 + a_3 \omega^3$$

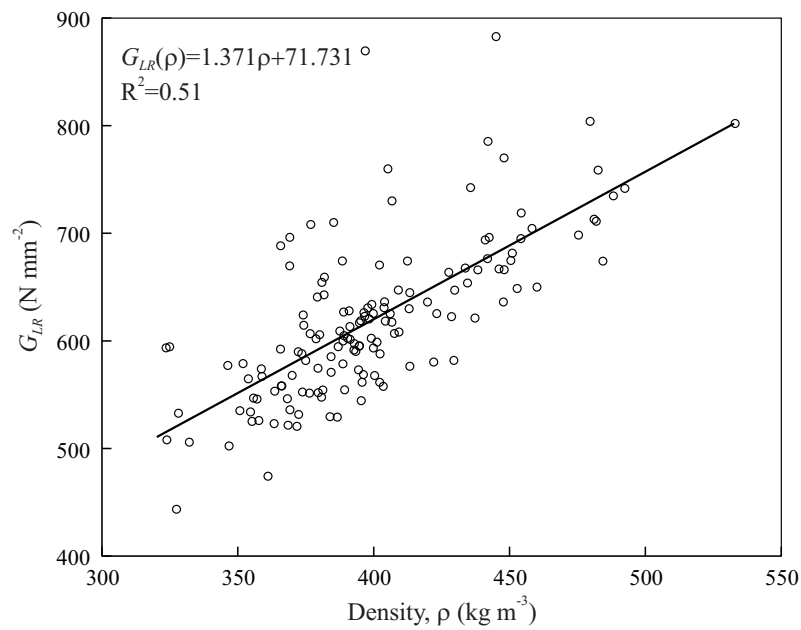
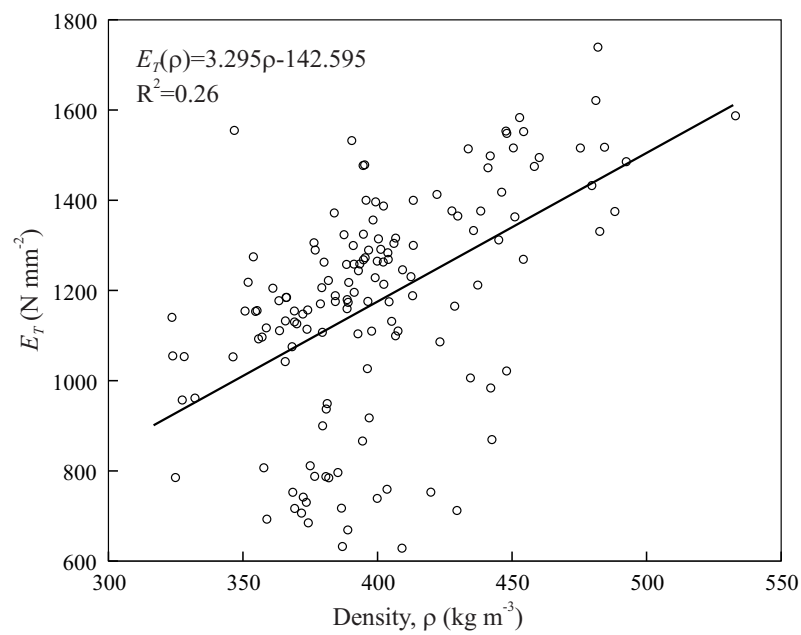
$$0 \leq \omega \leq 0.28$$

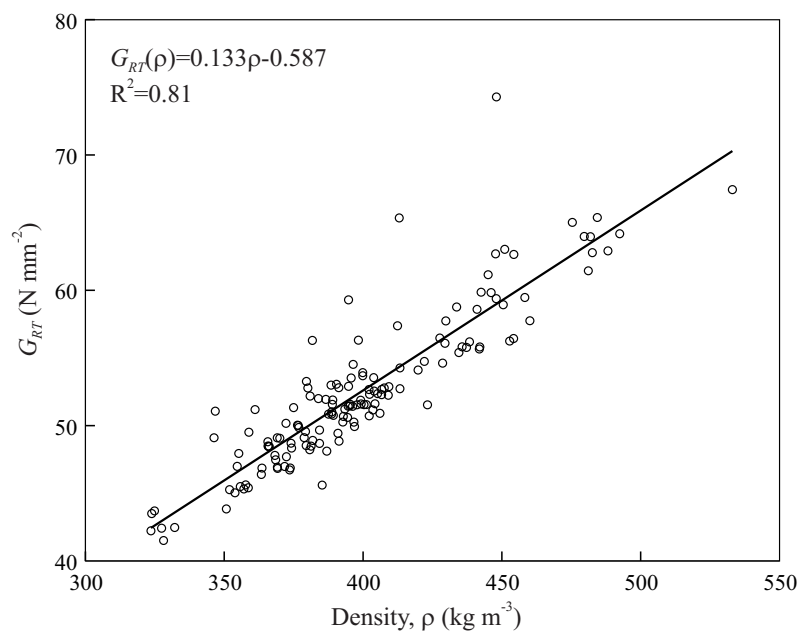
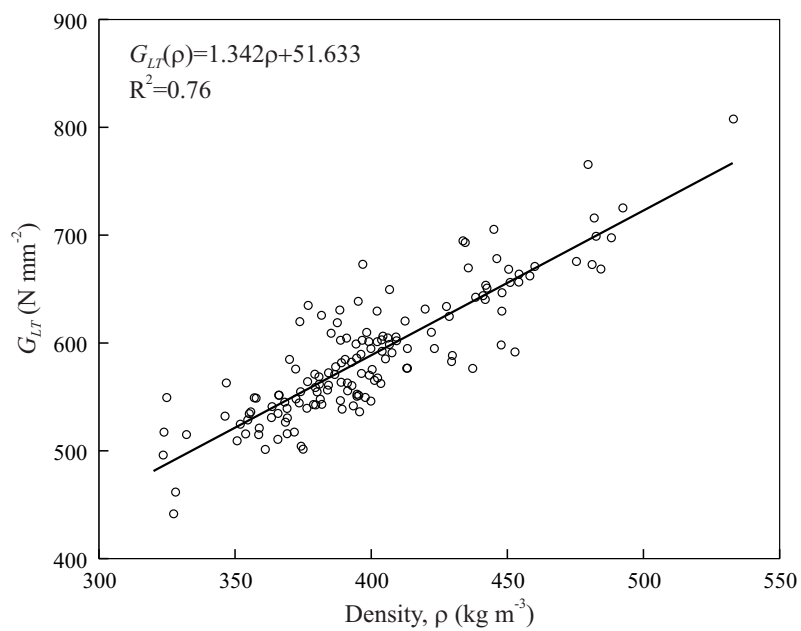
Elastic coefficient	a_0	a_1 $\times 10^{-6} \text{ mm}^2 \text{ N}^{-1}$	a_2	a_3
s_{11}	2029.51	-46.008	7.609	-0.1101
s_{22}	78.37	-0.195	0.066	-0.0012
s_{33}	1022.73	-15.176	3.207	-0.0457
s_{44}	1360.83	-36.805	6.279	-0.1270
s_{55}	16657.64	89.000	43.029	-0.1865
s_{66}	1142.99	-5.172	1.951	-0.0096
$-s_{12}$	24.05	-4.124	0.669	-0.0145
$-s_{21}$	44.12	3.240	0.002	-0.0014
$-s_{13}$	390.16	-30.061	6.799	-0.1590
$-s_{31}$	349.36	-7.640	4.546	-0.0995
$-s_{23}$	47.82	0.024	0.196	-0.0050
$-s_{32}$	19.06	-1.196	0.295	-0.0075

Relation between density and elastic parameters

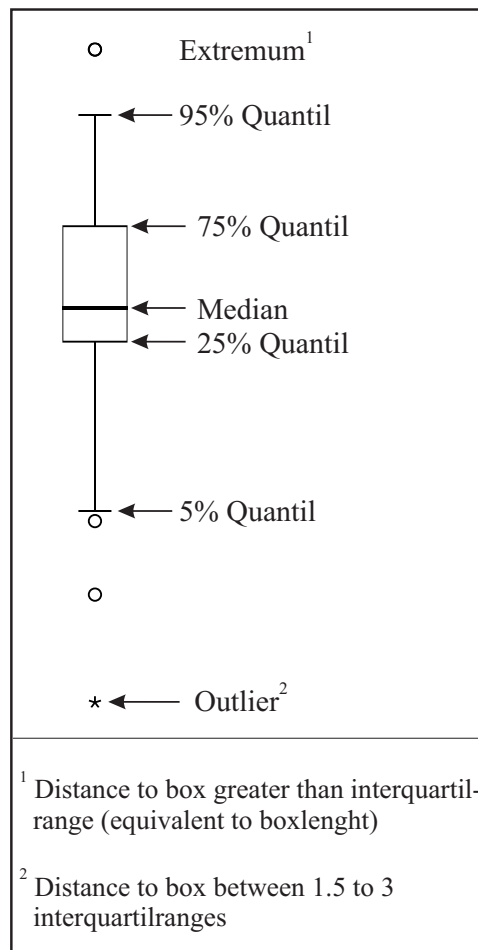
The relationships between density and moduli E_L , E_R , E_T , G_{LR} , G_{LT} , G_{RT} , as analysed according to the data of Keunecke et al. [55], are displayed (Norway spruce).







Definition of boxplots



Curriculum vitae

Name	Thomas Gereke
Date of birth	October 8, 1980 in Magdeburg (Germany) Citizen of Germany
01/2008 – 02/2008 02/2007 – 03/2007	Guest researcher in wood material modelling under the supervision of Prof. Dr. P. J. Gustafsson and Dr. K. Persson, Lund University, Division of Structural Mechanics, Lund (Sweden)
01/2006 – 06/2009	PhD thesis under the supervision of Prof. Dr.-Ing. habil. P. Niemz: <i>Moisture-induced stresses in cross-laminated wood panels</i> , ETH Zurich, Department of Civil, Environmental and Geomatic Engineering, Institute for Building Materials, Wood Physics Group, Zurich (Switzerland)
06/2005 – 09/2005	Diploma thesis (Master level) in wood science under the supervision of Prof. Dr.-Ing. habil. M. Kaliske: <i>Materialmodellierung gekoppelter Versagensmoden zur Untersuchung hölzerner Strukturen</i> (in German), University of Leipzig, Institute for Structural Mechanics, Leipzig (Germany)
04/2003 – 12/2004 (part time)	Student assistant, University of Leipzig, Institute for Structural Mechanics
10/2000 – 09/2005	Diploma studies in Civil Engineering, University of Leipzig
07/1999 – 04/2000	Basic military service, Rotenburg/Wümme (Germany)
06/1999	German university entrance diploma, Albert-Einstein-Gymnasium, Magdeburg (Germany)